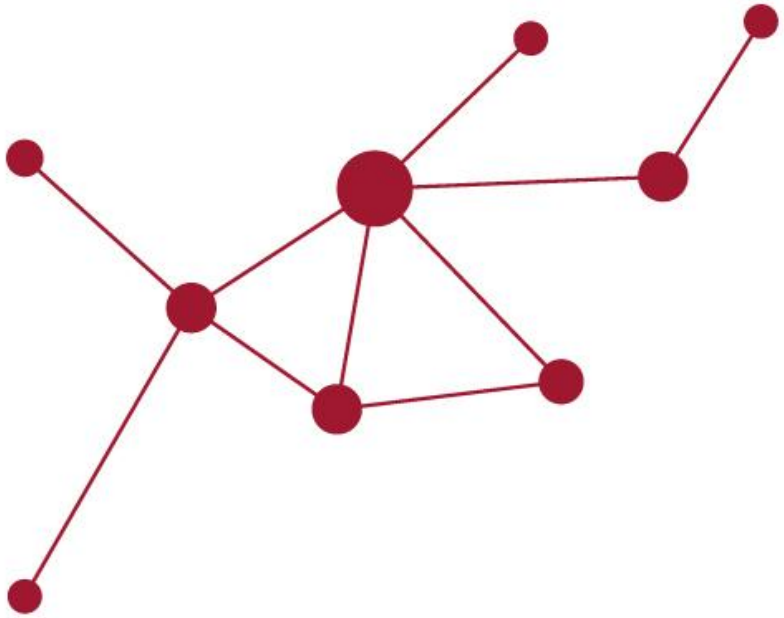
ZLC

TALENT HUB FOR SUPPLY CHAIN

Zaragoza Logistics Center

MZ 510 – Analytical Methods

September 14, 2026



MIT GLOBAL SCALE NETWORK

INTRODUCTION

Behind it all: mathematics!

Underneath the development of all technology and science lurks a belief in a perfect world that can be described in a precise language.

This language is **mathematics**…

Every truth we claim to possess about nature can be formulated as a mathematical statement.

By expressing our beliefs in terms of numbers we are able to *quantify and qualify*. Hence we can master the uncertain and wrest more knowledge from nature.

"Math adds character to a logistics professional"

– A very renowned academic and logistician, 2002

Slide borrowed from Dr. Prashant Yadav

HBR editorial thoughts on Supply Chain Management

- **In our increasingly interconnected and interdependent global economy, the process of delivering supplies and finished goods from one place to another is accomplished by means of**
  - mind-boggling new technology,
  - clever new applications of old ideas,
  - **seemingly magical mathematics,**
  - powerful software,
  - and old-fashioned concrete, steel, and labour.
    - Source: Harvard Business Review

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Introduction to Operations Research

- **Operations research (management science)** is a scientific approach to decision making that seeks to best design and operate a system, usually under conditions requiring the allocation of scarce resources.
  - Term coined during WW II when leaders asked scientists and engineers to analyze several military problems.
- A **system** is an organization of interdependent components that work together to accomplish the goal of the system.
- A **mathematical model** is a mathematical representation of the actual situation that may be used to make better decisions or clarify the situation.
  - *OR* requires the use of one or more mathematical models.

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Brief History of Operations Research & Management Science

- During World War II, enormous quantities of resources (personnel, supplies, equipment, …) had to be deployed.
- WWII - British government brought scientists from various disciplines together to work on problems of military operations
- Military operations research (OR) teams were formed to deal with the complexity of the deployment.
- After the war, operations researchers found their way back to universities, industry, government, and consulting firms.
- OR helps operations managers make decisions when problems are complex and wrong decisions are costly.

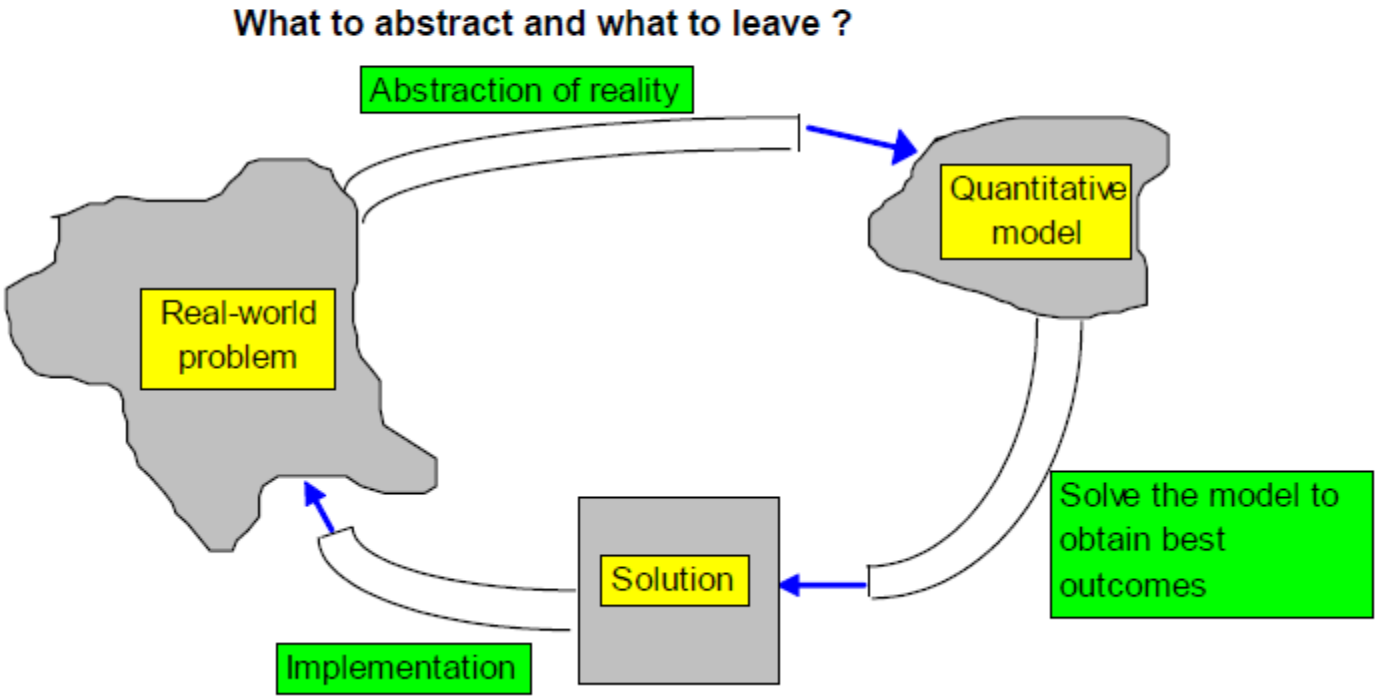


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This course?

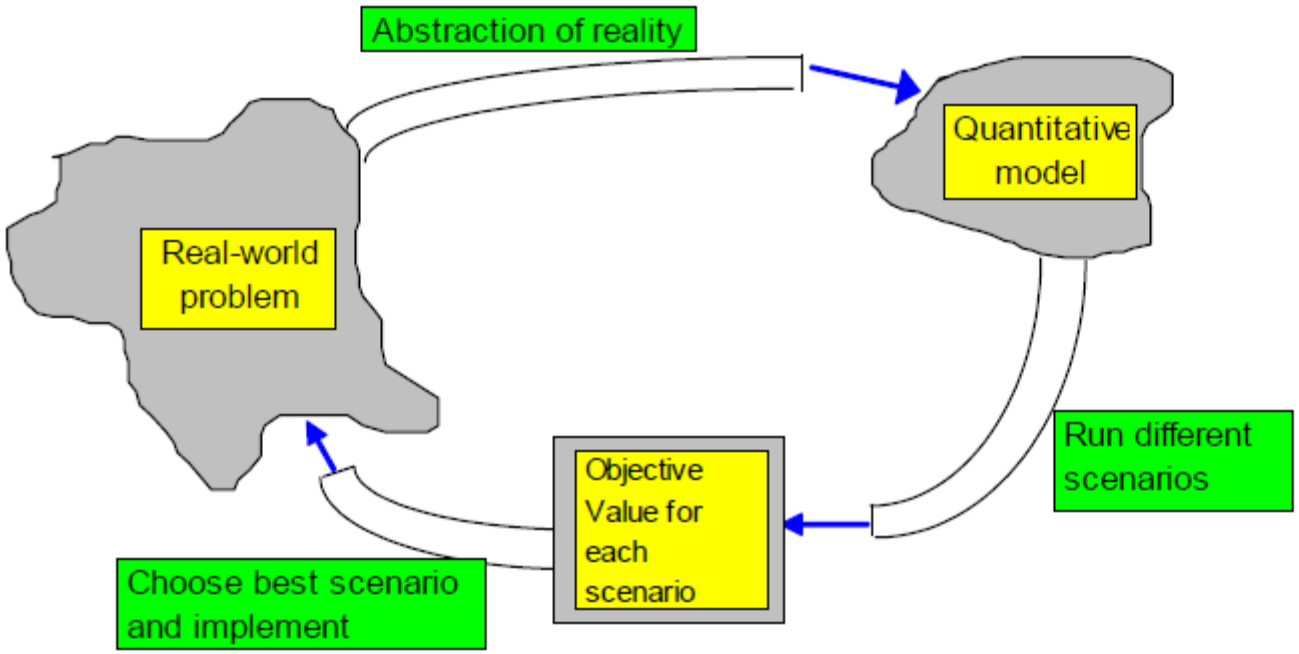
- **A taste of mathematical programming and problem formulation**
  - *some simple problems with relevance to logistics (e.g., transportation problem)*
- **Decision analysis**
  - *capacity expansion decisions*
- **Queueing**
  - *distribution centers, theme parks, call centers*
- **Simulation**
  - *simulation of inventory models*
- **familiarity with solution methods**
  - learn to appreciate the pros and cons of each
- **"reasonable" excel spreadsheet modeling**
- ***hopefully provides you with the tools to do the analysis in your thesis project ☺***

Model based problem solving



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Model based problem solving 2



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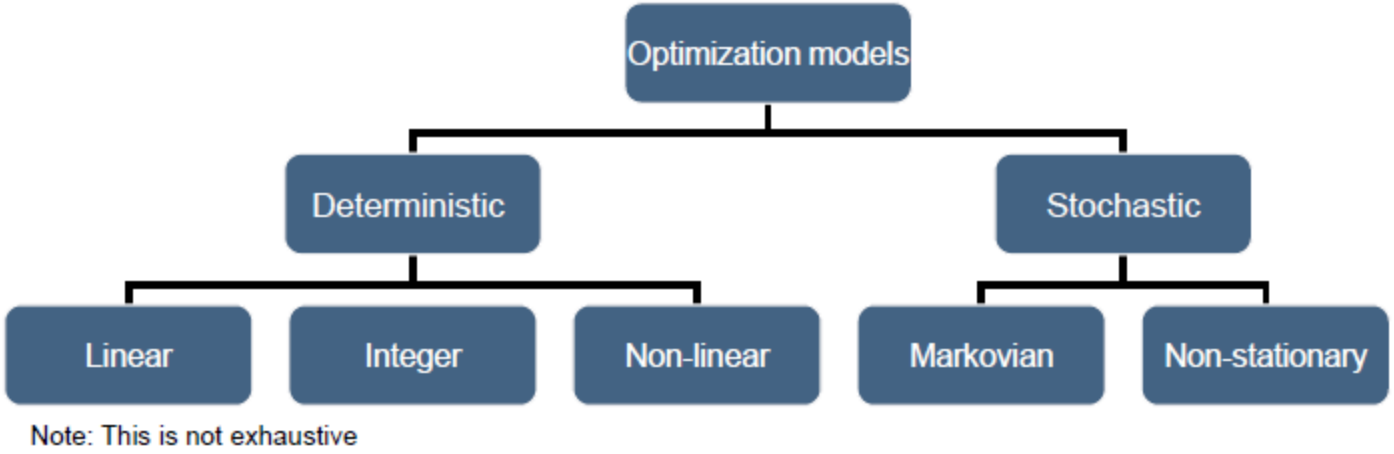
Optimization Models

Maximize or minimize an objective subject to constraints

- **Deterministic model**
  - All data and parameters are known with certainty.
  - needs sensitivity analysis to see the effect of changes in data on the objective
- **Stochastic model**
  - Data or parameters are not known with certainty.
  - build probability distributions for the uncertain data or parameters

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Classification of Optimization Models



Note: This is not exhaustive

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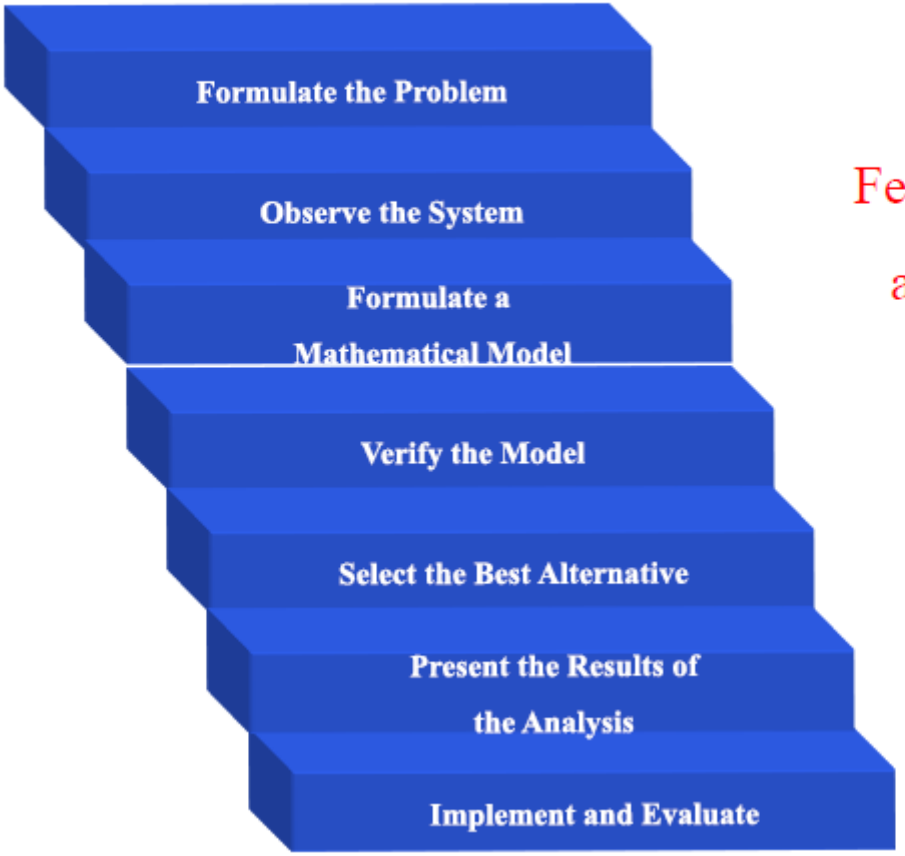
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Optimization Models

- A **static model** is one in which the decision variables do not involve sequences of decisions over multiple periods.
- A **dynamic model** is a model in which the decision variables do involve sequences of decisions over multiple periods.
- If the objective functions and constraints are represented entirely by linear equations, then the model is regarded as a **linear model**. If one or more of the objective functions or constraints are represented with a nonlinear equation, then the model is known as a **nonlinear model**.
  - In general, nonlinear models are much harder to solve.
- If one or more decision variables must be integer, then an optimization model is an **integer model**. If all the decision variables are free to assume fractional values, then the optimization model is a **non-integer model**.
  - Integer models are much harder to solve than non-integer models.

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The Seven Steps to a Good OR Analysis



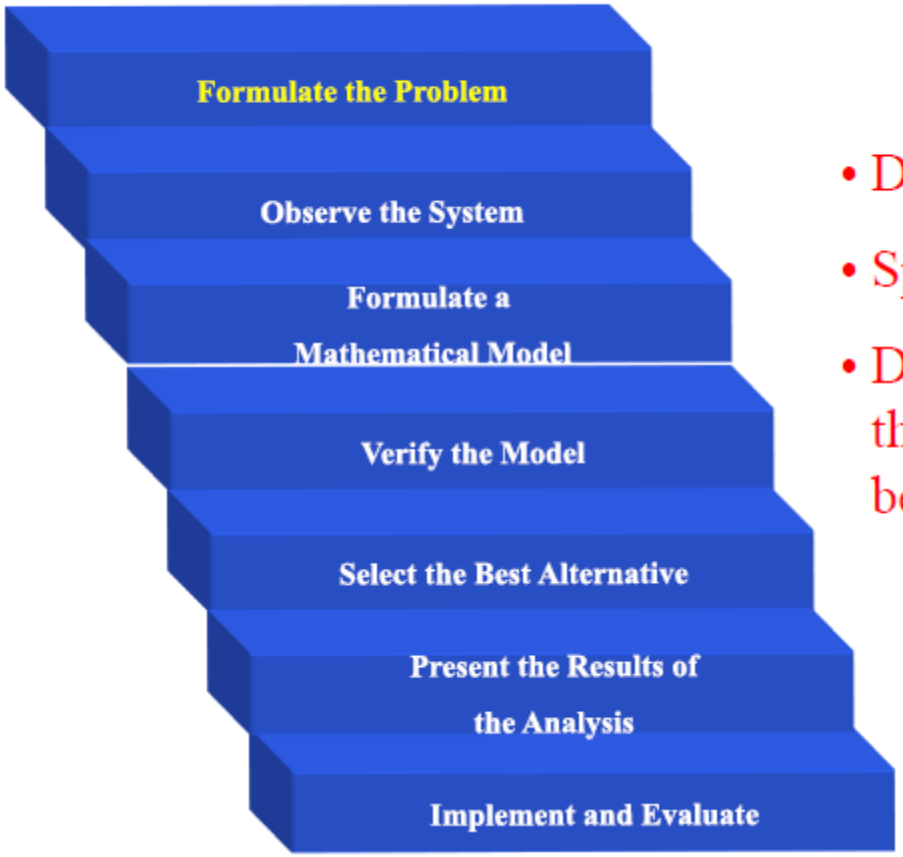
Feedback loops

at all levels!

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The Seven Steps to a Good OR Analysis

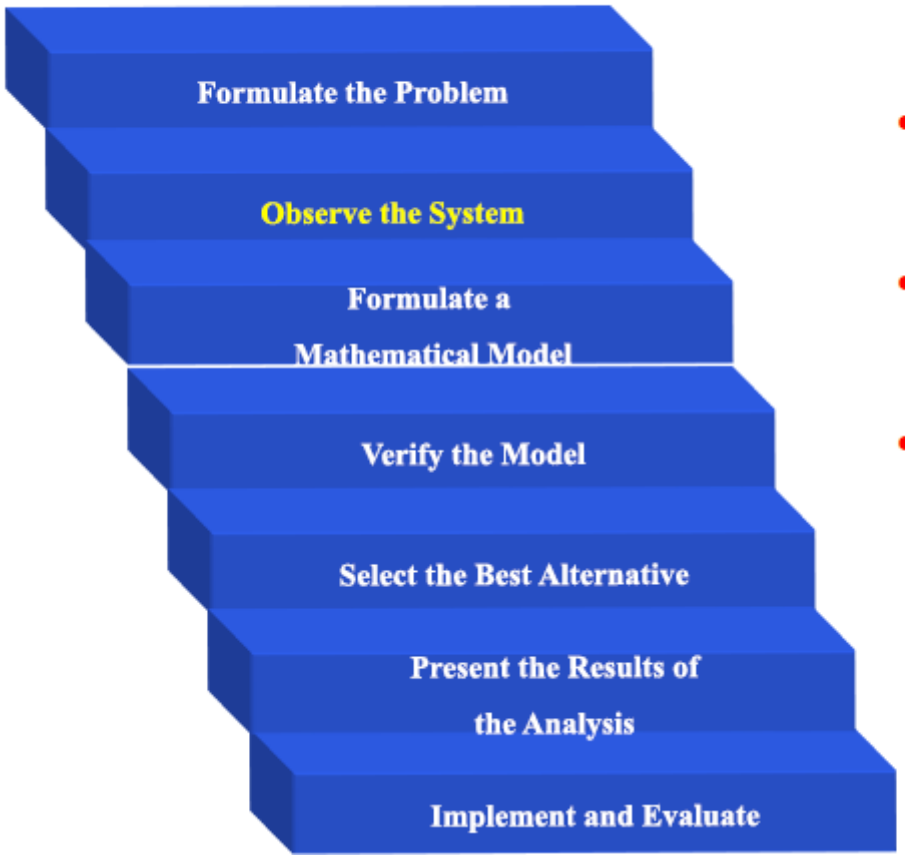


- Define the problem
- Specify objectives
- Determine parts of the organization to be studied

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The Seven Steps to a Good OR Analysis

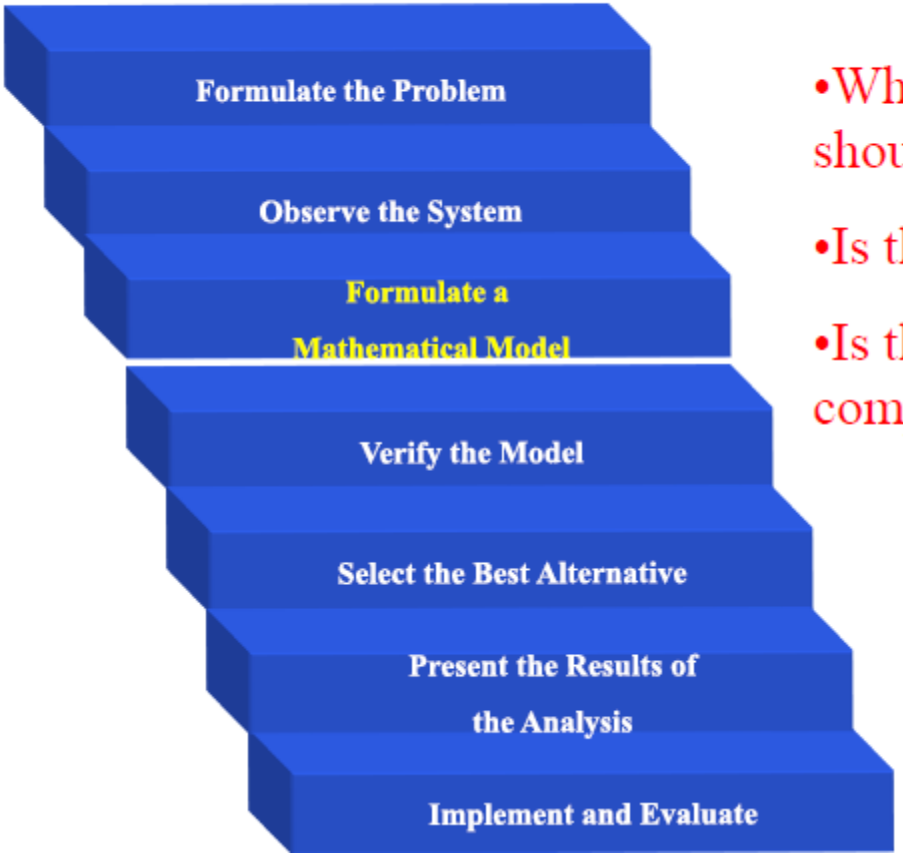


- What data should be collected?
- How will data be collected?
- How do different components of the system interact with each other?

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The Seven Steps to a Good OR Analysis

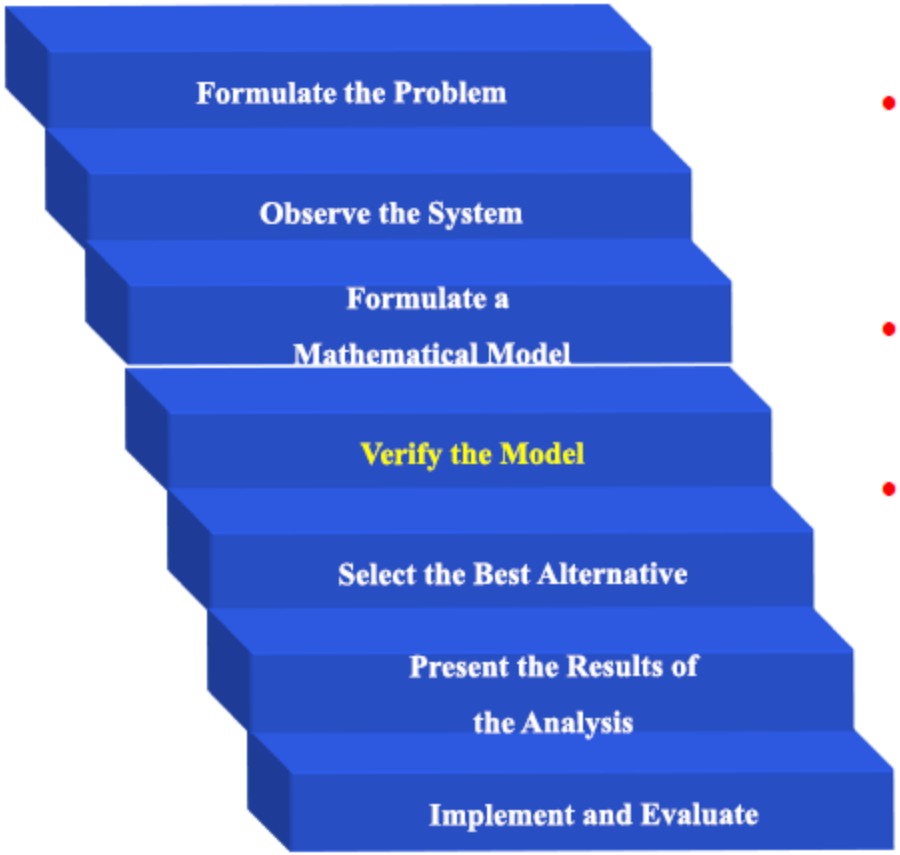


- What kind of model should be used?
- Is the model accurate?
- Is the model too complex?

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The Seven Steps to a Good OR Analysis

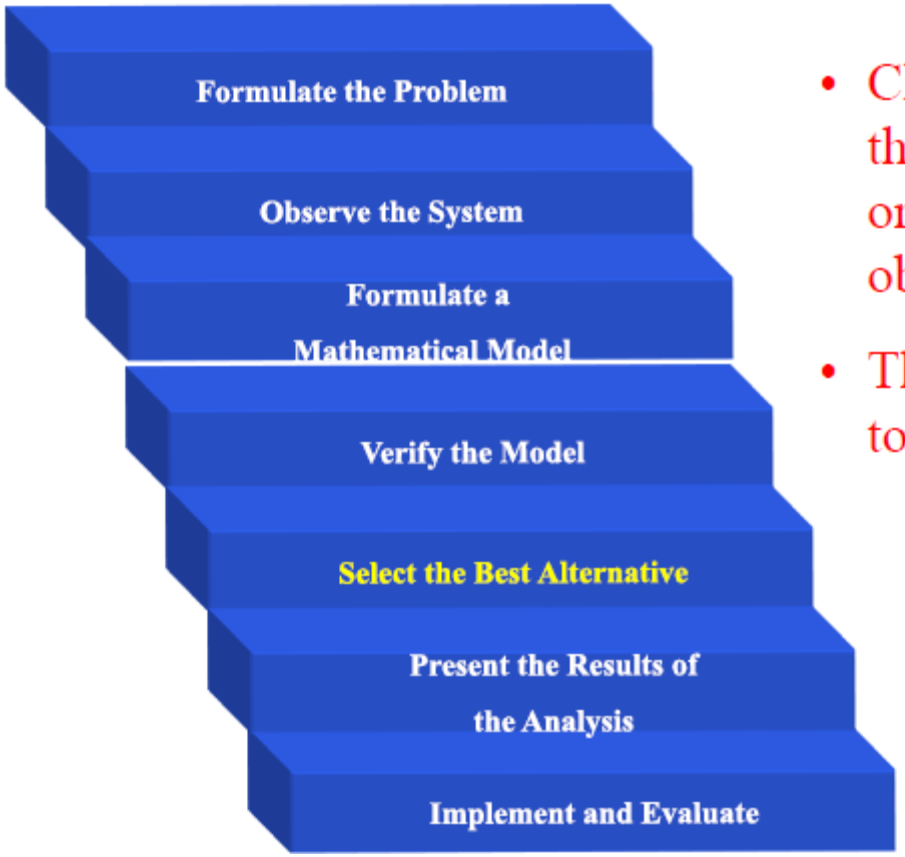


- Do outputs match current observations for current inputs?
- Are outputs reasonable?
- Could the model be erroneous?

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The Seven Steps to a Good OR Analysis

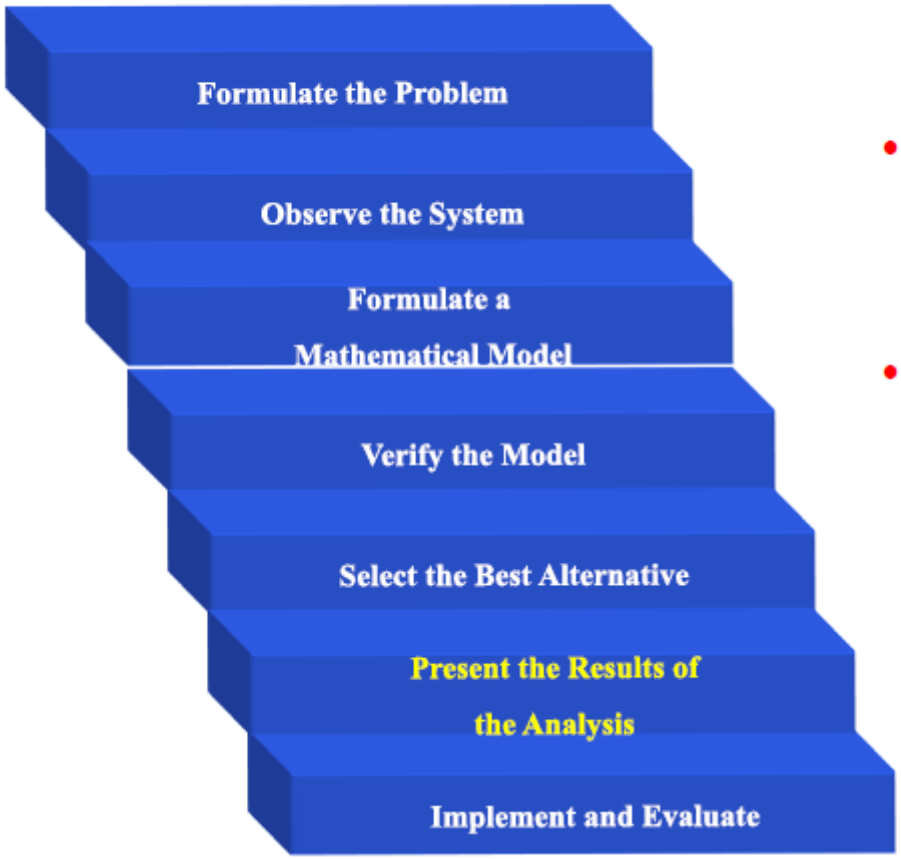


- Choose the alternative that best meets the organization's objectives.
- This is where software tools will help us!

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The Seven Steps to a Good OR Analysis

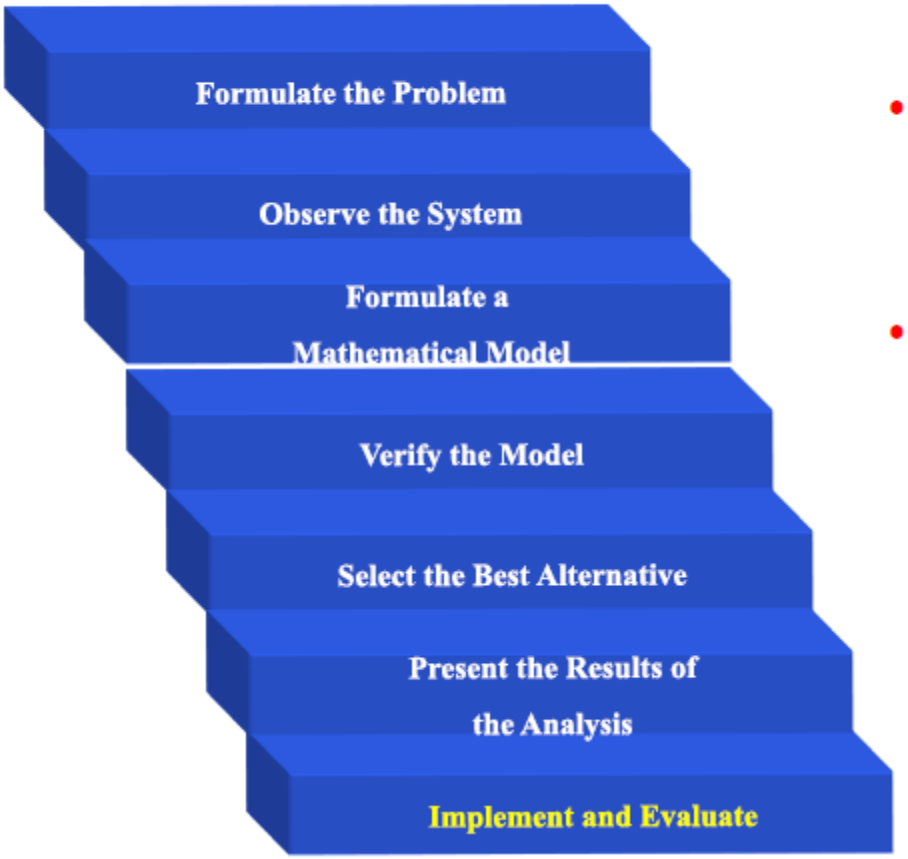


- Present the results to the decision makers
- System must be user friendly!

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The Seven Steps to a Good OR Analysis



- Users must be trained on the new system.
- System must be observed over time to ensure it works properly.

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CITGO Petroleum

- Klingman et al. (1987) applied a variety of management-science techniques to CITGO Petroleum. Their work saved the company an estimated $70 million per year.
- A mathematical model – supply, distribution and marketing (SDM) system – used to develop an 11- week supply, distribution and marketing plan for the entire business .

- Source: Klingman, D., N. Phillips, D. Steiger, and W. Young, "The Successful Development of Management Science Throughout Citgo Petroleum Corporation," *Interfaces*, 17, pp.4-25, 1987.

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CITGO Petroleum

- The Supply Distribution Marketing (SDM) System
  - **Step 1** (Formulate the Problem) CITGO wanted a mathematical model that could be used to make supply, distribution, and marketing decisions such as:
    - Where should the crude oil be purchased?
    - Where should products be sold?
    - What price should be charged for products?
    - How much of each product should be held in inventory?
  - The goal was to maximize profitability associated with these decisions.

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CITGO Petroleum

- **Step 2** (Observe the System) A database that kept track of sales, inventory, trades, and exchanges of all refined goods was installed. Also regression analysis was used to develop forecasts of for wholesale prices and wholesale demand for each CITGO product.
- **Step 3** (Formulate a Mathematical Model of the Problem) and **Step 5** (Select Suitable Alternative Solutions) A minimum-cost network flow model (MCNFM) is used a determine an 11-week supply, marketing, and distribution strategy.
  - The model makes all decisions discussed in Step 1.
  - A typical model run involved 3,000 equations and 15,000 decision variables required only 30 seconds on an IBM 4381.

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CITGO Petroleum

- **Step 4** (Verify the Model and Use the Model for Prediction)
  - The forecasting models are continuously evaluated to ensure that they continue to give accurate forecasts.
- **Step 6** (Present the Results and Conclusions) and **Step 7** (Implement and Evaluate Recommendations) Implementing the SDM required several organizational changes.
  - A new vice-president was appointed to coordinate the operation of the SDM and refinery LP model.
  - The product supply and product scheduling departments were combined to improve communications and information flow.

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San Francisco Police Department Scheduling

- Taylor and Huxley (1989) developed a police patrol scheduling system (PPSS).
  - All San Francisco (SF) police precincts use the PPSS to schedule their officers.
  - It is estimated that the PPSS saves the SF police more than $5 million annually.
  - Other cities have also adopted PPSS.
- Following the seven-step model here is a description of PPSS.

- Source: Taylor, P., and S. Huxley, "A Break from Tradition for the San Francisco Police: Patrol Officer Scheduling Using an Optimization-Based Decision Support Tool," *Interfaces*, 19, pp. 4-24, 1989.

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San Francisco Police Department Scheduling

- **Step 1** The SFPD wanted a method to schedule patrol officers in each precinct that would:
  - Quickly produce a schedule and graphically display it.
  - Minimize the sum of hourly shortages and surpluses
  - Would allow precinct captains to easily fine-tune to produce the optimal schedule.
- **Step 2** The SPFD had a sophisticated computer-aided dispatch (CAD) system to keep track of calls for police help, police travel time, police response time, and so on.

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San Francisco Police Department Scheduling

- **Step 3** An LP model was formulated
  - Primary objective was to minimize the sum of hourly shortages and surpluses.
  - The constraints in the PPSS model reflected the limited number of officers available and the relationship of the number of officers working each hour to the shortages and surpluses for that hour.
  - PPSS would produce a schedule that would tell the precinct captains how many officers should start work at each possible shift time.
    - The fact that this number was required to be an integer made it far more difficult to find an optimal solution.

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San Francisco Police Department Scheduling

- **Step 4** Before implementing PPSS, the SFPD tested the PPSS schedule against manually created schedules.
  - It produced approximately a 50% reduction in both shortages and surpluses.
- **Step 5** PPSS can be used to minimize the shortages and surpluses as well as being used to experiment with shift times and work rules.
  - Officers had wanted to switch to a 4/10 schedule versus a 5/8 schedule for years but were not permitted to because city felt it would reduce productivity.
  - PPSS showed that it would not reduce productivity but improve productivity.

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San Francisco Police Department Scheduling

- **Step 6 & Step 7**
  - Estimated that PPSS created an extra 170,000 productive hours per year, thereby saving the city of San Francisco $5.2 million dollars per year.
  - 96% of workers preferred PPSS generated schedules
  - PPSS enabled SFPD tot make strategic changes which made officers happier and increased productivity.
  - Response times to calls improved 20% after PPSS was adopted.
  - Precinct captains could fine-tune the computer-generated schedule and obtain a new schedule in 1 minute.

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Some Examples from theses projects

Inventory Optimization in Clinical Trial Supply Chain



Alvaro Lerma

Spain

Education

Bachelor of Science International Business and Management, University of Murcia, Murcia, Spain; Bachelor of Laws, University of Murcia, Murcia, Spain; Graduate Certificate in International Business and Operations, Lublin University of Technology, Lublin, Poland

Work Experience

Global Business Development Manager, Greiwing Truck and Trailer; Operations Project Manager, Greiwing Truck and Trailer; General Manager Assistant, Greiwing Truck and Trailer; Co-Founder & COO, BREEF Watches: Logistics and Finance Intern, ELOPAK



Paul Lichtenstein

Brazil

Education

Post-Graduation (MSc) in Business and Administration of Operations, FIA, São Paulo, Brazil; Bachelor in Agriculture Engineering, Unicamp, Campinas, Brazil.

Work Experience

National Supply Chain Planning Manager and Project Supply Chain Manager at M Dias Branco, Brazilian package-food company.



Dr. Mustafa Çagri Gürbüz

Turkey

Professor of Supply Chain Management at the MIT-Zaragoza Program

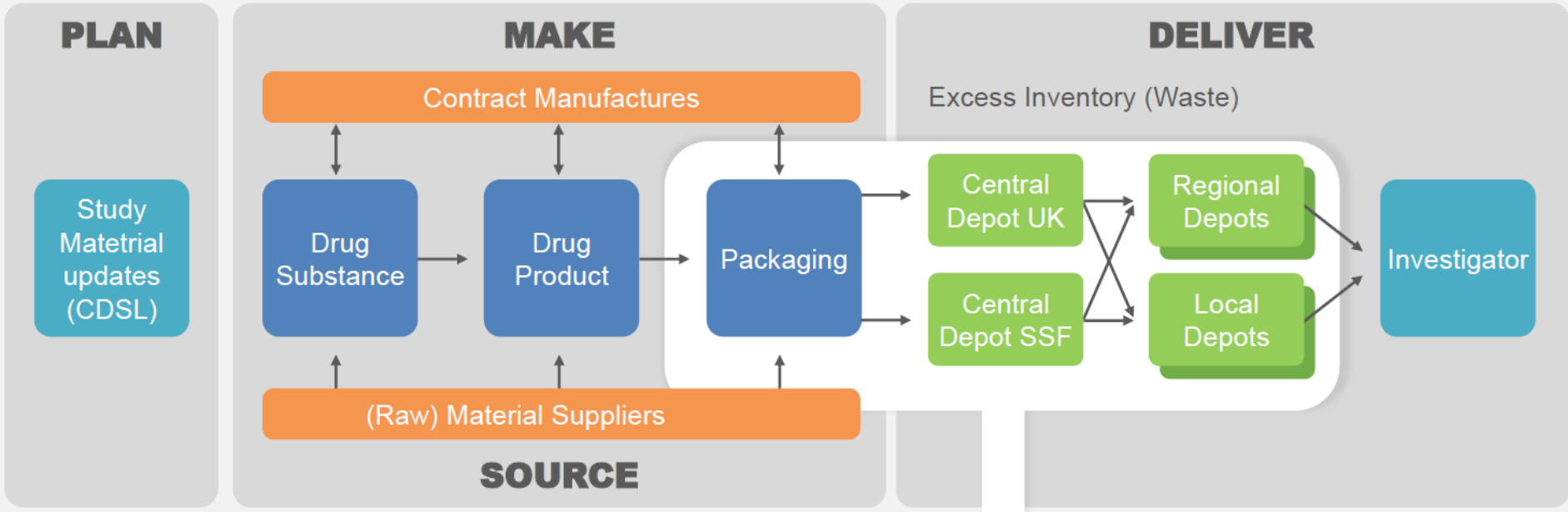
Dr. Mustafa Çagri Gürbüz is Professor of **Supply Chain Management** at the MIT-Zaragoza International Logistics Program. He is also a **Research Affiliate** at the MIT Center for Transportation and Logistics.

He completed his PhD at the Department of Management Science at the University of Washington in July 2006. He earned his BS and MS degrees from the Department of Industrial Engineering at Bilkent University, Turkey in 1999 and 2001, respectively.

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Scope and Introduction



IN SCOPE

CHF250M
Average daily world inventory

20-25%
Investment in R&D every year

48%
Clinical Trial Waste every year

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Attributes of CTSC







Patient Recruitment Process

- Get targeted number of patients (demand) enrolled in the clinical trial
- Open till target is reached
- Set by official institutions

Service Level

- Immediate fill rate: 99%
- Patient fill rate: above 99.99%

Cross-shipping Regulation

- Site-to-site shipping should remain the exception
- Quality controls and safety concerns

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Inputs

1) Localization : Country and Sites,

2) Study total time

3) Recruitment Rate, Lead Time,

4) Number of patients per site

5) Drug cost, fixed and variable costs

Outputs

1) Per depot : reorder point, quantity, number of shipments

2) Per clinical site: Initial stock

3) Study Expected Total Overage

4) Expected Total Supply Chain Study Cost

Site_ij	Clinical_Study	Country	i	j	Depot_i	End Study_Depot	Length Study_Depot	Lambda_Site_Real	Lambda_Site_Month_Real	Lead_Time_to_Depot	Lead_Time_to_Site	Total_Lead_Time_CW_Site	Demand_Depot_day	Demand_Site_day	Demand_LT_Depot	Demand_LT_Site	Drug Cost
1017680	WA29767	Switzerland	0	1	Switzerland	07/07/2016	106	0.0189	0.5739	1	4	5	1.19	1.12	1.19	4.48	152
1017830	WA29767	Switzerland	0	2	Switzerland	07/07/2016	106	0.0094	0.2869	1	4	5	1.19	0.56	1.19	2.24	152
1017631	WA29767	Argentina	1	1	Argentina	14/12/2016	98	0.0204	0.6207	44	7	51	3.43	0.98	150.86	6.86	152
1017652	WA29767	Argentina	1	2	Argentina	14/12/2016	98	0.0408	1.2415	44	7	51	3.43	0.28	150.86	1.96	152
1017659	WA29767	Argentina	1	3	Argentina	14/12/2016	98	0.0102	0.3104	44	7	51	3.43	0.42	150.86	2.94	152
1017664	WA29767	Argentina	1	4	Argentina	14/12/2016	98	0.0102	0.3104	44	7	51	3.43	0.70	150.86	4.90	152
1019098	WA29767	Canada	2	1	Canada	05/01/2017	167	0.0120	0.3643	25	7	32	0.75	0.70	18.86	4.90	152
1019646	WA29767	Canada	2	2	Canada	05/01/2017	167	0.0060	0.1821	25	7	32	0.75	0.42	18.86	2.94	152
1016495	WA29767	Poland	3	1	Germany	15/02/2017	399	0.0102	0.3114	23	7	30	2.46	0.56	56.58	3.92	152
1016502	WA29767	Poland	3	2	Germany	15/02/2017	399	0.0171	0.5191	23	7	30	2.46	0.70	56.58	4.90	152
1016537	WA29767	Poland	3	3	Germany	15/02/2017	399	0.0102	0.3114	23	7	30	2.46	0.98	56.58	6.86	152
1016545	WA29767	Hungary	3	4	Germany	15/02/2017	399	0.0317	0.9656	23	7	30	2.46	0.84	56.58	5.88	152
1016552	WA29767	Lithuania	3	5	Germany	15/02/2017	399	0.0133	0.4056	23	9	32	2.46	0.98	56.58	8.82	152
1016583	WA29767	Denmark	3	6	Germany	15/02/2017	399	0.0345	1.0489	23	7	30	2.46	0.42	56.58	2.94	152

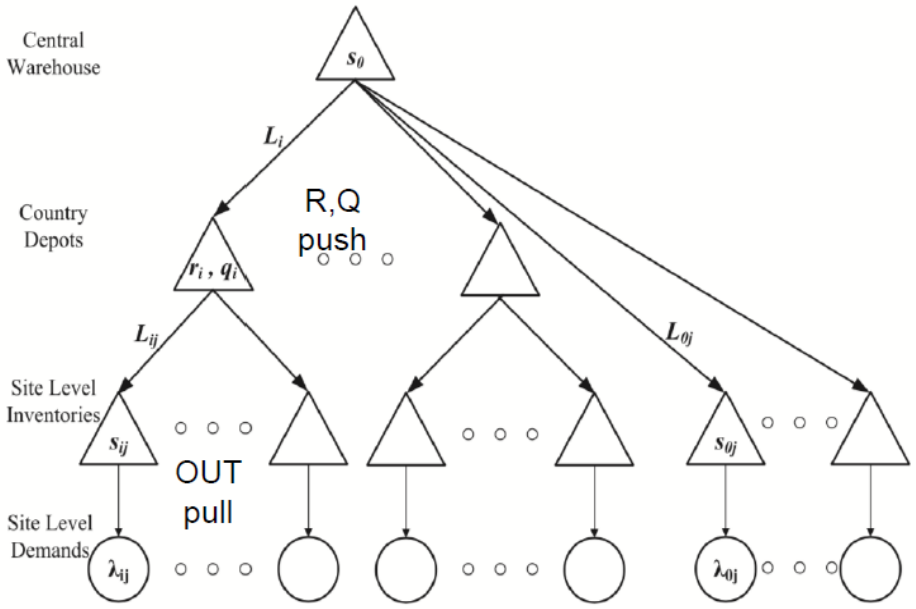
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Model: Mathematical Programming

Fleischhacker et al. (2014)

Figure 1 A General Clinical Trial Supply Chain



Source: Fleischhacker et al. (2014).

Decision Variables

$s0$: Stock at Central Warehouse
r_i : Reorder point of depot i
q_i : Batch size of depot i
s_{ij} : Stock level of site ij
s_{0j} : Stock level of site 0j
x_i : Expected number of shipments to depot i
y_i : Expected number of drugs handled through depot i

Objective Function

$$\textit{minimize Total Cost} =$$
$$+ c\left[s_0 + \sum_i \left(r_i + q_i + \sum_j s_{ij}\right) + \sum_j s_{0j} - S\right]$$
$$+ \sum_i \left[k_i + v_i\left(r_i + q_i + \sum_j s_{ij}\right)\right]$$
$$+ \sum_i [k_i x_i + v_i y_i]$$

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Model: Mathematical Programming

Fleischhacker et al. (2014)

Non-linear

$$
\begin{aligned}
\text{s.t.}\quad & \Pr\{s_{ij} - B_{ij}(r_i, q_i) \geq D_{ij}(L_{ij})\} \geq \zeta' && \forall\, i,j && (2)\\
& \Pr\{s_{0j} \geq D_{0j}(L_0)\} \geq \zeta' && \forall\, 0j && (3)\\
& q_i \leq \tilde{Q}^1 && \forall\, i && (4)\\
& y_i = x_i q_i && \forall\, i && (5)\\
& \left[S - (r_i + \min_j s_{ij})\right] \frac{\sum_j \lambda_{ij}}{\sum_{ij}\lambda_{ij} + \sum_j \lambda_{0j}} - q_i \leq x_i q_i && \forall\, i && (6)\\
& s_0 \geq S - \min\{\min_i\{r_i + q_i + \min_j s_{ij}\}, \min_j s_{0j}\} && && (7)\\
& s_0, r_i, s_{ij}, s_{0j}, q_i, x_i, y_i \text{ are nonnegative integers} && \forall\, i,j &&
\end{aligned}
$$

Source: Fleischhacker et al. (2014).

Linear

$$
\begin{aligned}
\text{s.t.}\quad & \bar{s}_i \geq u_{i1}\bar{s}_i^{(1)} + \cdots + u_{in_i}\bar{s}_i^{(n_i)}, && \forall\, i && (8)\\
& u_{i1} + \cdots + u_{in_i} = 1, && \forall\, i && (9)\\
& y_i = 2^0\bar{y}_{i0} + 2^1\bar{y}_{i1} + \ldots + 2^n\bar{y}_{in}, && \forall\, i && (10)\\
& \bar{y}_{ik} \leq S y_{ik}, \mathrm{k} = 1, \ldots, \mathrm{n}, && \forall\, i && (11)\\
& \bar{y}_{ik} \leq x_i, \mathrm{k} = 1, \ldots, \mathrm{n}, && \forall\, i && (12)\\
& \bar{y}_{ik} \geq x_i - \mathrm{S}(1 - y_{ik}), \mathrm{k} = 1, \ldots, \mathrm{n}, && \forall\, i && (13)\\
& \left[S - (r_i + s_{ij})\right] \frac{\sum_j \lambda_{ij}}{\sum_{ij}\lambda_{ij} + \sum_j \lambda_{0j}} - q_i \leq y_i && \forall\, i,j && (14)\\
& s_0 + \{r_i + q_i + s_{ij}\} \geq S, && \forall\, i,j && (15)\\
& s_0 + s_{0j} \geq S, && \forall\, 0j && (16)\\
& s_0, r_i, q_i, s_{ij}, x_i, y_i, \bar{y}_{ik} \text{ are nonnegative integers}, \mu_{ij}, y_{ik} \text{ are binary}, \forall\, i,j,k. && &&
\end{aligned}
$$

Linearization

- **Chance Constraints =>** Minimum Inventory Position (Graves, 1985)
- **Quadratic Term $x_i q_i$ =>** Binary representation of q_i
- **Min functions =>** Constraints for all *i* and for all *j*

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Results

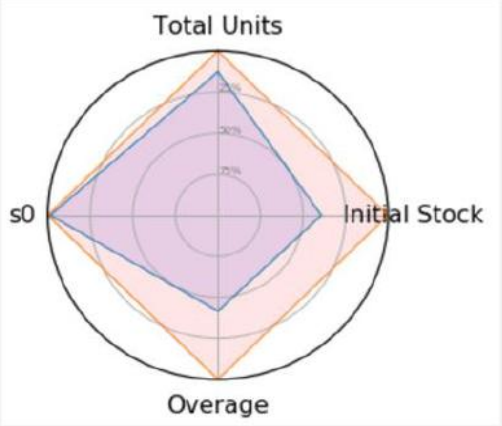
49 patients taking one drug every 3 weeks for 15 periods => S = 735 drug units

6 countries / 2 regional depots / 18 clinical sites

Clinical Trial 1 - BO29389

Depot	r	q	x	y	s_ij
Germany	41	7	52	364	[5, 4, 7, 5, 3, 4, 5, 5, 6, 6, 4]
USA	24	22	10	220	[7, 10, 6, 9, 4]
Switzerland	0	0	0	0	[3, 4]

Clinical Study	s0	Initial Stockage	Total Units	Overage	Overage %
Optimization	732	191	923	188	25.58
Baseline	743	314	1057	322	43.81



s0 1.48%

Init. Stock 39.17%

Total Units 12.68%

Overage 41.61%

Comparing both scenarios from their initial state, the optimized solution would have needed less inventory overage to buffer for same demand

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Results

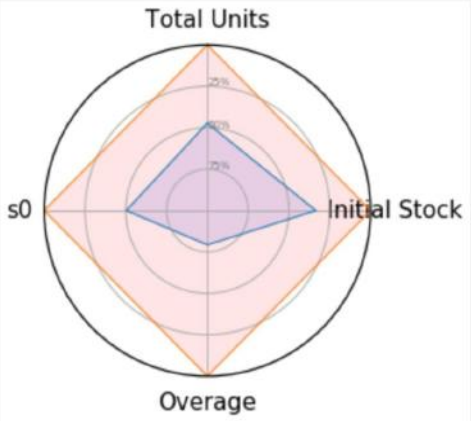
212 patients taking one drug every 3 weeks for 42 periods => S = 8,904 drug units

20 countries / 6 regional depots / 75 clinical sites

Clinical Trial 2 - WA29767

Depot	r	q	x	y	s_ij
Argentina	56	86	4	344	[21, 9, 11, 16]
Canada	7	34	5	170	[16, 11]
Germany	100	67	64	4288	[23, 27, 36, 32, 38, 18, 32, 32, 29, 32, 38, 2...
Japan	96	94	14	1316	[37, 31, 25, 43, 25, 31, 43, 49, 18, 55, 55]
Mexico	6	92	4	368	[27, 14, 30]
USA	98	98	16	1568	[12, 32, 20, 12, 32, 12, 20, 16, 32, 24, 32, 1...
Switzerland	0	0	0	0	[10, 6]

Clinical Study	s0	Initial Stockage	Total Units	Overage	Overage %
Optimization	8898	2718	11616	2712	30.46
Baseline	17903	4091	21994	13090	147.01



s0
50.30%

Init. Stock
33.56%

Total Units
47.19%

Overage
79.28%

The practice of producing between 60% and 150% overage based on high fill rates is not justified. 20%-40% will do just fine

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Sensitivity Analysis

Lead Time

WA29767	Optimized	Reduced LT	Increased LT
Total Cost	3,902,231	3,315,912	4,450,404
s0	8898	8900	8896
s0 [%]	76.62	81.38	70.38
Inital Stock	2718	2036	3744
Total Units	11616	10936	12640
Overage	2712	2032	3736
Overage [%]	30.46	22.82	41.96

Waste reduction per day:

6.42%

Cost reduction with 100% faster CTSC:

CHF 200K

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Future Work

CTSC

- Linear integer program chooses the optimal inventory positioning to meet the desired service level at each site
- Adequate inventory positioning might impact the overall costs in clinical trial supply chain, by enhancing its efficiency, reducing inventory levels and keeping high service levels, so that volumes, costs and risks are balanced in a more effective approach

Insights on how to improve efficiency and reduce waste

	Overview	Future Work
Operational Process	• Implications of results for future studies could address operational processes	• Better understanding of production runs, batch size, and **perishability** • Improve **forecast accuracy**
Efficiency Supply Chain	• Better visibility for operations planning	• Anticipate planning for packaging and labeling • Implement **Depot-to-Site distribution costs** • Reduce **variability in lead time**
Strategy Supply Chain	• Avoid risks and continuous implementation	• Immediate data analysis, optimal network design and boosted **supply chain visibility**.

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Supply Chain Design and Replenishment Policies for 2-hour Delivery: E-retailer in Vietnam



Tai Ho



Kareem Abdelmonem



Kate Lishchenko



Mustafa Çagri Gürbüz, PhD

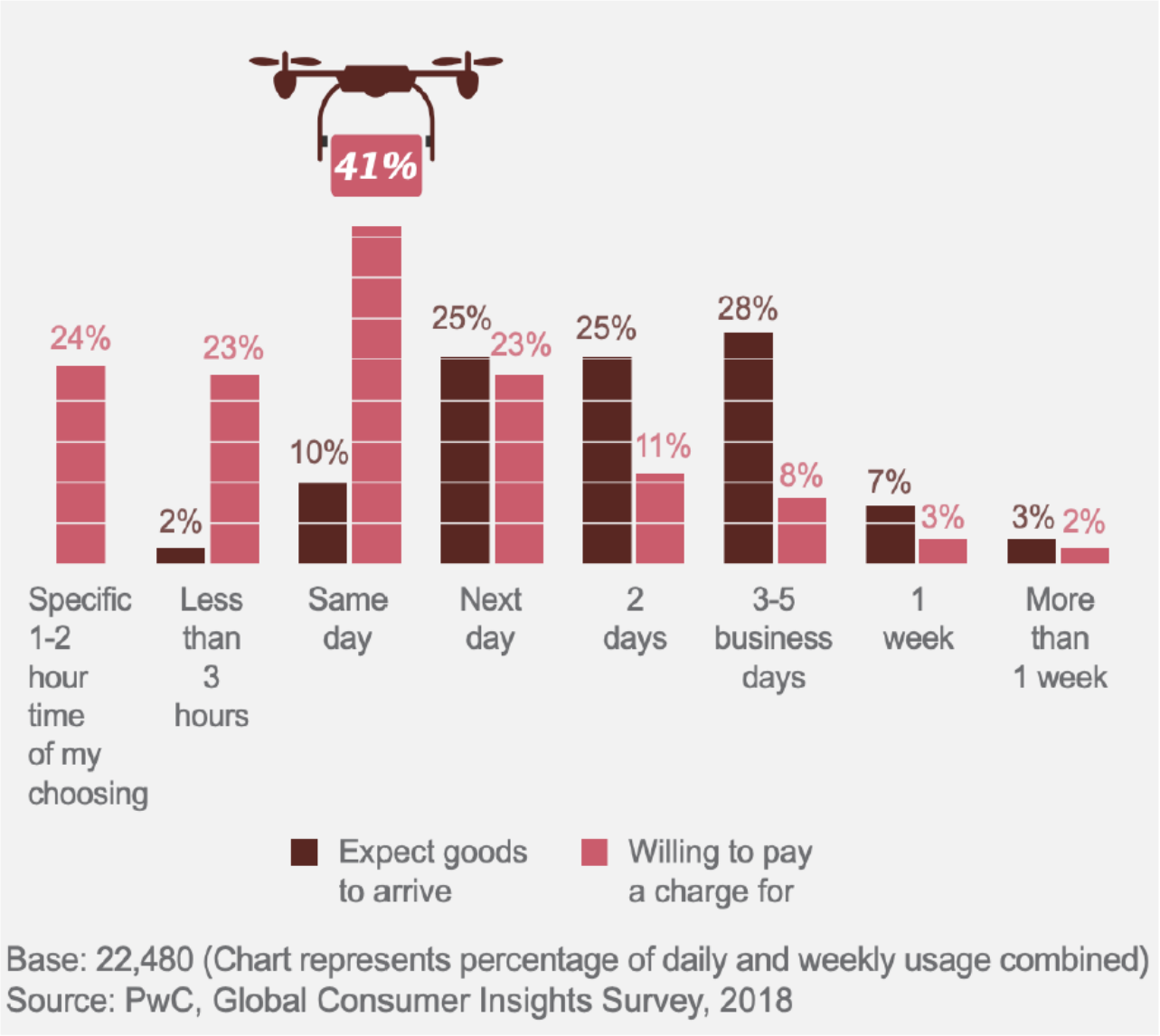
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Background







Base: 22,480 (Chart represents percentage of daily and weekly usage combined)
Source: PwC, Global Consumer Insights Survey, 2018

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Introduction

2-hours delivery



01

Network

Operational in 6 biggest cities

02

Volume

20% of total volume
1/3 of gross merchandise value

Challenges

How to increase coverage?

How to manage costs (operations, real estate, inventory)?

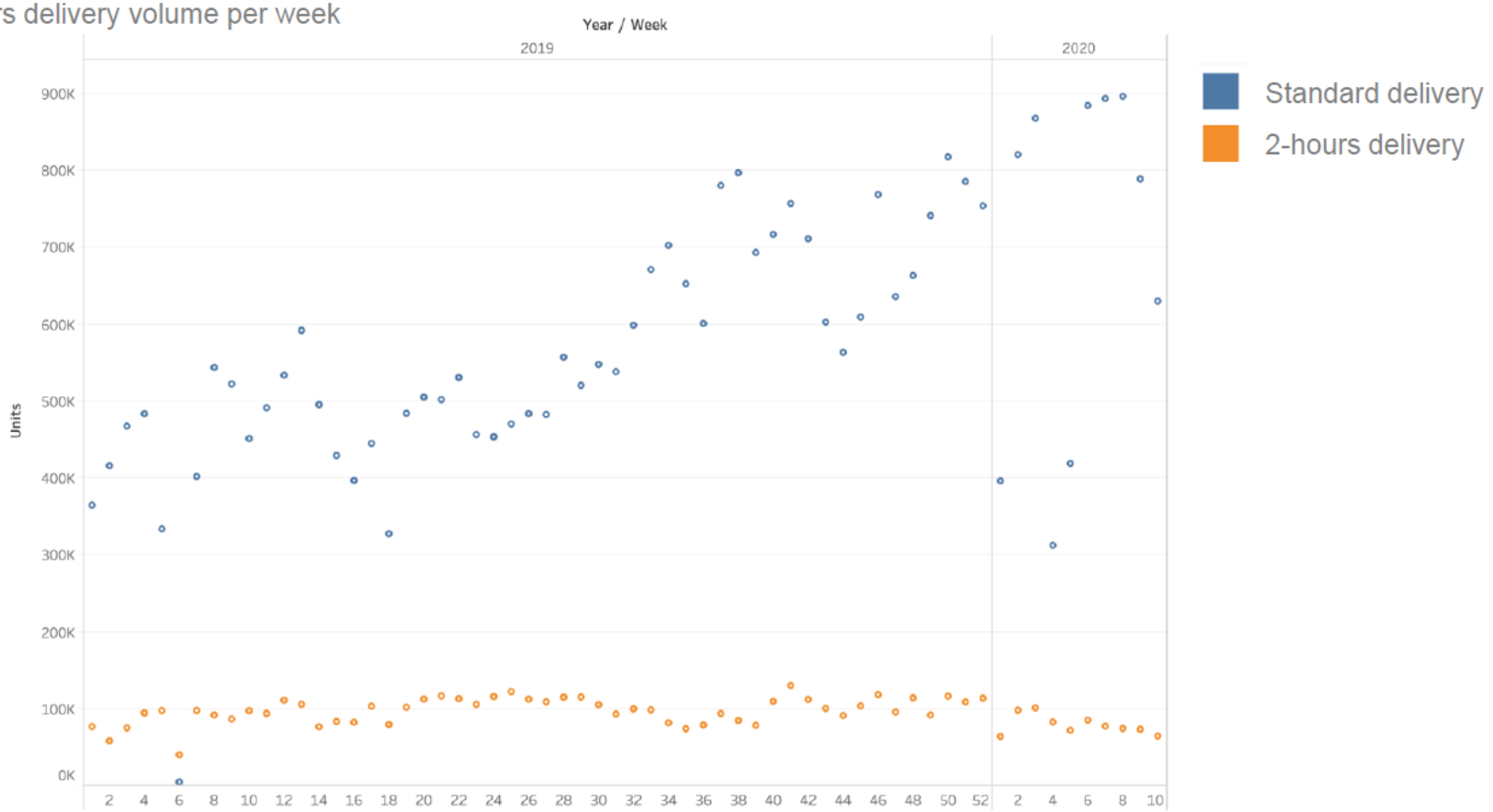
How to account for e-commerce specifics?

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Data analysis

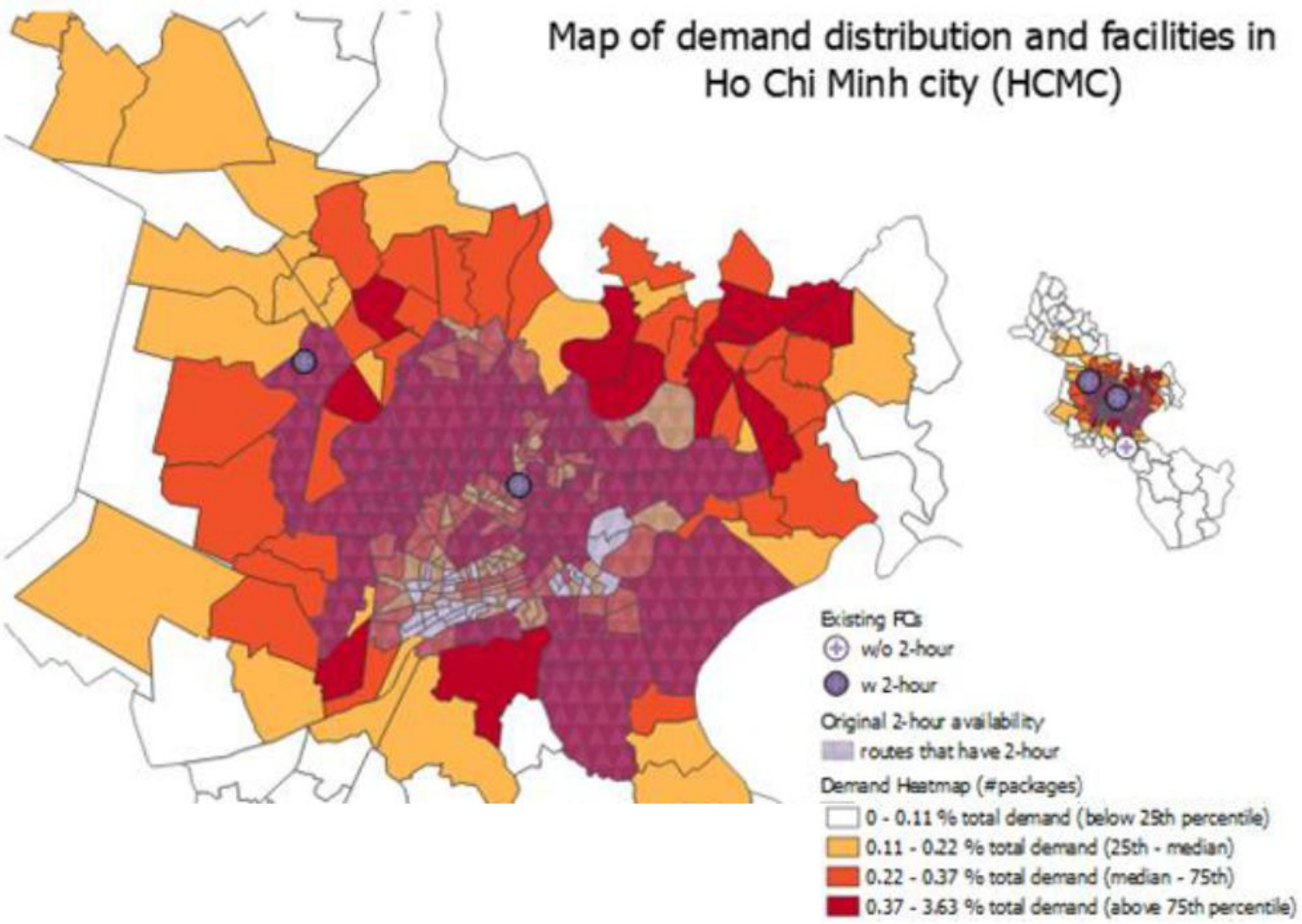
Standard vs 2-hours delivery volume per week



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Ho Chi Minh City



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Research Questions



Where?

Where to locate the fulfillment centers to have the 2-hour delivery network optimized?

How many facilities should be operational?



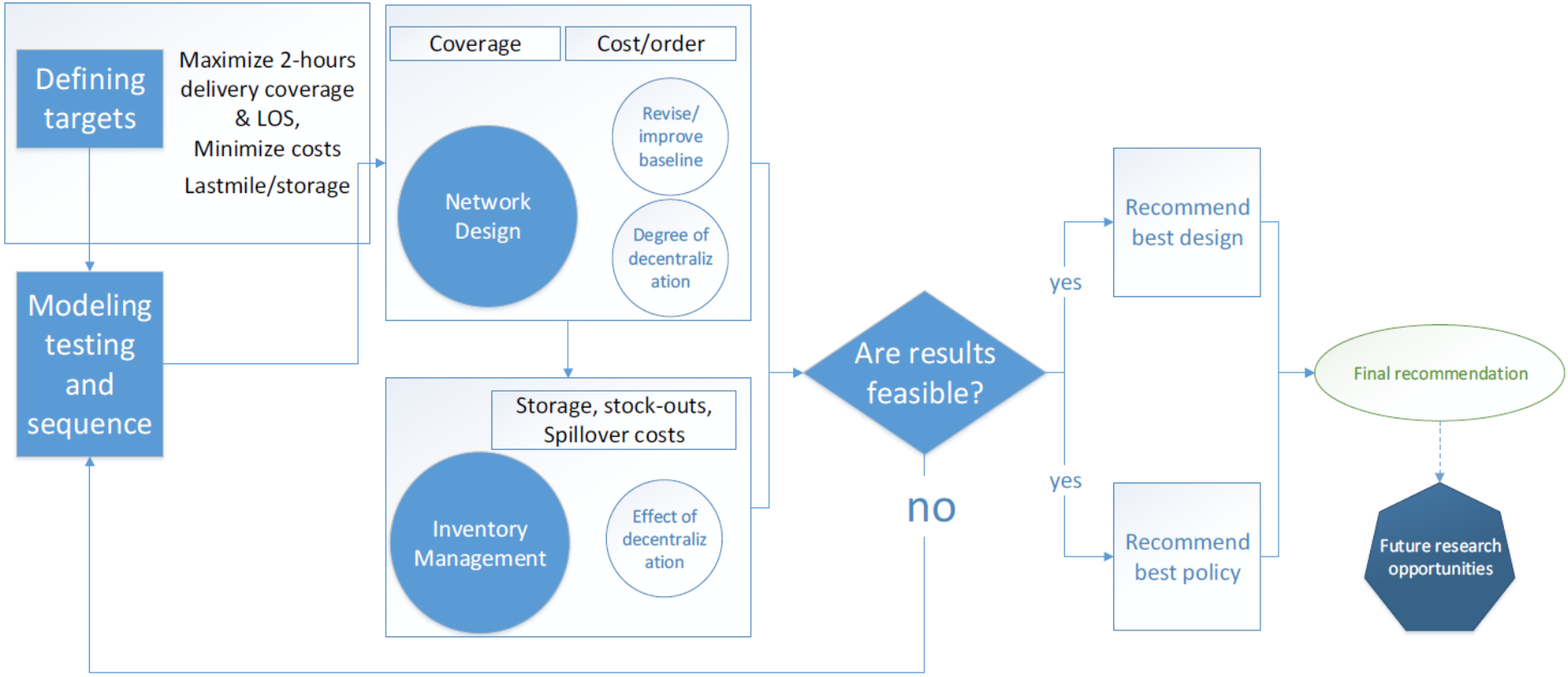
How to replenish?

Given there are more facilities, which inventory policies should be followed?

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Methodology



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Cost estimation

Current cost per order

Item	Name	Unit cost
1	Management cost	Irrelevant for benchmarking
2	Storage cost	115.8 VND per item per day 31,550 VND per cbm per day
3	Fulfillment cost	8,846 VND per item 14,943 VND per order
4	Middle-mile transport cost	Nominal rate card, 1,500 VND per kg 1,343.5 VND per item (estimated)
5	Last-mile cost	12,152 VND per order + 1,927 VND per km
6	Aftersales cost	Irrelevant for benchmarking
	Cost per order	**55,219 VND**

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Maximum coverage problem

Choice of locations

Maximize: $z = \sum_{i \epsilon J} w_j * y_j$

Subject to:

$$\sum_{i \epsilon N_j} x_i \geq y_j \; \forall \; j \epsilon J, N_j = \{dist.\, from\; x_i \; to \; y_j \leq D\}$$

$$\sum_{i \epsilon I} x_i = P$$

$$x_i = (0,1) \; \forall \; i \epsilon I$$

$$y_j = (0,1) \; \forall \; j \epsilon J$$

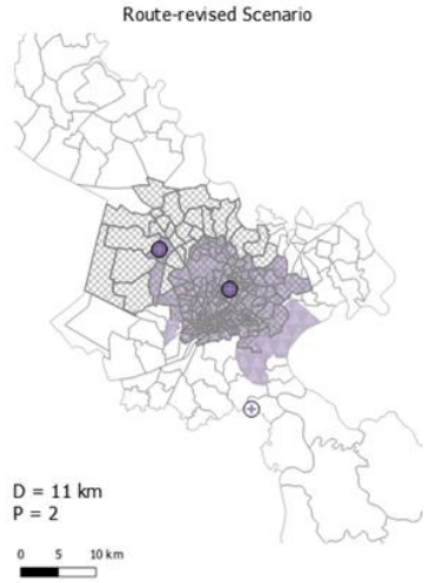
Notation:

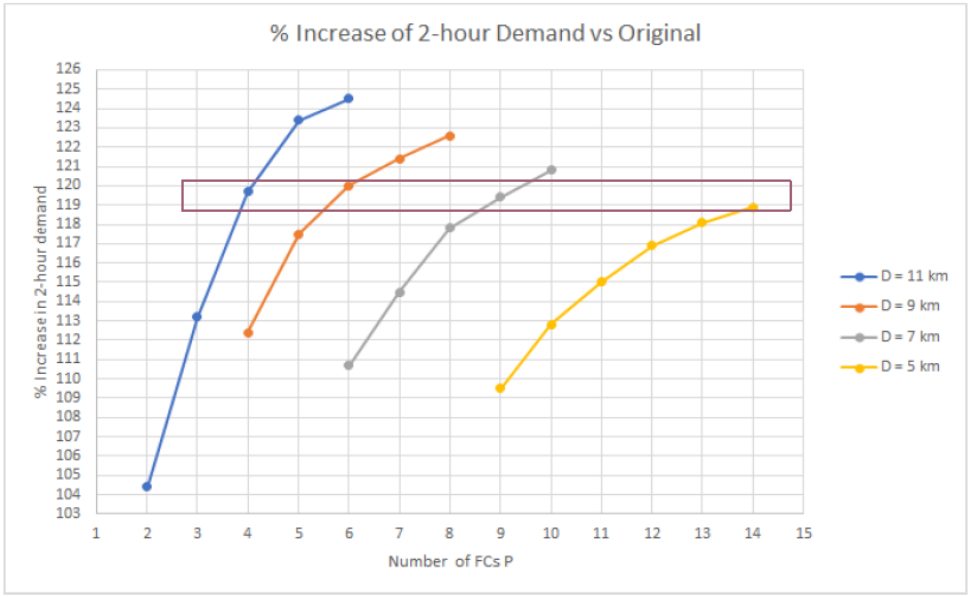
w_j	Weight as a fraction of demand for the region
D	Maximum distance covered
I	Number of locations
J	Number of customers
P	Number of sites open
x_i	Binary, $x \in \{0,1\}$, where 1 means the FC is in the location i^{th}, 0 – closed
y_j	Binary, $y \in \{0,1\}$, where 1 means the demand node j^{th} is covered, 0 – otherwise

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Maximum coverage results

Iterate over many combinations of D and P





- 4.4% increase in demand coverage by reconfiguration

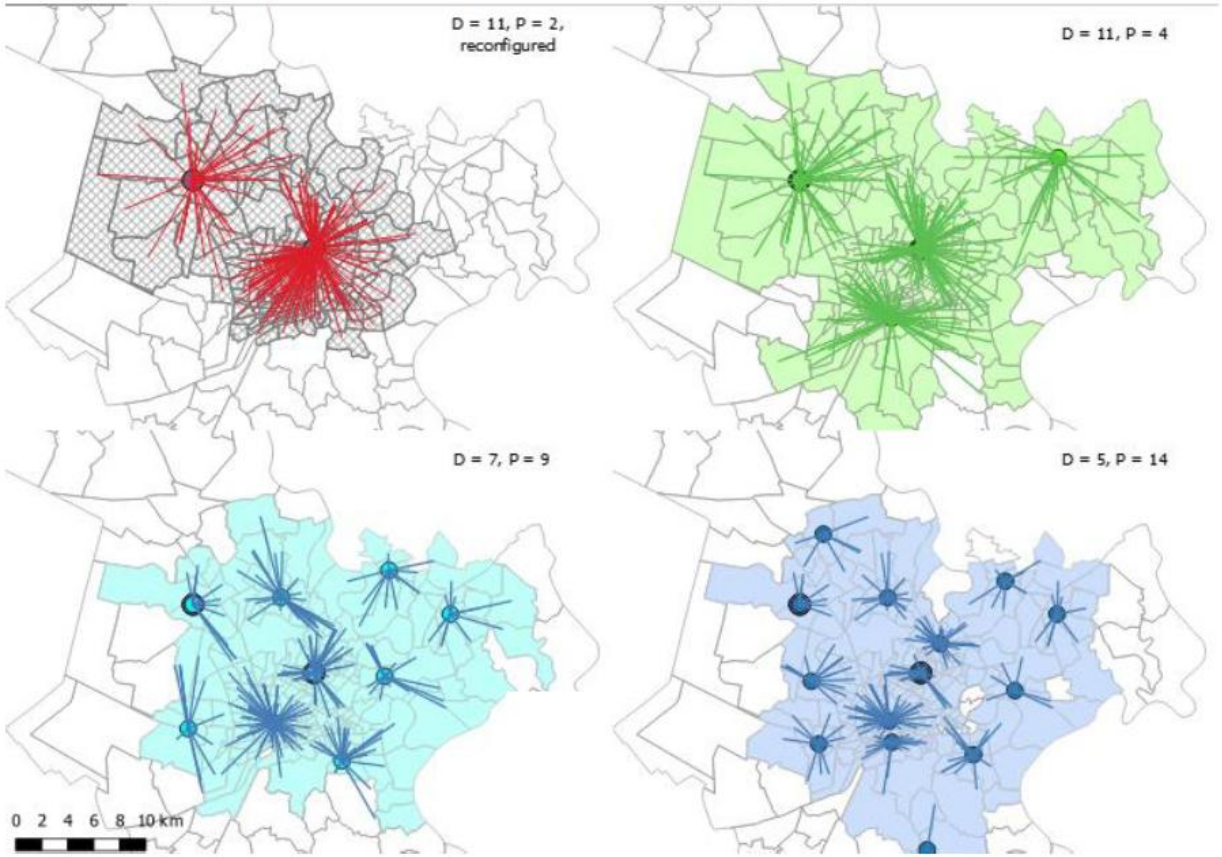
- Pick 3 scenarios (D = 11 P = 4, D = 7 P = 9, D = 5, P = 14) for further analysis
- All scenarios provide ~20% uplift in demand coverage

D: max distance allowed
P: number of facilities

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Facility allocation

Assign demand nodes to FCs and estimate their throughputs



D: max distance allowed

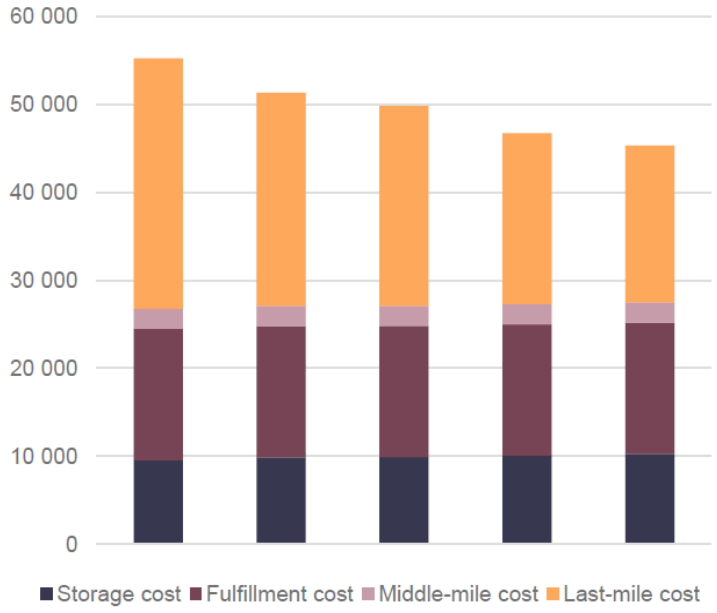
P: number of facilities

FC: fulfilment center

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Comparison of cost per order

Cost per order breakdown



	As-is	Route-revised	D11 P4	D7 P9	D5 P14
Cost per order, VND	55,219	51,304	49,837	46,726	45,309
% Savings vs As-is	-	7.1%	9.7%	15.4%	17.9%
% Demand added vs As-is	-	4.4%	19.7%	19.4%	18.9%

VND: Vietnam dong
D: max distance allowed
P: number of facilities

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Sensitivity analysis

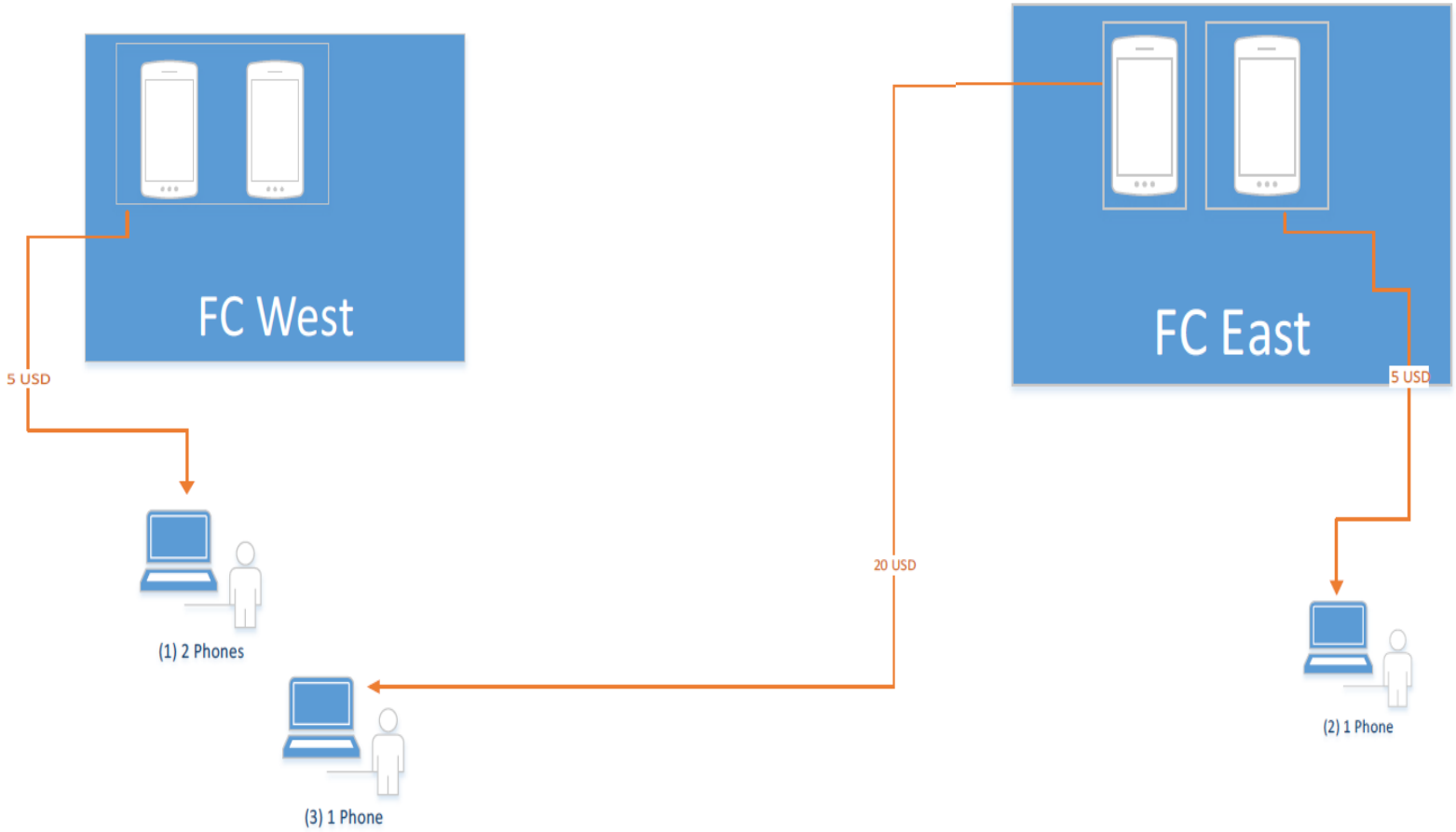
Test against change in rental conditions, cost per order in VND

		Price up 10%	Price up 20%	Price up 30%
Min Rental 200 sqm	D11P4	49.837	50.171	50.504
	D7P9	46.726	47.400	48.074
	D5P14	**45.309**	**46.116**	**46.923**
Min Rental 400 sqm	D11P4	49.837	50.171	50.504
	D7P9	46.846	47.531	48.217
	D5P14	**46.053**	**46.928**	**47.803**
Min Rental 600 sqm	D11P4	49.837	50.171	50.504
	D7P9	**47.319**	**48.047**	**48.775**
	D5P14	47.661	48.682	49.703

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Spillover effect on outbound costs



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Basestock vs Projected Basestock

- What is different?

Estimated IOH right before order arrival

$$\chi_{t'1} = \left(\chi_{t1} - d_{sys}.\lambda_1.L\right)^+ - \left(d_{sys}.\lambda_2.L - \chi_{t2}\right)^+ \quad (1)$$

Projected Basestock (PBS)

Base level

- Decide base level in FC1
- Decide base level in FC2

Review

- Check IOH both FCs
- Calculate order size FC1 accounting for expected spillover to other FCs
- Check IOH both FCs
- Calculate order size FC1 accounting for expected spillover to other FCs

IOH1-Demand over leadtime

Extra demand in FC2 expected to be spillover

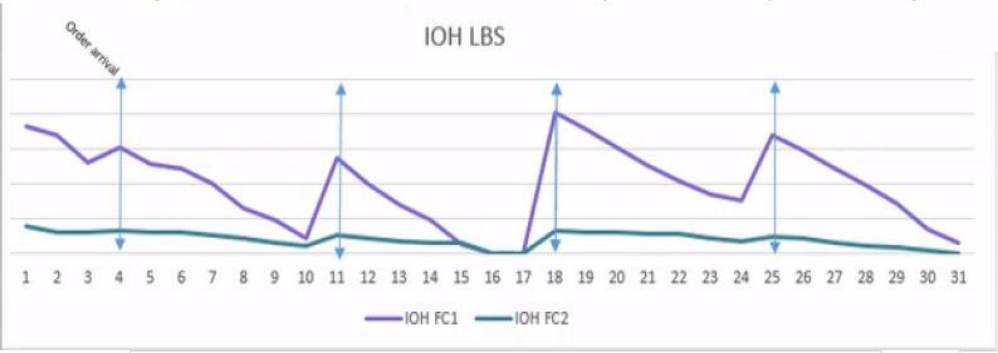
Place order and repeat

FC: fulfilment center
IOH: inventory on hand

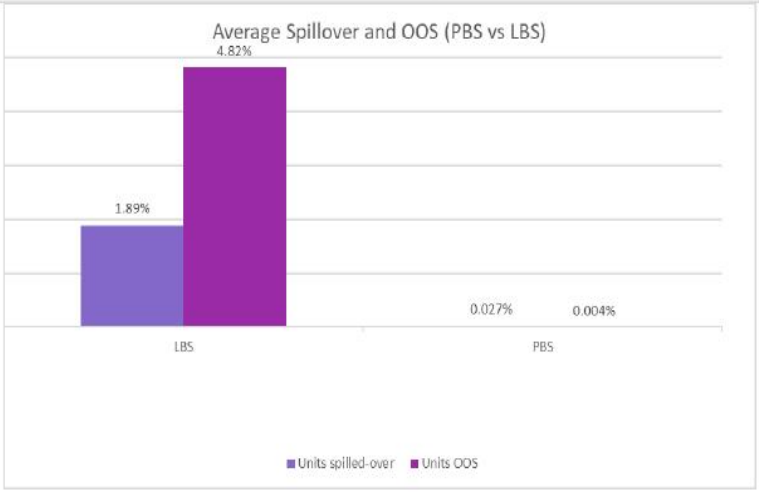
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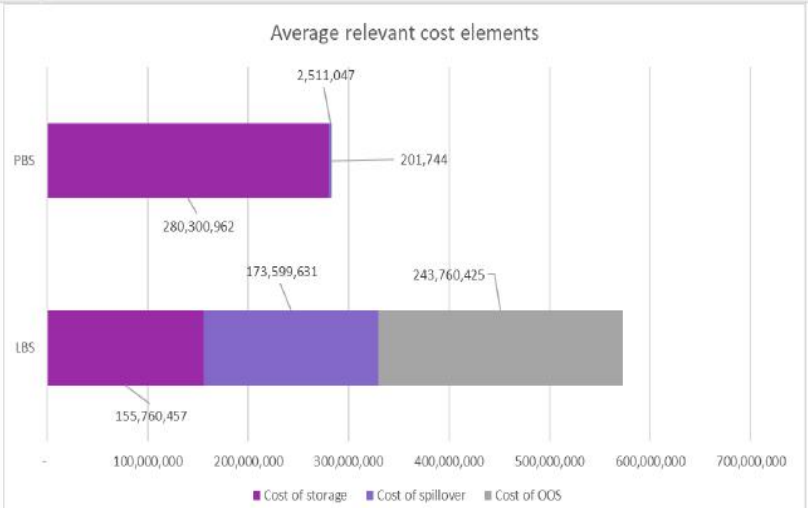
Simulation and Results

31 days, 3500 simulations runs * 2 FCs (LBS vs PBS) ~108K days iterations









FC: fulfilment center
IOH: inventory on hand
OOS: out of stock
OOS cost is the net lost profit cost

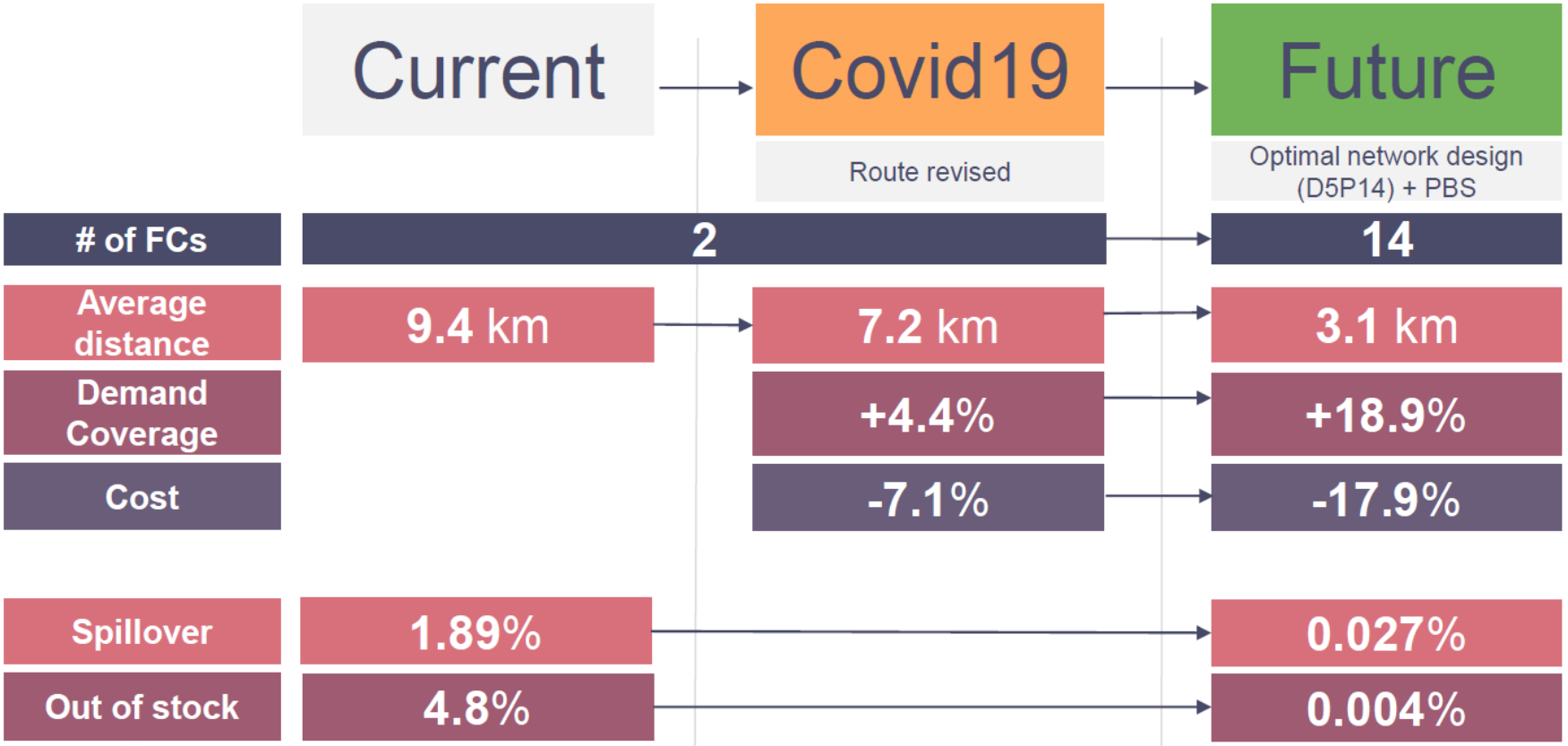
~290 Million VND savings per month

VND: Vietnam dong

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Final Recommendation



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Business Applications of OR/MS

- Airline industry – Flight scheduling
- Revenue management – Airfare, Hotel, Rental Car charges
- Telecommunications – Queuing theory, network algorithms
- Transportation industry – Routing, logistics
- Production – Inventory theory, simulation, analytical production line models, supply chain models
- Engineering and development – Scheduling, resource allocation
- Finance – Quantitative modeling

- ***Interfaces*** - Applications journal published by Institute for Operations Research and Management Sciences

Now: INFORMS Journal on Applied Analytics

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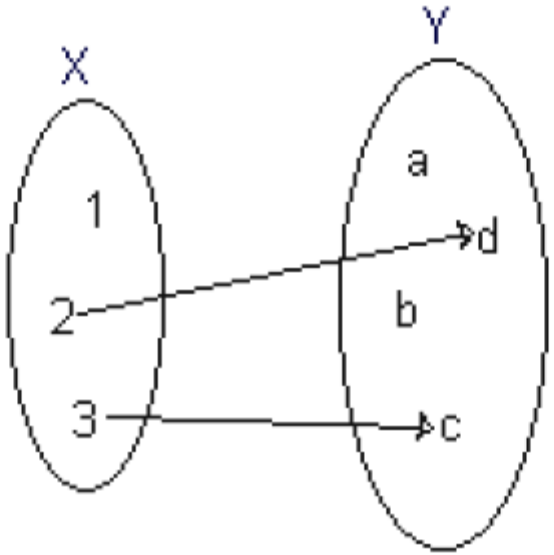
The notion of a function

- **In this program and in your professions you would have heard statements such as**
  - Demand is a function of the price
  - Transportation cost is a function of the distance
  - The time to solve a problem is a function of the size of the problem instance

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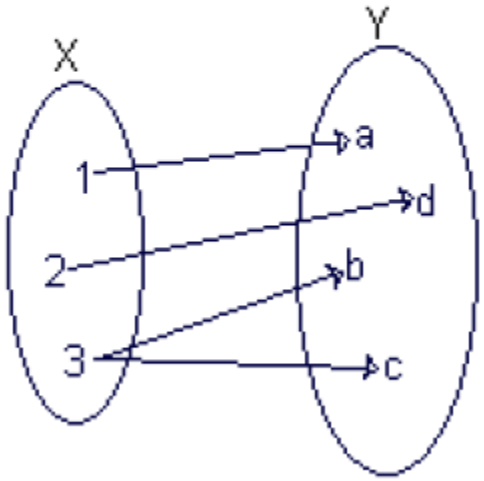
Quiz: Is this a function ?



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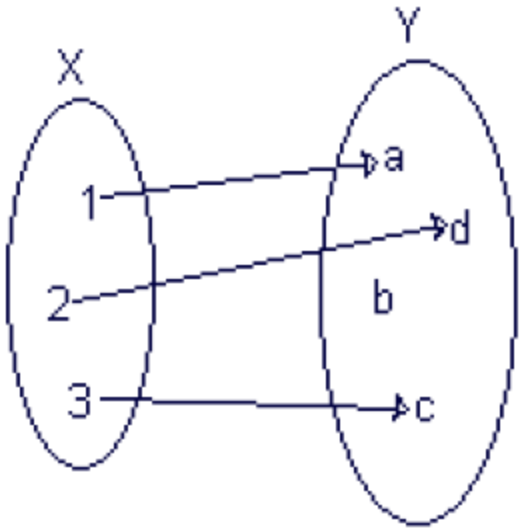
Is this a function?



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Is this a function ?



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Definition of function

- **Formally, a function *f* from a set *X* of input values to a set *Y* of possible output values (written as $f : X \rightarrow Y$) is a relation between *X* and *Y* which satisfies:**
  - *f* is *total*, or *entire*: for all *x* in *X*, there exists a *y* in *Y* such that *x f y* (*x* is *f*-related to *y*),
  - *f* is *many-to-one*, or *functional*: if *x f y* and *x f z*, then *y* = *z*. i.e., many input values can be related to one output value, but one input value cannot be related to many output values.

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Linear Function (especially important for the LP/IP models)

- **A function in which each term is in the form of 'a coefficient times a variable'**

  - Proportionality

    no products, quotients of variables; no exponents other than 1; no logarithm, exponential, trigonometry.

  - Which of these functions are linear?
    - $f(x,y)= 3x+5y$
    - $f(x_1,x_2,x_3)= p_1x_1+p_2x_2+p_3x_3$
    - $f(x,y)= 4x+12xy$
    - $f(x,y)= 9x^2+8y$
    - $f(x,y)= 2 \log x + 12 x/y$

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Differentiation

The derivative

- The derivative of a function at a point is the *instantaneous slope* of the function at that point. This is, the slope of the tangent line to the function at that point.
- Notation: for a function $y = f(x)$, the derivative of f with respect to x can be written as:

$$\frac{dy}{dx} = f'(x) = \frac{df(x)}{dx}$$

$$f'(x) = \lim_{\delta \to 0} \left(\frac{f(x+\delta) - f(x)}{\delta} \right)$$

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Differentiation

Higher order derivatives:

- The second derivative of $f(x)$ is the derivative of $f'(x)$. It is the rate of change of function $f'(x)$.
- Notation, for a function $y=f(x)$, the second order derivative with respect to x can be written as:

$$\frac{d^2y}{dx^2} = f''(x) = \frac{df'(x)}{dx} = \frac{d^2f(x)}{dx^2}$$

- Higher order derivatives are defined analogously.

$$f^{(n)}(x) = \frac{df^{(n-1)}(x)}{dx} = \frac{d^nf(x)}{dx^n}$$

- Example: Second order derivative of $f(x) = 3x^2-12x+6$

$$f(x) = x^3+x^{3/2}+5x$$

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Partial derivatives

- For functions of more than one variable, *f(x,y)*, the rate of change with respect to one variable is given by the partial derivative.

- The derivative with respect to *x* is denoted by $\frac{\partial f(x,y)}{\partial x}$

- The derivative with respect to *y* is denoted by $\frac{\partial f(x,y)}{\partial y}$

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Monotonic (Boring but helpful)

- The function *f* is strictly increasing on *I* if

$$\forall x_1 \in I, \forall x_2 \in I,\ x_1 < x_2 \implies f(x_1) < f(x_2)$$

- The function *f* is increasing on *I* if

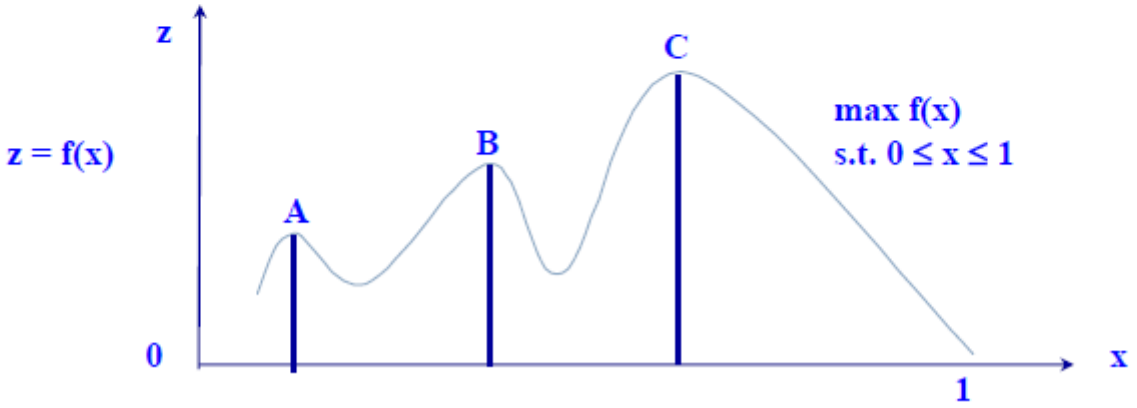
$$\forall x_1 \in I, \forall x_2 \in I,\ x_1 < x_2 \implies f(x_1) \leq f(x_2)$$

- The function *f* is (strictly) monotonous on *I* if it is either (strictly) increasing on *I* or (strictly) decreasing on *I*.

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Local vs. Global Optima

- Let x be a feasible solution, then
  - x is a *global max* if $f(x) \geq f(y)$ for every feasible y.
  - x is a *local max* if $f(x) \geq f(y)$ for every feasible y sufficiently close to x (i.e., $x_j - \varepsilon \leq y_j \leq x_j + \varepsilon$ for all j and some small ε).



- There may be several locally optimal solutions.

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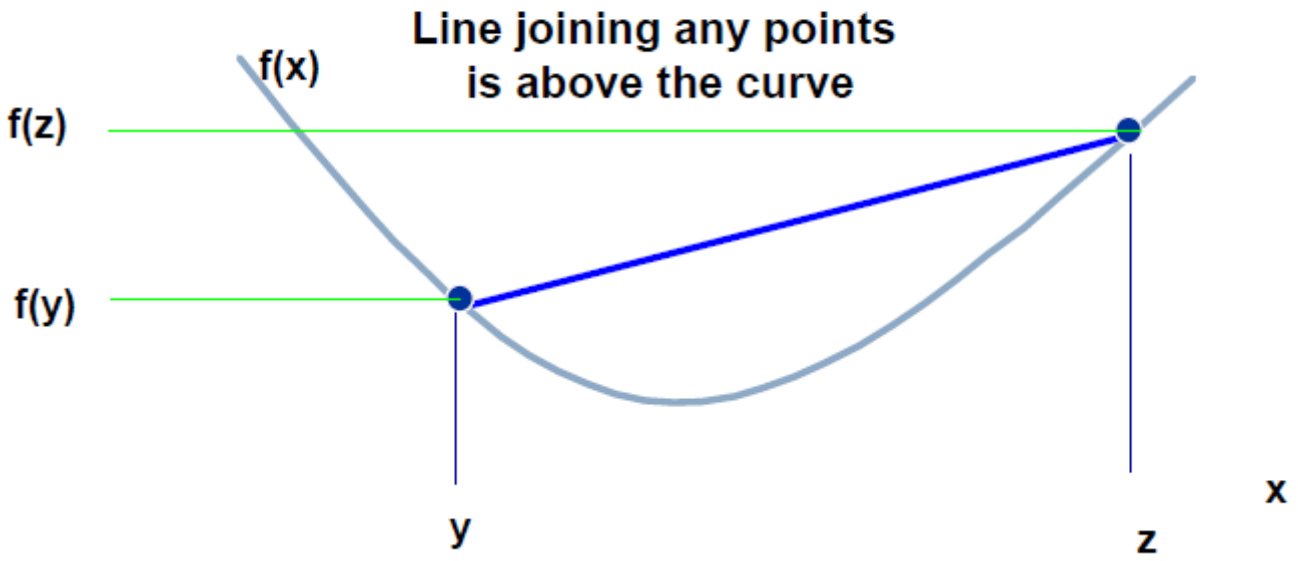
When local optimum is also global optimum

- **For maximization problems**
  - The objective function is concave.
  - The feasible region is convex.
- **For minimization problems**
  - The objective function is convex.
  - The feasible region is convex.

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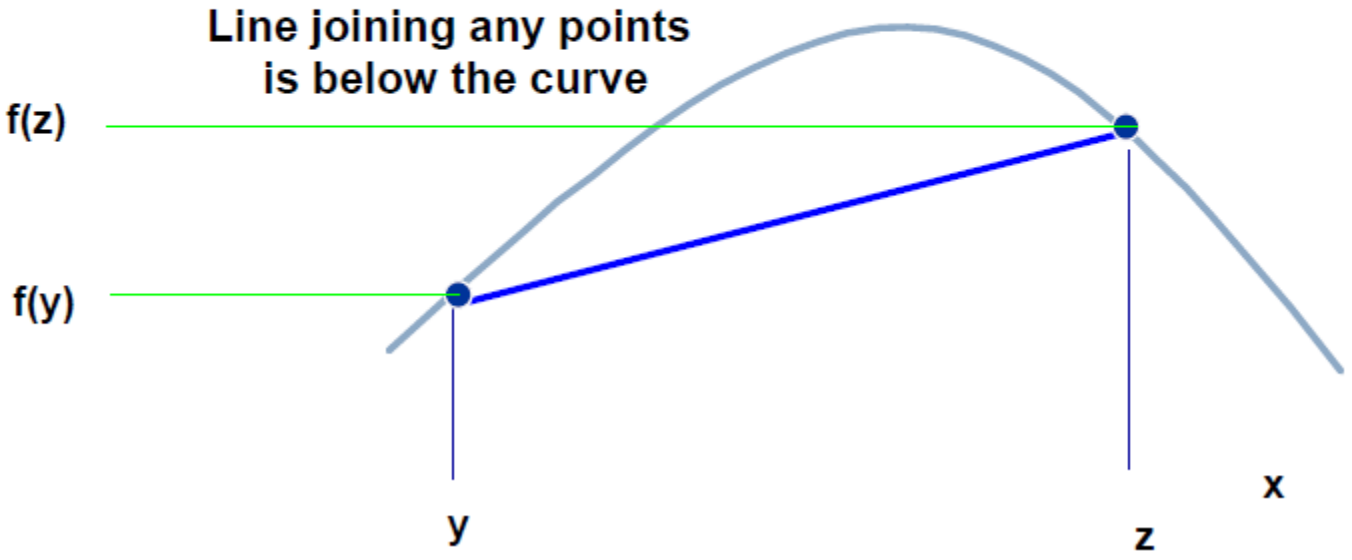
Convex Functions: $f(\lambda y + (1-\lambda)z) \leq \lambda f(y) + (1-\lambda)f(z)$ **for every** y **and** z **and for** $0 \leq \lambda \leq 1$.



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Concave Functions: $f(\lambda y + (1-\lambda)z) \geq \lambda f(y) + (1-\lambda)f(z)$ **for every** y **and** z **and for** $0 \leq \lambda \leq 1$.



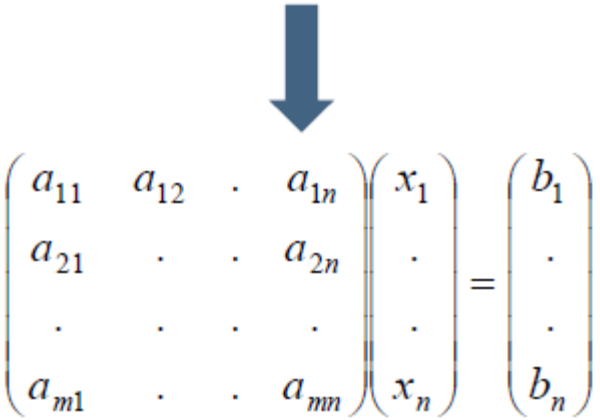
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Linear Algebra

- **A system of equations**

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots\ldots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots\ldots + a_{2n}x_n = b_2$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots\ldots + a_{mn}x_n = b_m$$



A system of linear equations has a unique solution
iff the number of equations equals the number of unknowns.

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Classification of problems

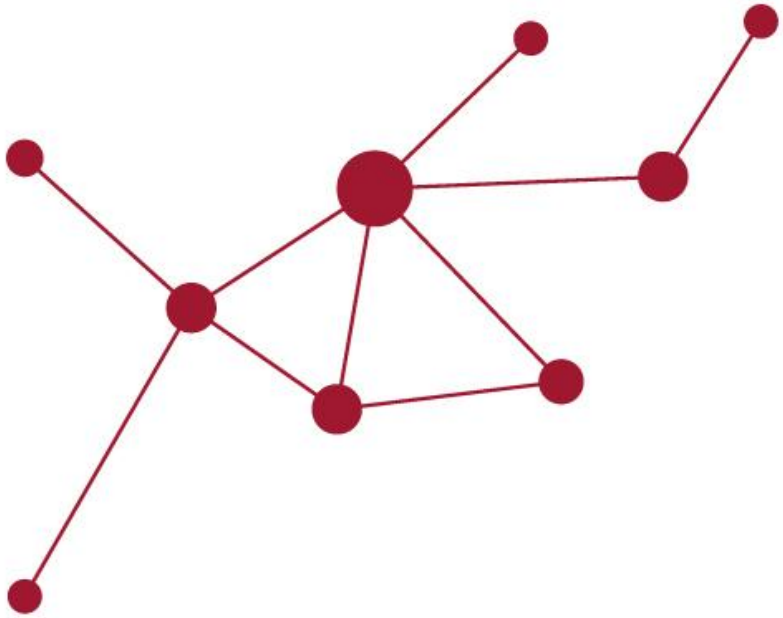
- **Polynomial-Time Algorithms**
- **Non-Polynomial Time algorithms**
  - Traveling Salesman Problem – NP Hard
  - Knapsack Problem with a lower bound on the objective value – NP Complete

ZLC

TALENT HUB FOR SUPPLY CHAIN

Zaragoza Logistics Center

Thanks!



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LINEAR PROGRAMMING

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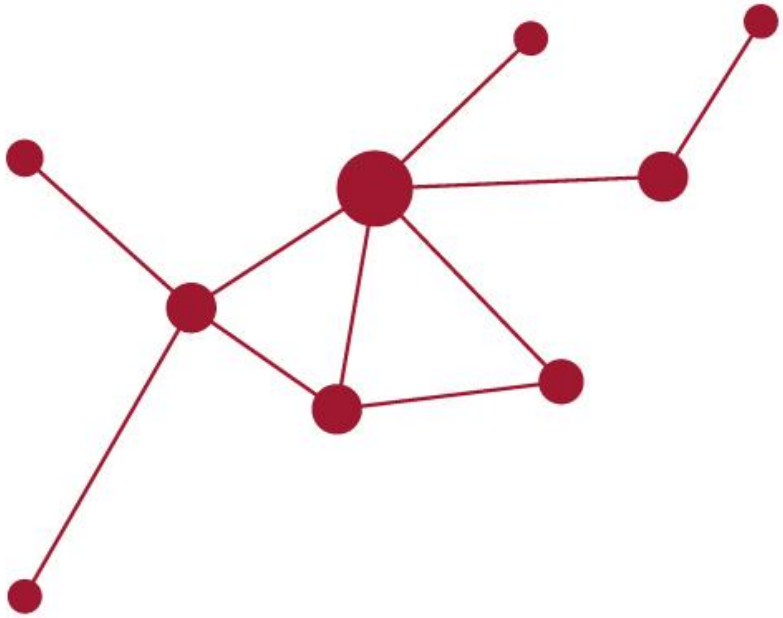
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Linear Programming Part 1

September 14, 2026

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Exercise #1:

Two products A and B are produced in a factory. The price, cost of raw materials and market demand for each product is as follows:

Product	A	B
Price ($/unit)	80	100
Cost of materials ($/unit)	20	25
Demand (units/week)	200	110
Time on bottleneck machine (Hrs/unit)	0.2	0.5

Suppose that there is only 60 hours of available production time per week (for the bottleneck machine and both products need to be processed on this machine). What should be the production plan?

Exercise #2:

Three products A, B, and C are produced in a factory. The price, cost of raw materials and market demand for each product is as follows:

Product	A	B	C
Price ($/unit)	80	100	90
Cost of materials ($/unit)	20	25	10
Demand (units/week)	200	110	70
Time on bottleneck machine (Hrs/unit)	0.2	0.5	0.4
Labor hours/unit required	0.05	0.10	0.05

Suppose that there is only 60 hours of available production time per week (for the bottleneck machine and all three products need to be processed on this machine) and 10 labor hours available per week. What should be the production plan?

Example: Giapetto's Woodcarving

- **Giapetto's, Inc., manufactures wooden soldiers and trains.**
  - Each soldier built:
    - Sell for $27 and uses $10 worth of raw materials.
    - Increase Giapetto's variable labor/overhead costs by $14.
    - Requires 2 hours of finishing labor.
    - Requires 1 hour of carpentry labor.
  - Each train built:
    - Sell for $21 and used $9 worth of raw materials.
    - Increases Giapetto's variable labor/overhead costs by $10.
    - Requires 1 hour of finishing labor.
    - Requires 1 hour of carpentry labor.





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Example: Giapetto's Woodcarving

- Each week Giapetto can obtain:
  - All needed raw material.
  - Only 100 finishing hours.
  - Only 80 carpentry hours.
- Demand for the trains is unlimited.
- At most 40 soldiers are bought each week.
- Giapetto wants to maximize weekly profit (revenues – costs).

- Formulate a mathematical model of Giapetto's situation that can be used maximize weekly profit.

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Example: Giapetto's Woodcarving

- **The Giapetto solution model incorporates the characteristics shared by all linear programming problems.**

  - **Decision variables** should completely describe the decisions to be made.
    - x1 = number of soldiers produced each week
    - x2 = number of trains produced each week

  - The decision maker wants to maximize (usually revenue or profit) or minimize (usually costs) some function of the decision variables. This function to maximized or minimized is called the **objective function**.

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Example: Giapetto's Woodcarving

- Giapetto's weekly profit can be expressed in terms of the decision variables x_1 and x_2:
  - Weekly profit = weekly revenue – weekly raw material costs – the weekly variable costs = $3x_1 + 2x_2$
- Thus, Giapetto's objective is to choose x_1 and x_2 to maximize weekly profit. The variable z denotes the objective function value of any LP.
- Giapetto's objective function is

  Maximize $z = 3x_1 + 2x_2$
- The coefficient of an objective function variable is called an **objective function coefficient.**

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Example: Giapetto's Woodcarving

- As x_1 and x_2 increase, Giapetto's objective function grows larger.

- For Giapetto, the values of x_1 and x_2 are limited by the following three restrictions (often called constraints):
  - Each week, no more than 100 hours of finishing time may be used. ($2x_1 + x_2 \leq 100$)
  - Each week, no more than 80 hours of carpentry time may be used. ($x_1 + x_2 \leq 80$)
  - Because of limited demand, at most 40 soldiers should be produced. ($x_1 \leq 40$)

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Example: Giapetto's Woodcarving

- The coefficients of the decision variables in the constraints are called the **technological coefficients**. The number on the right-hand side of each constraint is called the constraint's **right-hand side** (or rhs).

- To complete the formulation of a linear programming problem, the following question must be answered for each decision variable.
  - Can the decision variable only assume nonnegative values, or is the decision variable allowed to assume both positive and negative values?
    - If the decision variable can assume only nonnegative values, the sign restriction $x_i \geq 0$ is added.
    - If the variable can assume both positive and negative values, the decision variable x_i is **unrestricted in sign** (often abbreviated urs).

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Example: Giapetto's Woodcarving

$$\text{Max } z = 3x_1 + 2x_2 \quad \text{(objective function)}$$

Subject to (s.t.)

$$2x_1 + x_2 \leq 100 \quad \text{(finishing constraint)}$$

$$x_1 + x_2 \leq 80 \quad \text{(carpentry constraint)}$$

$$x_1 \leq 40 \quad \text{(constraint on demand for soldiers)}$$

$$x_1 \geq 0 \quad \text{(sign restriction)}$$

$$x_2 \geq 0 \quad \text{(sign restriction)}$$

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Linear Programming

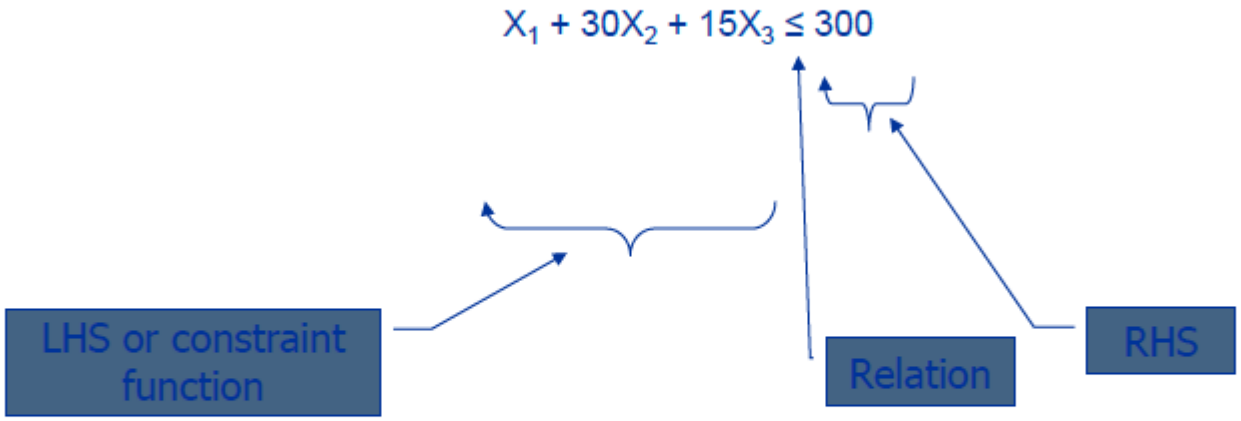
- **Uses a mathematical model to find the best level of various activities (e.g. production levels, staffing levels, $ invested),**
- **Goal is to:**
  - Maximize profit
  - Minimize cost
- **Decisions are subject to various constraints such as**
  - Limited resources
  - Minimum requirements
- **Some Examples:**
  - Diet Planning
  - Work Force Scheduling
  - Transportation/Distribution
  - Assignment Problem
  - Dynamic Pricing

Linear Programming

- **Decision Variables**
  - symbols used to represent an item that can take on any value (e.g., x_1=labor hours, x_2=# of workers)
- **Parameters/ Data**
  - known constant values that are defined for each problem (e.g., price of a unit, production capacity)
- **Constraints**
  - linear relationships of decision variables representing restrictions or rules
  - e.g., limited resources like labor or capital
- **Objective Function**
  - linear relationships of decision variables describing the problems objective
  - always consists of maximizing or minimizing some value
    - e.g., maximize Z = profit, minimize Z = cost

Formulation Points

- **Constraints –**



- **Solution Spaces**
  - Feasible solution – any value of the decision variables where all of the constraints and conditions are satisfied.
  - Optimal solution – a feasible solution that optimizes the objective function. May be more than one.

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Four Assumptions of Linear Programming:

- **Linearity**
  - All the equations/inequalities in the model are linear
  - Each unit is contributing equally to the profit/cost (no economies of scale)
- **Divisibility:**
  - Fractions are okay if
    - Production rate
    - Rounding is reasonable
- **Certainty**
  - All the parameter values are known quantities
- **Nonnegativity**
  - Negative decision variables are inconceivable in most LP problems
    - Minus 10 units of production or a negative consumption make no sense
  - The assumptions are written as follows
    - $x_1, x_2 \geq 0$

Why use Linear Programming?

- **Linear programs are easy (efficient) to solve**
  - Simplex method (Dantzig)
- **The best (optimal) solution is guaranteed to be found (if exists)**
- **Useful sensitivity analysis information is generated**
- **Many problems are essentially linear**

Diet Problem

- MREs (Meals Ready to Eat) is a totally self-contained operational ration consisting of a full meal packed in a flexible meal bag and is designed for individuals in heavy activity duty. Extreme care and planning is required to ensure that the MRE provides the soldier a balanced diet

- For simplicity assume that the ingredients that make up a particular MRE consist of : corn, beefsteak, soybeans, and fish.

- These ingredients contain the following nutrients: vitamins, protein, calcium, and fat in the following quantities:

	Nutrient			
Ingredient	Vitamins	Protein	Calcium	Fat
Corn	8	5	6	8
Beefsteak	2	5	10	8
Soybeans	6	12	6	4
Fish	8	18	6	5

Let a_{ij} = quantity of nutrient j per kg of ingredient i

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Restrictions (Constraints)

- The following "quality" bounds have been defined for a diet.

Vitamins		Protein		Calcium		Fat	
min	max	min	max	min	max	min	max
60	--	300	--	70	--	40	200

  - The above values represent bounds: L_j & U_j

- The army supply system can only guarantee the availability of the raw ingredients in the following quantities per meal

Supply (kg)	Corn	Beef	Soybeans	Fish
s_i	10	6	4	5

- Cost per kg of the raw ingredients is as follows:

	Corn	Beefsteak	Soybeans	Fish
cost/kg, c_i	60¢	400¢	75¢	350¢

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Diet problem- History & Future

- When first solved by the army in 1947 it took nine clerks using hand-operated desk calculators 120 man days to solve for the optimal solution of the linear program of 9 equations and 77 unknowns (for more details see July-Aug 1990 *Interfaces* article "The Diet Problem")
- The U.S. Department of Defense Combat Feeding Program has started tracking MRE combat rations from "vendor to foxhole" using RFID technology developed at the Massachusetts Institute of Technology in Cambridge, MA
  - *Meal quality index* is tracked in real-time
- Least-Cost Menu Planning Models in Food Systems Management
  - Used in many institutions with feeding programs: hospitals, nursing, homes, schools, prisons, etc.
  - Menu planning often extends to a sequence of meals
  - Generally models have reduced cost about 10%, meet nutritional requirements better.

Diet Problem

Notation

Indices/sets

$i \in I$ ingredients {corn, beef, soybeans, fish}

$j \in J$ nutrients {vitamins, protein, calcium, fat}

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Diet Problem

Data

a_{ij} number of nutrients j per kg of ingredient i

L_j lower bound on number of nutrients of type j

U_j upper bound on number of nutrients of type j

s_i supply of ingredient i (kg)

c_i cost per kg of ingredient i

Decision Variables

x_i amount (kg) of ingredient i to be used

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LP Formulation

$$\min \quad \sum_{i \in I} c_i x_i$$

$$\text{s.t.} \quad x_i \leq s_i \qquad \forall\, i \in I$$

$$\sum_{i \in I} a_{ij} x_i \leq U_j \qquad \forall\, j \in J,$$

$$\sum_{i \in I} a_{ij} x_i \geq L_j \qquad \forall\, j \in J$$

$$x_i \geq 0 \qquad \forall\, i \in I$$

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Diet Problem cont'd..

$$Min\ \ 60X_1 + 400X_2 + 75X_3 + 350X_4$$

subject to

$$X_1 \le 10$$

$$X_2 \le 6$$

$$X_3 \le 4$$

$$X_4 \le 5$$

$$8X_1 + 2X_2 + 6X_3 + 8X_4 \ge 60$$

$$5X_1 + 5X_2 + 12X_3 + 18X_4 \ge 300$$

$$6X_1 + 10X_2 + 6X_3 + 6X_4 \ge 70$$

$$8X_1 + 8X_2 + 4X_3 + 5X_4 \ge 40$$

$$8X_1 + 8X_2 + 4X_3 + 5X_4 \le 200$$

$$X_1, X_2, X_3, X_4 \ge 0$$

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Linear Program – Another Example

- **Problem**
  - You are the product manager for two lines of products and have been asked to determine how much of each product to make in order to maximize the expected profit.
- **Product Characteristics**
  - Product A - $400 profit per item
  - Product B – $300 profit per item
- **Resource Characteristics**
  - Labor – 7,000 hours
  - Raw Material – 5,000 lbs
  - Shipping Capacity – 7,000 cf
- **Resource Usage**

Resource	A	B
Labor	**30**	**60**
Raw Material	**40**	**25**
Capacity	**50**	**50**

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Linear Program Example - Notation

Max $400x_A + 300x_B$

s.t.

$30x_A + 60x_B \leq 7000$ (labor)

$40x_A + 25x_B \leq 5000$ (material)

$50x_A + 50x_B \leq 7000$ (capacity)

$x_A \geq 0$

$x_B \geq 0$

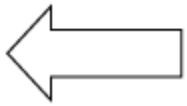


$$\max \quad \begin{bmatrix} c_A & c_B \end{bmatrix} \begin{bmatrix} x_A \\ x_B \end{bmatrix}$$

s.t.

$$\begin{bmatrix} a_{1A} & a_{1B} \\ a_{2A} & a_{2B} \\ a_{2A} & a_{2B} \end{bmatrix} \begin{bmatrix} x_A \\ x_B \end{bmatrix} \leq \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\begin{bmatrix} x_A \\ x_B \end{bmatrix} \geq 0$$



Max cx

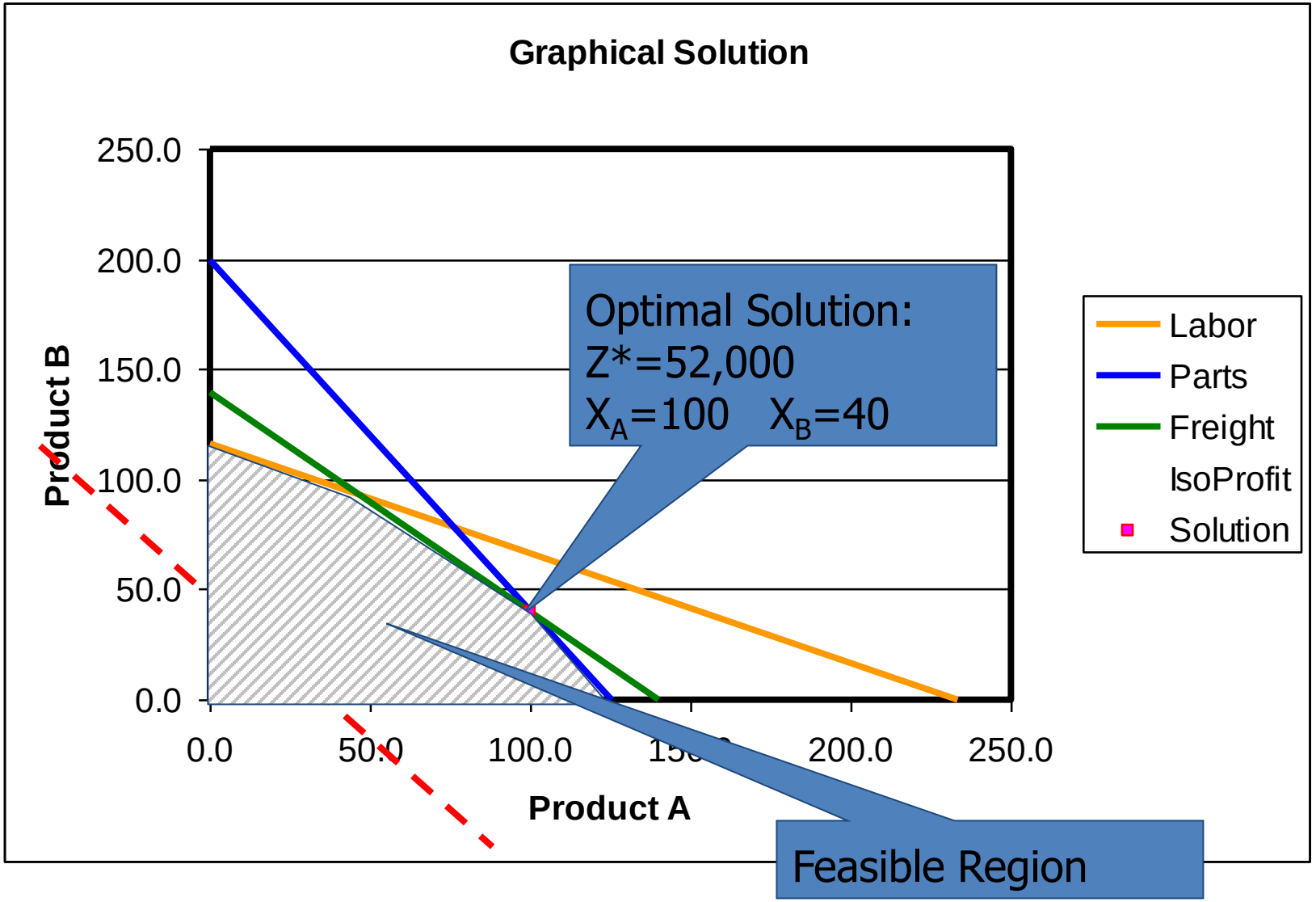
s.t.

Ax ≤ b

x ≥ 0

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Linear Program Example



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Graphical Interpretation

- Linear Inequalities form the Feasible Region:

  $a_{11}x + a_{12}y \geq b_1$

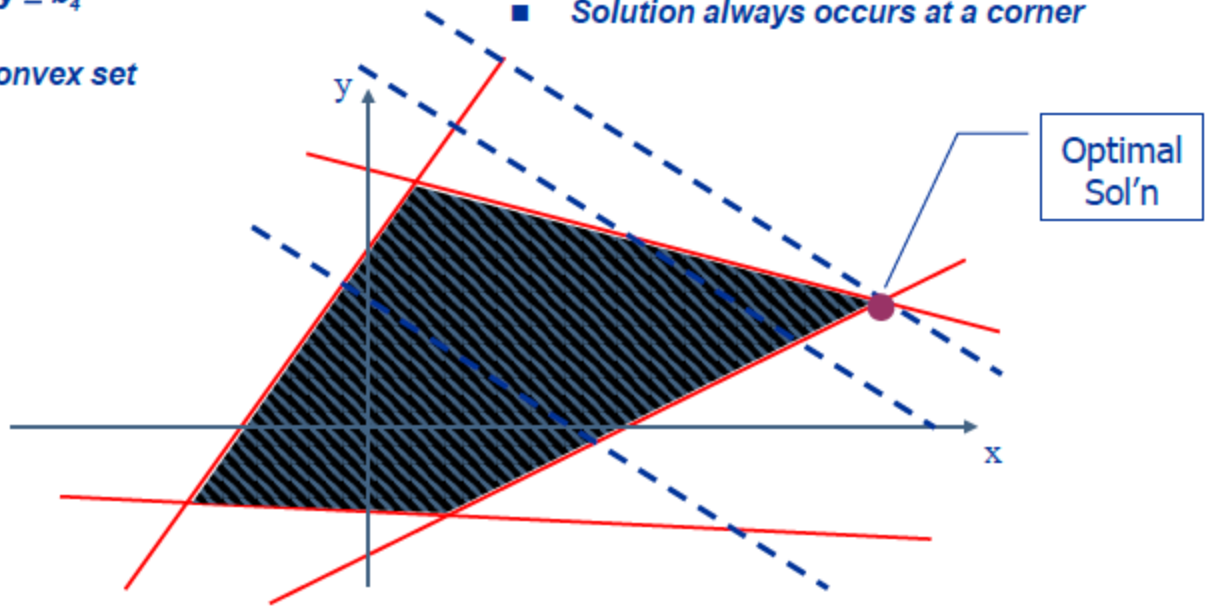
  $a_{21}x + a_{22}y \geq b_2$

  $a_{31}x + a_{32}y \geq b_3$

  $a_{41}x + a_{42}y \geq b_4$

- *FR is a convex set*

- *Objective Function can be expressed in linear form*
- *Forms isoprofit (isocost) lines*
- *Solution always occurs at a corner*



Linear Program

Observations

- Solution always occurs at a 'corner' (boundary of the feasible region)
- Four possible outcomes of linear programming:
  - A unique optimal solution is found
  - An infinite number of optimal solutions exist
  - No feasible solutions exist
  - The objective function is unbounded
- Limits to Graphical Solution
- Algorithmic Approaches
  - Simplex Method
  - Interior Point Method (for linear and nonlinear convex problems)

Example with Multiple Optimal Solutions

Maximize $Z = (\$15)C + (\$30)T$

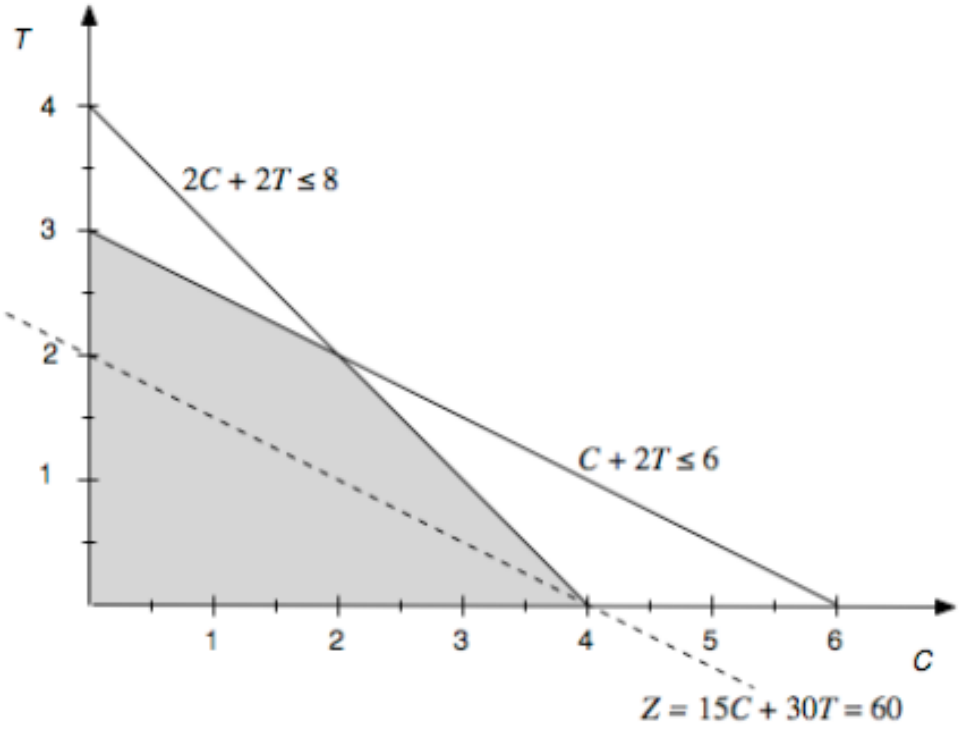
subject to

Large Bricks: $C + 2T \leq 6$

Small Bricks: $2C + 2T \leq 8$

and

$C \geq 0, \quad T \geq 0.$



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Example with No Feasible Solution

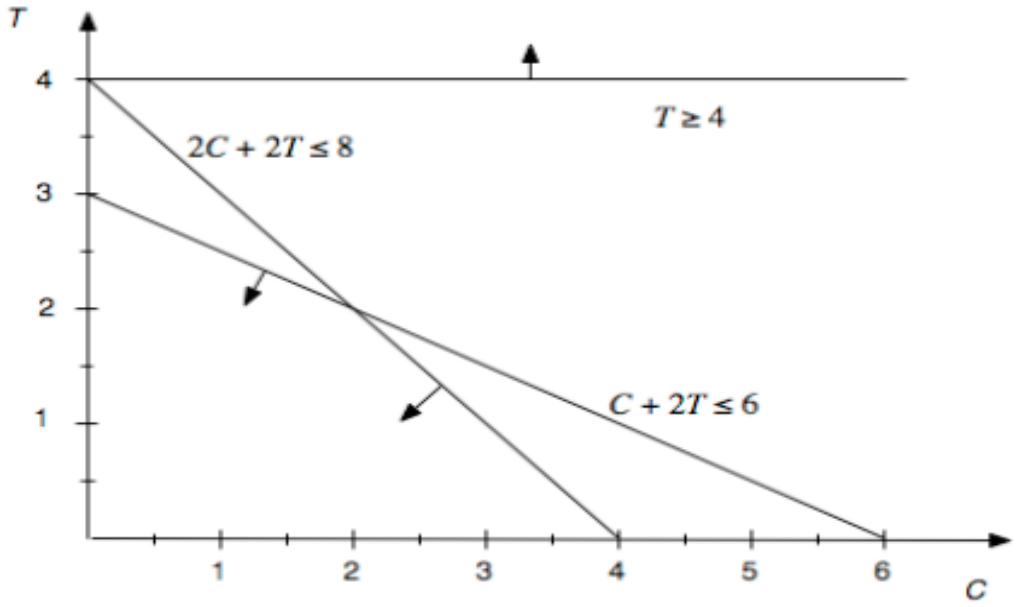
Maximize $Z = (\$15)C + (\$20)T$

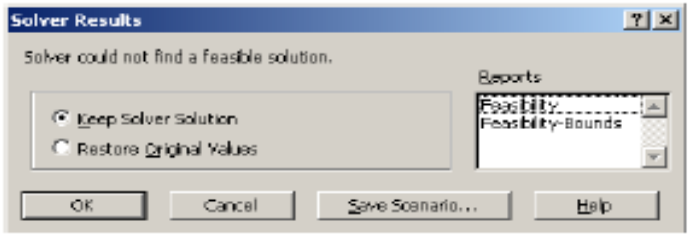
subject to

Large Bricks:	$C + 2T \leq 6$
Small Bricks:	$2C + 2T \leq 8$
Contract:	$T \geq 4$

and

$C \geq 0, \quad T \geq 0.$





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Example with Unbounded Solution

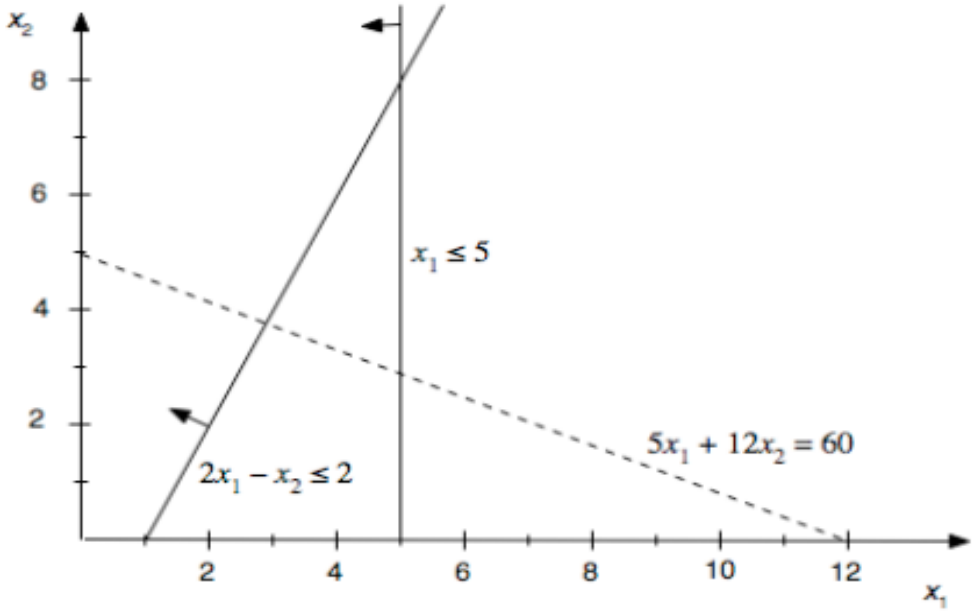
Maximize $Z = 5x_1 + 12x_2$

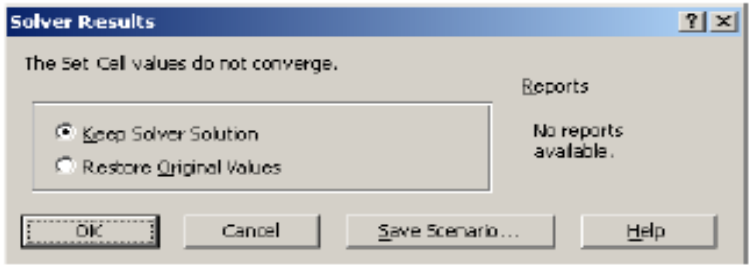
subject to

$$x_1 \leq 5$$
$$2x_1 - x_2 \leq 2$$

and

$$x_1 \geq 0, \quad x_2 \geq 0.$$





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Another Example: Scheduling Problem

Time Period	Number of Officers Needed
12 a.m. – 4 a.m.	5
4 a.m. – 8 a.m.	7
8 a.m. – 12 p.m.	15
12 p.m. – 4 p.m.	8
4 p.m. – 8 p.m.	12
8 p.m. – 12 a.m.	9

- Union contract requires all employees to work 8 consecutive hours
- Goal: Hire the minimum number of officers needed to cover all shifts

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LP Formulation

Define workshifts as:

Workshift 1: 12 a.m. – 8 a.m.

Workshift 2: 4 a.m. – 12 p.m.

Workshift 3: 8 a.m. – 4 p.m.

Workshift 4: 12 p.m. – 8 p.m.

Workshift 5: 4 p.m. – 12 a.m.

Workshift 6: 8 p.m. – 4 a.m.

Decision Variables:

X_i = # of employees assigned to workshift i

Objective Function:

minimize number of officers needed

Min $\sum_{i=1}^{6} X_i$

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Dynamic Pricing

- A popular retail store stocks red sweaters for the fall season. The weekly demand at full price of $ 60 is 100 units
- The new graduate from Parsons they hired as a buyer did some naïve calculations and purchased 2500 units of the red sweater.
- As a supply chain manager you do not feel all the 2500 units can be sold at a price of $ 60.
- The fall season lasts 15 weeks and any extra units at the end have to be sold at an outlet store at $ 25 each
- A market research firm estimated the following weekly demand based on markdowns

Price levels.	$60	$54	$48	$36	$25
Demand multipliers	1	1.5	1.75	2	INF
Avg. weekly demand	100	150	175	200	INF

- What discounting strategy ***maximizes*** revenue?

Source: Prof. G. van Ryzin

Rules of Thumb for LPs

- **Identify the right Decision Variables**
- **Formulate an Objective Function that makes sense**
- **Identify the right Constraints**
  - Type (Resource balance, Capacity, Quality, Policy, etc.)
  - Ensure linearity
  - Try to avoid strict equality constraints
- **Formulations are not unique – strive for simplicity and clarity**
- **Evaluate quality of the data**
- **Multiple model runs "what if" scenarios add to the robustness of the solution**
  - Use a table to monitor constraints and outcomes
  - Try to isolate effects by changing one thing at a time
  - Able to make comparisons across different runs

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Some of the slides borrowed from Dr. Prashant Yadav & Dr. Jianjun Xu

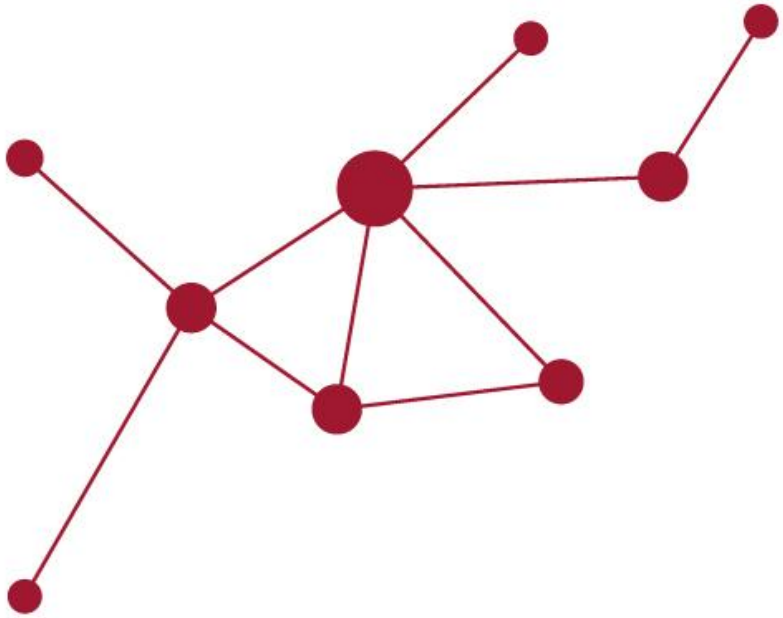
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Linear Programming Part 2

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Network Flow Problems

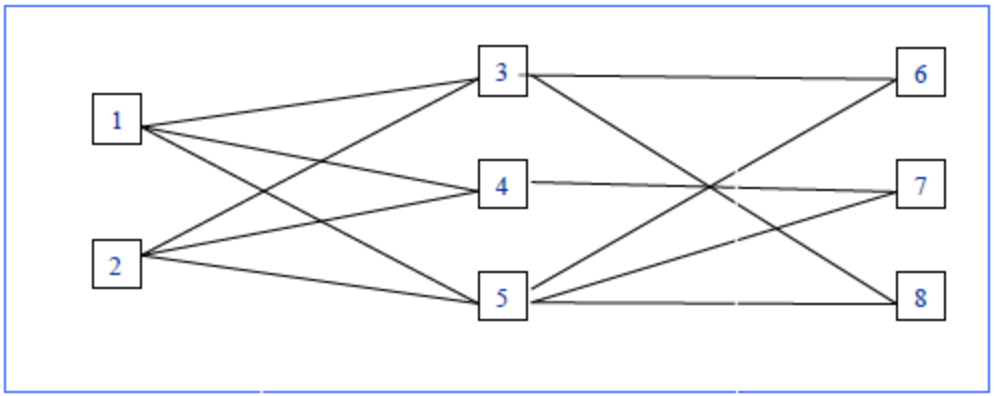
- **Logistics problems can be represented graphically as networks**
- **We can categorize them as**
  - Transportation Problems
  - Assignment Problems
  - Transshipment Problems

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Representing your SC

- **A simple SC can be represented as a collection of nodes connected by arcs.**
- **There are three types of nodes:**
  - Supply
  - Demand
  - Transshipment

A graphical representation



Supply Nodes

Transshipment Nodes

Demand Nodes

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Transportation Problem

- **If only supply and demand nodes: no transshipment**
- **More efficient solution techniques can be used to solve them (transportation method)**
  - *Data: price, shipping cost, demand, route capacity, etc.*
  - *Decision Variables: units sent from node i to node j*
  - *Objective: maximize profit or minimize cost function*
  - *Constraints: supply and demand, route capacity, etc.*

Transportation Problem

- A company has two plants producing a certain product that is to be shipped to three distribution centers. The unit production costs are the same at the two plants, and the shipping cost per unit is shown below.

		Distribution Center		
		1	2	3
Plant	A	$4	$6	$4
	B	$6	$5	$2

- Shipments are made once per week. During each week, each plant produces at most 70 units and each distribution center needs at least 40 units. How many units should be shipped from each plant to each distribution center, so as to minimize cost?

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Transportation Problem

- If

$$\sum_{i=1}^{i=m} s_i = \sum_{j=1}^{j=n} d_j$$

  then total supply equals to total demand, the problem is said to be a **balanced transportation** problem.

- If total supply exceeds total demand, we can balance the problem by adding **dummy demand point**. Since shipments to the dummy demand point are not real, they are assigned a cost of zero.

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Transportation Problem

- If a transportation problem has a total supply that is strictly less than total demand the problem has no feasible solution.
  - No doubt that in such a case one or more of the demand will be left unmet.
  - Generally in such situations a penalty cost is often associated with unmet demand and as one can guess the total penalty cost is desired to be minimum.

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Assignment Problem

- The assignment problem is a special case of the transportation problem, with the supply at each origin and the demand at each destination is equal to 1.
- In general, it is the problem of determining the optimal assignment of n agents or objects to n tasks.
- **Many different problems can be setup as an assignment problem**

  e.g. Manpower allocation

Machine Assignment Problem

- Machineco has four jobs to be completed.
- Each machine must be assigned to complete one job.
- The time required to setup each machine for completing each job is shown.

	Time (Hours)			
	Job1	Job2	Job3	Job4
Machine 1	14	5	8	7
Machine 2	2	12	6	5
Machine 3	7	8	3	9
Machine 4	2	4	6	10

- Machineco wants to minimize the total setup time needed to complete the four jobs.

Assignment Problem

- The coach of a swim team needs to assign swimmers to a 200-yard medley relay team (four swimmers, each swim 50 yards of one of the four strokes). Since most of the best swimmers are very fast in more than one stroke, it is not clear which swimmer should be assigned to each of the four strokes. The five fastest swimmers and their best times (in seconds) they have achieved in each of the strokes (for 50 yards) are

		Strokes			
		Backstroke	Breaststroke	Butterfly	Freestyle
Swimmer	Carl	37.7	43.4	33.3	29.2
	Chris	32.9	33.1	28.5	26.4
	David	33.8	42.2	38.9	29.6
	Tony	37.0	34.7	30.4	28.5
	Ken	34.4	41.8	33.6	31.1

How should the swimmers be assigned to make the fastest relay team?

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Assignment Problem Example

BALLSTON ELECTRONICS

- Five different electrical devices produced on five production lines, are needed to be inspected.
- The travel time of finished goods to inspection areas depends on both the production line and the inspection area.
- Management wishes to designate a separate inspection area to inspect the products such that the total travel time is minimized.

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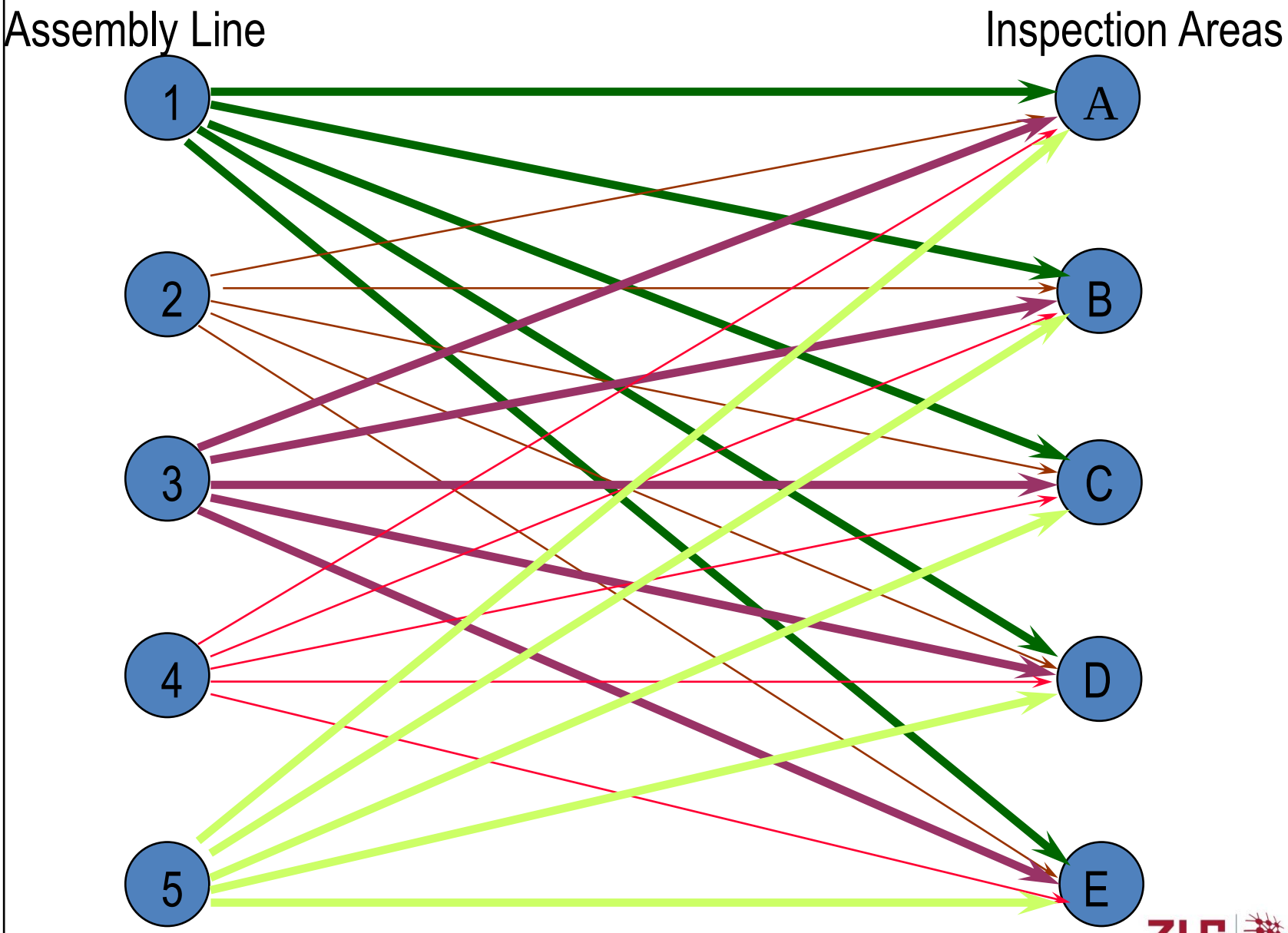
- **Data: Travel time in minutes from assembly lines to inspection areas.**

		Inspection Area				
		A	B	C	D	E
Assembly Lines	1	10	4	6	10	12
	2	11	7	7	9	14
	3	13	8	12	14	15
	4	14	16	13	17	17
	5	19	17	11	20	19

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NETWORK REPRESENTATION



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- **Assumptions and restrictions**

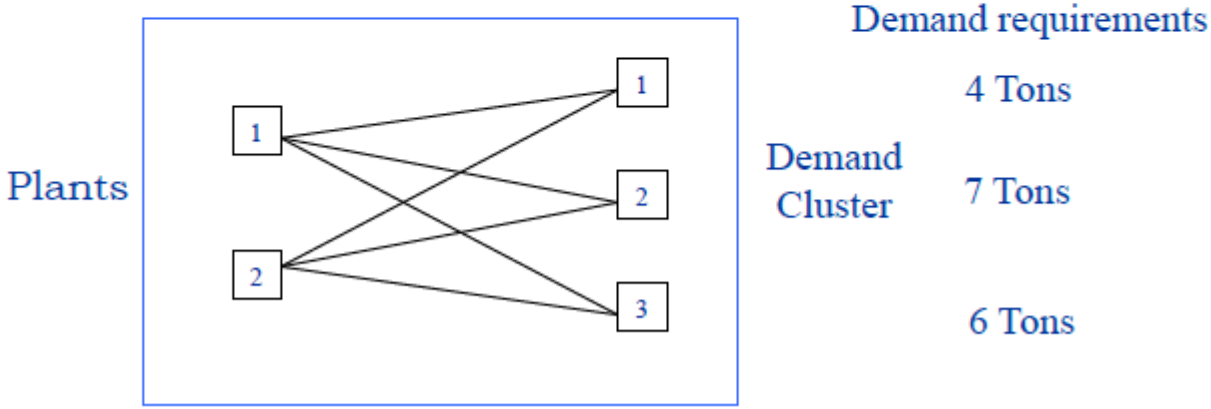
  - The number of workers equals the number of jobs.
  - Given a balanced problem, each worker is assigned exactly once, and each job is performed by exactly one worker.
  - For an unbalanced problem “dummy” workers (in case there are more jobs than workers) or “dummy” jobs (in case there are more workers than jobs) are added to balance the problem.

- **Computer solutions**
  - A complete enumeration is not efficient even for moderately large problems (with m=8, m! > 40,000 is the number of assignments to enumerate).
  - The *Hungarian method* provides an efficient solution procedure.

- **Special cases**
  - A worker is unable to perform a particular job.
  - A worker can be assigned to more than one job.

Sourcing Network (with piecewise linear objective)

UniLeser has two ice-cream factories in its Indian operations. There are three main demand clusters in the country where distribution centers have been setup to fulfill demand. The COO would like to determine optimal operating levels for their plants and the distribution pattern that will satisfy customer demand.



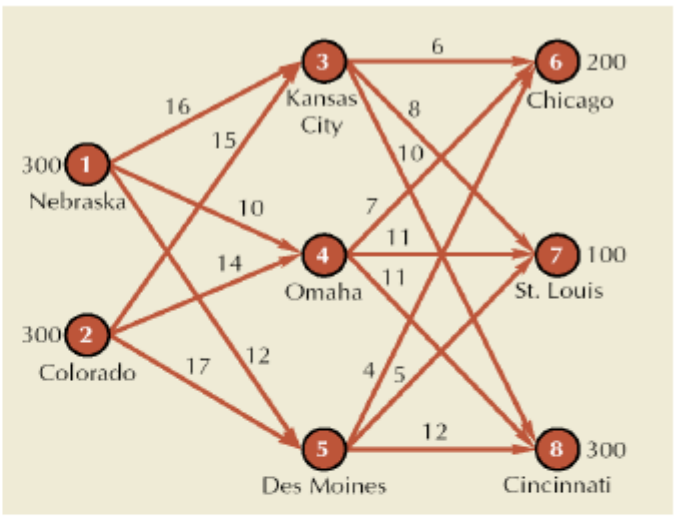
The two plants have the following variable cost of production:

Plant 1 Unit cost ($/Ton)	[0, 6 Tons] $10	[6Tons, 10Tons] $25
Plant 2 Unit cost ($/Ton)	[0, 5 Tons] $8	[5Tons, 11Tons] $28



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Transshipment Model



Farm	Distribution Centers		
	3. Kansas City	4. Omaha	5. Des Moines
1. Nebraska	$16	10	12
2. Colorado	15	14	17

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SOME OTHER MODELS

MONTPELIER Ski Company
Using a Transportation Model for Production Scheduling

- Montpelier is planning its production of skis for the months of July, August, and September.
- Production capacity and unit production cost will change from month to month.
- The company can use both regular time and overtime to produce skis.
- Production levels should meet both demand forecasts and end-of-quarter inventory requirement.
- Management would like to schedule production to minimize its costs for the quarter.

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MONTPELIER Ski Company (cont'd)

- **Data:**
  - Initial inventory = 200 pairs; Ending inventory required =1200 pairs
  - Production capacity for the overtime is equal to half the capacity for the regular time
  - Holding cost rate is 3% per month per ski.
  - Production capacity, and forecasted demand for this quarter (in pairs of skis), and production cost per unit (by months) are:

Month	Forecasted Demand	Production Capacity	Production Costs	
			Regular Time	Overtime
July	400	1000	25	30
August	600	800	26	32
September	1000	400	29	37

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MONTPELIER Ski Company (cont'd)

- **Demand**
  - Net demand to satisfy in July = 400 - 200 = 200 pairs
  - Net demand in August = 600
  - Net demand in September = 1000 + 1200 = 2200 pairs
- **Supply:**
  - Production capacities are thought of as supplies.
  - There are two sets of "supplies":
    - Set 1- Regular time supply (production capacity)
    - Set 2 - Overtime supply
- **Unit Cost**
  - Unit Cost=Unit production cost + (Unit holding cost per month)*(# of months in inventory)
  - Example:
    - A unit produced in July in regular time and sold in September costs: 25+(3%)(25)(2)=$26.50.

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Montpelier Ski Company (balanced transportation model formulation)

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Montpelier Ski Company (unbalanced transportation model formulation)

Montpelier Ski Company (balanced transportation model -- solution)

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Example: Sailco Inventory (from the textbook)

- Sailco Corporation must determine how many sailboats should be produced during each of the next four quarters (one quarter=3 months). The demand during each of the next four quarters is as follows: 40, 60, 75, and 25. Sailco must meet demands on time. At the beginning of the first quarter, Sailco has an inventory of 10 sailboats. At the beginning of each quarter, Sailco must decide how many sailboats should be produced during that quarter. For simplicity, we assume that sailboats manufactured during a quarter can be used to meet demand for that quarter. During each quarter, Sailco can produce up to 40 sailboats with regular-time labor at a total cost of $400 per sailboat. By having employees work overtime during a quarter, Sailco can produce additional sailboats with overtime labor at a total cost of $450 per sailboat.

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 3, page 101.

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Example: Sailco Inventory (from the textbook)

- At the end of each quarter (after production has occurred and the current quarter's demand has been satisfied), a carrying or holding cost of $20 per sailboat is incurred. Use linear programming to determine a production schedule to minimize the sum of production and inventory costs during the next four quarters.

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 3, page 101.

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Example: Sailco Inventory (Formulation without backlogging)

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 3, page 101.

Example: Sailco Inventory Formulation with Backlogging Allowed

Example: Red Tomato Tools

- The demand for Red Tomato's gardening tools from consumers is highly seasonal, peaking in the spring as people plant their gardens. Red Tomato has decided to use aggregate planning to overcome the obstacle of seasonal demand maximize profits. The options Red Tomato has for handling the seasonality are:
- Adding workers during the peak season
- Subcontracting out some of the work
- Building up inventory during the slow months
- Building up a backlog of orders that will be delivered late to customers.

Example: Red Tomato Tools

- The following table summarizes the demand forecast for Red Tomato Tools:

Month	*Demand Forecast*
January	1600
February	3000
March	3200
April	3800
May	2200
June	2200

Example: Red Tomato Tools

- The following table summarizes the unit costs for Red Tomato Tools:

Item	*Cost*
Material cost	$10/unit
Inventory holding cost	$2/unit/month
Marginal cost of backlog	$5/unit/month
Hiring/training costs	$300/worker
Layoff cost	$500/worker
Labor hours required	4/unit
Regular time cost	$4/hour
Overtime cost	$6/hour
Cost of subcontracting	$30/unit

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Example: Red Tomato Tools

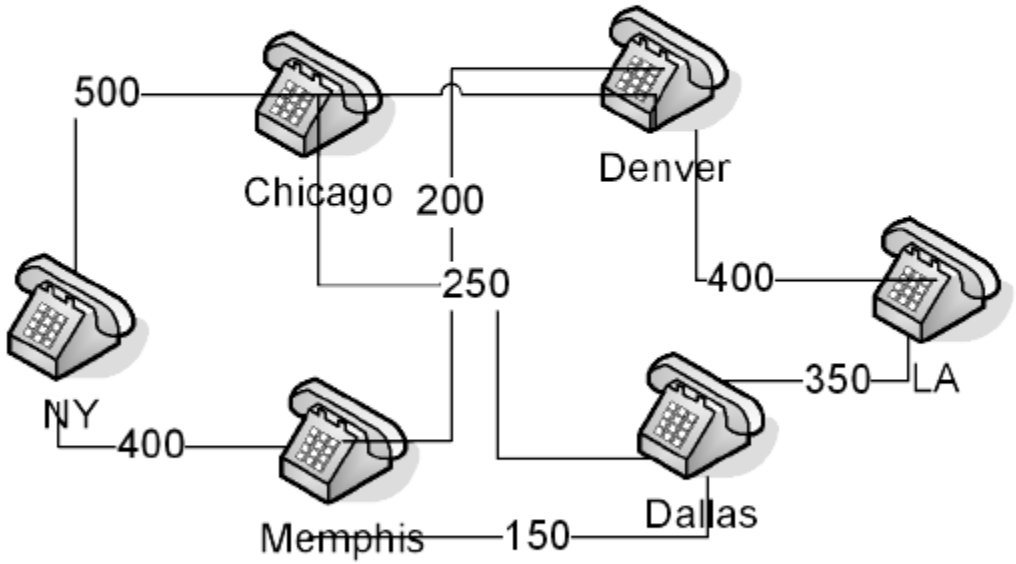
- **Red Tomato sells each tool to the retailers for $40. The company has a starting inventory in January of 1000 tools. At the beginning of January the company has a workforce of 80 employees. The plant has a total of 20 working days in each month, and each employee works eight hours per day on straight time and the rest on overtime. Because of labor rules, no employee works more than 10 hours of overtime per month.**
  - Red Tomato has no limits on subcontracting, inventories, and backlog.
  - All stockouts are backlogged and supplied from the following month's production.
  - Inventory costs are incurred on the ending inventory in the month.
  - The supply chain manager's goal is to end June with at least 500 units of inventory.
  - All demand has to be met (even backlogged).
  - Hiring/firing is only possible in the beginning of each month.

Formulate and LP and check your solution

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Network Maximal Flow LP

Telephone calls from New York to LA are transmitted as shown below. The arcs represent the number of lines that exist between the locations.



What is the maximum number of calls that can be transmitted from NY to LA?

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Network Maximal Flow LP

Maximize f

$s.t.$

$$\sum_{j=1}^{n} x_{1j} - \sum_{j=1}^{n} x_{j1} - f = 0$$

$$\sum_{j=1}^{n} x_{ij} - \sum_{j=1}^{n} x_{ji} = 0, i \neq 1, m$$

$$\sum_{j=1}^{n} x_{mj} - \sum_{j=1}^{n} x_{jm} + f = 0$$

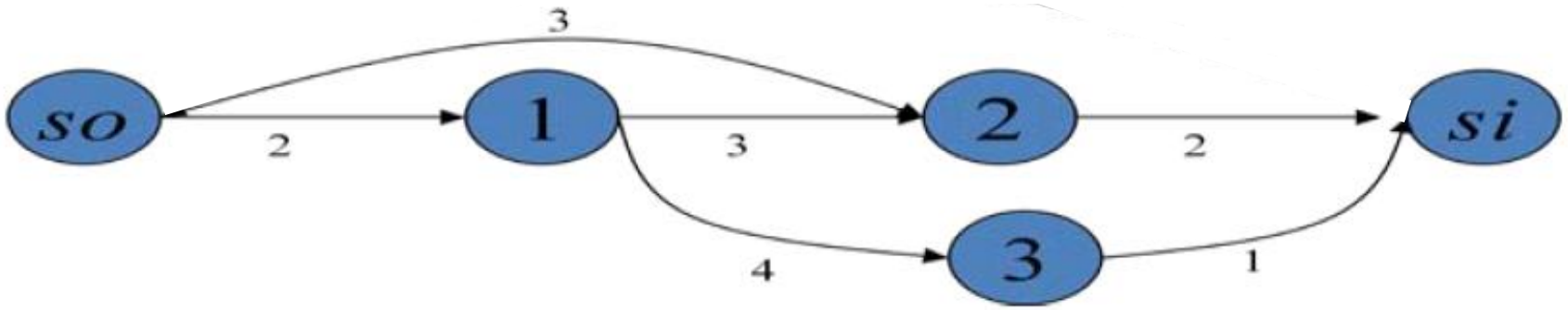
$$0 \leq x_{ij} \leq k_{ij}, \forall i, j$$

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Sunco Oil Example (Maximum Flow)

- Sunco Oil wants to ship the maximum possible amount of oil (per hour) via pipeline from node so to node s_i.
- The various arcs represent pipelines of different diameters.
- The maximum number of barrels of oil that can be pumped through each arc the **arc capacity**.

Arc	*Capacity*
(*so*,1)	2
(*so*,2)	3
(1,2)	3
(1,3)	4
(3,*si*)	1
(2,*si*)	2



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Sunco Oil Example (Maximum Flow) -- solution

Sunco Oil Example (Maximum Flow) – alternative formulation without the artificial arc

Other interesting problems

- Shortest path problem
- Minimum cut (dual of maximal flow)
- Minimal spanning tree

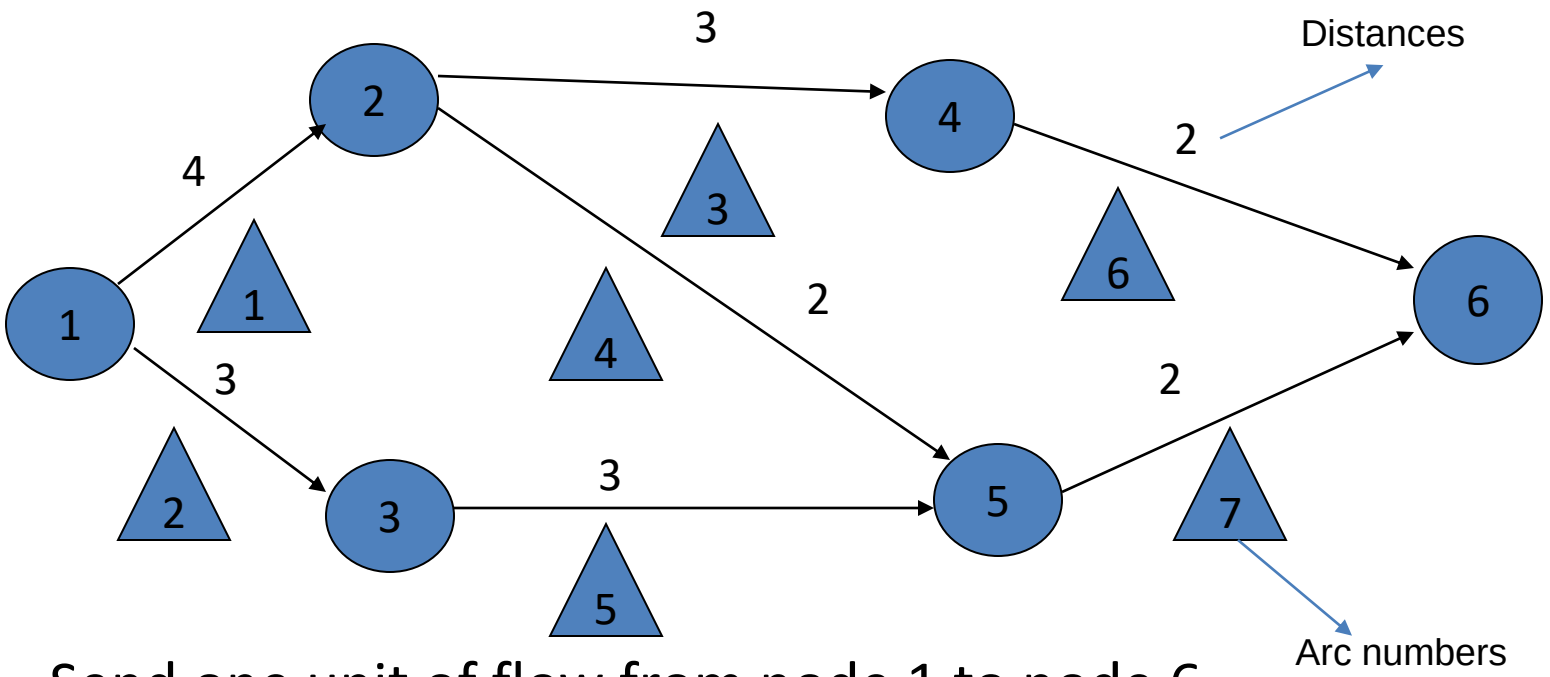
Shortest Path Problem

Borrowed from Dr. Jarrod Goentzel

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LP Formulation of the Shortest Path Problem

- Consider the following shortest path problem from node 1 to node 6
- Δ: denotes a link



- Send one unit of flow from node 1 to node 6

LP Formulation of the Shortest Path Problem

- Use flow conservation constraints
  - (Outflow from any node – inflow to that node) = 0
  - For origin = 1
  - For destination = -1
  - For all other nodes = 0
  - Let xj denote the flow along link j, j = 1, 2, .., 7, xj = 0 or 1
  - It turns out that this 0-1 constraints can be replaced by $0 \leq xj \leq 1$, which can in turn be replaced by $xj \geq 0$

LP Formulation of the Shortest Path Problem

- Min $4x1 + 3x2 + 3x3 + 2x4 + 3x5 + 2x6 + 2x7$

  S.t.

- $x1 + x2 = 1$
- $-x1 + x3 + x4 = 0$
- $-x2 + x5 = 0$
- $-x3 + x6 = 0$
- $-x4 - x5 + x7 = 0$
- $-x6 - x7 = -1$
- $Xj \geq 0$, $j = 1, 2, \ldots, 7$ (xj integers)

Equipment Replacement Problem

- Assume that a new car (or machine) has been purchased for $12,000 at time 0.
- The cost of maintaining the car during a year depends on the age of the car at the beginning of the year, as given in the table below.
- In order to avoid the high maintenance cost associated with an older car, we may trade in the car and purchase a new car. The trade-in depends on the age of the car at the time of trade-in.
- To simplify the computations we assume that at any time it costs $12,000 to purchase a new car.
- The goal is to minimize the net cost incurred during the next five years.

Age of Car (Years)	Annual Maintenance cost	Age of Car (Years)	Trade-in Price
0	$2,000	1	$7,000
1	$4,000	2	$6,000
2	$5,000	3	$2,000
3	$9,000	4	$1,000
4	$12,000	5	$0

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Equipment Replacement Problem

Equipment Replacement Problem

Some of the slides borrowed from Dr. Prashant Yadav & Dr. Jianjun Xu

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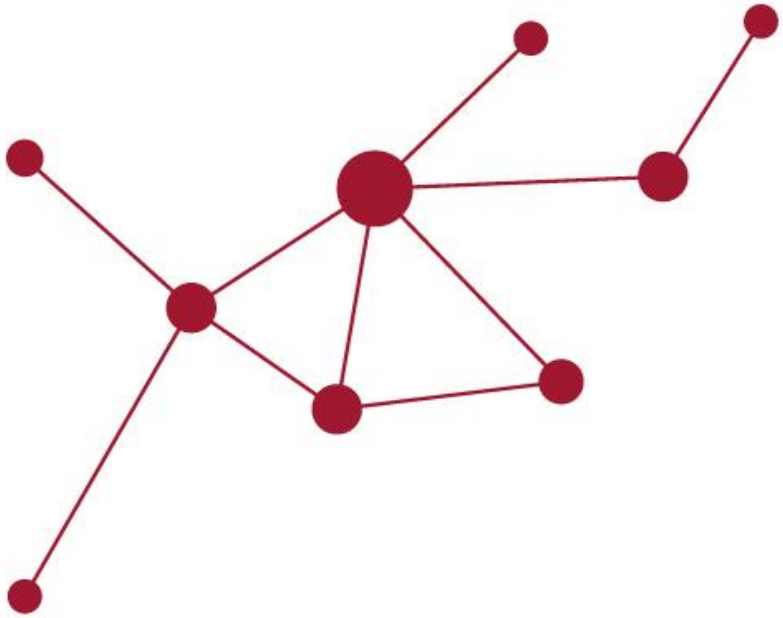
TALENT HUB FOR SUPPLY CHAIN

Zaragoza Logistics Center

Linear Programming Part 3

September 14, 2026

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Excel Solver Examples

Product Mix Simplest Model

	Resource requirements		
	Large Bricks	Small Bricks	Revenue
Product	(#/unit)	(#/unit)	($/unit)
Table	2	2	20
Chair	1	2	15

There are 6 large bricks and 8 small bricks available for production. Goal is to maximize the revenue given the raw materials requirements.

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Product Mix Simplest Model Solution

The Lego Production Problem						
	Tables	Chairs				
Profit	$20.00	$15.00				
	Bill of Materials		Total Used		Available	
Large Bricks	2	1	6	<=	6	
Small Bricks	2	2	8	<=	8	
	Tables	Chairs			Total Profit	
Production Quantity	2	2			$70.00	

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Diet Problem

- A prison is trying to decide what to feed its prisoners. They would like to offer some combination of milk, beans, and oranges. Their goal is to minimize cost, subject to meeting the minimum nutritional requirements imposed by law. The cost and nutritional content of each food, along with the minimum nutritional requirements are shown below.

	Milk (gallons)	Navy Beans (cups)	Oranges	Minimum daily requirement
Niacin (mg)	3.2	4.9	0.8	13.0
Thiamin (mg)	1.12	1.3	0.19	1.5
Vitamin C (mg)	32.0	0.0	93.0	45.0
Cost ($)	2.0	0.2	0.25	

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Diet Problem Solution

The Prison Diet Problem						
	Milk (gal.)	Beans (cups)	Oranges			
Cost	$2.00	$0.20	$0.25			
						Minimum
	Nutritional Contents (mg)			**Total**		**Requirement**
Niacin (mg)	3.2	4.9	0.8	13	>=	13
Thiamin (mg)	1.12	1.3	0.19	3.438215932	>=	1.5
Vitamin C (mg)	32	0	93	45	>=	45
						Total Cost
Quantity (per prisoner)	0	2.57406188	0.48387097			$0.64

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Security Force Scheduling Problem

Time Period	Number of Officers Needed
12 a.m. – 4 a.m.	5
4 a.m. – 8 a.m.	7
8 a.m. – 12 p.m.	15
12 p.m. – 4 p.m.	8
4 p.m. – 8 p.m.	12
8 p.m. – 12 a.m.	9

- Union contract requires all employees to work 8 consecutive hours
- Goal: Hire the minimum number of officers needed to cover all shifts

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Security Force Scheduling Problem Solution: *please remember that depending on how you define workshifts (i.e., number of workshifts a day and the starting-ending times of each shift), you will probably get a different schedule and total number of officers.*

Security Force Scheduling Problem

	Shift Works Time Period?								
	Shift 1	Shift 2	Shift 3	Shift 4	Shift 5	Shift 6	Number		Minimum
Time Period	12a–8a	4a–12p	8a–4p	12p–8p	4p–12a	8p–4a	Working		Required
12am – 4am	1					1	5	>=	5
4am – 8am	1	1					12	>=	7
8am – 12pm		1	1				15	>=	15
12pm – 4pm			1	1			8	>=	8
4pm – 8pm				1	1		12	>=	12
8pm – 12am					1	1	12	>=	9
							Total		
Number Working	5	7	8	0	12	0	32		

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Transportation Problem

- A company has two plants producing a certain product that is to be shipped to three distribution centers. The unit production costs are the same at the two plants, and the shipping cost per unit is shown below.

		Distribution Center		
		1	2	3
Plant	A	$4	$6	$4
	B	$6	$5	$2

- Shipments are made once per week. During each week, each plant produces at most 70 units and each distribution center needs at least 40 units. How many units should be shipped from each plant to each distribution center, so as to minimize cost?

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Transportation Problem Solution

A Distribution Problem

Cost	Distribution Center 1	Distribution Center 2	Distribution Center 3
Plant A	$4	$6	$4
Plant B	$6	$5	$2

Shipment Quantities	Distribution Center 1	Distribution Center 2	Distribution Center 3	Shipped		Available
Plant A	40	10	0	50	<=	70
Plant B	0	30	40	70	<=	70
Shipped	40	40	40			
	>=	>=	>=			Total Cost
Needed	40	40	40			$450

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Assignment Problem

- The coach of a swim team needs to assign swimmers to a 200-yard medley relay team (four swimmers, each swim 50 yards of one of the four strokes). Since most of the best swimmers are very fast in more than one stroke, it is not clear which swimmer should be assigned to each of the four strokes. The five fastest swimmers and their best times (in seconds) they have achieved in each of the strokes (for 50 yards) are

		Strokes			
		Backstroke	Breaststroke	Butterfly	Freestyle
Swimmer	Carl	37.7	43.4	33.3	29.2
	Chris	32.9	33.1	28.5	26.4
	David	33.8	42.2	38.9	29.6
	Tony	37.0	34.7	30.4	28.5
	Ken	34.4	41.8	33.6	31.1

How should the swimmers be assigned to make the fastest relay team?

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Assignment Problem Solution

An Assignment Problem

Best Times	Backstroke	Breastroke	Butterfly	Freestyle
Carl	37.7	43.4	33.3	29.2
Chris	32.9	33.1	28.5	26.4
David	33.8	42.2	38.9	29.6
Tony	37.0	34.7	30.4	28.5
Ken	34.4	41.8	33.6	31.1

Assignment	Backstroke	Breastroke	Butterfly	Freestyle			
Carl	0	0	0	1	1	<=	1
Chris	0	0	1	0	1	<=	1
David	1	0	0	0	1	<=	1
Tony	0	1	0	0	1	<=	1
Ken	0	0	0	0	0	<=	1
	1	1	1	1			
	=	=	=	=			Time
	1	1	1	1			126.2

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Sensitivity Analysis

- ***Benefits of sensitivity analysis:***

- **Typically many of the parameters are estimates of quantities**
  - What-if analysis reveals how close each of these estimates needs to be to avoid erroneous optimal solution
  - Pinpoints the sensitive parameters where extra care is needed to refine their estimates
- **Businesses operate in a dynamic environment**
  - Conditions may change
  - Indicates whether the current solution is still optimal
    - If yes, then there is no need to solve the problem again (important for large scale problems)
- **Shadow price analysis is used to help management do cost-benefit tradeoff.**

Sensitivity Analysis

The answers to following questions are given by the sensitivity analysis:

- **How does the solution change**
  - If the RHS of a constraint changes?
  - If an objective function coefficient changes?

- **What is the range of optimality and feasibility?**
  - Allowable change in the objective function coefficient
  - Allowable change in the RHS.

Sensitivity Analysis (Output)

- **Changing Cells (Objective Function Coefficients/Decision Variables)**

- **Final Value**
  - The value of the variable in the optimal solution
- **Reduced Cost**
  - Increase in the objective function value per unit increase in the value of a zero-valued variable (for small increases)
- **Allowable Increase/Decrease**
  - Defines the range of the cost coefficients in the objective function for which the current solution (values of the variables in the optimal solution) will not change

Sensitivity Analysis (Output)

- **Constraints**

- **Final Value**
  - The usage of the resource in the optimal solution
- **Shadow price**
  - The change in the value of the objective function per unit increase in the right hand side of the constraint:
  - ΔZ=(Shadow Price)*(Δ RHS) (this is for a maximization problem)
  - Only valid if change in RHS is within the allowable range of the RHS
  - Captures the incremental value of a scarce resource
  - All constraints have a SP (for non-binding constraints = 0)
- **Allowable Increase/Decrease**
  - Defines the range of values of the RHS for which the shadow price is valid and hence for which the new objective function value can be calculated (not the range for which the current solution will not change)

Sensitivity Analysis (Example: Ice Cream Problem)

- Ken and Larry, Inc., supplies its ice cream parlors with three flavors of ice cream: chocolate, vanilla, and banana. Due to extremely hot weather and a high demand for its products, the company has run short of its supply of ingredients: milk, sugar, and cream. Hence, they will not be able to fill all the orders received from their retail outlets, the ice cream parlors. Due to these circumstances, the company has decided to choose the amount of each flavor to produce that will maximize total profit, given the constraints on supply of the basic ingredients.

	Chocolate	Vanilla	Banana
Milk	0.45	0.50	0.40
Sugar	0.50	0.40	0.40
Cream	0.10	0.15	0.20

- The chocolate, vanilla, and banana flavors generate, respectively, $1, $0.90, and $0.95 of profit per gallon sold. The company has only 200 gallons of milk, 150 pounds of sugar, and 60 gallons of cream left in its inventory.

Sensitivity Analysis (Example: Ice Cream Problem)

- **The linear programming formulation for this problem is:**
  - C= gallons of chocolate ice cream produced
  - V= gallons of vanilla ice cream produced
  - B= gallons of banana ice cream produced

- **Maximize Profit= 1.00C + 0.90V + 0.95B**
- **Subject to**
  - Milk: 0.45C+0.50V+0.40B <= 200 gallons
  - Sugar: 0.50C+0.40V+0.40B<=150 pounds
  - Cream: 0.10C+0.15V+0.20B<=60 gallons
  - C,V,B >=0

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$G$1	Units Produced Total Profit	341.25	341.25

Objective function value

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$1	Units Produced Chocolate	0	0	Contin
$C$1	Units Produced Vanilla	300	300	Contin
$D$1	Units Produced Banana	75	75	Contin

Final value

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$E$7	Milk Used	180	$E$7<=$G$7	Not Binding	20
$E$8	Sugar Used	150	$E$8<=$G$8	Binding	0
$E$9	Cream Used	60	$E$9<=$G$9	Binding	0

Resource usage

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Sensitivity Analysis (Example: Ice Cream Problem)

- ***The optimal solution is:***

- **Final values:**
  - Chocolate: 0 gallons
  - Vanilla: 300 gallons
  - Banana: 75 gallons
- **Resource usage:**
  - Total gallons of milk used is 180
  - Total pounds of sugar used is 150
  - Total gallons of cream used is 60
- **Objective function value:**
  - Total profit is $341.25.

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Range of optimality:
[0.8875, 0.95]

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$12	Units Produced Chocolate	0	-0.0375	1	0.0375	1E+30
$C$12	Units Produced Vanilla	300	0	0.9	0.05	0.0125
$D$12	Units Produced Banana	75	0	0.95	0.021428571	0.05

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$7	Milk Used	180	0	200	1E+30	20
$E$8	Sugar Used	150	1.875	150	10	30
$E$9	Cream Used	60	1	60	15	3.75

Range of feasibility:
[120, 160]

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Sensitivity Analysis (Example: Ice Cream Problem)

- **Answer the following independent questions:**
  - What is the optimal solution and total profit?
  - Suppose the profit per gallon of banana changes to $1.00. Will the optimal solution change and what can be said about the effect on total profit?
  - Suppose the profit per gallon of banana changes to $0.92. Will the optimal solution change and what can be said about the effect on total profit?
  - Suppose the company discovers that three gallons of cream have gone sour and so must be thrown out. Will the optimal solution change and what can be said about the effect on total profit?
  - Suppose the company has the opportunity to buy an additional 15 pounds of sugar at a total cost of $15. Should they? Explain.
  - Fill in all the sensitivity report information for the milk constraint, given just the optimal solution for the problem. Explain how you were able to deduce each number.

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Sensitivity Analysis (Example: Ice Cream Problem)

What is the optimal solution and total profit?

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$G$1	Units Produced Total Profit	341.25	341.25

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$1	Units Produced Chocolate	0	0	Contin
$C$1	Units Produced Vanilla	300	300	Contin
$D$1	Units Produced Banana	75	75	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$E$7	Milk Used	180	$E$7<=$G$7	Not Binding	20
$E$8	Sugar Used	150	$E$8<=$G$8	Binding	0
$E$9	Cream Used	60	$E$9<=$G$9	Binding	0

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Sensitivity Analysis (Example: Ice Cream Problem)

Suppose the profit per gallon of banana changes to $1.00. Will the optimal solution change and what can be said about the effect on total profit?

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$12	Units Produced Chocolate	0	-0.0375	1	0.0375	1E+30
$C$12	Units Produced Vanilla	300	0	0.9	0.05	0.0125
$D$12	Units Produced Banana	75	0	0.95	0.021428571	0.05

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$7	Milk Used	180	0	200	1E+30	20
$E$8	Sugar Used	150	1.875	150	10	30
$E$9	Cream Used	60	1	60	15	3.75

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Sensitivity Analysis (Example: Ice Cream Problem)

Suppose the profit per gallon of banana changes to $0.92. Will the optimal solution change and what can be said about the effect on total profit?

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$12	Units Produced Chocolate	0	-0.0375	1	0.0375	1E+30
$C$12	Units Produced Vanilla	300	0	0.9	0.05	0.0125
$D$12	Units Produced Banana	75	0	0.95	0.021428571	0.05

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$7	Milk Used	180	0	200	1E+30	20
$E$8	Sugar Used	150	1.875	150	10	30
$E$9	Cream Used	60	1	60	15	3.75

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Sensitivity Analysis (Example: Ice Cream Problem)

Suppose the company discovers that three gallons of cream have gone sour and so must be thrown out. Will the optimal solution change and what can be said about the effect on total profit?

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$12	Units Produced Chocolate	0	-0.0375	1	0.0375	1E+30
$C$12	Units Produced Vanilla	300	0	0.9	0.05	0.0125
$D$12	Units Produced Banana	75	0	0.95	0.021428571	0.05

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$7	Milk Used	180	0	200	1E+30	20
$E$8	Sugar Used	150	1.875	150	10	30
$E$9	Cream Used	60	1	60	15	3.75

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Sensitivity Analysis (Example: Ice Cream Problem)

Suppose the company has the opportunity to buy an additional 15 pounds of sugar at a total cost of $15. Should they? Explain.

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$12	Units Produced Chocolate	0	-0.0375	1	0.0375	1E+30
$C$12	Units Produced Vanilla	300	0	0.9	0.05	0.0125
$D$12	Units Produced Banana	75	0	0.95	0.021428571	0.05

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$7	Milk Used	180	0	200	1E+30	20
$E$8	Sugar Used	150	1.875	150	10	30
$E$9	Cream Used	60	1	60	15	3.75

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The 100 Percent Rule (Sensitivity Analysis)

- ***The 100 Percent Rule for Simultaneous Changes in Objective Function Coefficients:***
- **For all objective function coefficients that are changed, sum the percentages of the allowable increases and the allowable decreases represented by the changes. If the sum of the percentage changes does not exceed 100%, the optimal solution will not change.**
- ***The 100 Percent Rule for Simultaneous Changes in Right-Hand Sides:***
- **For all right-hand sides that are changed, sum the percentages of the allowable increases and the allowable decreases. If the sum of the percentage changes does not exceed 100%, then the shadow prices will not change.**

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Sensitivity Analysis (Example: Ice Cream Problem)

Suppose the profit per gallon of banana changes to $0.96 and the profit per gallon of Vanilla changes to $0.92. Will the optimal solution change?

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$12	Units Produced Chocolate	0	-0.0375	1	0.0375	1E+30
$C$12	Units Produced Vanilla	300	0	0.9	0.05	0.0125
$D$12	Units Produced Banana	75	0	0.95	0.021428571	0.05

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$7	Milk Used	180	0	200	1E+30	20
$E$8	Sugar Used	150	1.875	150	10	30
$E$9	Cream Used	60	1	60	15	3.75

Sensitivity Analysis (Example: Ice Cream Problem)

Suppose the company discovers that 2 gallons of cream have gone sour and has the opportunity to buy an additional 5 pounds of sugar. Will the shadow prices change?

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$12	Units Produced Chocolate	0	-0.0375	1	0.0375	1E+30
$C$12	Units Produced Vanilla	300	0	0.9	0.05	0.0125
$D$12	Units Produced Banana	75	0	0.95	0.021428571	0.05

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$7	Milk Used	180	0	200	1E+30	20
$E$8	Sugar Used	150	1.875	150	10	30
$E$9	Cream Used	60	1	60	15	3.75

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Duality

- **Associated with every linear program, there is a dual linear programming problem**
- **Dual viewpoint often provides computational advantage and economic insights**
  - The variables of the dual problem can be interpreted as prices associated with the constraints of the original problem
  - Dual simplex is often used when finding the new optimal solution after a constraint is added to a LP or changing an LP's RHS

Primal

$$
\begin{array}{lll}
\min & \boldsymbol{c}'\boldsymbol{x} & \\
\text{s.t.} & \boldsymbol{a}_i'\boldsymbol{x} \geq b_i & i \in M_1 \\
& \boldsymbol{a}_i'\boldsymbol{x} \leq b_i & i \in M_2 \\
& \boldsymbol{a}_i'\boldsymbol{x} = b_i & i \in M_3 \\
& x_j \geq 0 & j \in N_1 \\
& x_j \leq 0 & j \in N_2 \\
& x_j \gtrless 0 & j \in N_3
\end{array}
$$

Dual

$$
\begin{array}{lll}
\max & \boldsymbol{p}'\boldsymbol{b} & \\
\text{s.t.} & p_i \geq 0 & i \in M_1 \\
& p_i \leq 0 & i \in M_1 \\
& p_i \gtrless 0 & i \in M_3 \\
& \boldsymbol{p}'\boldsymbol{A}_j \leq c_j & j \in N_1 \\
& \boldsymbol{p}'\boldsymbol{A}_j \geq c_j & j \in N_2 \\
& \boldsymbol{p}'\boldsymbol{A}_j = c_j & j \in N_3
\end{array}
$$

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Duality

- **An example:**

$$
\begin{array}{lrcl}
\min & x_1 + 2x_2 + 3x_3 & & \\
\text{s.t.} & -x_1 + 3x_2 & = & 5 \\
& 2x_1 - x_2 + 3x_3 & \geq & 6 \\
& & x_3 \leq & 4 \\
& x_1 \geq 0 & & \\
& x_2 \leq 0 & & \\
& x_3 \text{ free}, & &
\end{array}
\qquad
\begin{array}{lrcl}
\max & 5p_1 + 6p_2 + 4p_3 & & \\
\text{s.t.} & p_1 \text{ free} & & \\
& p_2 \geq 0 & & \\
& p_3 \leq 0 & & \\
& -p_1 + 2p_2 & \leq & 1 \\
& 3p_1 - p_2 & \geq & 2 \\
& 3p_2 + p_3 & = & 3.
\end{array}
$$

Primal	min	max	dual
constraints	$\geq b_i$ $\leq b_i$ $= b_i$	≥ 0 ≤ 0 $\gtrless 0$	variables
variables	≥ 0 ≤ 0 $\gtrless 0$	$\leq c_j$ $\geq c_j$ $= c_j$	constraints

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Duality

- **Relations between primal and dual problems:**
  - Here primal is a minimization problem
  - And dual is a maximization problem

	Finite opt.	Unbounded	Infeasible
Finite opt.	*		
Unbounded			*
Infeasible		*	*

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Diet Problem

- A prison is trying to decide what to feed its prisoners. They would like to offer some combination of milk, beans, and oranges. Their goal is to minimize cost, subject to meeting the minimum nutritional requirements imposed by law. The cost and nutritional content of each food, along with the minimum nutritional requirements are shown below.

	Milk (gallons)	Navy Beans (cups)	Oranges	Minimum daily requirement
Niacin (mg)	3.2	4.9	0.8	13.0
Thiamin (mg)	1.12	1.3	0.19	1.5
Vitamin C (mg)	32.0	0.0	93.0	45.0
Cost ($)	2.0	0.2	0.25	

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Diet Problem

Primal :

$$Min\ Z = 2x_1 + 0.2x_2 + 0.25x_3$$

$$s.t.$$

$$3.2x_1 + 4.9x_2 + 0.8x_3 \geq 13$$

$$1.12x_1 + 1.3x_2 + 0.19x_3 \geq 1.5$$

$$32x_1 + 0 + 93x_3 \geq 45$$

$$x_1, x_2, x_3 \geq 0$$

Dual :

$$Max\ Z = 13y_1 + 1.5y_2 + 45y_3$$

$$s.t.$$

$$3.2y_1 + 1.12y_2 + 32y_3 \leq 2$$

$$4.9y_1 + 1.3y_2 + 0y_3 \leq 0.2$$

$$0.8y_1 + 0.19y_2 + 93y_3 \leq 0.25$$

$$y_1, y_2, y_3 \geq 0$$

Imagine a pharmaceutical firm that produces in pill from each nutrient

Tries to convince the dietician to buy pills :

* supply the nutrients directly rather than purchase of various foods (milk, etc.)

Problem faced by the pharmaceutical company :

* Determine prices of nutrients to maximize profit
* Be competitive with real food.

--> the cost of a unit food i made synthetically from pure nutrients bought from the druggist must be no greater than c_i

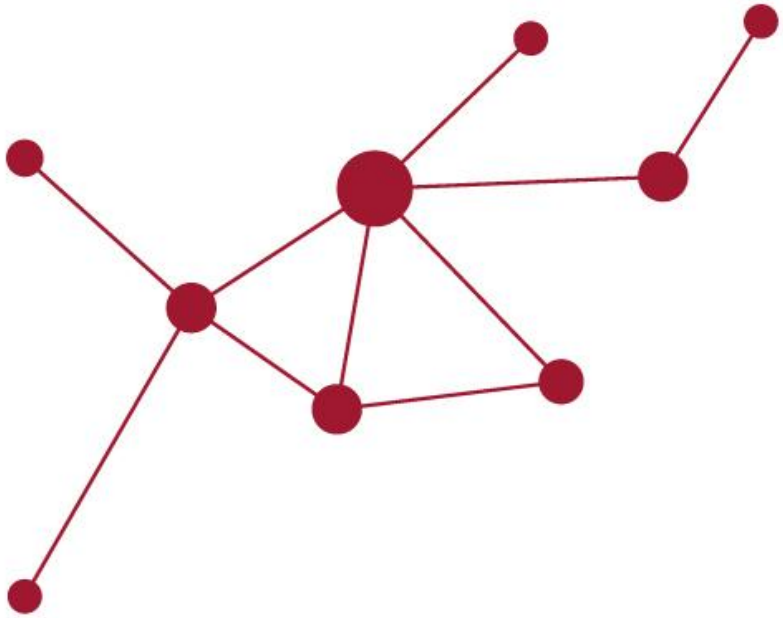
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TALENT HUB FOR SUPPLY CHAIN

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Thanks!



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INTEGER PROGRAMMING

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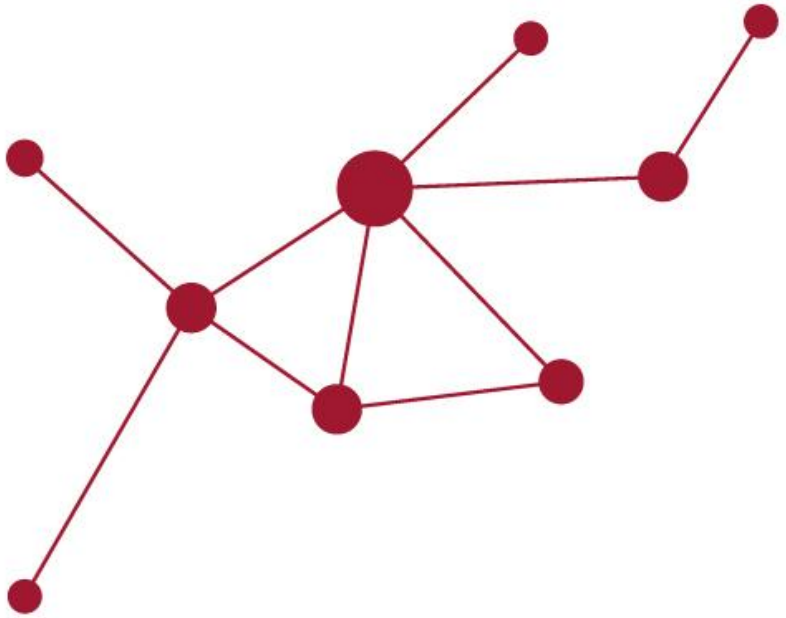
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Integer Programming

September 14, 2026

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The Knapsack Problem

- **A thief breaks into a museum. Fabulous paintings, sculptures, and jewels are everywhere. The thief has a good eye for the value of these objects, and knows that each will fetch hundreds or thousands of dollars on the clandestine art collector's market.**
  - *Problem: S/he only has brought a single knapsack to the scene of robbery!!*
    - *Cannot take away more than what the knapsack can hold!*
  - **What items should the thief take to maximize the haul?**

Some slides borrowed from Dr. Prashant Yadav

The Knapsack Problem

- **More formally, the *binary knapsack problem*:**
  - The thief must choose among n items, where the ith item worth v_i dollars and weighs w_i pounds
  - Carrying at most W pounds, maximize value
    - Note: assume v_i, w_i, and W are all integers
    - [0,1] each item must be taken or left in entirety
- **A variation, the *fractional knapsack problem*:**
  - Thief can take fractions of items
  - Think of items in 0-1 problem as gold ingots, in fractional problem as buckets of gold dust

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A logistical application of the knapsack problem!



https://www.pinterest.com/wdevoogd/loads/

- **When are non-integer solutions okay?**
  - Solution is naturally divisible
    - e.g. pounds, $, hours
  - Solution represents a rate
    - e.g. Units per week
  - Solution for only planning purposes
- **When is rounding okay?**
  - When numbers are large
    - e.g. 114.286 to 114 is probably okay.
  - Not okay
    - e.g. Family planning: have a child (yes or no)

Challenges of Rounding?

- **Rounded solution may not be feasible**
- **Rounded solution may not be close to optimal**
- **There can be many rounded solutions**
  - Consider a problem with 30 variables that are non-integer in the LP solution. How many possible rounded solutions are there?
    - Approximately 1 billion (2^{30})

Example: TBA Airlines

- **TBA Airlines is a small regional company that specializes in short flights in small airplanes. The management has decided to expand its operations. The basic issue is to purchase more small airplanes to add some new short flights or to start moving into the national market by purchasing some large airplanes. Each small airplane and large airplane will cost $5 and $50 million, respectively. And the net profit will be $1 and $5 million from small and large planes, respectively. The company can buy at most two more small planes, and the total budget for the new airplanes is $100 million. The LP formulation is:**

Max Profit = S + 5L

- Subject to
  - $5S + 50L \leq 100$
  - $S \leq 2$
  - $S, L \geq 0$

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Example: TBA Airlines

- **Solution to the LP problem is:**
  - S=2: Purchase 2 small planes
  - L=1.8: Purchase 1.8 large planes
- **The total profit is *$11 million*.**
- **Non-integer solution doesn't make sense here.**
- **What happens if we round 1.8 to 1?**
- **The integer problem, IP, is given below:**
  - Max Profit = S + 5L
    - Subject to
      - $5S + 50L \leq 100$
      - $S \leq 2$
      - $S, L \geq 0$ and ***S, L are integers***
- **Solution to IP?**

IP gives an optimal solution of S=0 and L=2 with a profit of *$10 million*.

Important Properties for Solving Integer Programs

- **LP solution is an upper (lower) bound on IP solution for a maximization (minimization) problem**
- **If LP is infeasible then so is IP**
- **If LP solution is integral (all variables have integer values), then it is the IP solution.**
- **Linear Programming Solution to IPs:**
  1. Some LP problems will always have integer solutions
     - transportation problem
     - assignment problem
     - min-cost network flow
  2. Solve as linear program and round. Can violate constraints, and be non-optimal. Works OK if
     - integer variables take on large values
     - accuracy of constraints is questionable

Applications of Binary variables

- **Making yes/no type of decisions:**
  - Build a factory?
  - Open/operate a warehouse?
  - Do a project?
- **Set-covering problems**
  - Make assignments that “cover” a set of requirements
- **Fixed Cost**
  - If a product produced → incur fixed cost
  - If a warehouse operated → incur fixed cost

Example: Distribution Channel Restructuring

- A specialty paper manufacturing firm currently has 5 warehouses in Seattle, Denver, St Louis, Atlanta and Philadelphia.
- The warehouse capacities and annual operating costs are as below

Warehouse	Annual Operating cost	Capacity
SEA	300000	500000
DEN	250000	500000
STL	220000	500000
ATL	220000	500000
PHI	240000	500000

Example: Distribution Channel Restructuring

- **The US market for paper is divided into 6 major regions :**
  - North West, South West, Upper Midwest, Lower Midwest, South East and North East
- **The annual demand in each market is as below**

Market	NW	SW	UMW	LMW	NE	SE
Annual demand Tons	320000	200000	160000	220000	350000	175000

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Example: Distribution Channel Restructuring

- The average cost of transporting a ton from each warehouse to each market is as tabulated below

Transportation cost	NW	SW	UMW	LMW	NE	SE
SEA	$ 1.00	$ 3.00	$ 3.00	$ 4.00	$ 6.00	$ 6.00
DEN	$ 2.00	$ 2.50	$ 1.50	$ 3.50	$ 4.00	$ 5.00
STL	$ 4.00	$ 2.00	$ 2.00	$ 1.00	$ 3.00	$ 3.00
ATL	$ 5.00	$ 3.00	$ 3.00	$ 2.00	$ 2.50	$ 1.00
PHI	$ 5.50	$ 4.00	$ 3.50	$ 3.00	$ 1.00	$ 2.50

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Example: Distribution Channel Restructuring

- **Goal:**
  - Evaluate whether it is economical to keep all its current warehouses operational.
  - Come up with a new distribution plan for the restructured network

Warehouse Location Problem: IP Model

Example: Capital Budgeting

- Mercer Development is considering the potential of four different development projects. Each project would be completed in at most three years. The required cash outflow for each project is given in the table below, along with the net present value of each project to Mercer, and the cash that is available for each year

	Cash outflow required ($ million)				*Cash Available*
	Project 1	Project 2	Project 3	Project 4	($ million)
Year 1	10	8	6	12	30
Year 2	8	5	4	0	15
Year 3	8	0	6	0	12
NPV	35	18	24	16	

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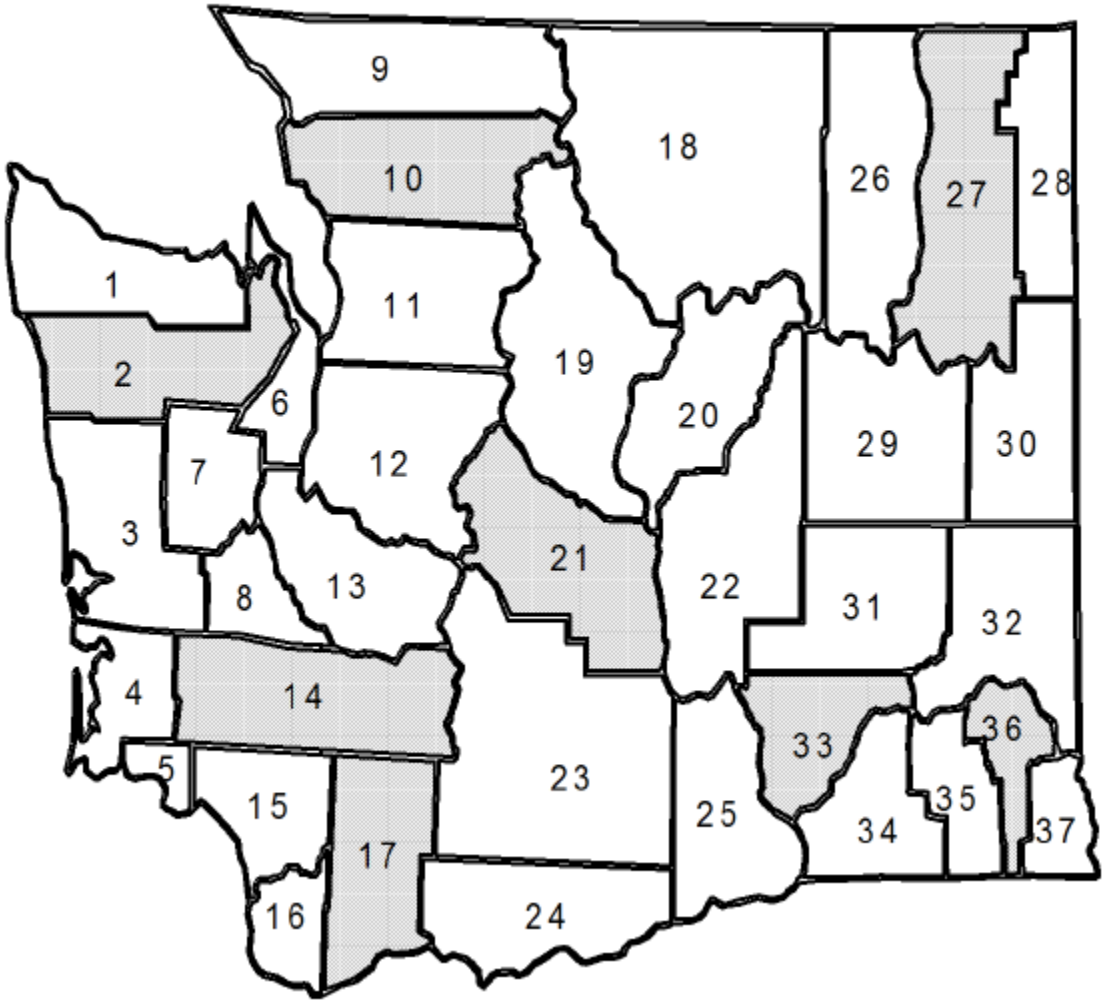
Example: Capital Budgeting

- At least one of projects 1, 2, or 3
- Project 3 cannot be done unless project 4 is done
- Either project 1 or project 2, but not both
- No more than 2 projects in total

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Example: Set Covering Problem

- The Washington State Legislature is trying to decide on locations at which to base search-and-rescue teams. The teams are expensive, and hence they would like as few as possible. However, since response time is critical, they would like every county to either have a team located in that county, or in an adjacent county. Where should the teams be located?



Counties

1. Clallum
2. Jefferson
3. Grays Harbor
4. Pacific
5. Wahkiakum
6. Kitsap
7. Mason
8. Thurston
9. Whatcom
10. Skagit
11. Snohomish
12. King
13. Pierce
14. Lewis
15. Cowlitz
16. Clark
17. Skamania
18. Okanogan
19. Chelan
20. Douglas
21. Kittitas
22. Grant
23. Yakima
24. Klickitat
25. Benton
26. Ferry
27. Stevens
28. Pend Oreille
29. Lincoln
30. Spokane
31. Adams
32. Whitman
33. Franklin
34. Walla Walla
35. Columbia
36. Garfield
37. Asotin

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Another Fixed Cost Example

- Gandy Cloth Company is capable of manufacturing three types of clothing: shirts, shorts, and pants. The manufacture of each type of clothing requires that Gandy have the appropriate type of machinery available. This machinery can be leased at the following rates: shirt machinery, $200/week; shorts machinery, $150/week; pants machinery, $100/week. Each type of clothing requires the amount of cloth and labor given in the table below. The sales price and variable cost are also given in the table.

	Labor Hours	Cloth (sq. yds.)	Sales Price	Variable Cost
Shirts	3	4	$12	$6
Shorts	2	3	$8	$4
Pants	6	4	$15	$8
Available:	150 hours	160 sq. yds.		

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Example: Fly-Right Airplane Company

- The Fly-Right Airplane Company builds small jet airplanes to sell to corporations for the use of their executives. To meet the needs of these executives, the company's customers sometimes order a custom design of the airplanes being purchased. When this occurs, a substantial start-up cost is incurred to initiate the production of these airplanes.

- Fly-Right has recently received purchase requests from three customers with short deadlines. However, because the company's production facilities are already almost completely tied up filling previous order, it will not be able to accept all three orders. Therefore, a decision now needs to be made on the number of airplanes the company will agree to produce (if any) for each of the three customers.

Example: Fly-Right Airplane Company

- The relevant data are given in the table below.
- The first row gives the start-up cost required to initiate the production of airplanes for each customer.
- Once production is under way, the marginal net revenue from each airplane is shown in the second row.
- The third row gives the percentage of the available production capacity that would be used for each airplane produced.
- The last row indicates the maximum number of airplanes requested by each customer (but less will be accepted).

	Customer 1	Customer 2	Customer 3
Start-up cost	$3 million	$2 million	0
Marginal net revenue	$2 million	$3 million	$0.8 million
Capacity used per plane	20%	40%	20%
Maximum order	3 planes	2 planes	5 planes

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Example: Fly-Right Airplane Company

- Fly-Right now wants to determine how many airplanes to produce for each customer (if any) to maximize the company's total profit (total net revenue minus start-up costs). Formulate the problem and solve it using Solver.

Example: Olympic Gymnastic Team

- You are the coach of the Olympic Gymnastic Team consisting of six gymnasts. Exactly three gymnasts are required to enter both the balance beam and floor exercises. You must also enter a total of four people in each event. The estimated score that each individual gymnast can attain in each event is shown in the table below. Formulate and solve a linear programming spreadsheet model with binary variables to determine how to maximize the total score attained.

	Balance Beam	Floor Exercise
Gymnast 1	9.1	8.4
Gymnast 2	8.2	8.8
Gymnast 3	8.7	7.8
Gymnast 4	7.4	8.6
Gymnast 5	9.0	8.4
Gymnast 6	9.3	8.2

IP with Piecewise Linear Functions

- **Suppose the objective function of the linear program is as follows:**

$$f(x) = \begin{cases} c_1 x + b_1, a_1 \le x \le a_2 \\ c_2 x + b_2, a_2 \le x \le a_3 \\ \ldots\ldots\ldots \\ c_{n-1} x + b_{n-1}, a_{n-1} \le x \le a_n \end{cases}$$

- **You can solve this as an LP as follows:**
  - Replace *f(x)* by: $f(x) = z_1 f(a_1) + z_2 f(a_2) + \ldots + z_n f(a_n)$
  - Add the following constraints to the problem:

$$z_1 \le y_1$$

$$z_2 \le y_1 + y_2$$

$$\ldots\ldots$$

$$z_{n-1} \le y_{n-2} + y_{n-1}$$

$$z_n \le y_n$$

$$\sum_{i=1}^{n-1} y_i = 1 \text{ and } \sum_{j=1}^{n} z_j = 1$$

$$x = \sum_{i=1}^{n} z_i a_i$$

$$y_i \in \{0,1\}, \forall i \in \{1,2,..,n-1\} \text{ and } z_i \ge 0, \forall i \in \{1,2,..,n\}$$

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Example: IP with Piecewise Linear Functions (from the textbook)

- Euing Gas produces two types of gasoline (gas 1 and gas 2) from two types of oil (oil 1 and oil 2). Each gallon of gas 1 must contain at least 50 percent oil 1, and each gallon of gas 2 must contain at least 60 percent oil 1. Each gallon of gas 1 can be sold for $0.12, and each gallon of gas 2 for $0.14. Currently, 500 gallons of oil 1 and 1000 gallons of oil 2 are available. As many as, 1500 more gallons of oil 1 can be purchased at the following prices: first 500 gallons, $0.25 per gallon; next 500 gallons, $0.20 per gallon; next 500 gallons, $0.15 per gallon. Formulate an IP that will maximize Euing's profits (revenues-purchasing costs).

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 9, page 492.

Euling Gas Example

Except for the fact that the cost of purchasing additional oil 1 is a piecewise linear function, this is a straightforward blending problem. With this in mind, we define

x = amount of oil 1 purchased

x_{ij} = amount of oil i used to produce gas j $\quad (i, j = 1, 2)$

Then (in cents)

Total revenue − cost of purchasing oil 1 = $12(x_{11} + x_{21}) + 14(x_{12} + x_{22}) - c(x)$

As we have seen previously,

$$c(x) = \begin{cases} 25x & (0 \le x \le 500) \\ 20x + 2{,}500 & (500 \le x \le 1{,}000) \\ 15x + 7{,}500 & (1{,}000 \le x \le 1{,}500) \end{cases}$$

Thus, Euing's objective function is to maximize

$$z = 12x_{11} + 12x_{21} + 14x_{12} + 14x_{22} - c(x)$$

Euing faces the following constraints:

Constraint 1 Euing can use at most $x + 500$ gallons of oil 1.

Constraint 2 Euing can use at most 1,000 gallons of oil 2

Constraint 3 The oil mixed to make gas 1 must be at least 50% oil 1.

Constraint 4 The oil mixed to make gas 2 must be at least 60% oil 1

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 9, page 492.

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Euling Gas Example

Constraint 1 yields

$$x_{11} + x_{12} \leq x + 500$$

Constraint 2 yields

$$x_{21} + x_{22} \leq 1{,}000$$

Constraint 3 yields

$$\frac{x_{11}}{x_{11} + x_{21}} \geq 0.5 \quad \text{or} \quad 0.5x_{11} - 0.5x_{21} \geq 0$$

Constraint 4 yields

$$\frac{x_{12}}{x_{12} + x_{22}} \geq 0.6 \quad \text{or} \quad 0.4x_{12} - 0.6x_{22} \geq 0$$



c(x) is nonlinear!

$$\begin{aligned}
\max z = {} & 12x_{11} + 12x_{21} + 14x_{12} + 14x_{22} - c(x) \\
\text{s t} \quad & x_{11} + x_{12} \leq x + 500 \\
& x_{21} + x_{22} \leq 1{,}000 \\
& 0.5x_{11} - 0.5x_{21} \geq 0 \\
& 0.4x_{12} - 0.6x_{22} \geq 0 \\
& x_{ij} \geq 0,\ 0 \leq x \leq 1{,}500
\end{aligned}$$

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 9, page 492.

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Euling Gas Example: transforming the problem into an IP

Step 1 Replace $c(x)$ by $c(x) = z_1c(0) + z_2c(500) + z_3c(1{,}000) + z_4c(1{,}500)$

Step 2 Add the following constraints:

$$x = 0z_1 + 500z_2 + 1{,}000z_3 + 1{,}500z_4$$

$$z_1 \leq y_1,\ z_2 \leq y_1 + y_2,\ z_3 \leq y_2 + y_3,\ z_4 \leq y_3$$

$$z_1 + z_2 + z_3 + z_4 = 1, \quad y_1 + y_2 + y_3 = 1$$

$$y_i = 0 \text{ or } 1\ (i = 1, 2, 3);\ z_i \geq 0\ (i = 1, 2, 3, 4)$$

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 9, page 492.

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Euling Gas Example: transforming the problem into an IP

$$\max z = 12x_{11} + 12x_{21} + 14x_{12} + 14x_{22} - z_1c(0) - z_2c(500) - z_3c(1{,}000) - z_4c(1{,}500)$$

$$\text{s.t.} \quad x_{11} + x_{12} \leq x + 500$$

$$x_{21} + x_{22} \leq 1{,}000$$

$$0.5x_{11} - 0.5x_{21} \geq 0$$

$$0.4x_{12} - 0.6x_{22} \geq 0$$

$$x = 0z_1 + 500z_2 + 1{,}000z_3 + 1{,}500z_4$$

$$z_1 \leq y_1$$

$$z_2 \leq y_1 + y_2$$

$$z_3 \leq y_2 + y_3$$

$$z_4 \leq y_3$$

$$y_1 + y_2 + y_3 = 1$$

$$z_1 + z_2 + z_3 + z_4 = 1$$

$$y_i = 0 \text{ or } 1 \quad (i = 1, 2, 3); \; z_i \geq 0$$

$$x_{ij} \geq 0$$

Source: "Operations Research: Applications and Algorithms" by Wayne L. Winston, Chapter 9, page 492.

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Plant & Warehouse Location Model: Notation

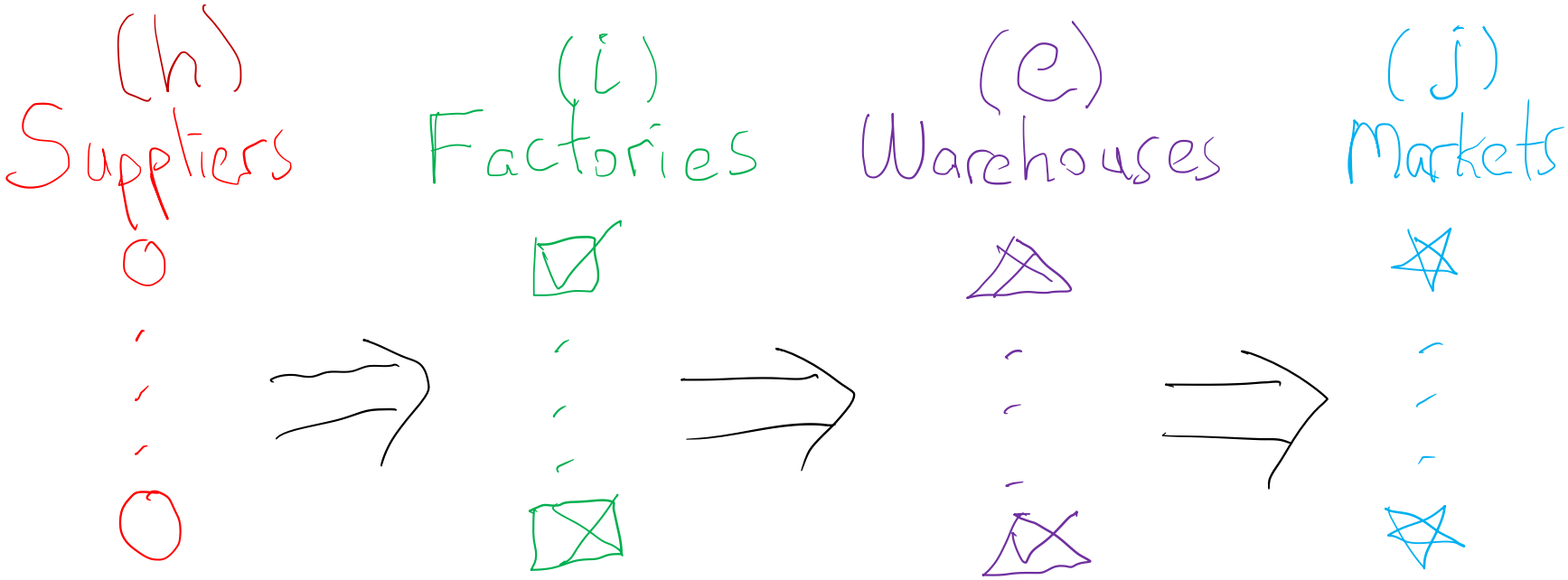
- m: number of markets or demand points
- n: number of potential factory locations
- l: number of suppliers
- t: number of potential warehouse locations
- D_j: annual demand from customer j
- K_i: potential capacity of factory at site i
- S_h: supply capacity at supplier h
- W_e: potential warehouse capacity at site e
- F_i: fixed cost of locating a plant at site i
- f_e: fixed cost of locating a warehouse at site e
- C_{hi}: cost of shipping one unit from supply source h to factory i
- C_{ie}: cost of producing and shipping one unit from factory i to warehouse e
- C_{ej}: cost of shipping one unit from warehouse e to customer j

Slide borrowed from Prof. Prashant Yadav

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Plant & Warehouse Location Model: Decision Variables

- y_i= 1 if factory is located at site i, 0 otherwise
- y_e=1 if warehouse is located at site e, 0 otherwise
- x_{hi}: quantity shipped from supplier h to factory at site i
- x_{ie}: quantity shipped from factory at site i to warehouse e
- x_{ej}: quantity shipped from warehouse e to market j



Slide borrowed from Prof. Prashant Yadav

Plant & Warehouse Location Model: Formulation

$$Min \sum_{i=1}^{n} F_i Y_i + \sum_{e=1}^{t} f_e y_e + \sum_{h=1}^{l} \sum_{i=1}^{n} c_{hi} x_{hi} + \sum_{i=1}^{n} \sum_{e=1}^{t} c_{ie} x_{ie} + \sum_{e=1}^{t} \sum_{j=1}^{m} c_{ej} x_{ej}$$

$$\sum_{i=1}^{n} x_{hi} \le S_h, \forall h = 1,..,l$$

$$\sum_{e=1}^{t} x_{ej} \ge D_j, \forall j = 1,..,m$$

$$\sum_{h=1}^{l} x_{hi} - \sum_{e=1}^{t} x_{ie} \ge 0, \forall i = 1,..,n \qquad \sum_{i=1}^{n} x_{ie} - \sum_{j=1}^{m} x_{ej} \ge 0, \forall e = 1,..,t$$

$$\sum_{e=1}^{t} x_{ie} \le K_i y_i, \forall i = 1,..,n \qquad \sum_{j=1}^{m} x_{ej} \le W_e y_e, \forall e = 1,..,t$$

$$y_i, y_e \in \{0,1\}, x_{ej}, x_{ie}, x_{hi} \ge 0$$

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Single Machine Scheduling Problem with Due Dates

Problem borrowed from the textbook by Winston (Chapter 9, page 538)

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Problem

- Four jobs must be processed on a single machine. The time required to perform each job and the due date for each job are shown in the table below:

Job (i)	Time to Perform Job (minutes)	Due Date of Job (d_i)
1	7	End of minute 14
2	5	End of minute 13
3	9	End of minute 18
4	11	End of minute 15

- Formulate an IP to solve this problem such that the total time jobs are delayed is minimized

IP Formulation

- Decision variables:

- Formulation:

A Mathematical Programming Approach to Shipment Consolidation & Dispatching

Notes borrowed from the book "Introduction to Logistics System Planning and Control" by Ghiani, Laporte, and Musmanno.

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Problem

- A producer has to choose the best way of delivering timely a set of orders to its customers over a planning horizon made up of *T* days.
- Decisions to be made are:
  - Best mode of transportation for each shipment
  - How orders will be consolidated
  - The features of vehicle schedules (start times, intermediate stops (if any), the order in which the stops are visited, etc.)

Notation

k : order ($k \in K$)

i_k : destination of order k

w_k : weight of order k

r_k : release time of order k (the day in which order k is ready for delivery)

d_k : deadline for order k

The company has two options to deliver:

1) renting "one-way" truck trips
   - a rented truck may follow any route r of a pre-established set R
   -with each route r are associated a set of stops S_r (visited in a given order), a (fixed) cost f_r, and a capacity on the total weight (q_r).
   - τ_{kr} is the number of days needed to deliver order k on route r.

2) using an LTL carrier
   - cost of transporting order k is g_k and it takes τ_k' days.

Decision Variables

$$x_{krt} = \begin{cases} 1, \text{if order k is assigned to route r starting on day t} \\ 0, \text{otherwise} \end{cases}$$

$$y_{rt} = \begin{cases} 1, \text{if route r is operated on day t} \\ 0, \text{otherwise} \end{cases}$$

$$v_k = \begin{cases} 1, \text{if order k is transported by the common carrier} \\ 0, \text{otherwise} \end{cases}$$

v_k is defined only if $r_k + \tau_k^{'} \le d_k$

Formulation

- Minimize

$$\sum_{r \in R} \sum_{t=1}^{T} f_r y_{rt} + \sum_{k \in K} g_k v_k$$

- Subject to

$$\sum_{k: r_k \le t \le d_k - \tau_{kr}, i_k \in S_r} w_k x_{krt} \le q_r y_{rt}, r \in R, t = 1,..,T. \quad (1)$$

$$\sum_{r: i_k \in S_r} \sum_{t: r_k \le t \le d_k - \tau_{kr}} x_{krt} + v_k = 1, k \in K \quad (2)$$

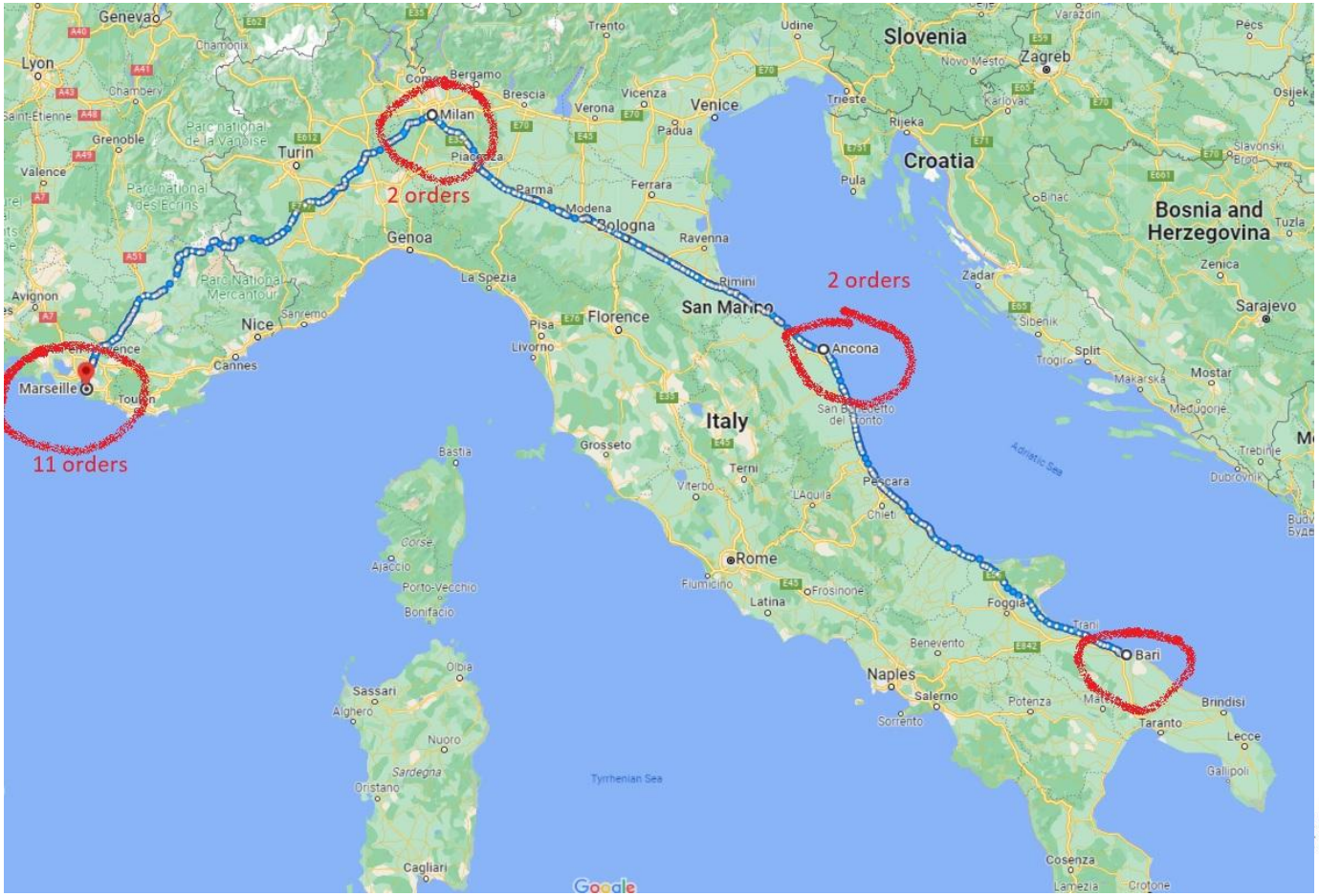
$$x_{krt} \in \{0,1\}, k \in K, r \in R, t = 1,..,T$$

$$y_{rt} \in \{0,1\}, r \in R, t = 1,..,T$$

$$v_k \in \{0,1\}, k \in K$$

Example

Oxximet manufactures semi-finished chemical products in a plant located close to Bari (Italy). The main customers are located in Marseille (France), Milan (Italy) and Ancona (Italy). Oxximet rents one-way truck trips visiting one or more customers. No common carrier is used. Trip fixed costs are reported in Table 6.12. Truck capacity is 260 hundred kilograms. Last 11 June, 15 customer orders were waiting to be satisfied (Table 6.13). Oxximet solved the model described before with a six-day horizon (T=6). No trips were allowed on the subsequent Saturday and Sunday. The input data and the optimal solution are reported in the following tables.



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Trip fixed costs (in euros) in the Oxximet problem

Trip	Fixed Cost
Bari-Marseille	800
Bari-Milan	750
Bari-Ancona	400
Bari-Ancona-Milan	780
Bari-Ancona-Marseille	830
Bari-Milan-Marseille	830
Bari-Ancona-Milan-Marseille	860

Customers orders (in hundreds of kilograms) in the Oxximet problem

Order	Customer Location	Release time	Deadline	Weight
1	Marseille	16-Jun	3-Jul	15.44
2	Ancona	14-Jun	15-Jun	242.65
3	Ancona	14-Jun	16-Jun	102.54
4	Milan	13-Jun	13-Jun	100.46
5	Marseille	14-Jun	15-Jun	154.79
6	Marseille	14-Jun	14-Jun	78.53
7	Marseille	13-Jun	13-Jun	56.89
8	Marseille	14-Jun	14-Jun	45.42
9	Marseille	13-Jun	13-Jun	39.55
10	Marseille	11-Jun	15-Jun	207.34
11	Marseille	11-Jun	16-Jun	19.05
12	Marseille	11-Jun	16-Jun	19.59
13	Marseille	11-Jun	16-Jun	35.23
14	Marseille	11-Jun	16-Jun	61.54
15	Milan	11-Jun	16-Jun	38.31

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Optimal trip schedule in the Oxximet problem

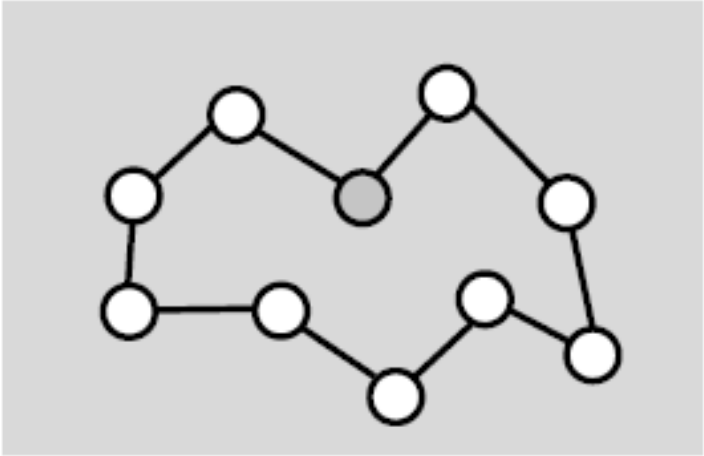
Day	Trip	Orders
11-Jun	Bari-Marseille	10,13
12-Jun	none	none
13-Jun	Bari-Milan-Marseille	4,7,9,11,15
13-Jun	Bari-Marseille	5,6,12
14-Jun	Bari-Ancona	2
14-Jun	Bari-Ancona-Marseille	3,8,14
17-Jun	Bari Marseille	1

Traveling Salesman Problem

Notes borrowed from the presentation by Massimo Paolucci (Vehicle Routing Problems)

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Problem



- TSP is a node routing problem without operational constraints
- A traveling salesman must visit his customers located in different cities and come back home
  - *Example: Picking up orders at different locations and bringing them back to the packaging area*
- Single vehicle
- Minimum cost/distance of traveling from city i to city j are given
- Solution: A minimum cost route which starts and ends at the origin and reaches each customer once
  - *Number of possible solutions is (n-1)!*

IP Formulation

$$x_{ij} = \begin{cases} 1 & \text{if vertex } j \text{ is visited immediately after vertex } i \\ 0 & \text{otherwise} \end{cases}$$

$$Min \sum_i \sum_j c_{ij} x_{ij}$$

$$s.t.$$

$$\sum_i x_{ij} = 1, \forall j$$

$$\sum_j x_{ij} = 1, \forall i$$

$$\sum_{i,j \in S} x_{ij} \leq |S| - 1, \forall S \subset V, |S| > 1)$$

* "V" is the set of all nodes (n nodes in total) and "S" is any subset of "V"
* Node 1 is the origin.

Last constraint can be changed by the following (Miller-Tucker-Zemlin Formulation):

$$u_1 = 1$$

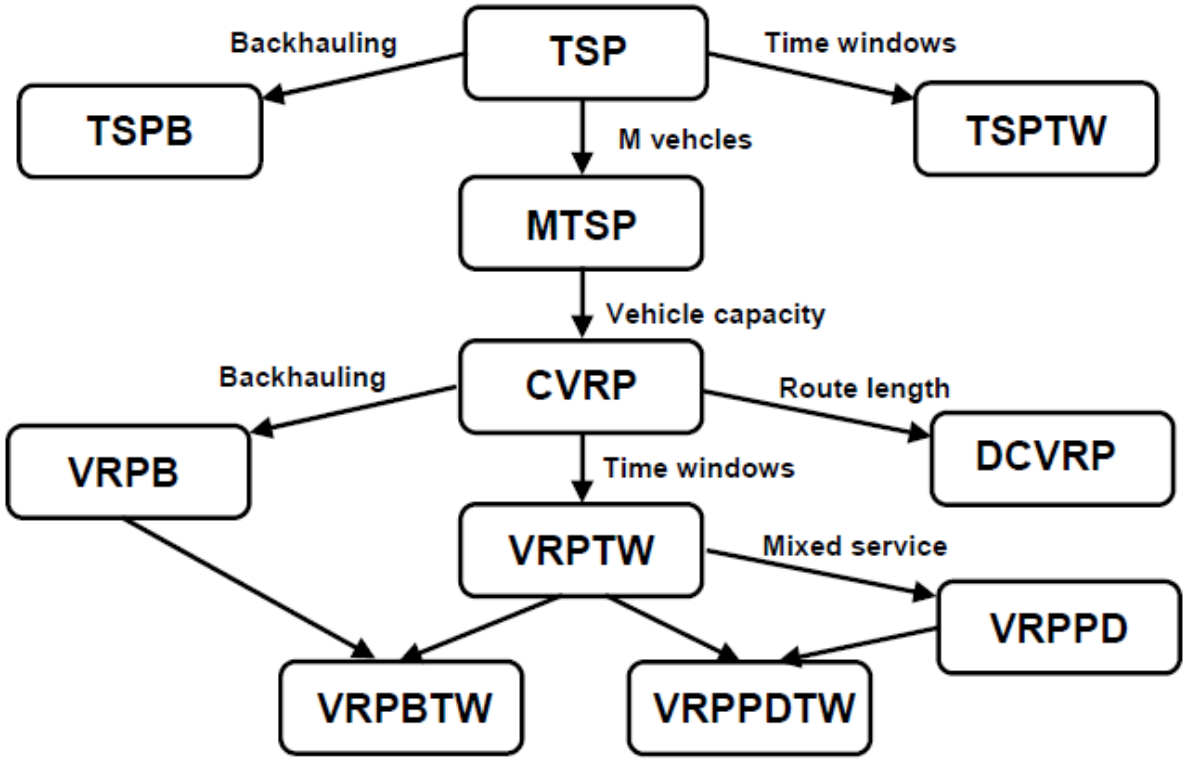
$$2 \leq u_i \leq n, \forall i \geq 2$$

$$u_i - u_j + 1 \leq (n-1)(1 - x_{ij}), \forall i, j \neq 1$$

Vehicle Routing Problem (VRP)

VRP – Node routing problems

- Relationships among the basic VRP (NRP) problems



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Vehicle Routing Problem (VRP)

VRP – Node routing problems

- Traveling Salesman Problem (TSP)
- Traveling Salesman Problem with Backhauls (TSPB)
- Traveling Salesman Problem with Time Windows (TSPTW)
- Multiple Traveling Salesman Problem (MTSP)
- Capacitated Vehicle Routing Problem (CVRP)
- Distance Constrained Vehicle Routing Problem (DCVRP)
- Vehicle Routing Problem with Backhauls (VRPB)
- Vehicle Routing Problem with Time Windows (VRPTW)
- Vehicle Routing Problem with Pickup and Delivery (VRPPD)

Integer Programming Algorithms

- **Use a linear program**
  - round to integer solution (what if not feasible?)
- **Search**
  - Branch and bound (integer ranges)
- **Cutting Planes**
  - Many variants

Branch and Bound

- **Lets first consider 0-1 programs.**
  - Exponential solution: try all $\{0,1\}^n$
- **Branch-and-bound solution:**
  - Solve the linear relaxation of the problem.
    - If the solution is integer, then we are done. Otherwise create two new subproblems by branching on a fractional variable.
  - A subproblem is not active when:
    - You used the subproblem to branch on,
    - All the variables in the solution are integer
    - The subproblem is infeasible
    - You can fathom the subproblem by a bounding argument
  - If it can be shown that a subproblem is not active, then prune it.
  - Choose an active subproblem and branch on a fractional variable. Repeat until there are no active subproblems.

Cutting Planes

- **Add constraints to a linear program until the optimal solution takes on integer values**
- **Careful not to change the problem by adding constraints!**
- **A "cut" is a special type of constraint satisfying the following:**
  - every feasible integer solution is feasible for the cut, and
  - the current fractional solution is not feasible for the cut.

Helpful tricks to formulate IPs: Logic Constraints

Either/Or Constraints

Either $f_1(x_1,\ldots,x_n) \le b_1$

or $f_2(x_1,\ldots,x_n) \le b_2$

IP Formulation :

Let $y \in \{0,1\}$

$f_1(x_1,\ldots,x_n) \le b_1 + My$

$f_2(x_1,\ldots,x_n) \le b_2 + M(1-y)$

$y = 0 \rightarrow$ first constraint must hold

$y = 1 \rightarrow$ second constraint must hold

Helpful tricks to formulate IPs: Logic Constraints

If - Then Constraints :

We want to ensure that if $f(x_1,\ldots,x_n) > 0$, then $g(x_1,\ldots,x_n) \geq 0$ must be satisfied.

IP Formulation :

Let $y \in \{0,1\}$

$f(x_1,\ldots,x_n) \leq M(1-y)$

$-g(x_1,\ldots,x_n) \leq My$

if f > 0, then $y = 0$

$y = 0 \rightarrow g \geq 0$

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Helpful tricks to formulate IPs: K out of N Constraints

N constraints :

$$f_1(x_1,\ldots,x_n) \le b_1$$
$$f_2(x_1,\ldots,x_n) \le b_2$$
$$\ldots\ldots$$
$$\ldots\ldots$$
$$f_N(x_1,\ldots,x_n) \le b_N$$

At least K out of N must be satisfied :

IP Formulation :

$$f_1(x_1,\ldots,x_n) \le b_1 + M(1-y_1)$$
$$f_2(x_1,\ldots,x_n) \le b_2 + M(1-y_2)$$
$$\ldots\ldots\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots\ldots\ldots$$
$$f_N(x_1,\ldots,x_n) \le b_N + M(1-y_N)$$
$$\sum_{i=1}^{N} y_i = K$$
$$y_i \in \{0,1\}, \forall i \in \{1,2,..,N\}$$

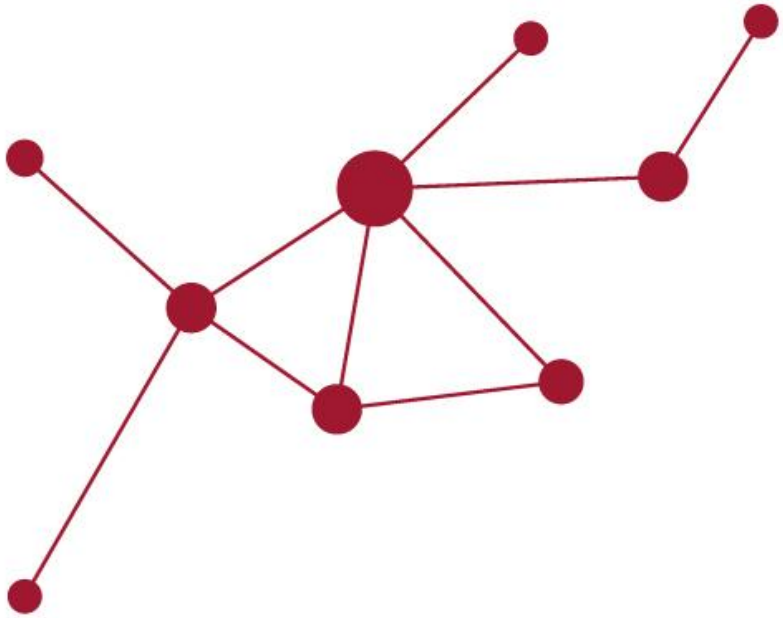
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TALENT HUB FOR SUPPLY CHAIN

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Thanks!



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GOAL PROGRAMMING

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TALENT HUB FOR SUPPLY CHAIN

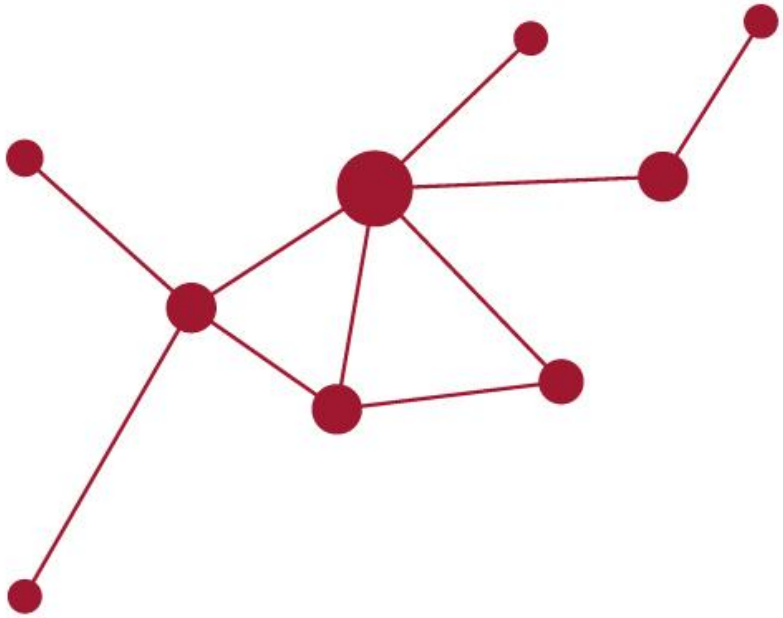
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Goal Programming

September 14, 2026

Some notes borrowed from Dr. Jianjun Xu

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Goal Programming used when:

- There are multiple objectives and there may be no point in the LP's feasible region satisfying all objectives

- Multiple optima to choose among these

Weighted Goal Programming

- A common characteristic of many management science models (linear programming, integer programming, nonlinear programming) is that they have a single objective function.
- It is not always possible to fit all managerial objectives into a single objective function. Managerial objectives might include:
  - Maintain stable profits.
  - Increase market share.
  - Diversify the product line.
  - Maintain stable prices.
  - Improve worker morale.
  - Maintain family control of the business.
  - Increase company prestige.
- Weighted goal programming provides a way of striving toward several objectives simultaneously.

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Weighted Goal Programming

- With weighted goal programming, the objective is to
  - Minimize W = weighted sum of deviations from the goals.
  - The weights are the penalty weights for missing the goal.
- Introduce new changing cells, Amount Over and Amount Under, that will measure how much the current solution is over or under each goal.
- The Amount Over and Amount Under changing cells are forced to maintain the correct value with the following constraints:

$$\textit{Level Achieved} - \textit{Amount Over} + \textit{Amount Under} = \textit{Goal}$$

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Example:

An advertisement agency needs to come up with a TV advertising schedule for an auto company, which has three goals:

- Goal #1: Ads should be seen by at least 40 million High-Income-Men (HIM)
- Goal #2: Ads should be seen by at least 60 million Low-Income-People (LIP)
- Goal #3: Ads should be seen by at least 35 million High-Income-Women (HIW)

The company can spend at most $600,000 on ads, to be shown during football games and soap operas. Average viewership information is as follows:

	Millions of Viewers			
	HIM	LIP	HIW	Cost per ad
Football Game	*7*	*10*	*5*	*$100,000*
Soap Opera	*3*	*5*	*4*	*$60,000*

Formulate the LP to find solutions meeting all 3 goals

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Example cont'd (advertising):

No feasible solution exists to the formulation on the previous slide → cannot meet all three goals at the s same time. Therefore, a solution needs to be find where some goals are met and some others are not. Define the penalties (of not meeting the goals) as follows:

Each million exposures by which they fall short of HIM goal → $200,000 (lost sales related costs)
Each million exposures by which they fall short of LIP goal → $100,000 (lost sales related costs)
Each million exposures by which they fall short of HIW goal → $50,000 (lost sales related costs)

NEW OBJECTIVE: Minimize the "deviation" from the goals:

Introduce "deviational variables":
S_i^+: amount by which we numerically exceed the ith goal (over goal)
S_i^-: amount by which we are numerically under the ith goal (under goal)

What would the LP look like in this case?

Example cont'd (advertising):

No feasible solution exists to the formulation on the previous slide → cannot meet all three goals at the s same time. Therefore, a solution needs to be find where some goals are met and some others are not. Define the penalties (of not meeting the goals) as follows:

Each million exposures by which they fall short of HIM goal → $200,000 (lost sales related costs)
Each million exposures by which they fall short of LIP goal → $100,000 (lost sales related costs)
Each million exposures by which they fall short of HIW goal → $50,000 (lost sales related costs)

NEW OBJECTIVE: Minimize the "deviation" from the goals:

Introduce "deviational variables":
S_i^+: *amount by which we numerically exceed the* i^{th} *goal (over goal)*
S_i^-: *amount by which we are numerically under the* i^{th} *goal (under goal)*

Alternative formulation (Correct, but I would not recommend)

Example cont'd (advertising):

No feasible solution exists to the formulation on the previous slide → cannot meet all three goals at the s same time. Therefore, a solution needs to be find where some goals are met and some others are not. Define the penalties (of not meeting the goals) as follows:

Each million exposures by which they fall short of HIM goal → $200,000 (lost sales related costs)
Each million exposures by which they fall short of LIP goal → $100,000 (lost sales related costs)
Each million exposures by which they fall short of HIW goal → $50,000 (lost sales related costs)

NEW OBJECTIVE: Minimize the "deviation" from the goals:

Introduce "deviational variables":
S_i^+: amount by which we numerically exceed the ith goal (over goal)
S_i^-: amount by which we are numerically under the ith goal (under goal)

What would the LP look like if the budget restriction is a goal as well, and there is a $1 penalty for each dollar by which this goal is unmet?

What goes into the objective function in general?

If failure to meet goal happens when "attained value" < "target value" → S_i^-
If failure to meet goal happens when "attained value" > "target value" → S_i^+
If failure to meet goal happens when "attained value" ≠ "target value" → both S_i^+ and S_i^-

What happens if you cannot precisely determine weights?

"**Preemptive Goal Programming**" technique can be used
Rank the goals in decreasing order of importance: $p_1 >> p_2 >> .... >> p_n$ (the "p" are the objective function coefficients of the goals, where the first goal is the most important)

Step 1: Solve to satisfy Goal #1 as much as possible
Step 2: Among all points satisfying Goal #1 (if possible), come as close as possible to satisfying Goal #2 (without hurting the higher priority goals – i.e., increasing the deviation from a higher priority goal)
Step 3: Continue until you are done with all the goals (or you end up with a unique solution at a particular step)

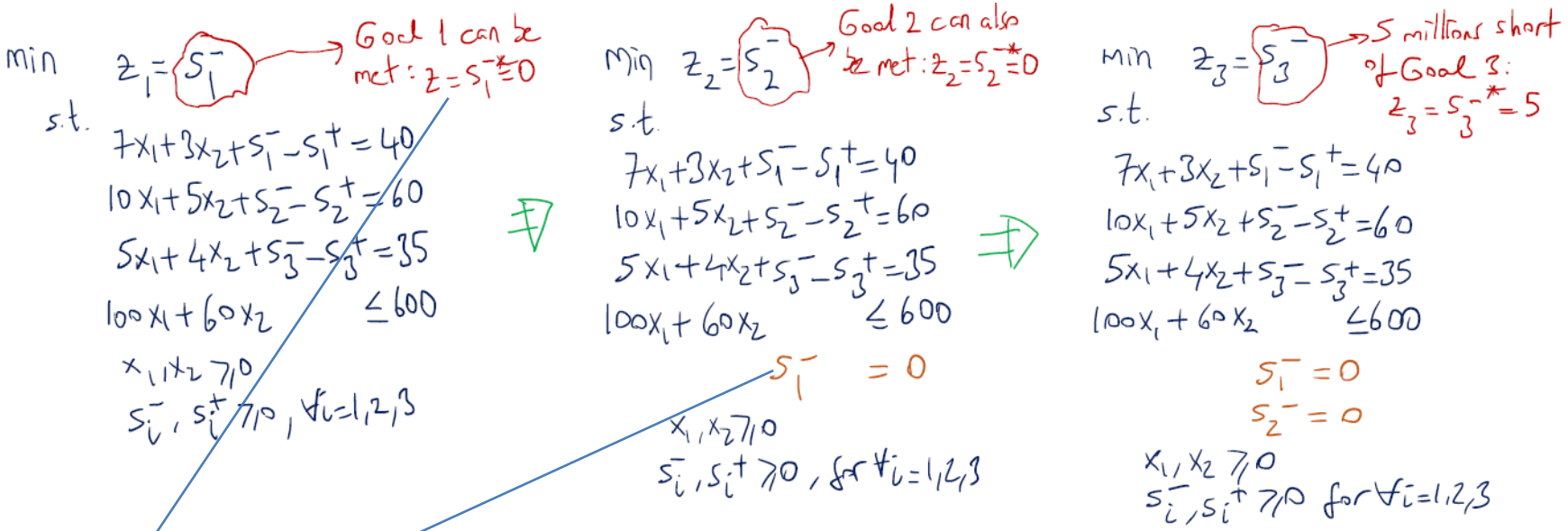
** Same methodology for IP and nonlinear formulations.*

Example cont'd (advertising) – using preemptive goal programming:

The ranking of the goals are as follows:

- Priorities: **HIM >> LIP >> HIW**

Step 1: Minimize the "deviation" from the most important goal (HIM target) with the original constraints: Solution to the first LP tells us that the first goal alone can be met (i.e., $s_1^{-*}=0$).
Step 2: Minimize the "deviation" from the second most important goal (LIP) adding the constraint that "$s_1^-=0$" to the formulation. Solution to the second LP tells us that the second goal can also be met (making sure that the first goal is still met via the additional constraint).
Step 3: Minimize the "deviation" from the third most important goal (HIW) adding the constraint that "$s_1^-=0$ and $s_2^-=0$" to the formulation. Solution to the second LP tells us that the second goal can also be met (making sure that the first and second goals are still met via the two additional constraints).



If the $s_1^-=2$ were the solution to the first LP, then the Constraint to be added would be $s_1^-=2$ to the second LP.

optimal solution:
$z = 250$
$x_1^* = 6$, $x_2^* = 0$
$s_1^{+*} = 2$, $s_3^{-*} = 5$
Goals 1 & 2 met
Fails to meet goal 3

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2nd Example:

Fruit Computer Company is ready to make its annual purchase of computer chips. Fruit can purchase chips (in lots of 100) from three suppliers. Each chip is rated of being excellent, good, or mediocre quality. During the coming year, Fruit will need 5000 excellent chips, 3000 good chips, and 1000 mediocre chips. The characteristics of the chips purchased from each supplier are shown in the table below. Each year, Fruit has budgeted $28000 to spend on chips. If Fruit does not obtain enough chips of a given quality, then the company may special-order additional chips at $10 per excellent chip, $6 per good chip, and $4 per mediocre chip. Fruit assesses a penalty of $1 for each dollar by which the amount paid to suppliers 1-3 exceeds the annual budget. Formulate an LP (using weighted goal programming) to help Fruit minimize the penalty associated with meeting the annual chip requirements. Also use preemptive goal programming to determine a purchasing strategy. Let the budget constraint have the highest priority, followed in order by the restrictions on excellent, good, and mediocre chips.

	Characteristics of a Lot of 100 Chips			Price per
Supplier	Excellent	Good	Mediocre	100 Chips
1	60	20	20	400
2	50	35	15	300
3	40	20	40	250

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Optimal Solution:

Preemptive GP Formulation:

3rd Example:

Goal Programming

- A Case Study: Dewright Co. Goal Programming
- Weighted Goal Programming
- Preemptive Goal Programming

* **Slides are from *Introduction To Management Science* by Frederick S. Hillier and Mark S. Hillier**

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The Dewright Company

- The Dewright Company is one of the largest producers of power tools in the United States.
- The company is preparing to replace its current product line with the next generation of products—three new power tools.
- Management needs to determine the mix of the company's three new products to best meet the following three goals:
  - Achieve a total profit (net present value) of at least $125 million.
  - Maintain the current employment level of 4,000 employees.
  - Hold the capital investment down to no more than $55 million.

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Penalty Weights

Goal	Factor	Penalty Weight for Missing Goal
1	Total profit	5 (per $1 million under the goal)
2	Employment level	4 (per 100 employees under the goal) 2 (per 100 employees over the goal)
3	Capital investment	3 (per $1 million over the goal)

Data for Contribution to the Goals

Factor	Unit Contribution of Product 1	2	3	Goal
Total profit (millions of dollars)	12	9	15	≥ 125
Employment level (hundreds of employees)	5	3	4	$= 40$
Capital investment (millions of dollars)	5	7	8	≤ 55

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Weighted Goal Programming Formulation

Let P_i = Number of units of product i to produce per day $(i = 1, 2, 3)$,
Under Goal i = Amount under goal i $(i = 1, 2, 3)$,
Over Goal i = Amount over goal i $(i = 1, 2, 3)$,

Minimize $W = 5(\text{Under Goal 1}) + 2(\text{Over Goal 2}) + 4(\text{Under Goal 2}) + 3(\text{Over Goal 3})$
subject to

	Level Achieved	Deviations	Goal
Goal 1:	$12P_1 + 9P_2 + 15P_3$	$-(\text{Over Goal 1}) + (\text{Under Goal 1})$	$= 125$
Goal 2:	$5P_1 + 3P_2 + 4P_3$	$-(\text{Over Goal 2}) + (\text{Under Goal 2})$	$= 40$
Goal 3:	$5P_1 + 7P_2 + 8P_3$	$-(\text{Over Goal 3}) + (\text{Under Goal 3})$	$= 55$

and

$P_i \geq 0$, Under Goal $i \geq 0$, Over Goal $i \geq 0$ $(i = 1, 2, 3)$

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Weighted Goal Programming Spreadsheet

	B	C	D	E	F	G	H	I	J	K	L	M	N	O
3					Goals				Deviations			Constraints		
4		Contribution per Unit Produced			Level				Amount	Amount		Balance		
5		Product 1	Product 2	Product 3	Achieved		Goal		Over	Under		(Level - Over + Under)		Goal
6	Goal 1 (Profit)	12	9	15	125	>=	125		0	0		125	=	125
7	Goal 2 (Employment)	5	3	4	48.333333	=	40		8.333333	0		40	=	40
8	Goal 3 (Investment)	5	7	8	55	<=	55		0	0		55	=	55
9														
10														
11		Product 1	Product 2	Product 3			Penalty		Over	Under		Weighted Sum		
12	Units Produced	8.33333333	0	1.66666667			Weights		Goal	Goal		of Deviations		
13							Profit			5		16.66666667		
14							Employment		2	4				
15							Investment		3					

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Weighted vs. Preemptive Goal Programming

- Weighted goal programming is designed for problems where all the goals are quite important, with only modest differences in importance that can be measured by assigning weights to the goals.
- Preemptive goal programming is used when there are *major* differences in the importance of the goals.
  - The goals are listed in the order of their importance.
  - It begins by focusing solely on the most important goal.
  - It next does the same for the second most important goal (as is possible without hurting the first goal).
  - It continues the following goals (as is possible without hurting the previous more important goals).

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Preemptive Goal Programming

- Introduce new changing cells, Amount Over and Amount Under, that will measure how much the current solution is over or under each goal.
- The Amount Over and Amount Under changing cells are forced to maintain the correct value with the following constraints:

  Level Achieved – Amount Over + Amount Under = Goal
- Start with the objective of achieving the first goal (or coming as close as possible):
  - *Minimize (Amount Over/Under Goal 1)*
- Continue with the next goal, but constrain the previous goals to not get any worse:
  - *Minimize (Amount Over/Under Goal 2)*
    subject to
    Amount Over/Under Goal 1 = (amount achieved in previous step)
- Repeat the previous step for all succeeding goals.

Preemptive Goal Programming for Dewright

- The goals in the order of importance are:
  - Achieve a total profit (net present value) of at least $125 million.
  - Avoid decreasing the employment level below 4,000 employees.
  - Hold the capital investment down to no more than $55 million.
  - Avoid increasing the employment level above 4,000 employees.
- Start with the objective of achieving the first goal (or coming as close as possible):
  - *Minimize (Under Goal 1)*
- Then, if for example goal 1 is achieved (i.e., Under Goal 1 = 0), then
  - *Minimize (Under Goal 2)*
    subject to
    (Under Goal 1) = 0

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Preemptive Goal Programming Formulation for the Dewright Co. Problem (Step 1)

Let P_i = Number of units of product i to produce per day $(i = 1, 2, 3)$,
Under Goal i = Amount under goal i $(i = 1, 2, 3)$,
Over Goal i = Amount over goal i $(i = 1, 2, 3)$,

Minimize (Under Goal 1)
subject to

	Level Achieved	Deviations	Goal
Goal 1:	$12P_1 + 9P_2 + 15P_3$	– (Over Goal 1) + (Under Goal 1)	= 125
Goal 2:	$5P_1 + 3P_2 + 4P_3$	– (Over Goal 2) + (Under Goal 2)	= 40
Goal 3:	$5P_1 + 7P_2 + 8P_3$	– (Over Goal 3) + (Under Goal 3)	= 55

and

$P_i \geq 0$, Under Goal $i \geq 0$, Over Goal $i \geq 0$ $(i = 1, 2, 3)$

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Preemptive Goal Programming Spreadsheet
Step 1: Minimize (Under Goal 1)

				Goals				Deviations			Constraints		
	Contribution per Unit Produce			Level				Amount	Amount		Balance		
	Product 1	Product 2	Product 3	Achieved		Goal		Over	Under		(Level-Over+Under)		Goal
Goal 1 (Profit)	12	9	15	125	>=	125		0	0		125	=	125
Goal 2 (Employment)	5	3	4	40	=	40		0	0		40	=	40
Goal 3 (Investment)	5	7	8	61,48148	<=	55		6,481481	0		55	=	55
	Product 1	Product 2	Product 3			**Penalty**		Over	Under		Weighted Sum		
Units Produced	3,703704	0	5,37037			**Weights**		Goal	Goal		of Deviations		
						Profit			5		19,44444444		
						Employment		2	4				
						Investment		3					

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Preemptive Goal Programming Formulation for the Dewright Co. Problem (Step 2)

Let P_i = Number of units of product i to produce per day $(i = 1, 2, 3)$,
Under Goal i = Amount under goal i $(i = 1, 2, 3)$,
Over Goal i = Amount over goal i $(i = 1, 2, 3)$,

Minimize (Under Goal 2)
subject to

	Level Achieved		Deviations		Goal
Goal 1:	$12P_1 + 9P_2 + 15P_3$	– (Over Goal 1) + (Under Goal 1)	=		125
Goal 2:	$5P_1 + 3P_2 + 4P_3$	– (Over Goal 2) + (Under Goal 2)	=		40
Goal 3:	$5P_1 + 7P_2 + 8P_3$	– (Over Goal 3) + (Under Goal 3)	=		55

(Under Goal 1) = (Level Achieved in Step 1)

and

$P_i \geq 0$, Under Goal $i \geq 0$, Over Goal $i \geq 0$ $(i = 1, 2, 3)$

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Preemptive Goal Programming Spreadsheet
Step 2: Minimize (Under Goal 2)

				Goals				Deviations			Constraints		
	Contribution per Unit Produce			Level				Amount	Amount		Balance		
	Product 1	Product 2	Product 3	Achieved		Goal		Over	Under		(Level-Over+Under)		Goal
Goal 1 (Profit)	12	9	15	125	>=	125		0	0		125	=	125
Goal 2 (Employment)	5	3	4	40	=	40		0	0		40	=	40
Goal 3 (Investment)	5	7	8	61,48148	<=	55		6,481481	0		55	=	55
	Product 1	Product 2	Product 3			**Penalty**		Over	Under		Weighted Sum		
Units Produced	3,703704	0	5,37037			**Weights**		Goal	Goal		of Deviations		
						Profit			5		19,4444444		
						Employment		2	4				
						Investment		3					
											Additional Constraints		
											Under Goal 1	=	Level Achieved in Previous Step
											0	=	0

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Preemptive Goal Programming Formulation for the Dewright Co. Problem (Step 3)

Let P_i = Number of units of product i to produce per day $(i = 1, 2, 3)$,
Under Goal i = Amount under goal i $(i = 1, 2, 3)$,
Over Goal i = Amount over goal i $(i = 1, 2, 3)$,

Minimize (Over Goal 3)
subject to

	Level Achieved	Deviations		Goal
Goal 1:	$12P_1 + 9P_2 + 15P_3$	– (Over Goal 1) + (Under Goal 1)	=	125
Goal 2:	$5P_1 + 3P_2 + 4P_3$	– (Over Goal 2) + (Under Goal 2)	=	40
Goal 3:	$5P_1 + 7P_2 + 8P_3$	– (Over Goal 3) + (Under Goal 3)	=	55

(Under Goal 1) = (Level Achieved in Step 1)
(Under Goal 2) = (Level Achieved in Step 2)

and

$P_i \geq 0$, Under Goal $i \geq 0$, Over Goal $i \geq 0$ $(i = 1, 2, 3)$

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Preemptive Goal Programming Spreadsheet Step 3: Minimize (Over Goal 3)

				Goals				Deviations			Constraints		
	Contribution per Unit Produce			Level				Amount	Amount		Balance		
	Product 1	Product 2	Product 3	Achieved		Goal		Over	Under		(Level-Over+Under)		Goal
Goal 1 (Profit)	12	9	15	125	>=	125		0	0		125	=	125
Goal 2 (Employment)	5	3	4	48,33333	=	40		8,333333	0		40	=	40
Goal 3 (Investment)	5	7	8	55	<=	55		0	0		55	=	55
	Product 1	Product 2	Product 3			Penalty		Over	Under		Weighted Sum		
Units Produced	8,333333	0	1,666667			**Weights**		Goal	Goal		of Deviations		
						Profit			5		16,66666667		
						Employment		2	4				
						Investment		3					
											Additional Constraints		
											Under Goal 1	=	Level Achieved in Previous Step
											0	=	0
											Under Goal 2	=	Level Achieved in Previous Step
											0	=	0

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Preemptive Goal Programming Formulation for the Dewright Co. Problem (Step 4)

Let P_i = Number of units of product i to produce per day $(i = 1, 2, 3)$,
Under Goal i = Amount under goal i $(i = 1, 2, 3)$,
Over Goal i = Amount over goal i $(i = 1, 2, 3)$,

Minimize (Over Goal 2)
subject to

	Level Achieved	Deviations		Goal
Goal 1:	$12P_1 + 9P_2 + 15P_3$	– (Over Goal 1) + (Under Goal 1)	=	125
Goal 2:	$5P_1 + 3P_2 + 4P_3$	– (Over Goal 2) + (Under Goal 2)	=	40
Goal 3:	$5P_1 + 7P_2 + 8P_3$	– (Over Goal 3) + (Under Goal 3)	=	55

(Under Goal 1) = (Level Achieved in Step 1)
(Under Goal 2) = (Level Achieved in Step 2)
(Over Goal 3) = (Level Achieved in Step 3)

and

$P_i \geq 0$, Under Goal $i \geq 0$, Over Goal $i \geq 0$ $(i = 1, 2, 3)$

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Preemptive Goal Programming Spreadsheet
Step 4: Minimize (Over Goal 2)

				Goals				Deviations			Constraints		
	Contribution per Unit Produced			Level				Amount	Amount		Balance		
	Product 1	Product 2	Product 3	Achieved		Goal		Over	Under		(Level-Over+Under)		Goal
Goal 1 (Profit)	12	9	15	125	>=	125		0	0		125	=	125
Goal 2 (Employment)	5	3	4	48,33333	=	40		8,333333	0		40	=	40
Goal 3 (Investment)	5	7	8	55	<=	55		2,91E-12	0		55	=	55
	Product 1	Product 2	Product 3			**Penalty**		Over	Under		Weighted Sum		
Units Produced	8,333333	0	1,666667			**Weights**		Goal	Goal		of Deviations		
						Profit			5		16,66666667		
						Employment		2	4				
						Investment		3					
											Additional Constraints		
											Under Goal 1	=	Level Achieved in Previous Step
											0	=	0
											Under Goal 2	=	Level Achieved in Previous Step
											0	=	0
											Over Goal 3	=	Level Achieved in Previous Step
											2,91145E-12	=	0

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Another approach:

You can also minimize the maximum deviation from any goal (worst-case focus):

Non-linear formulation

$$\min \quad z = \max(s_1^-, s_2^-, s_3^-)$$

s.t.

$$7x_1 + 3x_2 + s_1^- - s_1^+ = 40$$
$$10x_1 + 5x_2 + s_2^- - s_2^+ = 60$$
$$5x_1 + 4x_2 + s_3^- - s_3^+ = 35$$
$$100x_1 + 60x_2 \leq 600$$
$$x_1, x_2 \geq 0$$
$$s_i^-, s_i^+ \geq 0, \forall i = 1,2,3$$

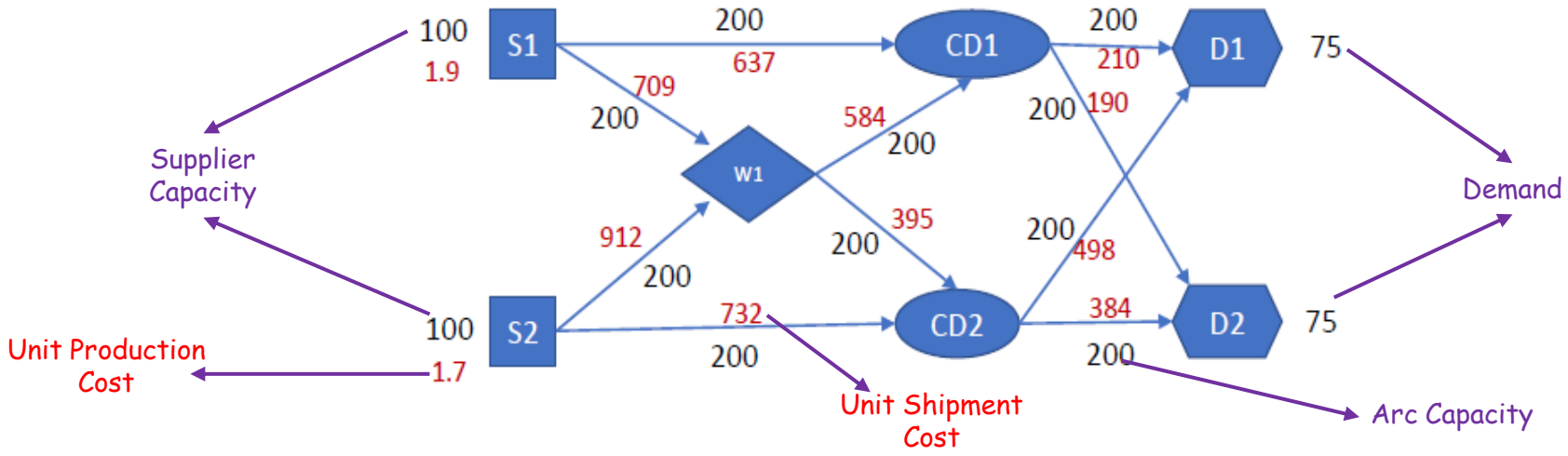
LP formulation

Minimize z

s.t.

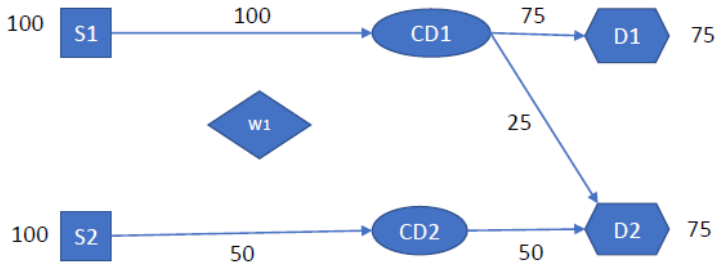
$$7x_1 + 3x_2 + s_1^- - s_1^+ = 40$$
$$10x_1 + 5x_2 + s_2^- - s_2^+ = 60$$
$$5x_1 + 4x_2 + s_3^- - s_3^+ = 35$$
$$100x_1 + 60x_2 \leq 600$$
$$s_1^- \leq z$$
$$s_2^- \leq z$$
$$s_3^- \leq z$$
$$x_1, x_2 \geq 0$$
$$s_i^-, s_i^+ \geq 0, \forall i = 1,2,3$$

Multiple Objective Linear/Integer Programming Example: Minimizing Cost and Maximizing Connectivity (against disruptions)



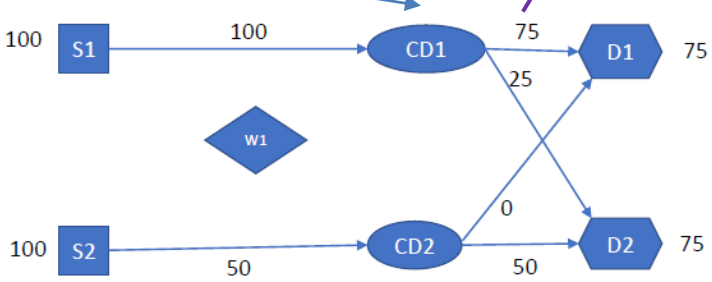
50: fixed cost of "opening" each arc
20: fixed cost of "opening" each supplier

These are the quantities shipped



These two solutions (left & right) are the Pareto optimal solutions)

These are the quantities shipped



Cost: $156,765
DWC=225 (2 node-disjoint paths for D2, and one path for D1)

Cost: $156,815
DWC=300 (2 node-disjoint paths for each demand node (D1, D2))

Source: "A multi-objective optimization model for designing resilient supply chain networks", J. T. Margolis et al., IJPE, 2018

Multiple Objective Linear/Integer Programming Example: Minimizing Cost and Maximizing Connectivity (against disruptions)

SETS:

N	Set of all nodes in G
E	Set of all arcs in G
P	Set of all commodities
D_p, S_p	Set of nodes that demand, supply product $p \in P$
FS(i)	Set of all nodes leading out from node i
RS(i)	Set of all nodes with arcs ending at node i

NETWORK DESIGN VARIABLES:

z_{ip}	Indicates whether production capacity of commodity $p \in P$ is high ($z_{ip} = 1$) or low ($= 0$) at node $i \in N$
y_{ij}	Indicates whether flow capacity of arc (i, j) is high ($y_{ij} = 1$) or low ($= 0$)

NETWORK OPERATION VARIABLES:

w_{ip}	Amount of product $p \in P$ produced at node $i \in N$
x_{ijp}	Flow of product $p \in P$ on arc $(i, j) \in E$

NETWORK CONNECTIVITY VARIABLES:

x'_{ijkp}	Indicates whether arc $(i, j) \in E$ is on a node-disjoint path from some supply node of commodity $p \in P$ (i.e., $s \in S_p$) to demand node $k \in D_p$
v_{kp}	Number of node-disjoint paths to demand node $k \in D_p$ associated with commodity $p \in P$

PARAMETERS:

d_{ip}	Demand for commodity $p \in P$ at node $i \in D_p$
b_{ijp}	Cost per unit flow of commodity $p \in P$ on arc $(i, j) \in E$
c_{ip}	Cost per unit of commodity $p \in P$ produced at node $i \in N$
u_{ij}^+, u_{ij}^-	Flow capacity of arc $(i, j) \in E$ at the high level, low level
r_{ip}^+, r_{ip}^-	Production capacity for commodity $p \in P$ at node $i \in N$ at the high level, low level
f_{ij}	Cost incurred if flow capacity of arc (i, j) is at the high level
g_{ip}	Cost incurred if production capacity of commodity $p \in P$ at node $i \in N$ is at the high level

Source: "A multi-objective optimization model for designing resilient supply chain networks", J. T. Margolis et al., IJPE, 2018

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Multiple Objective Linear/Integer Programming Example: Minimizing Cost and Maximizing Connectivity (against disruptions)

$$\max V = \sum_{p\in P}\sum_{k\in D_p} d_{kp}v_{kp}, \tag{1}$$

$$\min C = \sum_{p\in P}\sum_{i\in N}(c_{ip}w_{ip} + g_{ip}z_{ip}) + \sum_{(i,j)\in E}\left(f_{ij}y_{ij} + \sum_{p\in P} b_{ijp}x_{ijp}\right), \tag{2}$$

subject to the constraints

$$\sum_{j\in RS(i)} x_{jip} + w_{ip} = \sum_{j\in FS(i)} x_{ijp} + d_{ip}, \quad \forall\, p\in P,\ i\in N, \tag{3}$$

$$w_{ip} \le r_{ip}^{-} + (r_{ip}^{+} - r_{ip}^{-})z_{ip}, \quad \forall\, p\in P,\ i\in N, \tag{4}$$

$$\sum_{p\in P} x_{ijp} \le u_{ij}^{-} + (u_{ij}^{+} - u_{ij}^{-})y_{ij}, \quad \forall\,(i,j)\in E, \tag{5}$$

$$x'_{ijkp} \le 1 - 1_{\{u_{ij}^{-}=0\}}(1-y_{ij}), \quad \forall\, p\in P,\ k\in D_p,\ (i,j)\in E, \tag{6}$$

$$\sum_{j\in FS(i)} x'_{ijkp} \le 1 - 1_{\{r_{ip}^{-}=0\}}(1-z_{ip}), \quad \forall\, p\in P,\ k\in D_p,\ i\in S_p, \tag{7}$$

$$\sum_{j\in FS(i)} x'_{ijkp} \le 1, \quad \forall\, p\in P,\ k\in D_p,\ i\in N\backslash(S_p\cup\{k\}), \tag{8}$$

$$\sum_{j\in FS(i)} x'_{ijkp} - \sum_{j\in RS(i)} x'_{jikp} = 0, \quad \forall\, p\in P,\ k\in D_p,\ i\in N\backslash(S_p\cup\{k\}), \tag{9}$$

$$\sum_{i\in RS(k)} x'_{ikkp} = v_{kp}, \quad \forall\, p\in P,\ k\in D_p, \tag{10}$$

$$x_{ijp} \ge 0, \quad \forall\, p\in P,\ (i,j)\in E, \tag{11}$$

$$w_{ip} \ge 0, \quad \forall\, p\in P,\ i\in N, \tag{12}$$

$$y_{ij} \in \{0,1\}, \quad \forall\,(i,j)\in E, \tag{13}$$

$$z_{ip} \in \{0,1\}, \quad \forall\, i\in N,\ p\in P, \tag{14}$$

$$x'_{ijkp} \ge 0, \quad \forall\, p\in P,\ k\in D_p,\ (i,j)\in E, \tag{15}$$

$$v_{kp} \ge 0, \quad \forall\, p\in P,\ k\in D_p. \tag{16}$$

Formulation with the ε-constraint method

$$\min\ C - \alpha V,$$

$$\text{s.t.}\ \ V \ge \varepsilon,$$

Constraints (3)–(15).

Source: "A multi-objective optimization model for designing resilient supply chain networks", J. T. Margolis et al., IJPE, 2018

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Multiple Objective Linear Programming:

- Suppose you have 3 objectives
- You can solve 3 separate LPs (one for each objective only), and get the optimal solution for each LP.
- Then, you can use the weighted goal programming approach to minimize the sum of the deviations from the "optimal objective function values" (i.e., target values would be the optimal objective function value of each LP)

Multi-Objective Decision Making

- Many problems have multiple objectives:
  - Planning the national budget
    - save social security, reduce debt, cut taxes, build national defense
  - Admitting students to college
    - high SAT or GMAT, high GPA, diversity
  - Planning an advertising campaign
    - budget, reach, expenses, target groups
  - Choosing taxation levels
    - raise money, minimize tax burden on low-income, minimize flight of business
  - Planning an investment portfolio
    - maximize expected earnings, minimize risk
- Techniques
  - Preemptive goal programming
  - Weighted goal programming

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DECISION ANALYSIS

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TALENT HUB FOR SUPPLY CHAIN

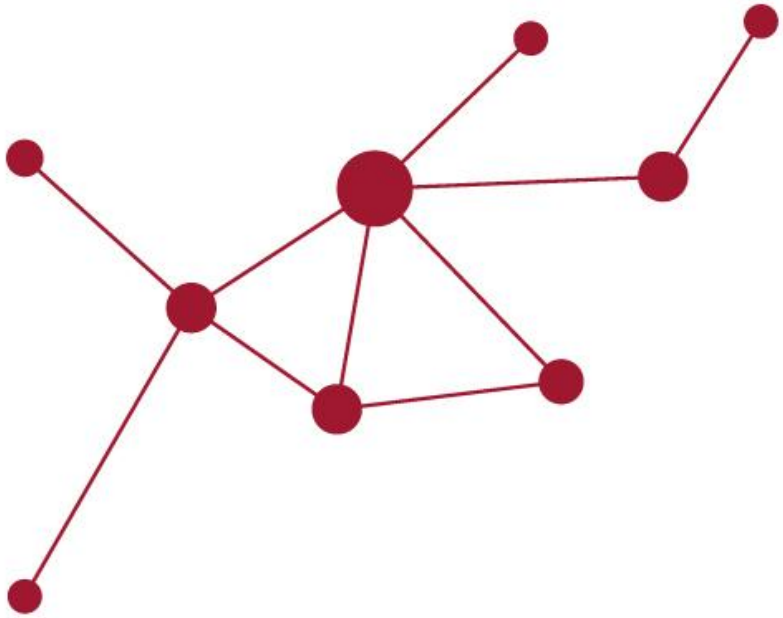
Zaragoza Logistics Center

Decision Analysis

September 14, 2026

Some notes borrowed from Dr. Mark Hillier

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Introduction

- **Decisions Under Certainty**
  - Each decision yields a certain outcome (one result)
  - Select decision that yields the best result
  - Examples: Diet Problem, Product Mix, Distribution, …

- **Decisions Under Uncertainty (Risk)**
  - Each decision yields an uncertain outcome (several possible results)
  - Decision criteria?
    - Select decision that…?

  - Examples:
    - Drilling for oil → oil found? How much? How deep? Selling price?
      - Decision: Drill or not!
    - Developing a new product → R&D Cost? Sales?
      - Decision: Develop or not!
    - Newsvendor Problem → Demand?
      - Decision: Order Quantity!

Oil Drilling Problem

- Consider the problem faced by an oil company that is trying to decide whether to drill an exploratory oil well on a given site. Drilling costs $200,000. If oil is found, it is worth $800,000. If the well is dry, it is worth nothing. However, the $200,000 cost of drilling is incurred, regardless of the outcome of the drilling.

Payoff Table	State of Nature	
Decision	*Wet*	*Dry*
Drill	600k	-200k
Don't drill	0	0

- Which decision is best?
- Optimist → maximax
- Pessimist → maximin
- Minimax regret
- Equally likely

The Maximin Criterion



Papers Ordered	Papers Demanded 6	7	8	9	10	min (r_{ij})
6	30	30	30	30	30	30
7	10	35	35	35	35	10
8	-10	15	40	40	40	-10
9	-30	-5	20	45	45	-30
10	-50	-25	0	25	50	-50



Half empty

maximin criterion: choose the action with the largest guaranteed return

maximize $_i$ min $\{r_{ij} : j = 6 \text{ to } 10\}$ = 30.

Borrowed from the class (15.053) notes of Prof. James Orlin

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The Maximax Criterion



Papers Ordered	Papers Demanded 6	7	8	9	10	max (r_{ij})
6	30	30	30	30	30	30
7	10	35	35	35	35	35
8	-10	15	40	40	40	40
9	-30	-5	20	45	45	45
10	-50	-25	0	25	50	50



Half full

maximax criterion: choose the action with the largest possible return

maximize $_i$ max $\{r_{ij} : j = 6 \text{ to } 10\}$ = 50.

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The Minimax Regret Criterion



Papers Ordered \ Papers Demanded	6	7	8	9	10
6	30	30	30	30	30
7	10	35	35	35	35
8	-10	15	40	40	40
9	-30	-5	20	45	45
10	-50	-25	0	25	50
$\max_i r_{ij}$	30	35	40	45	50



Half empty

$$\text{regret}(i, j) = \max_k r_{kj} - r_{ij}$$

regret (1, 1) = 0 regret (1, 2) = 5

The Minimax Regret Criterion



Papers Ordered \ Papers Demanded	6	7	8	9	10	max regret
6	0	5	10	15	20	20
7	20	0	5	10	15	20
8	40	20	0	5	10	40
9	60	40	20	0	5	60
10	80	60	40	20	0	80
$\max_i r_{ij}$	30	35	40	45	50	



Half empty

minimax regret : choose the action that minimizes the maximum possible regret

minimax regret = $\min_i \max_j$ regret (i, j) = 20

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Expected Value



Papers Ordered	Papers Demanded 6	7	8	9	10	Expected Value
6	30	30	30	30	30	30
7	10	35	35	35	35	30
8	-10	15	40	40	40	25
9	-30	-5	20	45	45	15
10	-50	-25	0	25	50	0

Suppose that p(j) = 1/5 for j = 6 to 10.

The expected value of ordering i papers is $E(i) = \sum_j 1/5\ r_{ij}$

The maximum expected value is 30.

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Bayes' Decision Rule (Oil Drilling Problem cont'd)

- **Suppose that the oil company estimated that the probability that the site is *Wet* is 40%.**

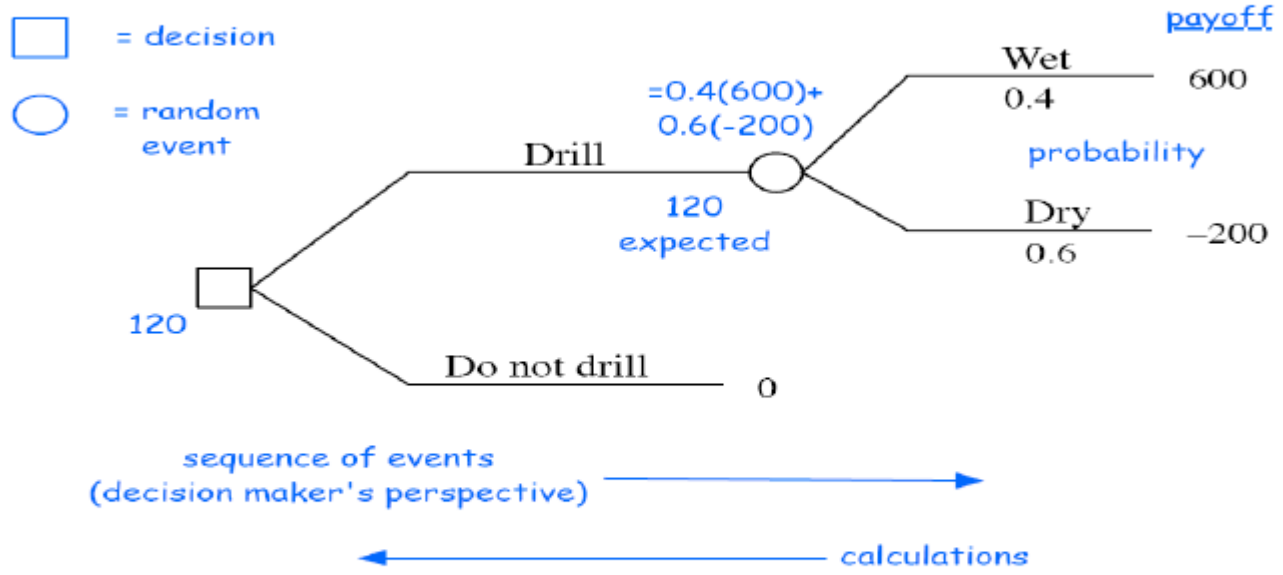
Payoff Table	State of Nature	
Decision	*Wet*	*Dry*
Drill	600k	-200k
Don't drill	0	0
Prior Probability	**0.4**	**0.6**

  - **Expected Value of payoff (Drill)=120k**
  - **Expected Value of payoff (Do not drill)=0**

- ***Accounts not only for the set of outcomes, but also their probabilities***
- ***Represents the average monetary outcome if the situation were repeated indefinitely***
- ***Can handle complicated situations involving multiple and related risks.***

Using a Decision Tree for the Oil Drilling Problem

Decision Tree



Folding back

- At each event node (circle): calculate expected value (SUMPRODUCT of payoffs and probabilities for each branch).
- At each decision node (square): choose *best* branch (maximum value).

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Juan Lee's Decision Problem

It is January 10th, and Juan Lee is currently a fourth year undergraduate in Management Science at Sloan. He has decided to seek out a job as a consultant. He already has received an offer from A to Z consulting for $72,000 per year. He has until February 1st to decide whether to accept the offer. An old classmate of his, Mary Kumar, has told him that she has recommended him highly to her consulting firm, and feels that there is an excellent chance that they would give him an offer for $80,000. However, they are not prepared to make any decision until February 15th. If they made him an offer, he would need to decide by March 1. He also has the option of taking part in the consulting job fair in the middle of March. He is fairly certain that he could get a consulting job at that time, but is uncertain as to what he would be paid.

Juan's probabilities

- Juan does not know how to evaluate his "excellent chances" at Mary's firm, nor does he know what to expect from the consulting fair.
- Juan's best guesses
  - Probability of getting Job at Mary's firm: 60%
  - Different possibilities for consulting fair offers.
    - $90,000: 10%
    - $70,000: 50%
    - $60,000: 40%

Juan's Goal

- Assume here that Juan views all of these consulting jobs as excellent opportunities, and that the only differentiating factor is money.
- Juan has decided to maximize his expected salary.

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If the probabilities are updated after Mary's offer…

Any change?

If the probabilities are updated after Mary's offer…

Now, any change?

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Making Sequential Decisions

- Consider a pharmaceutical company that is considering developing an anticlotting drug. They are considering two approaches. A biochemical approach would require less R&D and would be more likely to meet with at least some success. Some, however, are pushing for a more radical, biogenetic approach. The R&D would be higher, and the probability of success lower. However, if a biogenetic approach were to succeed, they would likely capture a much larger portion of the market, and generate much more profit. Some initial data estimates are given below:

R&D Choice	Investment	Outcomes	Profit	Probability
Biochemical	*$10M*	*Large Success*	*$90M*	*0.7*
		Small Success	*$50M*	*0.3*
Biogenetic	*$20M*	*Success*	*$200M*	*0.2*
		Failure	*$0M*	*0.8*

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Example: DriveTek Research Institute

DriveTek Research Institute discovers that a computer company wants a new tape drive for a proposed new computer system. Since the computer company does not have research people available to develop the new drive, it will subcontract the development to an independent research firm. The computer company has offered a fee of $250,000 for the best proposal for developing the new tape drive. The contract will go to the firm with the best technical plan and the highest reputation for technical competence and the $250,000 will be paid only when the product is successfully delivered.

DriveTek Research Institute wants to enter the competition. Management estimates a cost of $50,000 to prepare a proposal with a fifty-fifty chance of winning the contract.

However, DriveTek's engineers are uncertain about how they will develop the tape drive if they are awarded the contract. Three alternative approaches can be tried. The first approach is a mechanical method with a cost of $120,000, and the engineers are certain they can develop a successful model with this approach. A second approach involves electronic components. The engineers estimate that the electronic approach will cost only $50,000 to develop a model of the tape drive, but with only a 50 percent chance of satisfactory results. A third approach uses magnetic components; this costs $80,000, with a 70 percent chance of success.

DriveTek Research can work on only one approach at a time and has time to try only two approaches. If it tries either the magnetic or electronic method and the attempt fails, the second choice must be the mechanical method to guarantee a successful model.

The management of DriveTek Research needs help in incorporating this information into a decision to proceed or not.

Solution: DriveTek Research Institute

Incorporating New Information

- **Often, a preliminary study can be done to better determine the true state of nature.**

- **Examples**
  - Market Surveys
  - Test-marketing
  - Seismic testing (for oil)

- **Question:**
  - What is the value of this information?

  - *Improve probability estimates*
  - *Questions:*
    - *How do we incorporate information into tree?*
    - *How much is the information worth?*

Oil Drilling Problem with Perfect Information: A test tells us whether the site is dry or wet before the decision of drilling

Oil Drilling Problem with Imperfect Information

- **Suppose a seismic test is available that would better indicate whether or not the site was wet or dry.**

Record of 100 Past Seismic Test Sites: There was oil (wet) in 50 sites. Of these 50 sites, the test had good (that there is oil) in 30 tests, and it predicted dry in 20 tests. The site was dry in 50 sites also. Of these 50 sites, 40 tests correctly predicted that they were dry, and 10 test results were good (incorrect result).

Revising Probabilities

Suppose partners don't have the "Record of Past 100 Seismic Test Sites"

Vendor of test certifies:

Wet sites test "good" three quarters of the time
Dry sites test "bad" two thirds of the time.

$P(G \mid W) = 3/4$

$P(B \mid D) = 2/3$

Is this the information needed in the decision tree?

No. Need $P(W \mid G)$, $P(D \mid B)$, and $P(G)$

Depends on prior probabilities: $P(W)$, $P(D)$

Bayes' Theorem

$$P(A \mid B) = \frac{P(A \,\&\, B)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

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Revising Probabilities

Given Information (Test Performance, Prior Probabilities)

Seismic Result	Actual State of Nature: Wet (W)	Actual State of Nature: Dry (D)
Good (G)	$P(G \mid W) = 3/4$	$P(G \mid D) = 1/3$
Bad (B)	$P(B \mid W) = 1/4$	$P(B \mid D) = 2/3$
Prior	$P(W) = 0.4$	$P(D) = 0.6$

(Posterior Probabilities): $P(A|B) = P(A\&B) / P(B)$

	Actual State of Nature: Wet (W)	Actual State of Nature: Dry (D)
Good (G)	$P(W \mid G) = 0.3/0.5 = 0.6$	$P(D \mid G) = 0.2/0.5 = 0.4$
Bad (B)	$P(W \mid B) = 0.1/0.5 = 0.2$	$P(D \mid B) = 0.4/0.5 = 0.8$

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Risk attitude and Utility Functions

- **Decision tree analysis can still be used if the person is risk averse, risk seeking, or risk neutral.**
- **Through developing a mathematical representation of a person's attitude towards risk: a utility curve**
  - Creating a utility function:
    - Equivalent Lottery Method
    - Exponential Utility Function

Additional Exercises

Exercise #1

Shawna is the procurement manager of StayFit and needs to purchase recently designed dumbbells from a supplier in Thailand. The lead time is long compared to the selling season, and therefore there is a single opportunity to place an order. Each unit costs StayFit $60. The selling price is $140, and the leftover inventory at the end of the season can be salvaged at a markdown price of $20.

Shawna wants to maximize expected profit, and has one of the two following options:

1. She could place an order now, assuming that the demand follows a normal distribution with a mean of 15000 units and a standard deviation of 5000 units.

2. She could delay the ordering decision by a month and attend a trade fair anticipating some larger customers placing orders before the season starts. Although the total size of these orders is marginal compared to the total demand, incorporating these initial orders helped improve forecasts for similar product lines in the past. The feedback from the fair has been "high demand prediction (i.e., Normal(μ=22000, σ=1000)", "medium demand prediction (i.e., Normal (μ=15000, σ=500)", and "low demand prediction (i.e., Normal(μ=8000, σ=1000))" in 80%, 15%, and 5% of the cases where demand was actually High. Similarly, the feedback has been "high demand prediction", "medium demand prediction", and "low demand prediction" in 10%, 85%, and 5% of the cases where demand was actually Medium. Finally, the outcome of the fair has been "high demand prediction", "medium demand prediction", and "low demand prediction" in 5%, 25%, and 70% of the cases where demand was actually Low. The actual demand would be considered High (Low) if it is more (less) than one standard deviation above (below) the mean. Otherwise, it would be considered Medium. StayFit would incur a fixed cost of $20,000 (cost associated with the fair as well as the cost of reserving capacity at the supplier in Thailand) and the unit cost of each dumbbell would increase by 3% approximately and be equal to $62 (premium for shorter lead time).

What would be your recommendation for Shawna?

Solution #1: Newsvendor quantities and expected profit

Solution #1 (cont'd): Posterior Probabilities

Solution #1 (cont'd): Decision tree

Exercise #2

Suppose you have accepted an order for 10 air conditioners that you produce. There exist 2 modes of shipment that you can choose from. Carrier #1 guarantees to deliver the ACs to the customer within a day. The cost of shipment would be $300 per shipment in this case. Alternatively, you can ship the ACs with Carrier #2 at a cost of $240 per shipment. However, the transportation time with Carrier #2 is random, following an exponential distribution with a mean of 1 day. There are penalty fees associated with "late deliveries". The penalties are as follows:

Delivery time	Penalty
ACs delivered within a day	None
ACs delivered between 1 and 2 days	$10 per AC
AC delivered after 2 days	$15 per AC

What would you do?

Solution #2

Exercise #3

Suppose the demand for your product follows a discrete uniform distribution, and the values can be 6, 7, and 8. Also assume a single period setting. You have 3 options as follows:

Option 1: produce in-house (assume that this option allows you to match the demand perfectly). The cost of production is $3 per unit.

Option 2: Purchase the units from a supplier at $2.5 per unit.

Option 3: Purchase the units from a supplier at $2 per unit, but share 25% of the revenues with the supplier.

Assume a selling price of $5, and a salvage value of $1.5 for all three options. Moreover, the lead time for options 2 and 3 are long, and therefore you need to place your order before the demand realizes. What would you do?

Solution #3

Exercise #4

There are two products (A and B). The payoffs and the corresponding probabilities of marketing either product is given as follows:

Pr(gaining $100 with product A)=0.6 and Pr(losing $40 with product A)=0.4.
Pr(gaining $50 with product B)=0.5 and Pr(gaining $60 with product B)=0.5.

1. Assuming that you can only market one of these two products, which one would you choose?
2. Assume that now there is a test which tells you whether the market for Product A is good or bad, and this information is perfect. The probability of the test producing positive result is 0.6 (when the market is good you make $100, and when it is bad you lose $40). What would be the value of the perfect information? How much would you be willing to pay for this test?
3. Suppose now that you have a similar test, but with imperfect information. That is, the market itself is good with a likelihood of 5/6 when the test result is positive. Also, the market is bad with a likelihood of 3/4 when the test result is negative. The probability that the test result will be positive is still the same, 0.6. What would be the value of this imperfect information?

Solution #4 – no test

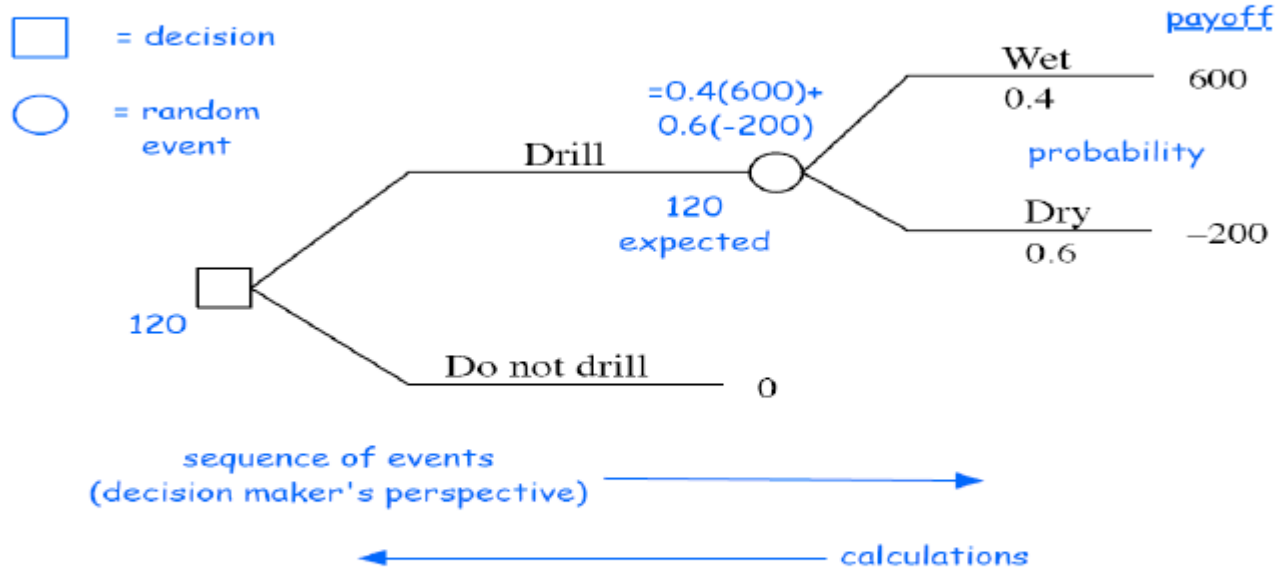
Solution #4 – perfect information

Solution #4 – imperfect information

Using the Decision Tree Add-in in Excel

Using a Decision Tree for the Oil Drilling Problem

Decision Tree



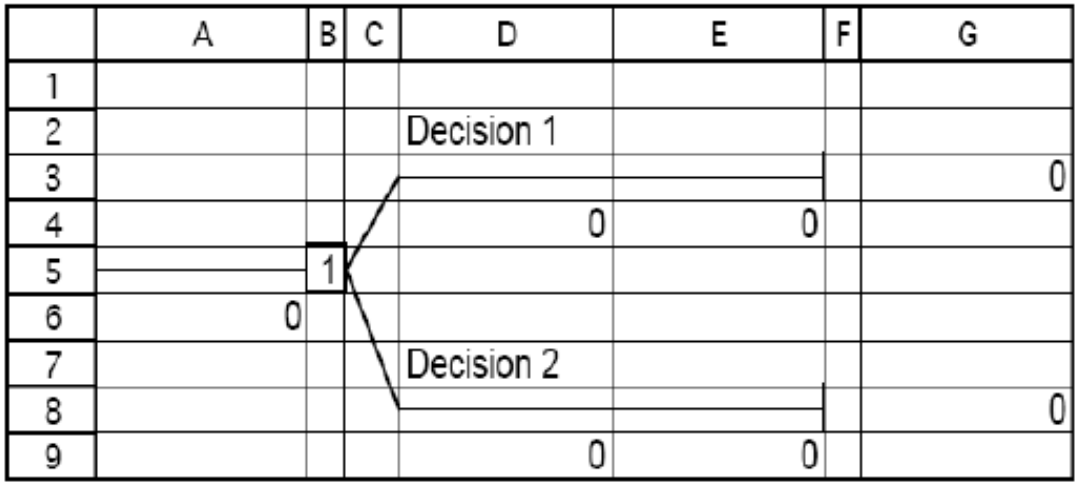
Folding back

- At each event node (circle): calculate expected value (SUMPRODUCT of payoffs and probabilities for each branch).
- At each decision node (square): choose *best* branch (maximum value).

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Using TreePlan to Analyze Oil Drilling Problem

1. Open TreePlan (the tree170a.xla file in Software/DecisionToolkit Academic on textbook CD).
2. Choose Decision Tree on the Add-In tab (Excel 2007) or under the Tools menu (earlier versions).
3. Click on "New Tree" and it will draw a default tree with a single decision node and two branches, as shown below.

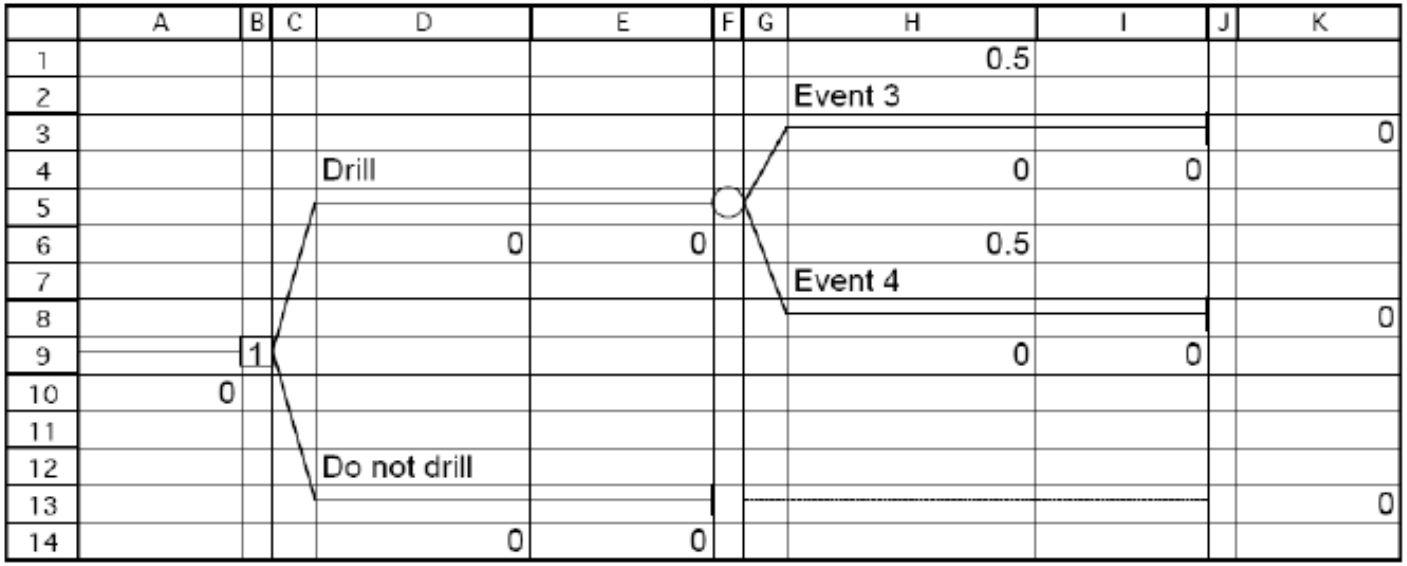


4. Label each branch. Replace "Decision 1" with "Drill" (cell D2). Replace "Decision 2" with "Do not drill" (cell D7).

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Using TreePlan to Analyze Oil Drilling Problem (cont'd)

5. To replace the terminal node of the drill branch with an event node, click on the terminal node (cell F3) and then choose Decision Tree on the Add-In tab (Excel 2007) or under the Tools menu (earlier versions). Click on "Change to event node," choose two branches, then click OK. TreePlan draws the tree below.



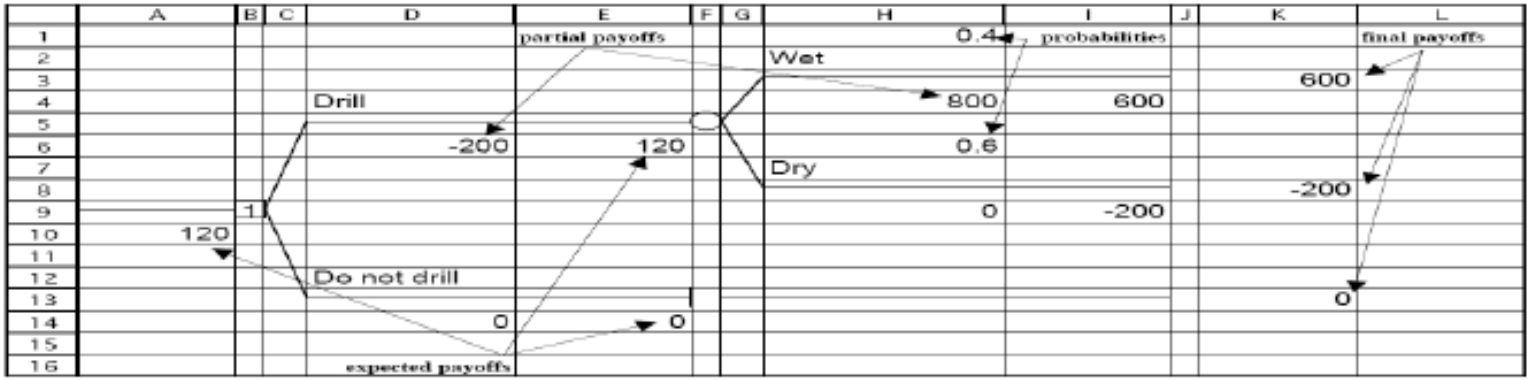
6. Change the labels "Event 3" and "Event 4" to "Wet" and "Dry", respectively.

7. Change the default probabilities (cells H1 and H6) from 0.5 and 0.5 to the correct values of 0.4 and 0.6.

8. Enter the partial payoffs under each branch: (-200) for "Drill" (D6), 0 for "Do not Drill" (D14), 800 for "Wet" (H4), and 0 for "Dry" (H9). The terminal value cash flows are calculated automatically from the partial cash flows.

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Final Decision Tree



Decision: 1 = first branch ; 2 = second branch ; etc.

Features of TreePlan

- Terminal values (payoff) are calculated automatically from the partial payoffs (K3=D6+H4, K8=D6+H9, K13=D14). Alternatively, they can be entered directly (in which case the partial payoffs are ignored).
- Foldback values are calculated automatically (I4=K3, I9=K8, E6=H1*I4+H6*I9, E14=K13, A10=Max(E6, E14)).
- Optimal decisions are indicated inside decision node squares (labeled by branch number from top to bottom, e.g., branch #1 = Drill, branch #2 = Do not drill).
- Changes in the tree can be made by clicking on a node, and choosing Decision Tree on the Add-In tab (Excel 2007) or under the Tools menu (earlier versions).
- Clicking "Options..." in the Decision Tree dialogue box allows the choice of Maximize Profit or Minimize Cost.

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QUEUEING MODELS

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TALENT HUB FOR SUPPLY CHAIN

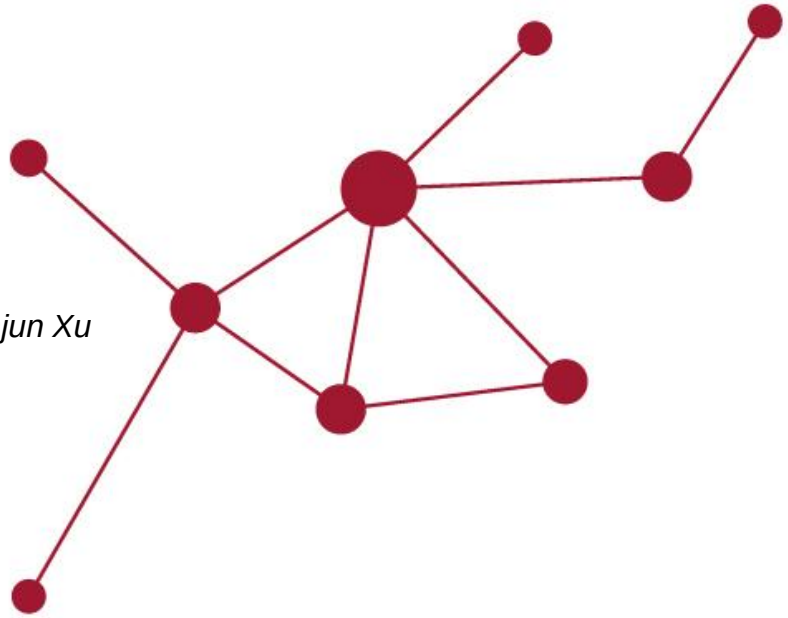
Zaragoza Logistics Center

Waiting Line Models (Queues)

September 14, 2026

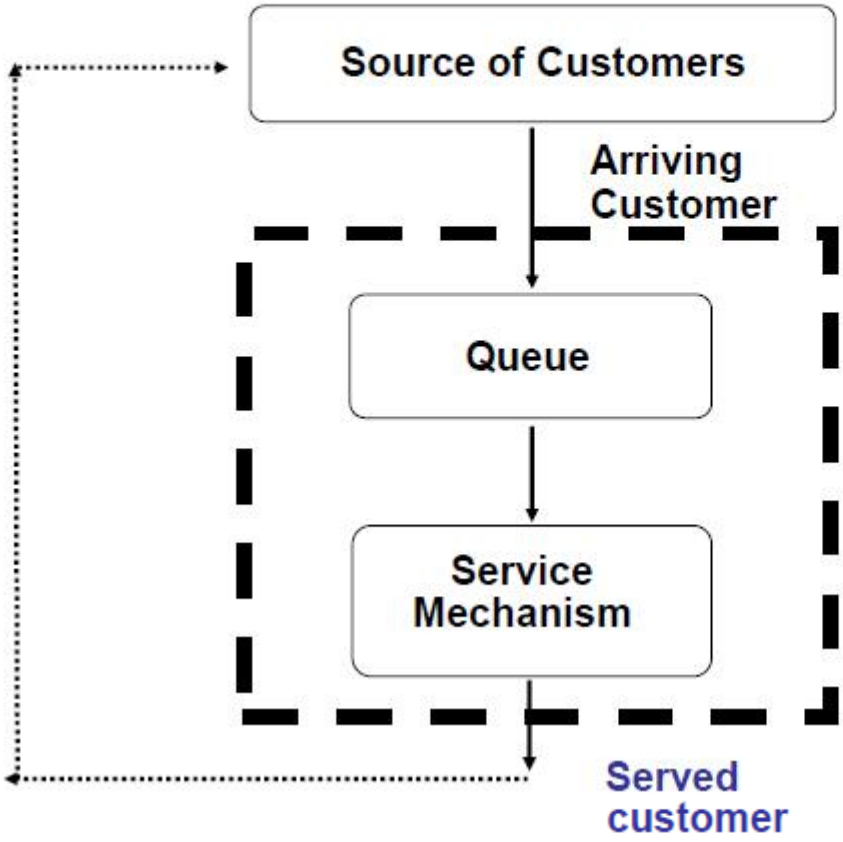
Some slides borrowed from Prof. Kamran Moinzadeh and Prof. Jianjun Xu

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Examples of Queuing Systems

Queuing Model



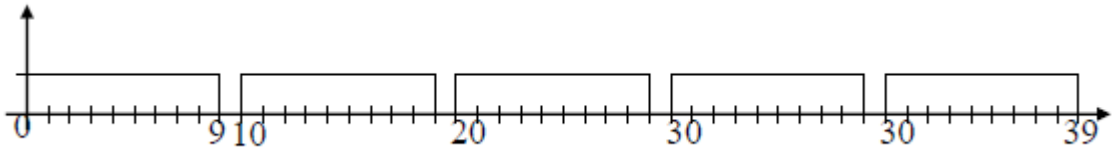


Queuing System

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Why does a queue form?

- Deterministic arrivals every 10 sec
- Deterministic service times 9 sec

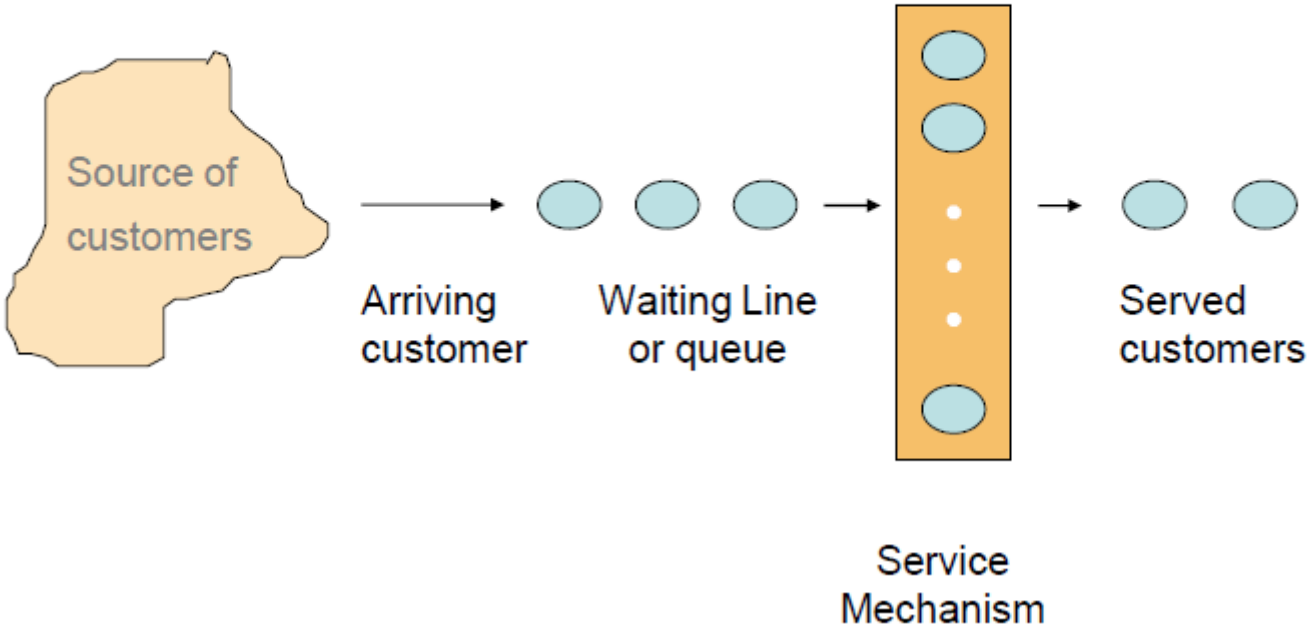


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The Psychology of Waiting Lines

Some "Creative" Solutions

Components Of Queuing System



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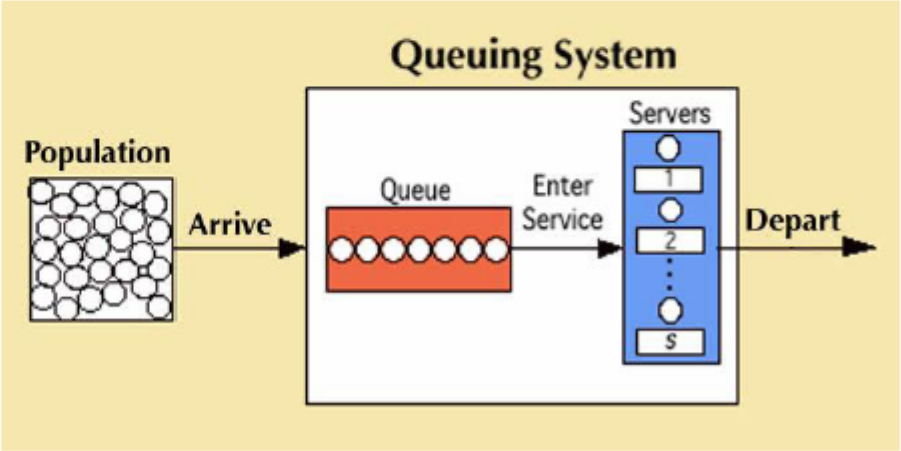
Components of the Queuing Model

1. Calling population
2. Arrival distribution
3. Number of servers
4. Service time distribution
5. Waiting line structure
6. Queue discipline
7. Maximum allowable queue length

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Components of Waiting Lines

1. CALLING POPULATION
   - **Finite**
   - **Infinite**



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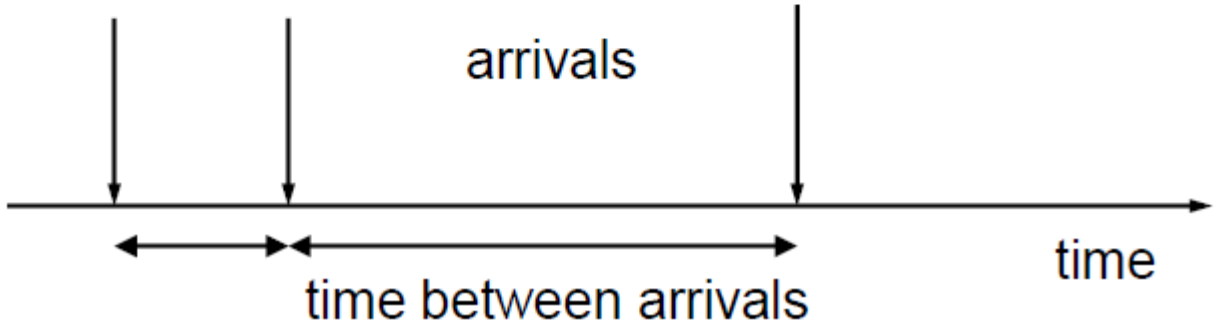
Components of Waiting Lines

2. DISTRIBUTION OF ARRIVALS

Terminology

Arrival rate (λ) = Mean number of arrivals per time period

Interarrival time = Time between consecutive arrivals



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Components of Waiting Lines

Poisson Arrivals

- Random (discretely distributed)
- Counted within some time period T

Probability of n arrivals in period T $= \frac{(\lambda T)^n e^{-\lambda T}}{n!}$

where n = 0,1, 2, 3, ...

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Poisson Arrivals

- Good assumption for many real life situations
- Allows simple quantitative models to be built
- Assumes that arrival process is “memoryless”, i.e., knowing when last arrival occurred, doesn’t help in predicting when next arrival will occur
  - Time between arrivals has an exponential distribution
    - Let t be the time between two consecutive arrivals:

$$f(t) = \begin{cases} 0 & t < 0 \\ \lambda e^{-\lambda t} & t \geq 0 \end{cases}$$

Poisson Arrivals (conditions)

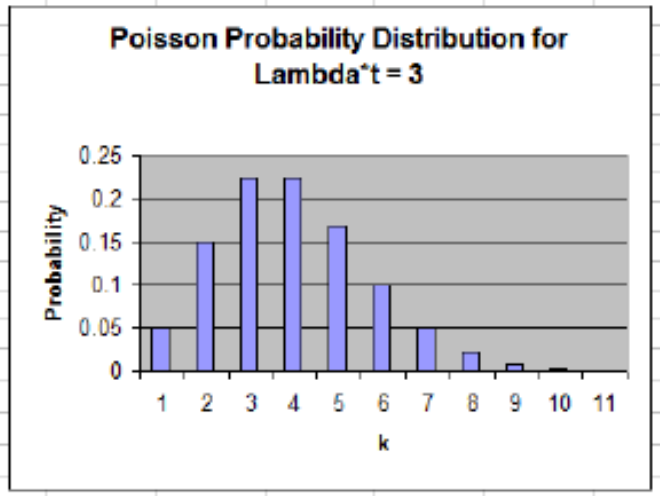
- Under three conditions the arrivals can be modeled as a Poisson process
  - ***Orderliness*** : one customer, at most, will arrive during any time instant.
  - ***Stationarity*** : for a given time frame, the probability of arrivals within a certain time interval is the same for all time intervals of equal length.
  - ***Independence*** : the arrival of one customer has no influence on the arrival of another.

HANK's Hardware -- Arrival

- Customers arrive at Hank's Hardware according to a Poisson distribution.
- Between 8:00 and 9:00 A.M. an average of 6 customers arrive at the store.
- What is the probability that k customers will arrive between 8:00 and 8:30 in the morning (k = 0, 1, 2,…)?

	A	B	C	D	E	F	G	H	I	J	K	L
1	**Poisson Probabilities**											
2												
3	**Lamba*t =**	3										
4												
5	**k**	**P(X=k)**										
6	0	0.049787										
7	1	0.149361										
8	2	0.224042										
9	3	0.224042										
10	4	0.168031										
11	5	0.100819										
12	6	0.050409										
13	7	0.021604										
14	8	0.008102										
15	9	0.002701										
16	10	0.00081										
17												
18	**P(X > 10) =**	0.000292										
19												
20	**P(X >= 2) =**	0.800852										
21												
22												
23												
24												
25												

=POISSON(A6,$B$3,FALSE)
Drag to cells B7:B16



=1-POISSON(10,$B$3,TRUE)

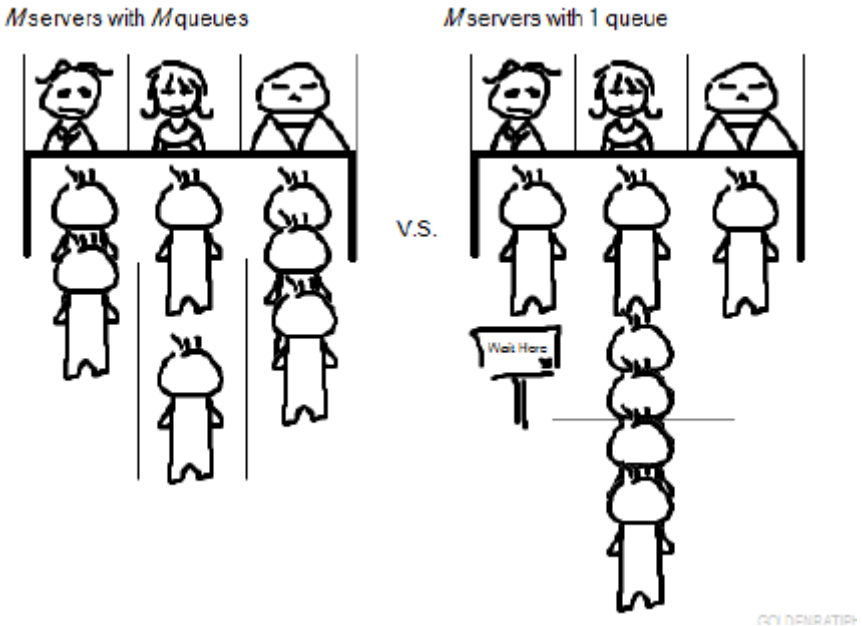
=1-POISSON(1,$B$3,TRUE)

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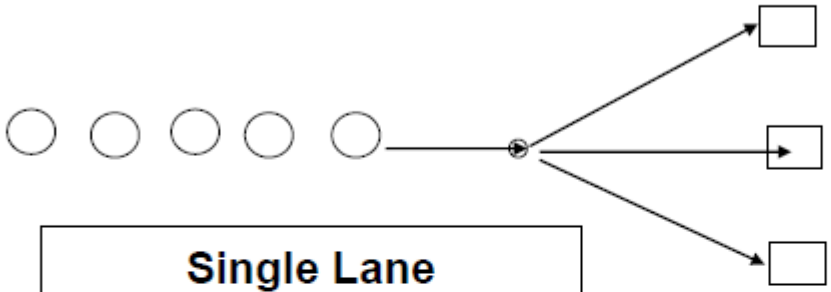
Components of Waiting Lines

3. NUMBER OF SERVERS (s)

   - Single server lines: s = 1
   - Multiple server lines: s > 1



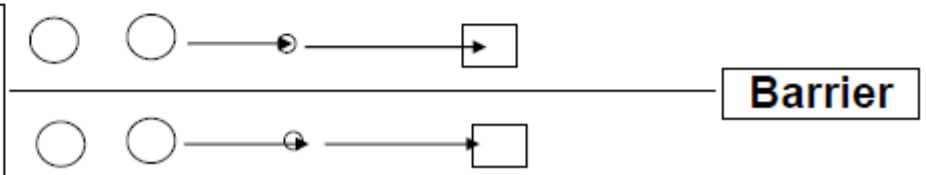
Tunnel Traffic Toll



Single Lane
Multiple Toll Booths

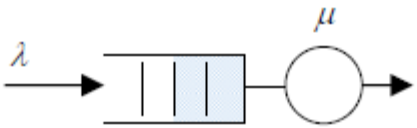
Multiple Lanes:

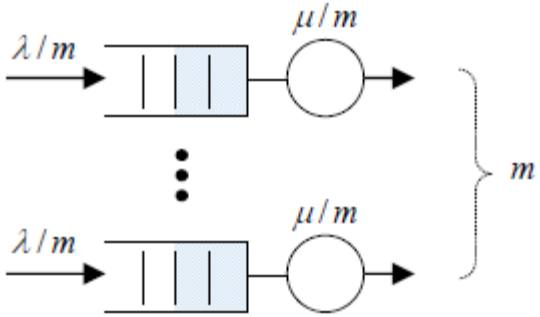
1 booth per server



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Multiplexing choices





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Components of Waiting Lines

4. SERVICE TIME

- Distribution of Service Time
  - Exponential
  - Deterministic
  - General

- Service Rate = capacity of a server in number of customers per time period = μ
  - **Average service time = $1/\mu$**

- NOTE: In order to have a stable system, the following condition must be satisfied:
  - **$s\mu > \lambda_e$**

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Exponential Service Time

- The memoryless property.
  - No additional information about the time left for the completion of a service, is gained by recording the time elapsed since the service started.

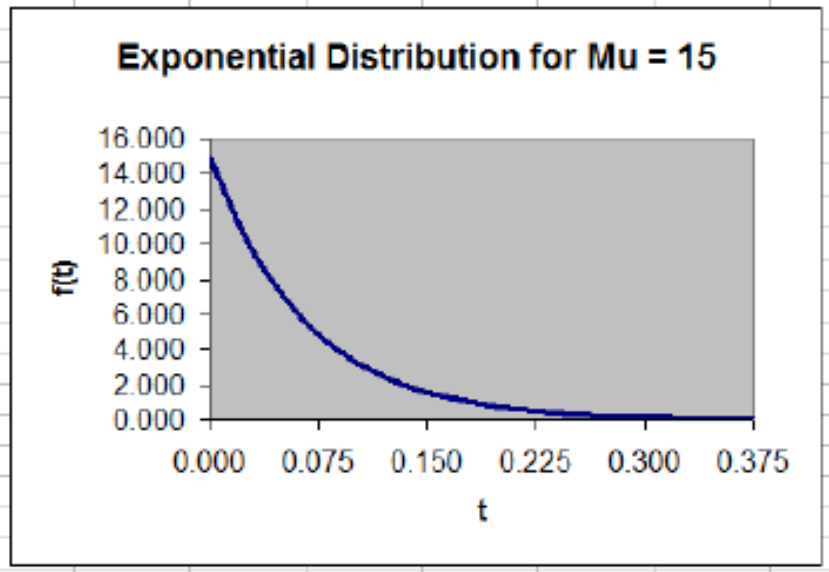
- The Exponential and the Poisson distributions are related to one another.
  - If customer arrivals follow a **Poisson distribution with mean rate λ,** their interarrival times are **exponentially distributed with mean time $1/\lambda$.**

HANK's Hardware -- Service

- Hank's estimates the average service time to be $1/\mu$ = 4 minutes per customer. Service time follows an exponential distribution.
- What is the probability that it will take less than 3 minutes to serve the next customer?

	A	B
1	Exponential Distribution	
2		
3	Mu	15
4	t =	0.05
5		
6	P(X <= t) =	0.527633
7		
8	Density Function	
9	t	f(t)
10	0.000	15.00000
11	0.025	10.30934
12	0.050	7.08550
13	0.075	4.86979
14	0.100	3.34695
15	0.125	2.30032
16	0.150	1.58099
17	0.175	1.08660
18	0.200	0.74681
19	0.225	0.51327
20	0.250	0.35277
21	0.275	0.24245
22	0.300	0.16663
23	0.325	0.11453
24	0.350	0.07871
25	0.375	0.05410
26	0.400	0.03718

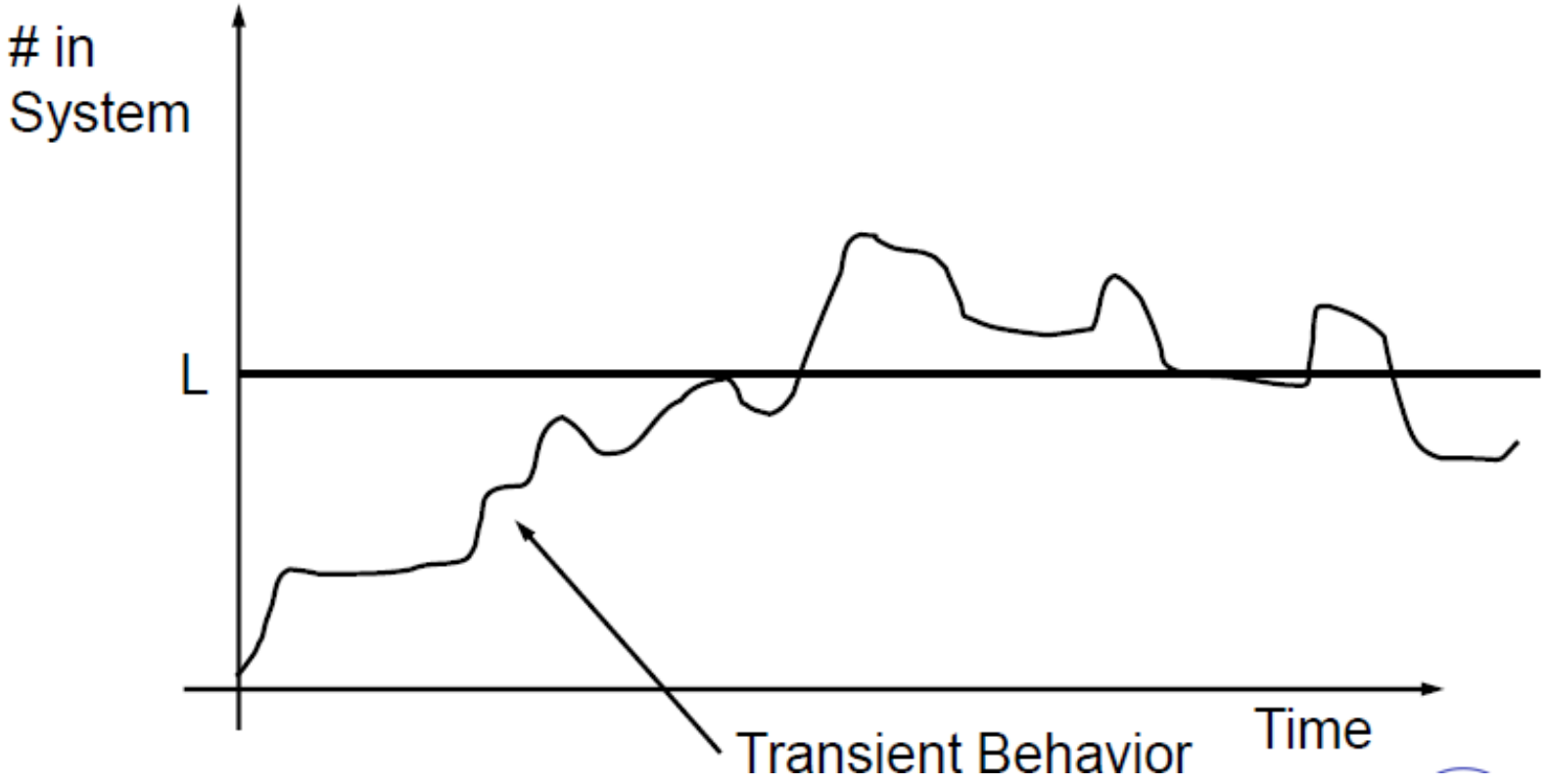
=EXPONDIST(B4,B3,TRUE)



=EXPONDIST(A10,$B$3,FALSE)
Drag to B11:B26

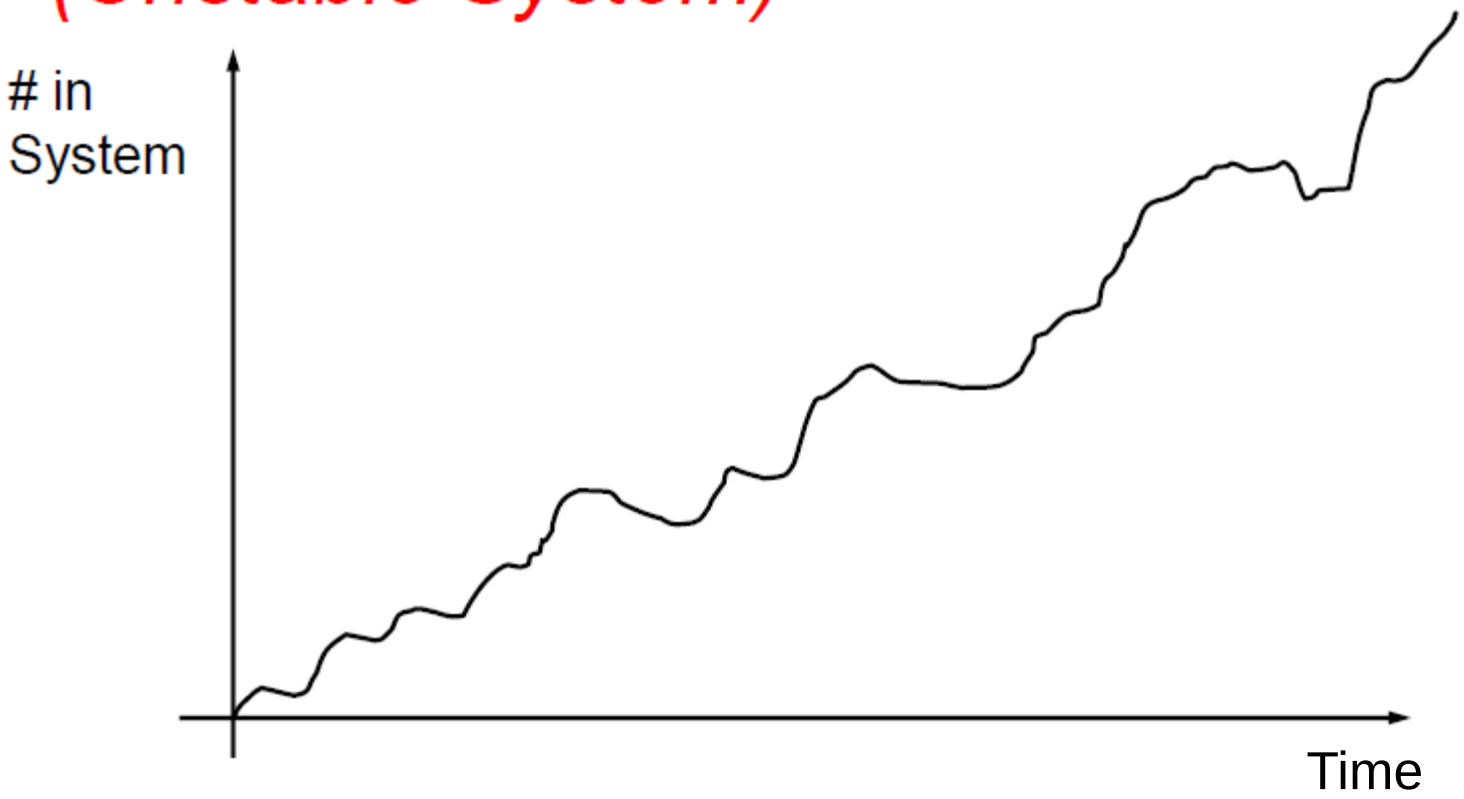
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A System with a Steady State (Stable System)



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A System without a Steady State (Unstable System)



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Steady State Results

- If a system has a steady state (i.e. in a stable system), then

  Rate of Entry = Rate of Exit (i.e. inflow=outflow)

  – Throughput of the system is the minimum of effective arrival rate or service rate

- A multichannel, single phase system has a steady state if and only if

  $$\lambda_e < s\,\mu$$

  – In such a system,

  λ_e = effective rate of entry and

  $s\,\mu$ =maximum rate customers can exit

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LAA

- Lack of Anticipation Assumption (LAA): Future inter-arrival times and service times of previously arrived customers are independent

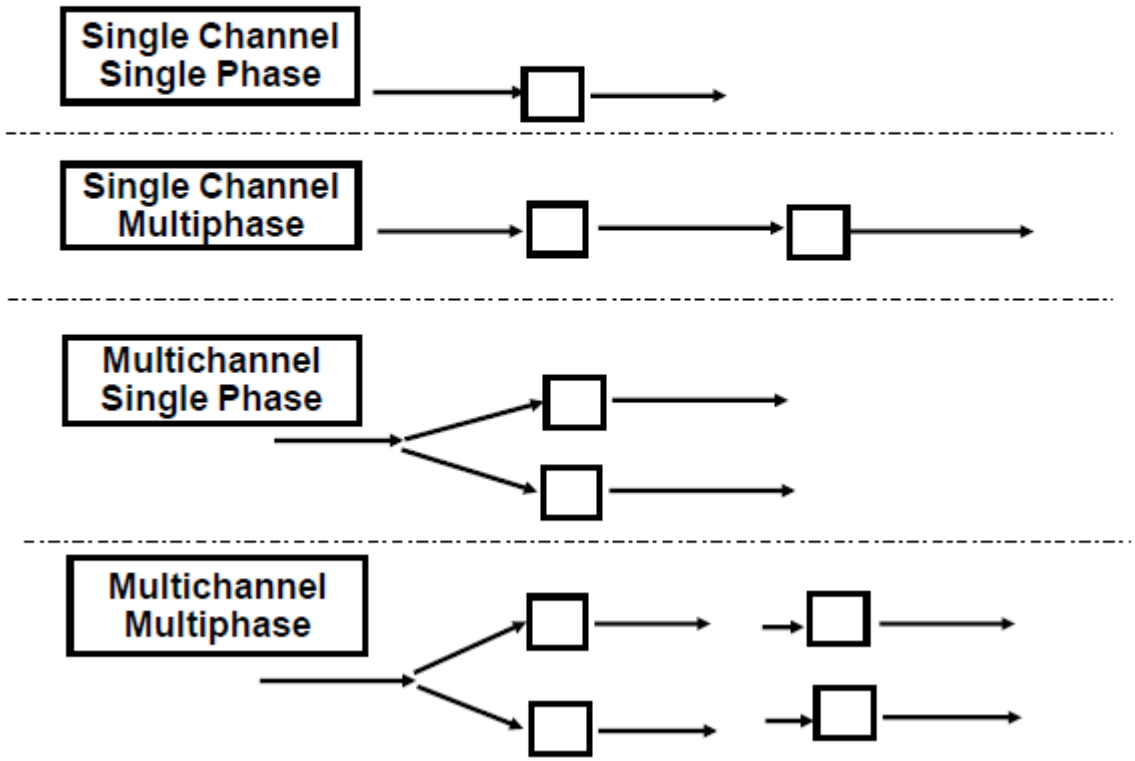
Components of Waiting Lines

5. WAITING LINE STRUCTURES

- Single Channel, Single Phase
- Single Channel, Multiphase
- Multichannel, Single Phase
- Multichannel, Multiphase
- Mixed

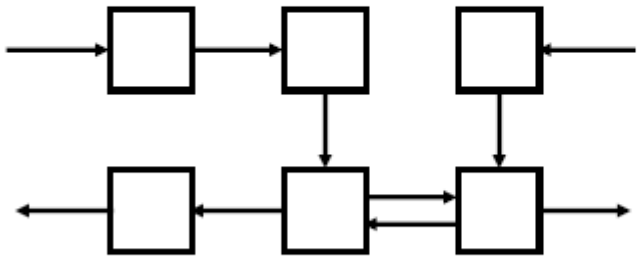
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WAITING LINE STRUCTURES



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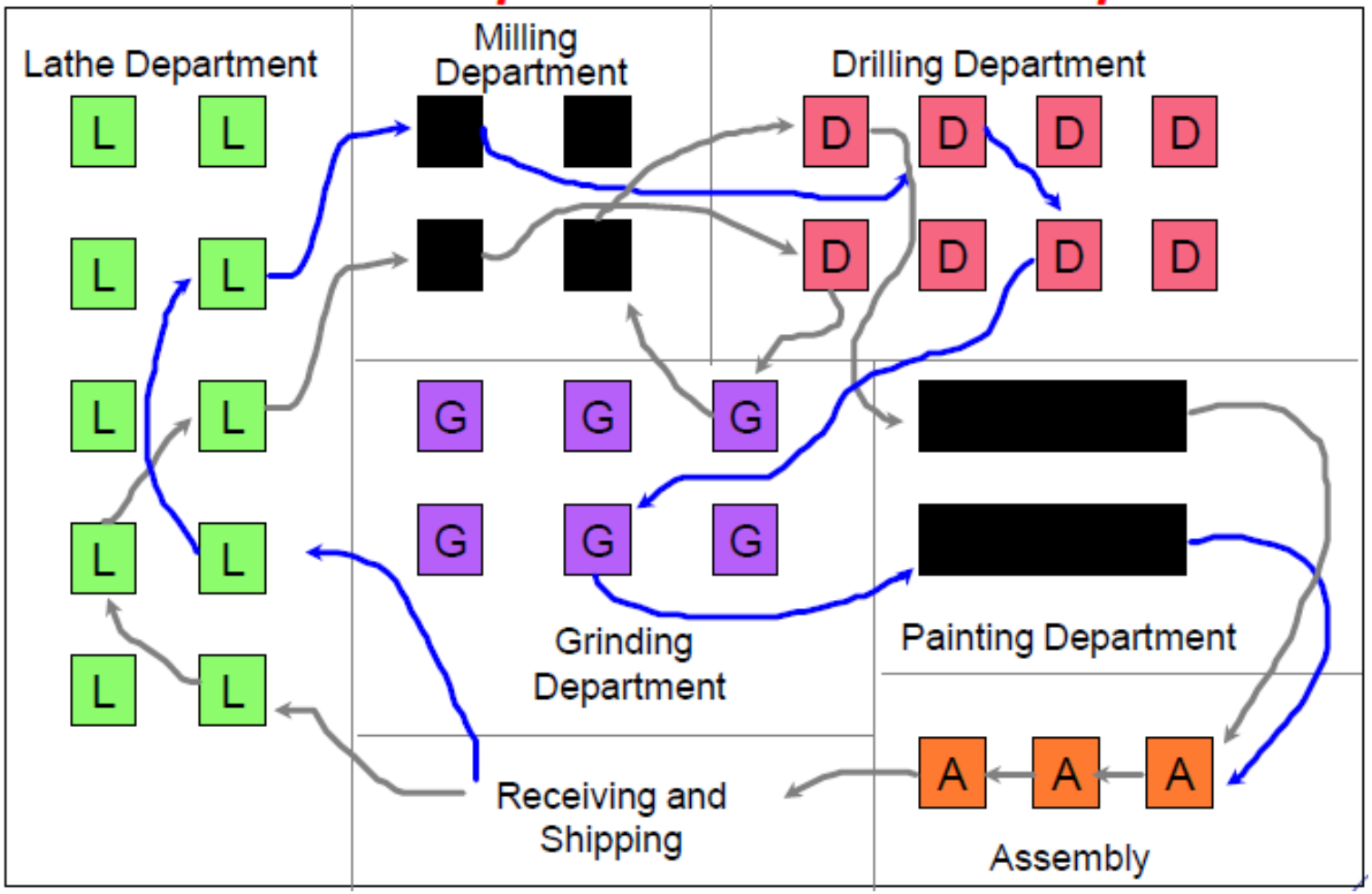
WAITING LINE STRUCTURES



Alternate paths

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Mixed Example: A Job Shop



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Components of Waiting Lines

6. QUEUE DISCIPLINE

- FCFS
- Shortest service time first
- Reservations first
- Emergencies first
- Highest profit customers first
- Largest order first
- Shared Services

Components of Waiting Lines

Queue Disciplines:

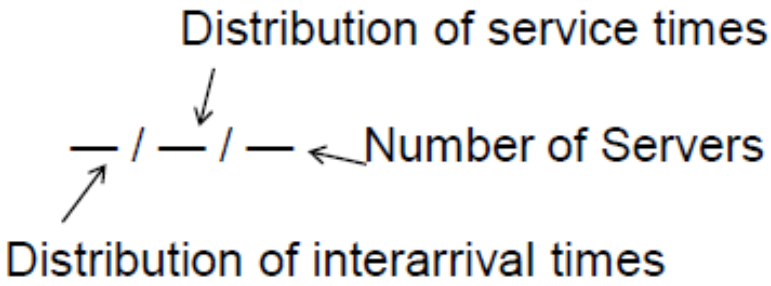
- *Jockeying:* moving from line to line in hopes of reducing waiting time
- *Balking:* refusing to enter the system because the lines are too long
- *Reneging:* leaving the system without receiving service after joining a line because the wait has been too long
- *Preemption:* displacing a customer being served by a more important customer
- *Resume:* continue processing the job from where it was left..

Components of Waiting Lines

7. MAXIMUM ALLOWABLE QUEUE LENGTH: number of customers a queue can hold
   - Finite
     - Customers will not enter if queue is full (not the same as balking)
       - Customers/jobs will be denied entry
     - but certain balking behavior can be approximated by a capacitated queue.
   - Infinite

Labels for Queueing Models

•To identify which probability distribution is being assumed for service times (and for interarrival times), a queueing model conventionally is labeled as follows:



•The symbols used for the possible distributions are

M = Exponential distribution (Markovian)

D = Degenerate distribution (constant times)

GI = General independent interarrival-time distribution (any distribution)

G = General service-time distribution (any arbitrary distribution)

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System Operating Characteristics

L: Average number of customers in the system (waiting to be served and are in service)

L_q: Average number of customers in the waiting line

W: Average time a customer spends in the system (waiting and being served)

W_q: Average time a customer spends waiting in line

- Operating characteristics are computed that describe system performance
- Steady state is constant, average value for performance characteristics that the system will reach after a long time

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Operating Characteristics

P_0:	Probability of no (zero) customers in the system
P_n:	Probability of *n* customers in the system
ρ:	Utilization rate; the proportion of time the system is in use

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Little's Law

- Let λ_e denote the average effective rate customers enter a queueing system
- Little's Laws states that
  - $L = \lambda_e W$ *and*
  - $L_q = \lambda_e W_q$
  - *Valid for queues with general arrival and service patterns..*

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Summary of Fundamental Laws

- Let μ = the service rate
- Then $1 / \mu$ = the average service time
  - $W = W_q + 1 / \mu$
  - $L = \lambda_e W$
  - $L_q = \lambda_e W_q$
- If you know <u>one</u> of L, W, L_q, and W_q you can find the other three by using the fundamental laws

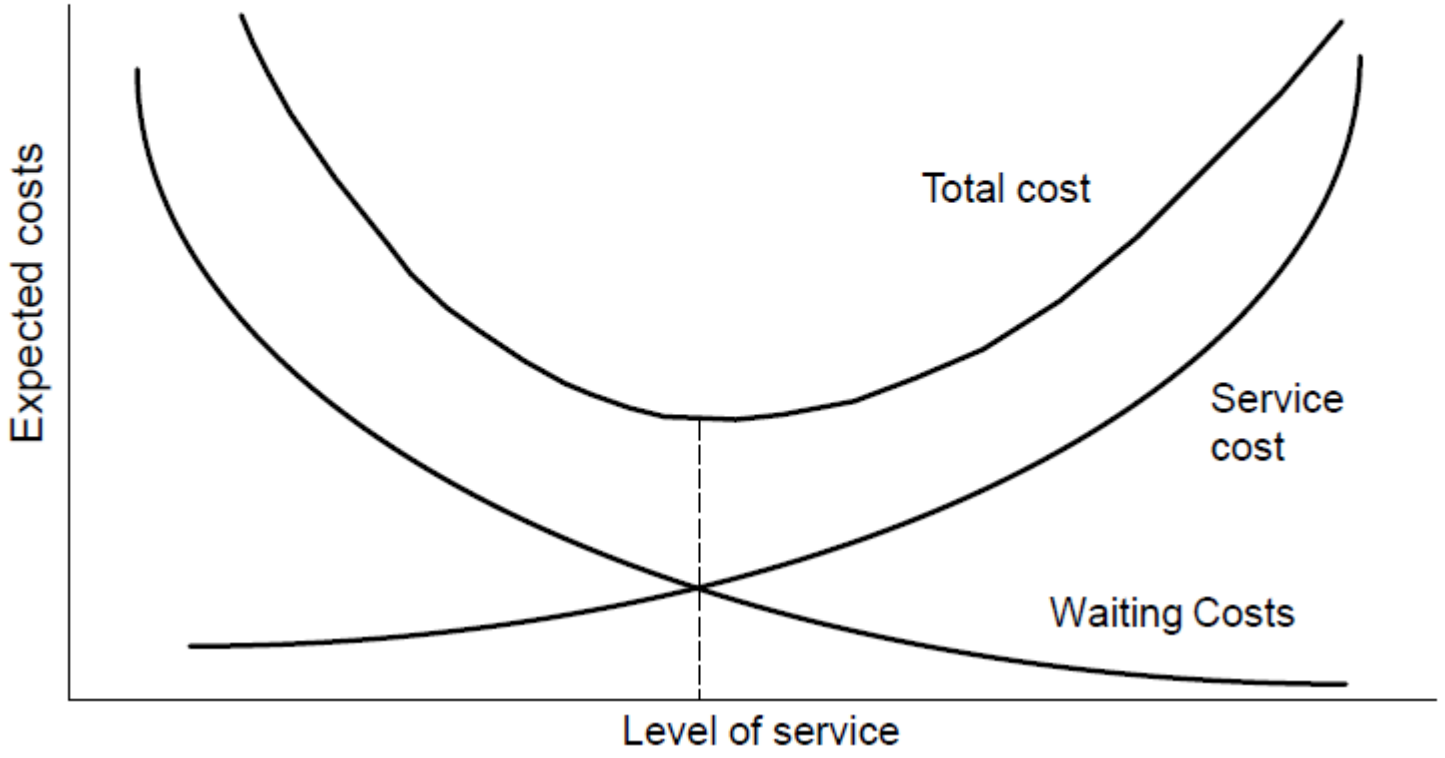
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Example of Using Fundamental Laws

- Customers arrive at McBurger at the rate of 30 per hour
- The average time to take and deliver a customer's order is 4 minutes
- The average time a customer spends in line and being served is 10 minutes
- Find the average time in line, the average number in line, and the average number in line and being served

Solution to the example:

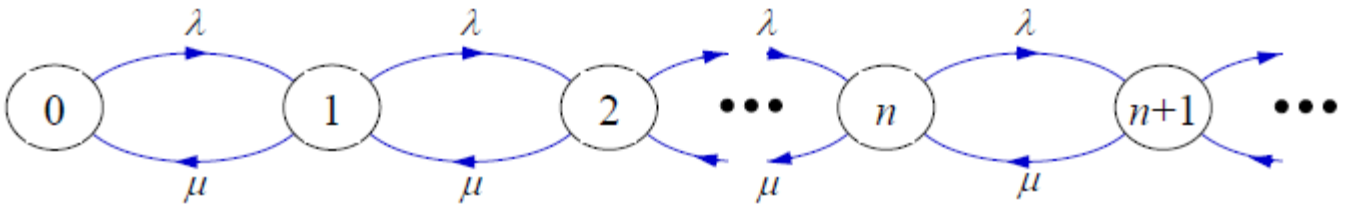
Cost Relationship In Waiting Line Analysis



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The M/M/1 Queue

- Arrival process: Poisson with rate λ (state independent)
- Service times: iid, exponential with parameter μ (state independent)
- Service times and interarrival times: independent
- Single server
- Infinite waiting room
- $N(t)$: Number of customers in system at time t (state)



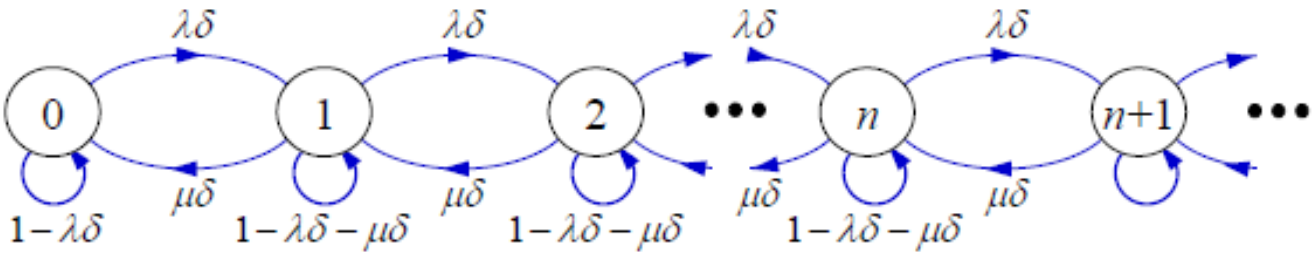
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M/M/1 Queue: Discrete-Time Formulation

- If we take snapshots at times 0, δ, 2δ,… (δ very small)
- Transition probabilities are

$$
\begin{aligned}
P_{00} &= 1-\lambda\delta+o(\delta) \\
P_{ii} &= 1-\lambda\delta-\mu\delta+o(\delta), \quad i \geq 1 \\
P_{i,i+1} &= \lambda\delta+o(\delta), \quad i \geq 0 \\
P_{i,i-1} &= \mu\delta+o(\delta), \quad i \geq 0 \\
P_{ij} &= o(\delta), \quad |i-j| > 1
\end{aligned}
$$

- If we ignore the higher order terms $o(\delta)$, the discrete time Markov chain would look like



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M/M/1 Queue: Discrete-Time Formulation

- The stationary distribution solves to

$$[\mu\delta + o(\delta)]\pi_n = [\lambda\delta + o(\delta)]\pi_{n-1} \Rightarrow$$

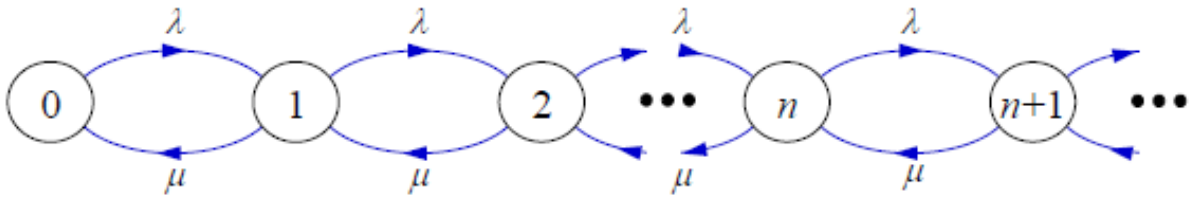
$$\pi_n = \frac{\lambda\delta + o(\delta)}{\mu\delta + o(\delta)}\pi_{n-1} = \cdots = \left[\frac{\lambda\delta + o(\delta)}{\mu\delta + o(\delta)}\right]^n \pi_0$$

- Taking the limit $\delta \rightarrow 0$:

$$\lim_{\delta\to 0}\pi_n = \lim_{\delta\to 0}\left[\frac{\lambda\delta + o(\delta)}{\mu\delta + o(\delta)}\right]^n \lim_{\delta\to 0}\pi_0 \Rightarrow p_n = \left(\frac{\lambda}{\mu}\right)^n p_0$$

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M/M/1 Queue: Stationary Distribution



- Birth-death process

$$\mu p_n = \lambda p_{n-1} \Rightarrow$$

$$p_n = \frac{\lambda}{\mu} p_{n-1} = \rho p_{n-1} = \ldots = \rho^n p_0$$

- Normalization constant

$$\sum_{n=0}^{\infty} p_n = 1 \Leftrightarrow p_0 \left[1 + \sum_{n=1}^{\infty} \rho^n \right] = 1 \Leftrightarrow p_0 = 1 - \rho \ , \ \text{if } \rho < 1$$

- Stationary distribution $p_n = \rho^n (1-\rho), \quad n = 0, 1, \ldots$

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The M/M/1 Queue

- Average number of customers in system

$$L = \sum_{n=0}^{\infty} np_n = (1-\rho)\sum_{n=0}^{\infty} n\rho^n = (1-\rho)\rho\sum_{n=0}^{\infty} n\rho^{n-1}$$

$$\Rightarrow L = (1-\rho)\rho\frac{1}{(1-\rho)^2} = \frac{\lambda}{\mu-\lambda}$$

- Little's Theorem: average time in system

$$W = \frac{L}{\lambda} = \frac{1}{\lambda}\frac{\lambda}{\mu-\lambda} = \frac{1}{\mu-\lambda}$$

- Average waiting time and number of customers in the queue – excluding service

$$W_q = W - \frac{1}{\mu} = \frac{\rho}{\mu-\lambda} \text{ and } L_Q = \lambda W_q = \frac{\rho^2}{1-\rho}$$

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The M/M/1 Queue

- The probability that a customer waits more than "t" units of time in system has an exponential distribution with the rate μ-λ:

$$P(\text{Wait in system} > t) = e^{-(\mu-\lambda)t}$$

- Similarly The probability that a customer waits more than "t" units of time in queue is:

$$P(\text{Wait in queue} > t) = \rho e^{-(\mu-\lambda)t}$$

Example:

Auto XYZ is an automotive manufacturer with most of its Tier-1 suppliers located within a 10 miles radius. Ready assemblies from the suppliers are delivered Just In Time(JIT) and Just in Sequence(JIS) to line side trucking docks by a TPL. The management estimates that trucks from the TPL will arrive at a rate of 15 per hour in its two shift operation. The conveyor system involved in unloading and transporting the assemblies to line side bins can unload at an average rate of one every three minutes.

Assuming Poisson arrivals and exponential service, find:

- *utilization of the truck unloading material handling system*
- *average number of trucks waiting in the receiving area*
- *average waiting time for a truck*

Example (cont'd)

Because of limited space availability in the receiving area and a desire to provide an acceptable level of service, the management would like to ensure, with 95 % confidence, that not more than three trucks will be in the system at any one time.

- *What is the present level of service for the three-truck limit?*
- *At what level of utilization of the unloading conveyor can the 95 % level of service be assured?*
- *What must be the service rate of the conveyor to assure the 95 % level of service?*

Example (cont'd)

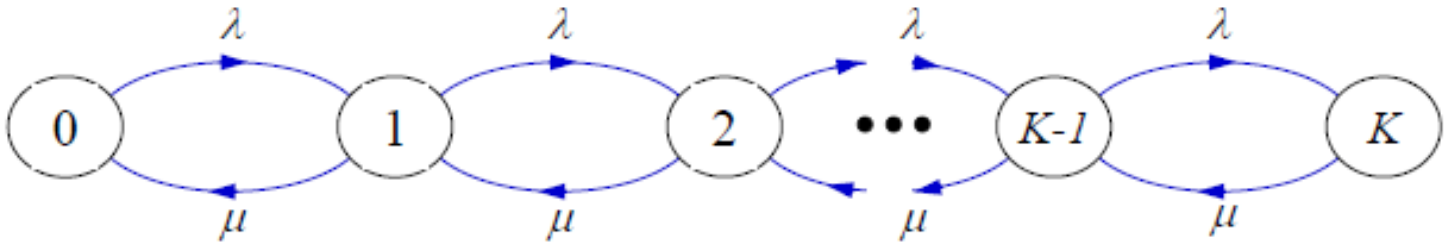
Additionally, the new plant manager in the future wants to invest in a new conveyor system which will drastically speed up the truck unloading process. The table below is a summary the engineering group has prepared for monthly costs of different options of speeding up the conveyor and auxiliary equipment

cost of speeding up conveyor	
minutes per truck	cost
3	$ -
2.8	$ 1,200
2.7	$ 1,700
2.5	$ 2,300
2	$ 3,300

•The management group had anticipated that a truck would have to wait in line no more than 12 minutes. To obtain better rates from the TPL, they decided to include a demurrage clause in the contract with the TPL. So if a truck has to wait more than 12 minutes in the receiving area , then the company pays $2.0 per truck to the TPL.

• *What will be your recommendation to the management group in this matter. Assume all other factors associated with making these changes stay the same.*

M/M/1/K Queue



- M/M/1 with finite waiting room
  - At most K customers in the system
  - Customer that upon arrival finds K customers in system is dropped
- Stationary distribution

$$p_n = \rho^n p_0, n = 1, 2, \ldots, K$$

$$p_0 = \frac{1-\rho}{1-\rho^{K+1}}$$

- *Is this system stable?*
  - If ρ=1, all states are equally likely: p_n=1/(K+1) for n=0,1,..,K and L=K/2.

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M/M/1/K Queue

- The effective arrival rate is:

$$\lambda_{eff} = \lambda P(\text{Number in system} < \text{K}) + 0\text{P}(\text{Number in system} = \text{K})$$
$$= \lambda(1 - P_K)$$

- The average number in the system is:

$$L = \frac{\rho}{1-\rho} - \frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}}$$

- The Little's Law:

$$W = L / \lambda_{eff}$$

$$W_q = L_q / \lambda_{eff}$$

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Example:

A popular attraction at the New Jersey Shore is a street artist who will paint a caricature in about five minutes. However, because the times requires for each drawing vary considerably, they are accurately described by an exponential distribution. People are willing to wait their turn, but when there are more than 10 waiting for a picture, customers are turned away and asked to return at a later time. At peak times one can expect as many as 20 customers per hour. Assume that customers arrive completely random at the peak arrival rate.

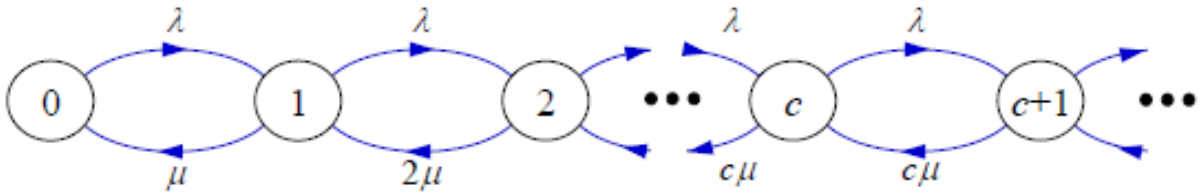
(a) What proportion of the time is the queue at maximum capacity?

(b) How many customers are being turned away, on average? Determine the measures of performance for this queueing system.

(c) If the waiting area were doubled in size, how would that affect your answer to part (a)?

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M/M/c Queue



- Poisson arrivals with rate λ
- Exponential service times with parameter μ
- *c* identical servers
- Arriving customer finds *n* customers in system
  - $n < c$: it is routed to any idle server
  - $n \geq c$: it joins the waiting queue – all servers are busy

➡ Birth-death process with state-dependent death rates

$$\mu_n = \begin{cases} n\mu, & 1 \leq n \leq c \\ c\mu, & n \geq c \end{cases}$$

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M/M/c Queue

- The probability of having "n" customers at any point in time is:

$$p_n = \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n p_0, 0 \le n \le c$$

$$p_n = \frac{1}{c!c^{n-c}}\left(\frac{\lambda}{\mu}\right)^n p_0, n > c$$

where

$$p_0 = \left\{\sum_{n=0}^{c-1}\frac{(c\rho)^n}{n!} + \frac{(c\rho)^c}{c!}(1-\rho)^{-1}\right\}^{-1}$$

$$\rho = \lambda / c\mu$$

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M/M/c Queue

- The performance measures are:

$$L_q = \frac{c^c \rho^{c+1}}{c!(1-\rho)^2} p_0$$

$$W = W_q + 1/\mu$$

$$W_q = L_q/\lambda$$

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Example:

Tony's Barbershop is run, owned, and operated by Anthony Jones, who has been cutting hair for more than 20 years. Anthony does not take appointments, so the arrival pattern of customers is essentially random. Traditionally, the arrival rate had been about one customer every 50 minutes. Two months ago, the local partner ran an article about Anthony that improved business substantially. Currently, the arrival rate is closer to one customer every 35 minutes. Haircuts require an average of 25 minutes, but the time vary considerably depending on customer needs. A trim might require as little as 5 minutes, but a shampoo and styling could take as long as an hour or more. For this reason, the exponential distribution seems to provide a reasonably good fit for the service time distribution. Due to the improvement in business, Anthony is considering to take his cousin Marvin into the business. Assume Marvin cuts hair at the same rate as Anthony.

- *How much has the quality of service declined since more customers have started using the shop (without Marvin)?*
- *How much improvement in the performance of the system are customers likely to see with an additional barber in the shop?*

M/G/1 Queue

- Pollaczek-Khintchine (*P-K*) formula:

$$L_q = \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)}$$

σ^2 : Variance of the service time

$\rho = \lambda/\mu$, where $1/\mu$: expected service time

- For an M/D/1 model:

$$L_q = \frac{\rho^2}{2(1-\rho)}$$

Half of the average number in line of an M/M/1queue

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M/G/∞ Queue

- Assume the arrival rate is λ (Poisson arrivals) and service rate is μ. Then the number in system is Poisson with rate λ / μ:

$$p_n = \frac{e^{-(\lambda/\mu)}(\lambda/\mu)^n}{n!}, n = 0,1,2,....$$

$$L_q = 0, W_q = 0$$

- A common inventory control policy is (S-1,S) (one-for-one policy). Suppose that the demands are generated by a stationary Poisson process with rate λ, and that the lead time required for replenishment is a random variable with arbitrary distribution with mean 1/ μ. This problem is an M/G/∞ model. The number of customers in the system is equal to the number of outstanding orders, and it has a Poisson distribution with rate λ / μ. *(The number of outstanding orders can be used to determine an expression for the expected number of holding, stockout costs).*

SIMULATION

ZLC

TALENT HUB FOR SUPPLY CHAIN

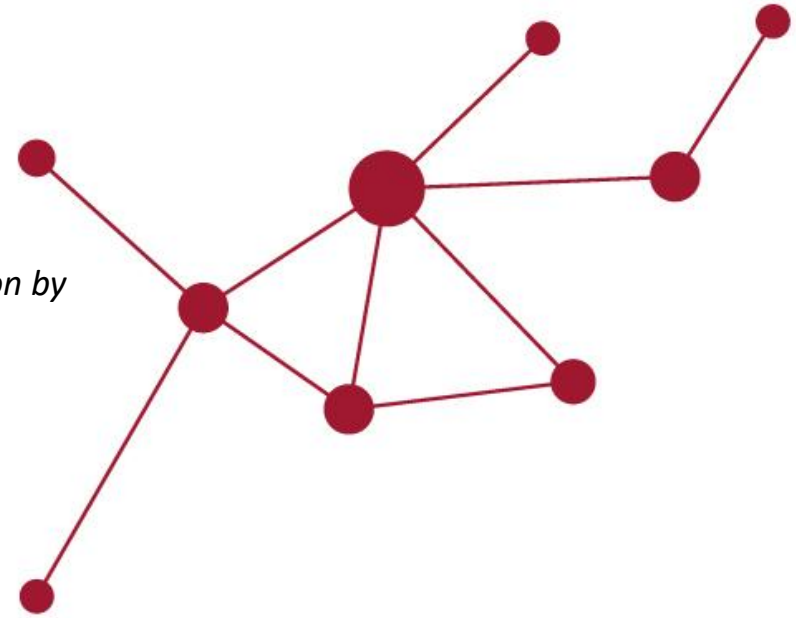
Zaragoza Logistics Center

~~Simulation~~

September 14, 2026

Notes mainly borrowed from the book Discrete-Event Simulation by Jerry Banks, John Carson, Barry Nelson, David Nicol

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Simulation

- Imitation of the operation of a real-world process/system over time
  - Generation of an **artificial history** of the system
  - Observation of this artificial history to gain some insights
  - Simulation model:
    - Takes the form of a set of assumptions concerning the operation of a system
    - Assumptions are expressed in *mathematical, logical, symbolic relationships* between the *entities* (objects) of the system
  - Used to investigate what-if questions

- Use of simulation:
  - Predict the impact of potential changes to an existing system
  - Study systems in the design stage, before they are built

Some applications

- Capacity (e.g., production, storage) expansion decisions
- Runway schedule determination through simulation optimization
- Modeling and simulation of a telephone call center
- Estimating maximum capacity in an emergency room
- Simulation of freeway merging and diverging behavior
- Segmenting the customer base for maximum returns
- Inventory analysis
- Impact of multi-tasking in project management
- Estimating the average wait of a customer in a queueing system
- Pricing policies
- Health care systems
- Urban traffic control

Simulation is appropriate for

- Studying (experimenting with) the internal interactions of a complex system
  - How variables interact, which variables are more important
- Simulating informational, organizational, and environmental changes and observing their impact on the model's behavior
  - Experimenting with new designs/policies before implementation to prepare for what might happen
- Verifying analytic solutions
- Determining the requirements on a machine through simulating different capabilities for it
- Designing tools to train people without the cost and disruption of on-the-job instruction
- New hardware designs, physical layouts, transportation systems can be tested without committing resources for their acquisition
- Animating the system so that the plan can be visualized

Simulation shouldn't be used when

- The problem can be solved by common sense
  - Determining the minimum number of servers given the arrival and service rate (for a stable system)
- The problem can be solved analytically
- It is easier to perform direct experiments
  - Impact of staging a person taking order using a hand-held terminal and voice communication→ impact of adding another order station on customer waiting time..
- Model building requires training/time but the results must be obtained soon
- When the simulation is expensive (cost more than the savings)
- If the system behavior is too complex or can't be defined

Components of a system

Entity: is an object of interest in the system

Example: Health Center

Patients,

Doctors & Nurses,

Rooms & beds

Lab equipment, X-Ray machine, etc.



Components of a system

Attribute: is a common characteristic of some entities, but with a specific value that can differ from one entity to another.

Example: Patient

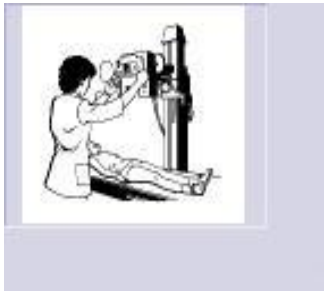
Type of illness,

Age,

Sex,

Temperature,

Blood Pressure



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Components of a system

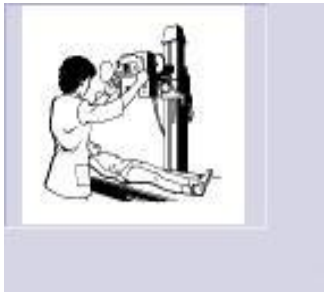
Activity: represents a time period of specified length.

Example: Health Center

Surgery,

Checking temperature,

X-Ray.



Components of a system

Variable: A piece of information that reflects some characteristic of the system

Example: Health Center

Number of patients in the system,

Number of idle doctors,

Current time



Components of a system

- ***State***: A collection of variables that contains all the information necessary to describe the system at any time

Example: Health Center

{Number of patients in the system,

Status of doctors (busy or idle),

Number of idle doctors,

Status of Lab equipment, etc}



Components of a system

- ***Event***: An instantaneous occurrence that changes the state of the system

Example: Health Centre

Arrival of a new patient,

Completion of service (i.e., examination)

Failure of medical equipment, etc.

Endogenous

Exogenous



Terminology (cont'd)

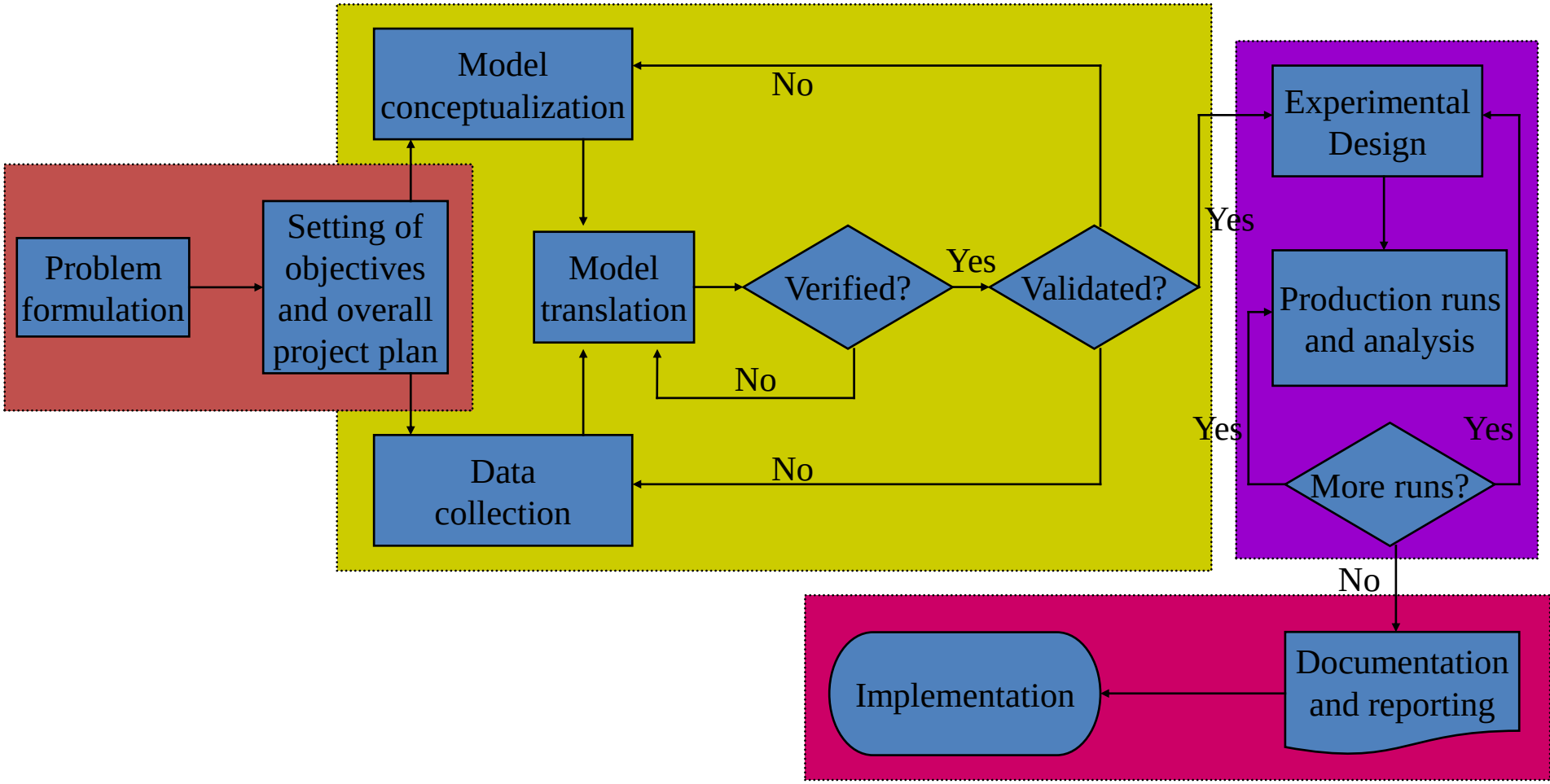
- **System**: group of objects that are joined together in some regular interaction or interdependence toward the accomplishment of some purpose
  - Production system: Machines, component parts, workers operating jointly along an assembly line to manufacture automobiles

- In modeling systems, it is necessary to define the boundary between the system and its environment
  - The boundary may depend on the purpose of the study
    - Bank system: there could be a limit on the maximum interest rate that can be paid
      - For the study of a single bank→ constraint imposed by the environment
      - For the study of the effects on monetary laws on the banking industry→ setting of this limit would be an activity of the system

More definitions…

- Discrete system: a system in which the state variables change only at a discrete set of points in time
  - Example: bank→ the state variable, number of customers, only change when an arrival or service completion happens.
- Continuous system: a system in which state variables change continuously over time
  - Example: the state variable, head of water behind a dam, changes continuously due to drawing water for flood control and to make electricity and evaporation.
- Model: representation of a system for the purpose of studying the system
  - Only those aspects that affect the problem under investigation must be considered
  - Must be detailed enough to permit valid conclusions to be drawn about the real system
- Static simulation model: represents a system at a particular point in time (sometimes called a Monte Carlo simulation)→ *Risk analysis, NPV analysis*
- Dynamic simulation model: represents systems as they change over time
  - Simulation of a bank from 9:00AM to 4:00PM.
- Deterministic simulation models: the models that contain no random variables
  - Deterministic arrivals to a dentist's office if all patients arrive at the scheduled time
- Stochastic simulation models: one or more random variables as inputs.

Steps in a simulation study



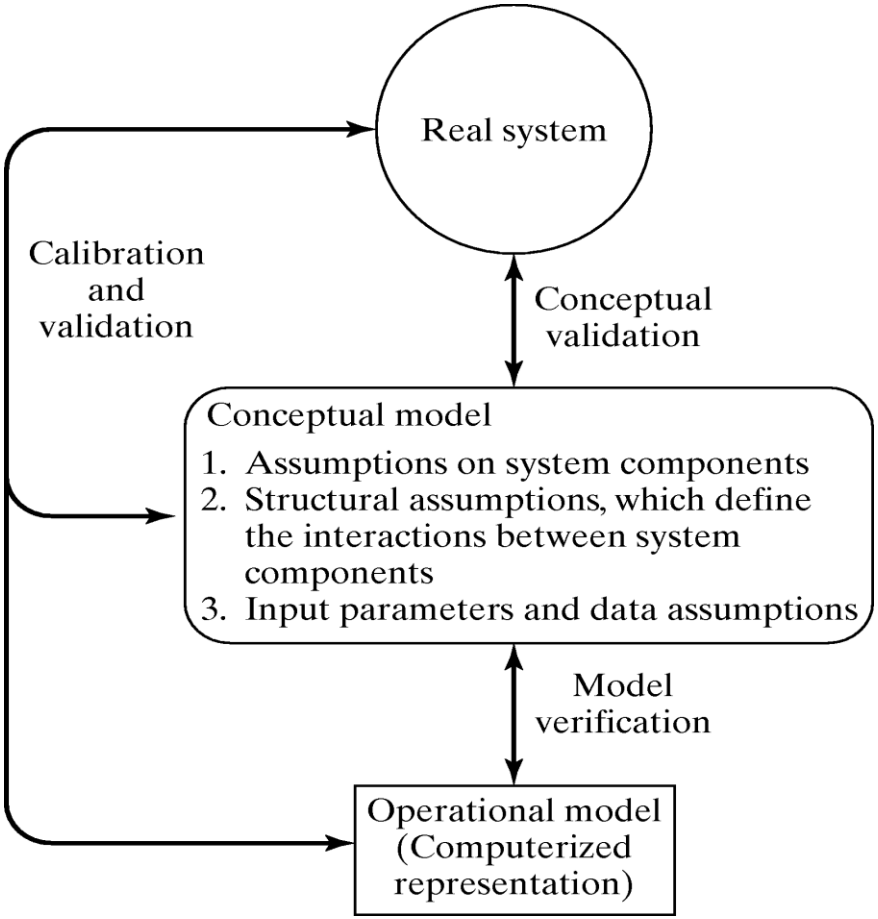
Steps in a simulation study

- **Problem formulation**
  - Statement of the problem
  - Communication between the policy makers and analyst
- **Setting the objectives and project plan**
  - Questions to be answered
  - Number of people involved, time required, results expected
- **Model conceptualization**
  - Better to start with a simple model
    - Select and modify basic assumptions that characterize the system
  - Later enrich and elaborate toward greater complexity
  - Operational rules, assumptions, events, facilities, decision variables, state variables…
  - Logical model: mathematical formulas, variables and their relationships, statistical measures

Steps in a simulation study

- **Data collection**
  - Might change as the complexity of the model changes
- **Model translation**
  - Programming or using simulation software
- **Verification**
  - Is the computer program functioning properly?
    - Input parameters and logical structures
  - Difficult with complex models to translate successfully in its entirety without a good deal of debugging
- **Validation**
  - Comparing the model against the actual system behavior
  - Using the discrepancies between the two to improve the model

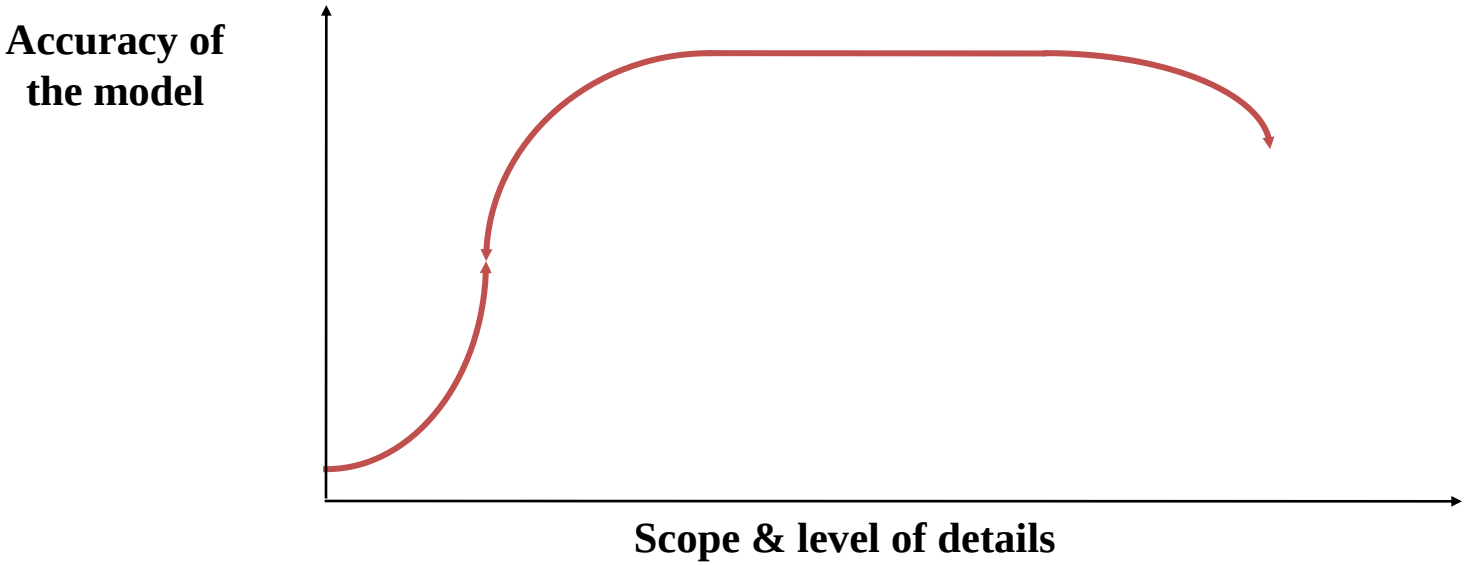
Model-Building, Verification & Validation

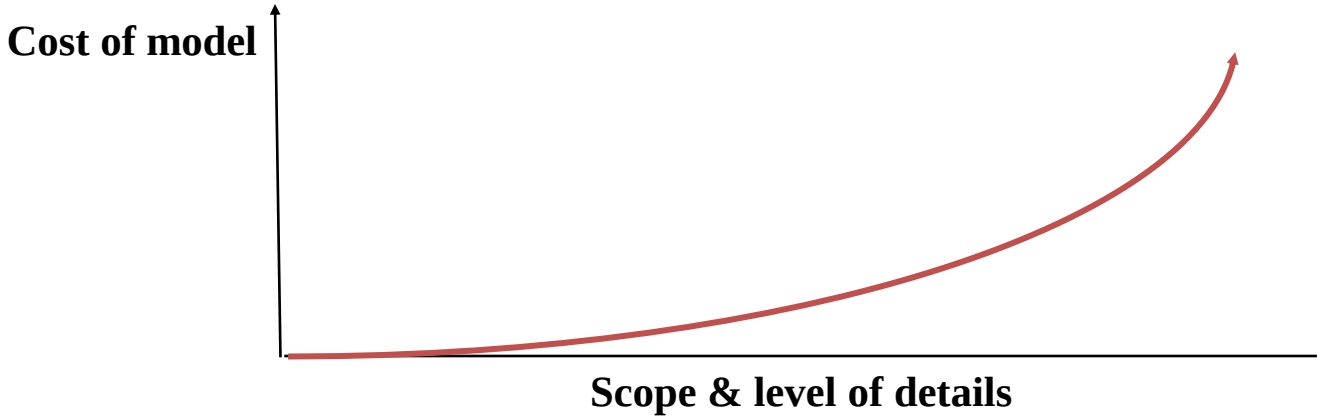


Steps in a simulation study

- **Experimental design**
  - Alternatives that are to be simulated
  - Length of simulation runs, number of replications to be made for each alternative
- **Production runs and analysis**
  - More runs?
  - Longer runs?
- **Documentation and reporting**
  - Program and progress
  - Written history of the simulation study
  - Makes it easier for a different analyst to change parameters, model, etc.
- **Implementation**

Cost vs. Accuracy





Concepts in discrete event simulation

- System is modeled in terms of:
  - State at each point in time
  - Entities that pass through the system
  - Entities that represent system resources
  - Events/activities causing the system state to change
- Model: an abstract representation of the system
  - Contains structural, logical, mathematical relationships describing the system
- Event notice: A record of an event to occur at the current or some future time
- Event list: A list of event notices for future events, ordered by time of occurrence
  - Also known as future event list (FEL)
- Activity: A duration of time of specified length which is known when it begins (although it may be defined in terms of a statistical distribution)
- Delay: A duration of time of unspecified indefinite length, which is not known until it ends (e.g., a customer's delay in a last-in-first-out waiting line in which, when it begins, depends on future arrivals)
- Clock: A variable representing simulated time

Discrete event simulation (cont'd)

- Entities in a list are always ordered by some rule (FIFO, LIFO, by priority or due date)
- Future event list (FEL) is always ranked by the event time recorded in the event notice
- Completion of an activity is called a primary event, and managed by placing an event notice in the FEL.
- Completion of a delay is called a secondary event. They do not appear in the FEL. Managed by placing the associated entity on another list.
- A given snapshot at a given time t includes:
- System state at time t
  - FEL
    - List of all activities currently in progress and when each such activity will end
  - Status of all entities
  - Current values of cumulative statistics
  - Counters to calculate the summary statistics at the end of simulation

Advancing simulation time

- Mechanism for advancing simulation time and guaranteeing all events occur in correct chronological order is based on FEL
  - FEL includes all event notices for events that have been scheduled to occur at a future time

- Scheduling an event:
  - At the instant an activity begins, its duration is computed or drawn from a statistical distribution
  - End-activity event together with its event time is placed on FEL

Advancing simulation time

- t=CLOCK (current value of simulated time)
- $t < t_1 \leq \ldots \leq t_n$
- Event associated with t_1 is called the imminent event
- After the system snapshot at time t has been updated→
  - CLOCK=t_1
  - and imminent event is removed from the FEL
  - Event is executed
  - New snapshot at t_1 is created based on the old snapshot at t and the nature of the imminent event.
  - And maybe new events are generated.
- Process repeats until the simulation is over.
- Sequence of actions that a simulator must perform to advance the clock and build a new system snapshot is called "*event-scheduling/time-advance algorithm*".

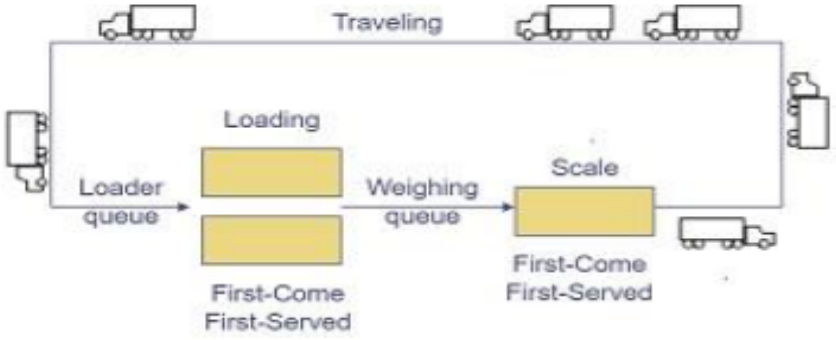
Generating a stopping event

- Schedule a stop simulation event
  - Either specify a time T_E
  - Or specify a stopping condition
    - Time of the 100th service completion
    - Shipment of the last truck in a distribution center
    - Simulation of a bank from 9:00AM to 5:00PM (T_E =8 hours)

SIMULATION EXAMPLES

Dump-Truck Problem

Six dump trucks are used to haul coal from the entrance of a small mine to the railroad. The figure below provides a schematic of the dump-truck operation.



Source: https://slideplayer.com/slide/9929809/

Each truck is loaded by one of the two loaders. After a loading, the truck immediately moves to the scale, to be weighed as soon as possible. Both the loaders and the scale have a first-come-first-served waiting line for trucks. Travel time from a loader to the scale is considered negligible. After being weighed, the truck begins a travel time (during which time the truck unloads) and then afterward returns to the loader queue. The distributions of loading time, weighing time, and travel time, together with the "randomly generated" activity times are given in the following table:

Distributions of loading, weighing, and travel times					
Loading Time	Probability	Weighing Time	Probability	Travel Time	Probability
5	0.3	12	0.7	40	0.4
10	0.5	16	0.3	60	0.3
15	0.2			80	0.2
				100	0.1

Randomly generated activity times in the near future
Loading Time: 10, 5, 5, 10, 15, 10, 10
Weighing Time: 12, 12, 12, 16, 12, 16
Travel time: 60, 100, 40, 40, 80

Assume that there are 3 trucks in the loader queue, two trucks being loaded, and one truck is at the scale (being weighed). Simulate the system for the next 24 minutes. Estimate the loader and scale utilizations (percentage of time busy).

Dump-Truck Problem

Dump-Truck Problem: solution

Random number generation

- Generation of random numbers (via linear or combined linear congruential methods) R_1, R_2,... that:
  - have uniformity (U(0,1))
  - are independent
- Subsequent testing for randomness -- *Kolmogorov-Smirnov or Chi-Square test for uniformity and Autocorrelation test for independence*
- Desired properties of routines to generate random numbers:
  - Should be fast
  - Should be portable (different programming languages)
  - Should have a sufficiently long cycle length
    - Cycle length: length of the random number sequence before previous numbers begin to repeat themselves..
    - If 1000 events are to be generated, the period (cycle length) should be many times that long.
  - Should be replicable: helpful for debugging purposes and is a means of facilitating comparisons between systems
  - Uniformity and independence

Random variate generation

- Random numbers (R) are used to generate random variates
- Inverse Transform Method
  - Uniform, Exponential, Weibull, Triangular, Empirical, standard normal, discrete random variates (discrete uniform, geometric)
- Acceptance-Rejection Method
  - Uniform (1/4,1), Poisson, nonhomogeneous Poisson, Gamma
  - Efficiency of this method depends heavily on being able to minimize the number of rejections
    - e.g., in general, it takes $K(1+\alpha)$ random numbers to generate K random variates with this method when the mean of the Poisson variate is α.
- Direct Transformation
  - Normal
- Convolution
  - Erlang, Binomial

Input Modeling

- Development of a useful model of input data:
  - Collect data from the real system of interest
  - Identify a probability distribution to represent input process
    - Histograms
  - Choose parameters that determine a specific instance of the distribution family
  - Evaluate the chosen distribution and parameters for goodness of fit
    - Graphically
    - Statistically (chi-square or Kolmogorov-Smirnov)
    - If you fail the tests→ empirical form may be used
  - ExpertFit, Stat:Fit, Arena's Input Processor, @Risk's BestFit are some simulation software input analyzers…

Data collection

- Collect data while pre-observing; devise forms for this purpose; watch for unusual circumstances and the way they are handled
- Try to analyze data as they are collected to find out if the data collected is adequate; find out if any data collected are useless to the simulation
- Check data for homogeneity in successive time periods (same time period on successive days)
- Be aware of the possibility of data censoring (e.g., the process might have started prior to the observation)
- Check to see if there is a relation between variables, detect dependent data
- Check for autocorrelation
- Keep in mind the difference between the input data and output data, and be sure to collect input data

Input modeling when there is no data available

- Engineering data
  - e.g., performance ratings of a product provided by the manufacturer
- Expert opinion
  - Optimistic, most likely, pessimistic values
  - Variance may be defined
- Physical limitations
  - Lower/upper limits on how long a process can take
- Nature of the process
  - Description of distributions can be used to justify the choice even when no data are available
- *Uniform, triangular, beta are often used in such cases!*

Output Analysis of a Single System

- Objective: Estimate system performance via simulation
- If q is the system performance, the precision of the estimator can be measured by: $\hat{\theta}$
  - The standard error of $\hat{\theta}$.
  - The width of a confidence interval (CI) for q.
- Purpose of statistical analysis:
  - To estimate the standard error or CI .
  - To figure out the number of observations required to achieve desired error/CI.
- Potential issues to overcome:
  - Autocorrelation
    - e.g. inventory cost for subsequent weeks lack statistical independence.
  - Initial conditions
    - e.g. inventory on hand and # of backorders at time 0 would most likely influence the performance of week 1.

Type of Simulations

- Terminating versus non-terminating simulations
- Terminating simulation:
  - Runs for some duration of time T_E, where E is a specified event that stops the simulation.
  - Starts at time *0* under well-specified initial conditions.
  - Ends at the stopping time T_E.
    - Bank example: Opens at 8:30 am (time *0*) with no customers present and *8* of the *11* teller working (initial conditions), and closes at 4:30 pm (Time T_E = *480* minutes).
  - The simulation analyst chooses to consider it a terminating system because the object of interest is one day's operation.

Type of Simulations

- Non-terminating simulation:
  - Runs continuously, or at least over a very long period of time.
    - Examples: assembly lines that shut down infrequently, telephone systems, hospital emergency rooms.
  - Initial conditions defined by the analyst.
  - Runs for some *analyst-specified* period of time T_E.
  - Study the steady-state (long-run) properties of the system, properties that are not influenced by the initial conditions of the model.
- Whether a simulation is considered to be terminating or non-terminating depends on both
  - The objectives of the simulation study
    - *Bank example: terminating if the interest is the interaction between teller and customer in one day. Nonterminating if interest is in the flow of money or operation of ATMs.*
  - The nature of the system.

Stochastic Nature of Output Data

- Model output consist of one or more random variables (r. v.) because the model is an input-output transformation and the input variables are r.v.'s.
- *M/G/1* queueing example:
  - Poisson arrival rate = *0.1* per minute;
    service time ~ *N(m = 9.5, s =1.75)*.
  - System performance: long-run mean queue length, $L_Q(t)$.
  - Suppose we run a single simulation for a total of 5,000 minutes
    - Divide the time interval [*0, 5000*) into *5* equal subintervals of *1000* minutes.
    - Average number of customers in queue from time *(j-1)1000* to *j(1000)* is Y_j .

Stochastic Nature of Output Data

- M/G/1 queueing example (cont.):
  - Batched average queue length for 3 independent replications:

Batching Interval (minutes)	Batch, j	Replication 1, Y_{1j}	2, Y_{2j}	3, Y_{3j}
[0, 1000)	1	3.61	2.91	7.67
[1000, 2000)	2	3.21	9.00	19.53
[2000, 3000)	3	2.18	16.15	20.36
[3000, 4000)	4	6.92	24.53	8.11
[4000, 5000)	5	2.82	25.19	12.62
[0, 5000)		3.75	15.56	13.66

  - Inherent variability in stochastic simulation both within a single replication and across different replications.
  - The average across 3 replications, $\overline{Y}_{1.}, \overline{Y}_{2.}, \overline{Y}_{3.}$, can be regarded as independent observations, but averages within a replication, Y_{11}, …, Y_{15}, are not.

C.I. with Specified Precision: Terminating Simulations

- The half-length *H* of a *100(1 – α)%* confidence interval for a mean *q*, based on the *t* distribution, is given by:

$$H = t_{\alpha/2,R-1}\frac{S}{\sqrt{R}}$$

- Suppose that an error criterion e is specified with probability *1 - α*, a sufficiently large sample size should satisfy:

$$P\left(\left|\overline{Y}_{\cdot\cdot} - \theta\right| < \varepsilon\right) \geq 1-\alpha$$

C.I. with Specified Precision: Terminating Simulations

- Assume that an initial sample of size R_0 (independent) replications have been observed.
- Obtain an initial estimate S_0^2 of the population variance s^2.
- Then, choose sample size R such that $R \geq R_0$:
  - Since $t_{\alpha/2,\, R-1} \geq z_{\alpha/2}$, an initial estimate of R:

$$R \geq \left(\frac{z_{\alpha/2} S_0}{\varepsilon} \right)^2, \quad z_{\alpha/2} \text{ is the standard normal distribution.}$$

  - R is the smallest integer satisfying $R \geq R_0$ and $R \geq \left(\frac{t_{\alpha/2,R-1} S_0}{\varepsilon} \right)^2$
- Collect $R - R_0$ additional observations.
- The *100(1-a)%* C.I. for *q*:

$$\bar{Y}_{..} \pm t_{\alpha/2,R-1} \frac{S}{\sqrt{R}}$$

Initialization Bias (Steady-State Simulations)

- Methods to reduce the point-estimator bias caused by using artificial and unrealistic initial conditions:
  - Intelligent initialization.
  - Divide simulation into an initialization phase and data-collection phase.

- Intelligent initialization
  - Initialize the simulation in a state that is more representative of long-run conditions.
  - If the system exists, collect data on it and use these data to specify more nearly typical initial conditions.
  - If the system can be simplified enough to make it mathematically solvable, e.g. queueing models, solve the simplified model to find long-run expected or most likely conditions, use that to initialize the simulation.

Initialization Bias (Steady-State Simulations)

- Divide each simulation into two phases:
  - An initialization phase, from time 0 to time T_0.
  - A data-collection phase, from T_0 to the stopping time T_0+T_E.
  - The choice of T_0 is important:
    - After T_0, system should be more nearly representative of steady-state behavior.
  - System has reached steady state: the probability distribution of the system state is close to the steady-state probability distribution (bias of response variable is negligible).

Initialization Bias (Steady-State Simulations)

- M/G/1 queueing example: A total of 10 independent replications were made.
  - Each replication beginning in the empty and idle state.
  - Simulation run length on each replication was T_0+T_E = 15,000 minutes.
  - Response variable: queue length, $L_Q(t,r)$ (at time t of the rth replication).
  - Batching intervals of 1,000 minutes, batch means
- Ensemble averages:
  - To identify trend in the data due to initialization bias
  - The average corresponding batch means *across* replications:

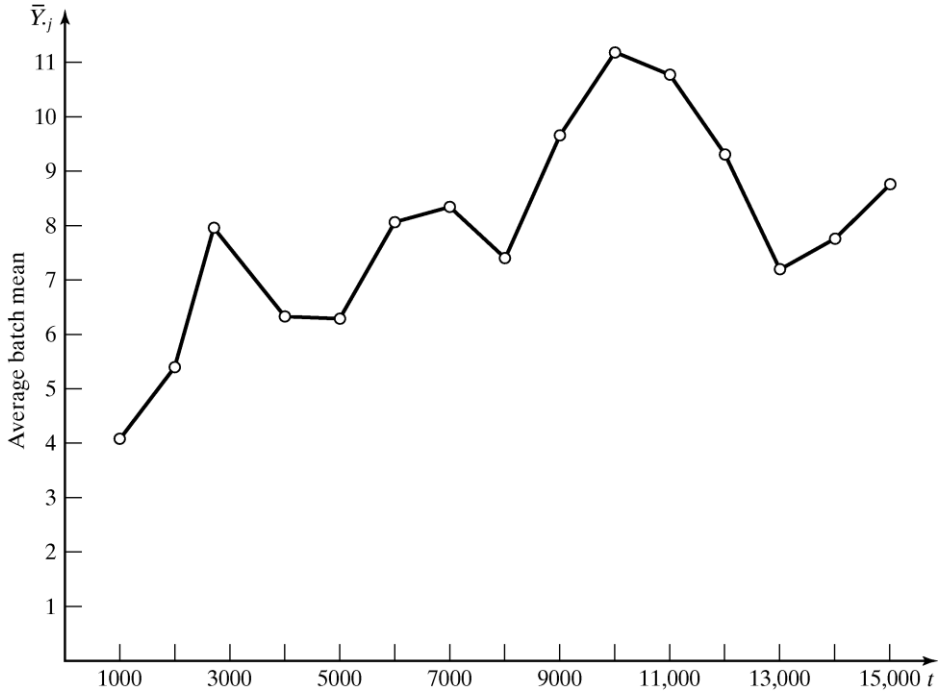
$$\overline{Y}_{.j} = \frac{1}{R}\sum_{r=1}^{R} Y_{rj}$$

R replications

  - The preferred method is to determine deletion point.

Initialization Bias (Steady-State Simulations)

- A plot of the ensemble averages, $\bar{Y}_{\cdot j}$, versus *1000j*, for *j = 1,2, …,15*.

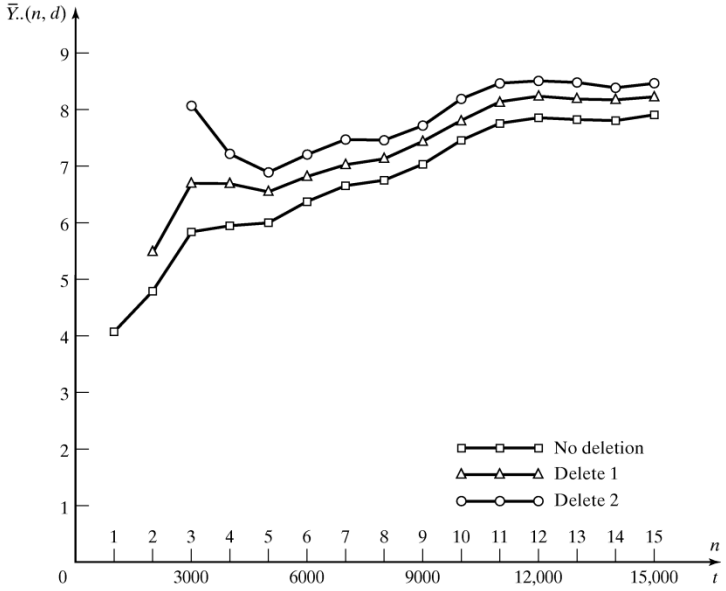


  - Illustrates the downward bias of the initial observations.

Initialization Bias (Steady-State Simulations)

- Cumulative average sample mean (after deleting *d* observations):

$$\bar{Y}_{\cdot\cdot}(n,d) = \frac{1}{n-d}\sum_{j=d+1}^{n}\bar{Y}_{\cdot j}$$



  - It is apparent that downward bias is present and this bias can be reduced by deletion of one or more observations.

Initialization Bias (Steady-State Simulations)

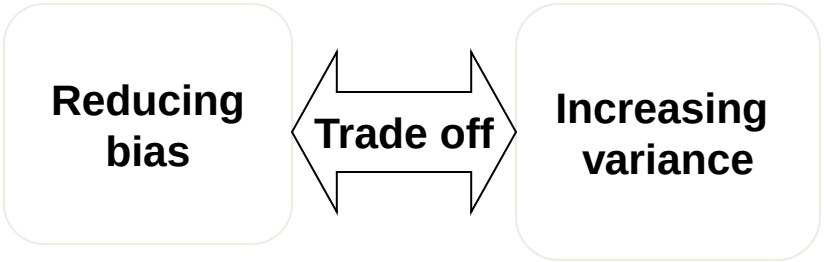
- No widely accepted, objective and proven technique to guide how much data to delete to reduce initialization bias to a negligible level.
- Plots can, at times, be misleading but they are still recommended.
  - Ensemble averages reveal a smoother and more precise trend as the # of replications, R, increases.
  - Ensemble averages can be smoothed further by plotting a moving average.
  - Cumulative average becomes less variable as more data are averaged.
  - The more correlation present, the longer it takes to approach steady state.
  - Different performance measures could approach steady state at different rates.
    - Initialization period must be determined separately for different variables.

Replication Method: Steady-State Simulations

- Rough rule: Length of each replication (n) beyond deletion point (d):

$$(n - d) > 10d$$

- Number of replications (R) should be as many as time permits
- For a fixed total sample size (n), as fewer data are deleted ($d\downarrow$):
  - C.I. shifts: greater bias.
  - Standard error of $\overline{Y}_{\cdot\cdot}(n,d)$ decreases: decrease variance.



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Comparison and Evaluation of Alternative System Designs

- Approach: discuss a few of many statistical methods that can be used to compare two or more system designs.
- Statistical analysis is needed to discover whether observed differences are due to:
  - Differences in design or,
  - The random fluctuation inherent in the models.

Comparison of Two System Designs

- Goal: compare two possible configurations of a system
  - e.g., two possible ordering policies in a supply-chain system, two possible scheduling rules in a job shop.
- Approach: the method of replications is used to analyze the output data.
- The mean performance measure for system *i* is denoted by q_i ($i = 1,2$).
- To obtain point and interval estimates for the difference in mean performance, namely $\theta_1 - \theta_2$.

Comparison of Two System Designs

- Vehicle-safety inspection example:
  - The station performs 3 jobs: (1) brake check, (2) headlight check, and (3) steering check.
  - Vehicles arrival: Possion with rate = 9.5/hour.
  - Present system:
    - Three stalls in parallel (one attendant makes all 3 inspections at each stall).
    - Service times for the 3 jobs: normally distributed with means 6.5, 6.0 and 5.5 minutes, respectively.
  - Alternative system:
    - Each attendant specializes in a single task, each vehicle will pass through three work stations in series
    - Mean service times for each job decreases by 10% (5.85, 5.4, and 4.95 minutes).
  - Performance measure: mean response time per vehicle (total time from vehicle arrival to its departure).

Comparison of Two System Designs

- From replication r of system i, the simulation analyst obtains an estimate Y_{ir} of the mean performance measure θ_i .
- Assuming that the estimators Y_{ir} are (at least approximately) unbiased:

  $\theta_1 = E(Y_{1r}),\ r = 1, \ldots, R_1; \qquad \theta_2 = E(Y_{2r}),\ r = 1, \ldots, R_2$
- Goal: compute a confidence interval for $\theta_1 - \theta_2$ to compare the two system designs
- Confidence interval for $\theta_1 - \theta_2$ (c.i.):
  - If c.i. is totally to the left of *0*, strong evidence for the hypothesis that $\theta_1 - \theta_2 < 0$ ($\theta_1 < \theta_2$).
  - If c.i. is totally to the right of *0*, strong evidence for the hypothesis that $\theta_1 - \theta_2 > 0$ ($\theta_1 > \theta_2$).
  - If c.i. is totally contains *0*, no strong statistical evidence that one system is better than the other
    - If enough additional data were collected (i.e., R_i increased), the c.i. would most likely shift, and definitely shrink in length, until conclusion of $\theta_1 < \theta_2$ or $\theta_1 > \theta_2$ would be drawn.

Comparison of Two System Designs

- A two-sided 100(1-α)% c.i. for $\theta_1 - \theta_2$ always takes the form of:

$$\bar{Y}_{.1} - \bar{Y}_{.2} \pm t_{\alpha/2,\upsilon} s.e.(\bar{Y}_{.1} - \bar{Y}_{.2})$$

where $\bar{Y}_{.i}$ is the sample mean performance measure for system i over all replications, and υ is the degress of freedom,

  - 3 techniques discussed assume that the basic data Y_{ir} are approximately normally distributed.

Comparison of Two System Designs

- Statistically significant versus practically significant
  - Statistical significance: is the observed difference $\bar{Y}_{.1} - \bar{Y}_{.2}$ larger than the variability in $\bar{Y}_{.1} - \bar{Y}_{.2}$?
  - Practical significance: is the true difference $\theta_1 - \theta_2$ large enough to matter for the decision we need to make?
    - Confidence intervals do not answer the question of practical significance directly, instead, they bound the true difference within a range.

Common Random Numbers (CRN): Comparison of 2 systems

- It is never enough to simply use the same seed for the random-number generator(s):
  - The random numbers must be synchronized: each random number used in one model for some purpose should be used for the same purpose in the other model.
    - e.g., if the ith random number is used to generate a service time at work station 2 for the 5th arrival in model 1, the ith random number should be used for the very same purpose in model 2.
  - Dedicate random number streams to a specific purpose (same random number streams for arrivals in both systems)

Comparison of Several System Designs

- To compare K alternative system designs
  - Based on some specific performance measure, θ_i, of system i , for $i = 1, 2, \ldots, K$.
- Procedures are classified as:
  - Fixed-sample-size procedures: predetermined sample size is used to draw inferences via hypothesis tests of confidence intervals.
    - e.g., Interval estimation of a mean performance measure
    - Known or easily estimated cost in terms of computer time before running the experiment (appropriate when the time is limited)
    - Confidence intervals can be too large, strong conclusions may not be possible.
  - Sequential sampling (multistage): more and more data are collected until an estimator with a prespecified precision is achieved or until one of several alternative hypotheses is selected.
- Some goals/approaches of system comparison:
  - Estimation of each parameter θ_i.
  - Comparison of each performance measure θ_i, to control θ_1 *(existing system performance)*.
  - All pairwise comparisons, θ_i - $\theta_{i'}$, for all i not equal to j
  - Selection of the best θ_i.

Optimization via Simulation

- Optimization usually deals with problems with certainty, but in stochastic discrete-event simulation, the result of any simulation run is a random variable.
- Let $x_1,x_2,\ldots,x_m$ be the m controllable design variables & $Y(x_1,x_2,\ldots,x_m)$ be the observed simulation output performance on one run:
  - To optimize $Y(x_1,x_2,\ldots,x_m)$ with respect to $x_1,x_2,\ldots,x_m$ is to maximize or minimize the mathematical expectation (long-run average) of performance, $E[Y(x_1,x_2,\ldots,x_m)]$.

- Example: select the best value of the order-up-to level in an inventory system that minimizes the cost.

Additional Slides for Reference

Testing for randomness

- Appropriate tests have already been conducted for generators used in simulation softwares
  - Kolmogorov-Smirnov or Chi-Square test for uniformity
  - Autocorrelation test for independence
- Tests should be applied to many samples of numbers from the generator
- Other commonly used tests:
  - Good's serial test
  - Median-Spectrum test
  - Runs test
  - Variance heterogeneity test
  - Spectral test

Tests for uniformity

- Kolmogorov-Smirnov is the more powerful of the two tests and recommended.
  - Kolmogorov-Smirnov can be applied to small sample size
  - For the Chi-Square to be valid, N≥50.

Identify distribution with data

- Histograms
  - Number of intervals depend on the number of observations and on the amount of scatter/dispersion in the data
  - Recommended to have the number of classes equal to the square root of sample size

- Selecting the family of distributions
  - Purpose of histogram is to infer a known pdf or pmf
  - Exponential, Normal, and Poisson distributions are frequently encountered
  - Beta, Gamma, Weibull provide a wide array of shapes and should not be overlooked
  - Keep the physical process in mind to help narrow down the family of distributions from which to choose:
    - Binomial: k successes out of n trials
    - Exponential: interarrival times
    - Poisson: number of occurences in a given time period
    - Is the process naturally discrete or continuous?
    - Is it bounded or is there no natural bound?

- Quantile-Quantile Plots
  - Histogram is useful but depends on the bin-width and hard to compare to a pdf
  - Q-Q plots doesn't suffer from this
  - They are useful tools for evaluating distribution fit
  - A plot of y_j (y_1,.., y_n are the ordered data) versus $F^{-1}((j-1/2)/n)$ will be approximately a straight line if F is a member of an appropriate family of distributions.
  - The line will have a slope of 1 if it is an appropriate family of distributions and has appropriate parameter values
  - Modern data analysis software often has tools for generating Q-Q plots

Goodness of fit tests

- Keep in mind that there is no single correct distribution!
- If very little data are available, then goodness-of-fit test is unlikely to reject any candidate distribution
- If a lot of data are available, then it is likely that the test will reject all distributions
  - Chi-Square test formalizes on the intuitive idea of comparing the shape of the histogram to shape of pdf or pmf of the candidate distribution
    - If a continuous distribution assumption is being tested, class intervals that are equal in probability rather than equal in width of interval should be used
    - Valid for large sample size
  - Kolmogorov-Smirnov test formalizes the idea behind examining Q-Q plot (chi-square is affected by the number of classes and interval width)
    - Particularly useful when:
      - Sample sizes are small
      - No parameters have been estimated from the data
      - Modified version can be used when parameter estimates are available
  - Best Fit:
    - Candidate distributions are ranked according to their p-values
    - p-value: significance level at which one would declare the distribution not valid for the given value of the test statistic (lower p-value indicates a lower probability of falsely rejecting the candidate distribution)

Verification & Validation

- Verification & Validation is required to reduce the skepticism by the engineers/managers who use the model outputs…
- The goal of the validation process is:
  - To produce a model that represents true behavior closely enough for decision-making purposes
  - To increase the model's credibility to an acceptable level
- Verification & Validation are an integral part of model development
  - Verification – building the model correctly (correctly implemented with the software)
    - Are the input parameters and logical structure of the model represented correctly?
  - Validation – building the correct model (an accurate representation of the real system)

Verification - Debugging

- Purpose: ensure the conceptual model is reflected accurately in the computerized representation.

- Many common-sense suggestions, for example:
  - Have someone else check the model.
  - Make a flow diagram that includes each logically possible action a system can take when an event occurs.
  - Closely examine the model output for reasonableness under a variety of input parameter settings. (Often overlooked!)
    - Have the implemented model display a wide variety of output statistics
      - e.g., Not just the total cost, but the holding, ordering, and shortage costs as well..

Verification (cont'd)

- 2 statistics as examples of output variables to print over time:
  - *current contents*: # of items in each component of the system at a given time
  - *total count*: total # of items that have entered each component of the system by a given time
    - If current contents in some portion of the system are high and this number tends to grow→ a large number of entities are delayed and sign of an unstable queue
    - If total count for a component is zero → no items have entered that subsystem!! A highly suspect occurrence!
  - Careful evaluation of these statistics over various run lengths aid in detection of errors in model logic and data specifications!

- Print the input parameters at the end of the simulation, make sure they have not been changed inadvertently.

Verification (cont'd)

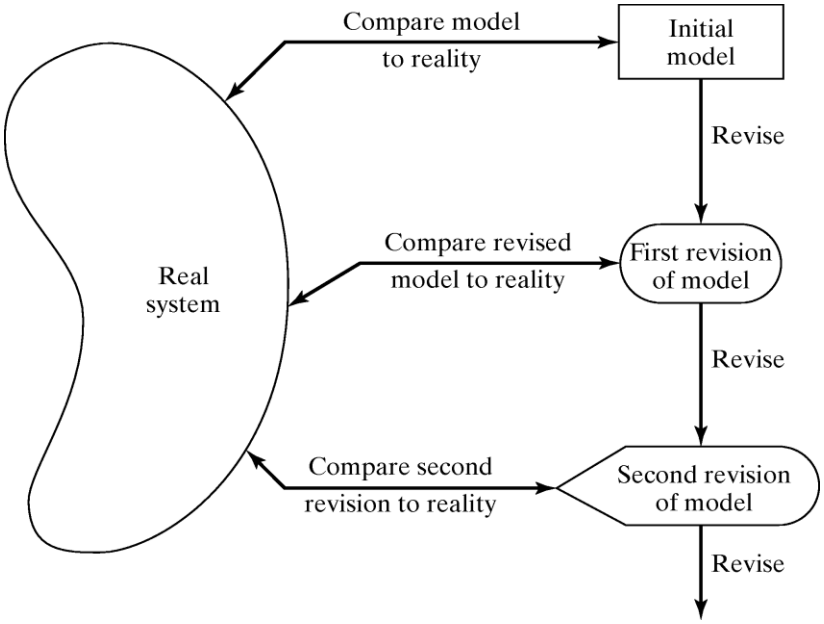
- Compare simulation results to analytical results if some measures can be computed analytically
  - Increases the credibility of the model
  - e.g., average queue length for an M/G/1 model
- Documentation
  - A means of clarifying the logic of a model and verifying its completeness
- Animation
  - Errors can be observed while animating
    - e.g., entities that follow wrong paths

Verification (cont'd)

- Use of a trace
  - A detailed printout of the state of the simulation model over time.
    - *the end output of a single-server queue from time 0 to 16 yields L(q)=0.4875 (looks okay)*
    - *But trace shows that at some time t, number of customers=1 and status of the server was idle!! → logical error in coding!*
    - *Examining the output only doesn't detect the error here. But trace does..*
  - A simulation with trace is restricted to a very short time period cause the purpose is to verify through paper-and-pencil calculations.
  - Selective trace is also possible wit some software:
    - *Trace whenever the queue length is larger than 5…*

Calibration and Validation

- Validation: the overall process of comparing the model and its behavior to the real system.
- Calibration: the iterative process of comparing the model to the real system and making adjustments.



Calibration and Validation

- No model is ever a perfect representation of the system
  - Tradeoff between the possible increase in model accuracy versus the cost of increased validation effort.

- Three-step approach:
  - Build a model that has high face validity.
  - Validate model assumptions.
  - Compare the model input-output transformations with the real system's data.

High Face Validity

- The model should appear reasonable to model users and others who are knowledgeable about the system.
  - Especially important when it is impossible to collect data from the system
- Ensure a high degree of realism:
  - Potential users should be involved in model construction (from its conceptualization to its implementation).
- Sensitivity analysis can also be used to check a model's face validity.
  - Example: In most queueing systems, if the arrival rate of customers were to increase, it would be expected that server utilization, queue length and delays would tend to increase.

Validate Model Assumptions

- General classes of model assumptions:
  - Structural assumptions: how the system operates.
  - Data assumptions: reliability of data and its statistical analysis.

- Bank example: customer queueing and service facility in a bank.
  - Structural assumptions, e.g., customer waiting in one line versus many lines, served FCFS versus priority.
  - Input data assumptions, e.g., interarrival time of customers, service times for commercial accounts.
    - Verify structural/data reliability with bank managers.
    - Test correlation and goodness of fit for data

Validate Input-Output Transformation

- Goal: Validate the model's ability to predict future behavior
  - The only objective test of the model.
  - The structure of the model should be accurate enough to make good predictions for the range of input data sets of interest.

- One possible approach: use historical data that have been reserved for validation purposes only
  - Assumption is that some version of the system exists.
  - Complete I-O validation is impossible for systems in the planning stages
  - Partial I-O validation is possible if a similar version of a subsystem exists.

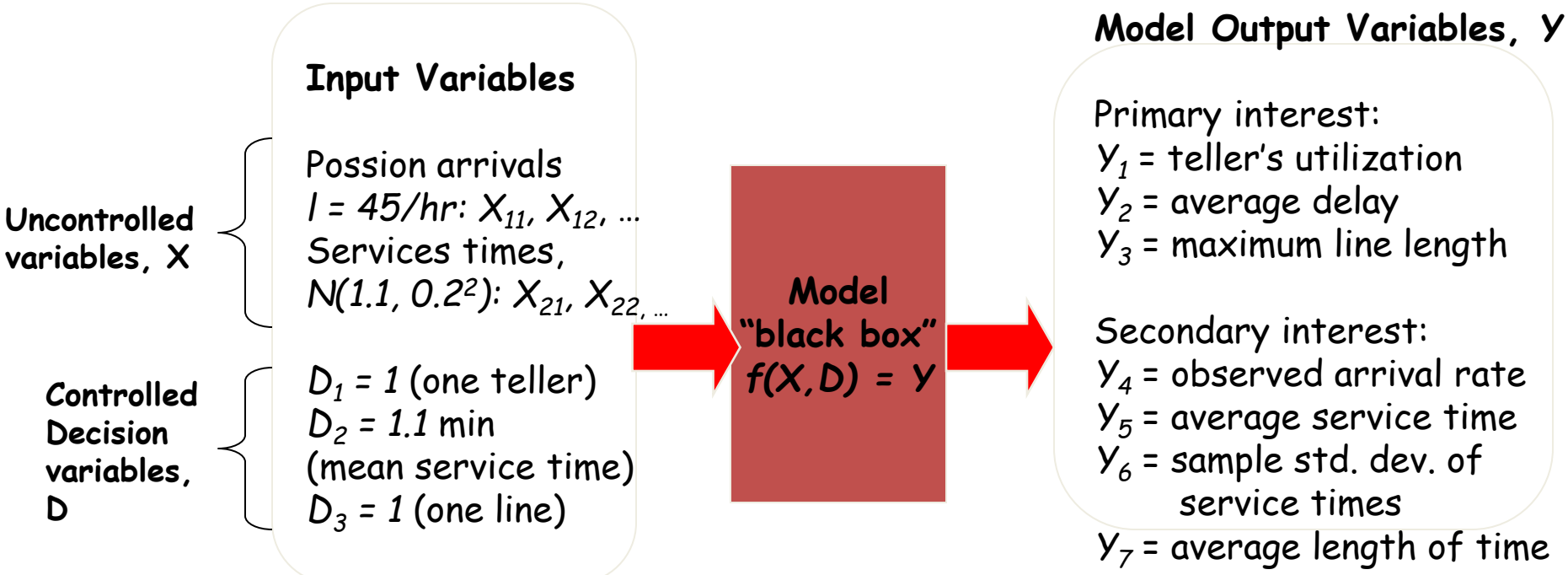
- Primary criteria: use the main system responses of interest.
  - Waiting time in queues, throughput in production systems

Bank Example: Validate I-O Transformation

- One drive-in window serviced by one teller, only one or two transactions are allowed.
  - Data collection: 90 customers during 11 am to 1 pm.
    - Observed service times $\{S_i, i = 1,2, \ldots, 90\}$.
    - Observed interarrival times $\{A_i, i = 1,2, \ldots, 90\}$.
  - Data analysis led to the conclusion that:
    - Interarrival times: exponentially distributed with rate $l = 45$
    - Service times: $N(1.1, 0.2^2)$

The Black Box Bank Example: Validate I-O Transformation

- A model was developed in close consultation with bank management and employees
- Model assumptions were validated



Comparison with Real System Data
Bank Example: Validate I-O Transformation

- Real system data are necessary for validation.
  - Average delays should have been collected during the same time period (from *11*am to *1*pm on the same Friday.)

- Compare the average delay from the model *Y* with the actual delay *Z*:
  - Average delay observed, $Z = 4.3$ minutes, consider this to be the true mean value $m_0 = 4.3$.
  - When the model is run with generated random variates X_{1n} and X_{2n}, *Y* should be close to *Z*.
  - Six statistically independent replications of the model, each of *2*-hour duration, are run.

Hypothesis Testing Bank Example: Validate I-O Transformation

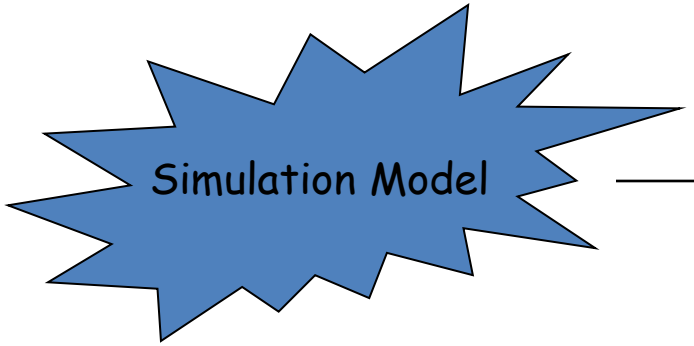
- Compare the average delay from the model *Y* with the actual delay *Z* (continued):
  - Null hypothesis testing: evaluate whether the simulation and the real system are *the same* (w.r.t. output measures):

$$H_0 : E(Y) = 4.3 \text{ minutes}$$

$$H_1 : E(Y) \neq 4.3 \text{ minutes}$$

    - If H_0 is not rejected, then, there is no reason to consider the model invalid
    - If H_0 is rejected, the current version of the model is rejected, and the modeler needs to improve the model

Hypothesis Testing Bank Example: Validate I-O Transformation



Average Delay Times

$Y_1, Y_2, \ldots, Y_6$ iid random variables

Replication	Average Delay
1	2.79
2	1.12
3	2.24

Replication	Average Delay
4	3.45
5	3.13
6	2.38

Hypothesis Testing Bank Example: Validate I-O Transformation

- Conduct the *t* test:
  - Choose level of significance ($a = 0.5$) and sample size ($n = 6$).
  - Compute the sample mean and sample standard deviation over the n replications:

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i = 2.51 \text{ minutes} \qquad S = \left[\frac{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}{n-1}\right]^{1/2} = 0.81 \text{ minutes}$$

  - Compute test statistics:

$$|t_0| = \left|\frac{\bar{Y} - \mu_0}{S/\sqrt{n}}\right| = \left|\frac{2.51 - 4.3}{0.82/\sqrt{6}}\right| = 5.24 > t_{\alpha/2,n-1} = 2.571$$

  - Hence, reject H_0. Conclude that the model is inadequate.
  - Check: the assumptions justifying a *t* test, that the observations (Y_i) are normally and independently distributed.

Bank Example: Validate I-O Transformation

- Once the model is claimed to be invalid, further investigation is done

- Model contained 2 unstated assumptions
  - When a car arrived to find the window immediately available, the teller began service immediately
  - There is no delay between one service ending and the next beginning, when a car is waiting

    - *It is found out assumption 1 is incorrect due to walk-in customers*
    - *Data about service/arrival of walk-in customers collected*
    - *New (revised) model is validated as a result of the new test!*

Using Historical Output Data: Validate I-O Transformation

- An alternative to generating input data:
  - Use the actual historical record.
  - Drive the simulation model with the historical record and then compare model output to system data.
  - In the bank example, use the recorded interarrival and service times for the customers $\{A_n, S_n, n = 1,2,\ldots\}$.

- Define Z_i to be the actual average delay for the ith data set, Y_i to be the simulated average delay for the ith data set, and .
  - Conduct the t-test for

$$d_i = Z_i - Y_i$$

$$H_0 : E(d) = 0$$

$$H_1 : E(d) \neq 0$$

Turing Test

- When no statistical tests are readily applicable
- Suppose 5 reports of system performance for 5 different days are prepared
- Simulation output data are used to produce 5 "fake" reports
- 10 reports shuffled randomly and given to the manager
  - If the manager identifies a substantial number of the fake reports, model builder questions the manager to improve the model
  - If the manager cannot distinguish between fake and real reports with any consistency, the modeler concludes there is no evidence of model inadequacy.

Replication Method: Steady-State Simulations

- Use to estimate point-estimator variability and to construct a confidence interval.
- Approach: make R replications, initializing and deleting from each one the same way.
- Important to do a thorough job of investigating the initial-condition bias:
  - Bias is not affected by the number of replications, instead, it is affected only by deleting more data (i.e., increasing T_0) or extending the length of each run (i.e. increasing T_E).
- Basic raw output data $\{Y_{rj}, r = 1, ..., R; j = 1, ..., n\}$ is derived by:
  - Individual observation from within replication r.
  - Batch mean from within replication r of some number of discrete-time observations.
  - Batch mean of a continuous-time process over time interval j.

Replication Method: Steady-State Simulations

- Each replication is regarded as a single sample for estimating θ. For replication r: $\bar{Y}_{r.}(n,d) = \frac{1}{n-d}\sum_{j=d+1}^{n} Y_{rj}$
- The overall point estimator:

$$\bar{Y}_{..}(n,d) = \frac{1}{R}\sum_{r=1}^{R}\bar{Y}_{r.}(n,d) \quad \text{and} \quad \mathrm{E}[\bar{Y}_{..}(n,d)] = \theta_{n,d}$$

- If d and n are chosen sufficiently large:
  - $\theta_{n,d} \sim \theta$.
  - $\bar{Y}_{..}(n,d)$ is an approximately unbiased estimator of θ.
- To estimate standard error of $\bar{Y}_{..}$, the sample variance and standard error:

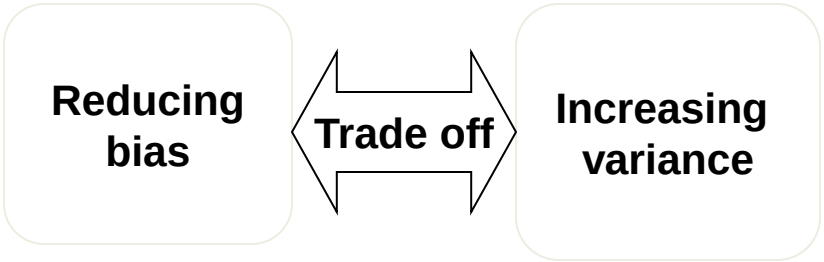
$$S^2 = \frac{1}{R-1}\sum_{r=1}^{R}(\bar{Y}_{r.} - \bar{Y}_{..})^2 = \frac{1}{R-1}\left(\sum_{r=1}^{R}\bar{Y}_{r.}^{\,2} - R\bar{Y}_{..}^{\,2}\right) \quad \text{and} \quad s.e.(\bar{Y}_{..}) = \frac{S}{\sqrt{R}}$$

Replication Method: Steady-State Simulations

- Rough rule: Length of each replication (n) beyond deletion point (d):

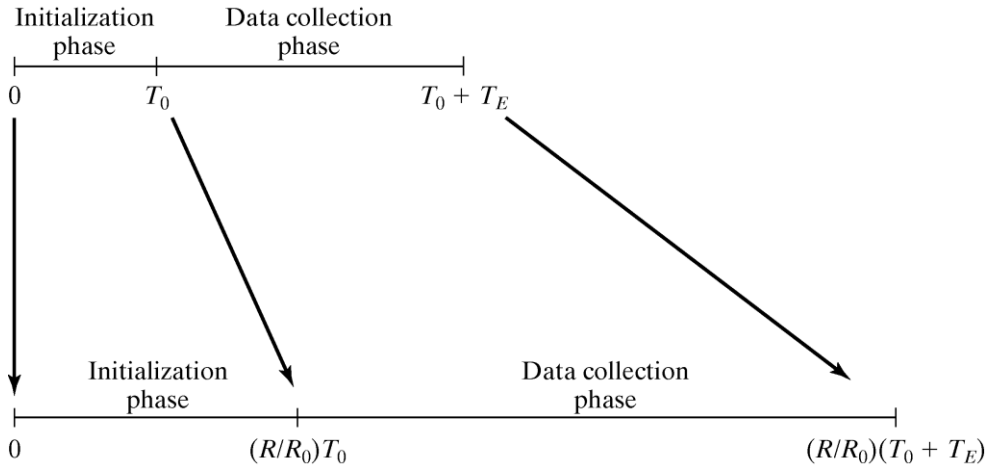
$$(n - d) > 10d$$

- Number of replications (R) should be as many as time permits
- For a fixed total sample size (n), as fewer data are deleted ($d\downarrow$):
  - C.I. shifts: greater bias.
  - Standard error of $\bar{Y}_{\cdot\cdot}(n, d)$ decreases: decrease variance.



Sample Size: Steady-State Simulations

- An alternative to increasing R is to increase total run length T_0+T_E within each replication.
  - Approach:
    - Increase run length from (T_0+T_E) to $(R/R_0)(T_0+T_E)$, and
    - Delete additional amount of data, from time 0 to time $(R/R_0)T_0$.
  - Advantage: any residual bias in the point estimator should be further reduced.



Batch Means for Interval Estimation: Steady-State Simulations

- Using a single, long replication:
  - Problem: data are dependent so the usual estimator is biased.
  - Solution: batch means.
- Batch means: divide the output data from *1* replication (after appropriate deletion) into a few large batches and then treat the means of these batches as if they were independent.

- *A* continuous-time process, $\{Y(t), T_0 \le t \le T_0+T_E\}$:
  - *k* batches of size $m = T_E/k$, batch means:

$$\bar{Y}_j = \frac{1}{m}\int_{(j-1)m}^{jm} Y(t+T_0)dt$$

- *A* discrete-time process, $\{Y_i, i = d+1,d+2, \ldots, n\}$:
  - *k* batches of size $m = (n - d)/k$, batch means:

$$\bar{Y}_j = \frac{1}{m}\sum_{i=(j-1)m+1}^{jm} Y_{i+d}$$

Batch Means for Interval Estimation: Steady-State Simulations

$$\underbrace{Y_1, \ldots, Y_d}_{\text{deleted}}, \underbrace{Y_{d+1}, \ldots, Y_{d+m}}_{\bar{Y}_1}, \underbrace{Y_{d+m+1}, \ldots, Y_{d+2m}}_{\bar{Y}_2}, \ \cdots \ , \underbrace{Y_{d+(k-1)m+1}, \ldots, Y_{d+km}}_{\bar{Y}_k}$$

- Starting either with continuous-time or discrete-time data, the variance of the sample mean is estimated by:

$$\frac{S^2}{k} = \frac{1}{k}\sum_{j=1}^{k}\frac{\left(\bar{Y}_j - \bar{Y}\right)^2}{k-1} = \sum_{j=1}^{k}\frac{\bar{Y}_j^2 - k\bar{Y}^2}{k(k-1)}$$

- If the batch size is sufficiently large, successive batch means will be approximately independent, and the variance estimator will be approximately unbiased.
- No widely accepted and relatively simple method for choosing an acceptable batch size *m*. Some simulation software does it automatically.
  - No need for more than 30 batches.
  - Less than 10 batches produce poor results.

Independent Sampling with Equal Variances: Comparison of 2 systems

- Different and independent random number streams are used to simulate the two systems
  - All observations of simulated system *1* are statistically independent of all the observations of simulated system *2*.
- The variance of the sample mean, $\overline{Y}_{.i}$, is:

$$V\left(\overline{Y}_{.i}\right) = \frac{V\left(Y_{.i}\right)}{R_i} = \frac{\sigma_i^2}{R_i}, \qquad i = 1,2$$

- For independent samples:

$$V\left(\overline{Y}_{.1} - \overline{Y}_{.2}\right) = V\left(\overline{Y}_{.1}\right) + V\left(\overline{Y}_{.2}\right) = \frac{\sigma_1^2}{R_1} + \frac{\sigma_2^2}{R_2}$$

Independent Sampling with Equal Variances: Comparison of 2 systems

- If it is reasonable to assume that $\sigma^2_1 = \sigma^2_2$ (approximately) or if $R_1 = R_2$, a two-sample-t confidence-interval approach can be used:
  - The point estimate of the mean performance difference is:

$$\bar{Y}_{.1} - \bar{Y}_{.2}$$

  - The sample variance for system i is:

$$S_i^2 = \frac{1}{R_i - 1}\sum_{r=1}^{R_i}\left(\bar{Y}_{ri} - \bar{Y}_{.i}\right)^2 = \frac{1}{R_i - 1}\sum_{r=1}^{R_i}\bar{Y}_{ri}^{\,2} - R_i\bar{Y}_{.i}^{\,2}$$

  - The pooled estimate of σ^2 is:

$$S_p^2 = \frac{(R_1 - 1)S_1^2 + (R_2 - 1)S_2^2}{R_1 + R_2 - 2}, \qquad \text{where } \upsilon = R_1 + R_2 \text{ -2 degrees of freedom}$$

  - C.I. is given by:

$$\bar{Y}_{.1} - \bar{Y}_{.2} \pm t_{\alpha/2,\upsilon}\, s.e.(\bar{Y}_{.1} - \bar{Y}_{.2})$$

  - Standard error:

$$s.e.\left(\bar{Y}_{.1} - \bar{Y}_{.2}\right) = S_p\sqrt{\frac{1}{R_1} + \frac{1}{R_2}}$$

Independent Sampling with Unequal Variances: Comparison of 2 systems

- If the assumption of equal variances cannot safely be made, an approximate *100(1-α)%* c.i. for can be computed as:

$$\bar{Y}_{.1} - \bar{Y}_{.2} \pm t_{\alpha/2,\upsilon} s.e.(\bar{Y}_{.1} - \bar{Y}_{.2})$$

- where

$$s.e.\left(\bar{Y}_{.1} - \bar{Y}_{.2}\right) = \sqrt{\frac{S_1^2}{R_1} + \frac{S_2^2}{R_2}}$$

$$\upsilon = \frac{\left(S_1^2/R_1 + S_2^2/R_2\right)^2}{\left[\left(S_1^2/R_1\right)^2/\left(R_1 - 1\right)\right] + \left[\left(S_2^2/R_2\right)^2/\left(R_2 - 1\right)\right]}, \quad \text{round to an integer}$$

  - Minimum number of replications $R_1 > 5$ and $R_2 > 5$ is recommended.

Common Random Numbers (CRN): Comparison of 2 systems

- For each replication, the same random numbers are used to simulate both systems.
  - For each replication *r*, the two estimates, Y_{r1} and Y_{r2}, are correlated.
  - However, independent streams of random numbers are used on different replications, so the pairs (Y_{r1} ,Y_{s2}) are mutually independent.
- Purpose: induce positive correlation between $\bar{Y}_{.1}, \bar{Y}_{.2}$ (for each *r*) to reduce variance in the point estimator of $\bar{Y}_{.1} - \bar{Y}_{.2}$.

$$V\left(\bar{Y}_{.1} - \bar{Y}_{.2}\right) = V\left(\bar{Y}_{.1}\right) + V\left(\bar{Y}_{.2}\right) - 2\,\mathrm{cov}\left(\bar{Y}_{.1}, \bar{Y}_{.2}\right)$$

$$= \frac{\sigma_1^2}{R} + \frac{\sigma_2^2}{R} - \frac{2\rho_{12}\sigma_1\sigma_2}{R}$$

ρ_{12} is positive

  - Variance of $\bar{Y}_{.1} - \bar{Y}_{.2}$ arising from CRN is less than that of independent sampling (with $R_1 = R_2$).

Common Random Numbers (CRN): Comparison of 2 systems

- The estimator based on CRN is more precise, leading to a shorter confidence interval for the difference.

$$\bar{Y}_{.1} - \bar{Y}_{.2} \pm t_{\alpha/2,\upsilon} s.e.(\bar{Y}_{.1} - \bar{Y}_{.2})$$

- Sample variance of the differences: $\bar{D} = \bar{Y}_{.1} - \bar{Y}_{.2}$

$$S_D^2 = \frac{1}{R-1}\sum_{r=1}^{R}\left(\bar{D}_r - \bar{D}\right)^2 = \frac{1}{R-1}\left(\sum_{r=1}^{R} D_r^2 - R\bar{D}^2\right)$$

where $D_r = Y_{r1} - Y_{r2}$ and $\bar{D} = \frac{1}{R}\sum_{r=1}^{R} D_r$, with degress of freedom $\upsilon = \mathrm{R} - 1$

- Standard error:

$$s.e.(\bar{D}) = s.e.\left(\bar{Y}_{.1} - \bar{Y}_{.2}\right) = \frac{S_D}{\sqrt{R}}$$